MTH401 Quiz-3

Final TERM QUIZ 3

Lec 31 to 40

ORANGE MONKEY

1. **The solution of Bessel equation is denoted by _____.**

$J_v(x)$……confirm

is called Bessel's differential equation.

Solution of this equation is usually denoted by $J_v(x)$ and is known as Bessel's function.

2. **X=0 is a(an) _____ point of the differential equation y'' + xy' + 2y = 0**

Ordinary…….confirm

If $x = x_0$ is an ordinary point of the differential equation $y'' + P(x)y' + Q(x)y = 0$,

3. **Matrix form of the following system of non-homogenius differential**

$$\frac{dx}{dt} = -2x + 4y + e^t - 4t$$
$$\frac{dy}{dt} = 3x - 4y + 11t$$

equation is

$$X' = \begin{bmatrix} -2 & 4 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} -4 \\ 11 \end{bmatrix} t$$

……confirm

4. **If determinant** $\begin{vmatrix} D+1 & 2 \\ 0 & D-1 \end{vmatrix} = 0$ **then _____.**

D^2 - 1….confirm

5. **The equation** $\det(A - \lambda I) = 0$ **is known as the characteristics equation**

True……confirm

$$\det(A - \lambda I) = 0$$

This equation is called the characteristic equation of the matrix A.

6. $\frac{dA}{dt} = [\frac{da_{ij}}{dt}]_{m \times n}$ **Is the _____ of a matrix of a function**

columns

7. **Matrix of the form of the following system of non-homogenous**

$$\frac{dy}{dx} = -2x + 4y + e^t - 4t$$
$$\frac{dy}{dx} = 3x - 4y + 11t$$

differential equations is

$$X' = \begin{bmatrix} -2 & 4 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} -4 \\ 11 \end{bmatrix} t$$

…..confirm

8. **The matrix form of non-homogenous system of differential equationals**
   X’ = -3x + 4y – 9z
   X’= 6x-y
   X’=10x – 4y + 3z

$$X' = \begin{bmatrix} -3 & 4 & -9 \\ 6 & -1 & 0 \\ 10 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

…..confirm

9. **Bessel functions of index half an _____ integer are called spherical Bessel functions.**

Odd……confirm

Bessel functions of index half an odd integer are called Spherical Bessel functions.

10. **The regular singular point of the differential equation $(x^2 – 4)^2y + (x – 2)y’ + y = 0$ is _____.**

2……confirm

$x = 2$ is a regular singular point because power of $x-2$ in $P(x)$ is 1 and in $Q(x)$ is 2.

11. The irregular singular point of the differential equation $(x^2 - 4)^2y + (x - 2)y' + y = 0$ is _____.

-2……confirm

2. $x = -2$ is an irregular singular point because power of $x + 2$ in $P(x)$ is 2.

The 1st condition is violated.

12. If n is positive integer then the solution of legendre's differential equation is called a legendre's _____ of degree n

Legendre's polynomial…….confirm

function. If n is positive integer then the solution of Legendre's differential equation is called a Legendere's polynomial of degree n and is denoted by $P_n(x)$.

13. Bessel functions of index half an odd integer are called _____ functions.

Spherical Bessel……confirm

Bessel functions of index half an odd integer are called Spherical Bessel functions.

14. Spherical Bessel functions are used in many physical problems

True……confirm

spherical Bessel functions are used in many physical problems.

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - 49) = 0$$

15. The second order differential equation is a Bessel equation of degree

$$A = \begin{bmatrix} 1 & 5 & 3 \\ -2 & 4 & 7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2 & 1 \\ 3 & 5 & 1 \end{bmatrix}$$

16. Let matrix then A-B is _____.

$$\begin{bmatrix} 1 & 3 & 2 \\ -1 & -1 & 6 \end{bmatrix}$$

……confirm

17. A solution of a system of differential equation is a set of differentiable functions, x = f(t), y = g(t), x = h(t) those satisfy _ equation(s) of the system on same interval I.

Each……confirm

A solution of a system of differential equations is a set of differentiable functions

$$x = f(t), \quad y = g(t), \quad x = h(t), \ldots$$

those satisfy each equation of the system on some interval I.

18. The equation $(x^2-4)^2 d^2y/dx^2 + (x-2)dy/dx + y = 0$ has the singular point(s) at x = _____.

±i…….confirm

(b) The singular points need not be real numbers.

The equation $(x^2+1)y'' + 2xy' + 6y = 0$ has the singular points at the solutions of $x^2 + 1 = 0$, namely, $x = \pm i$.

19. Solution of Legendre's differential equation is called ___

Legendre's equation…confirm

is called Legendre's differential equation and any of its solution is called Legendre's function. If n is positive integer then the solution of Legendre's differential equation is called a Legendre's polynomial of degree n and is denoted by $P_n(x)$

20.To solve a differential equation $a_2(x)y + a_1(x)y' + a_0(x)y = 0$ about a regular singular point we employ the ____ theorem

Frobenius…..confirm

To solve a differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ about a regular singular point we employ the Frobenius' Theorem.

21.The differential equation $(x^2-16)\ y'' + 2xy' + 16y = 0$ has singularity at

X = ± 4

22.Bassel equation occurs frequently in advances studies in _____.

All options…….confirm

This equation occurs frequently in advanced studies in applied mathematics, physics and engineering.

23.Legendary polynomials are specific _____ degree polynomials.

n…..confirm

Legendre's Polynomials are specific n^{th} degree polynomials

24.Legendre polynomials are normal with respect to the weight function



……confirm

which shows that Legender polynomials are normal with respect to the weight function $w(x) = \frac{2n+1}{2}$ over the interval $-1 < x < 1$.

$$B = \begin{bmatrix} b_{11} \\ b_{21} \\ \cdot \\ \cdot \\ \cdot \\ b_{m1} \end{bmatrix}_{m\times 1} = [b_{i1}]_{m\times 1}$$

25.A matrix of the from $[b_{i1}]_{m\times 1}$ is called

Column matrix…..confirm

26.The method of frobenius always provide two solutions.

True….confirm

Remark: The method of Frobenius may not always provide 2 solutions.

27.Recurrence relation for polynomials relates

Polynomials of different degrees…….confirm

Recurrence relations that relate Legendre's polynomials of different degrees are also very important in some aspects of their application. We shall derive one such relation using the formula

28. The non-trival solution of the system of differential equation exist only when

$\det(A-\lambda I)=0$ ✓ **….confirm**

1. The non-trivial solution of the system of differential equation exist only when

- $\det(X-\lambda I)=0$
- $\det(A-\lambda I)\neq 0$
- $\det(A+\lambda I)=0$
- $\det(A-\lambda I)=0$ ✓

29. The Eigenvalues of a matrix are given by the roots of the characteristics equation.

True……confirm

This equation is called the characteristic equation of the matrix A. Thus the Eigenvalues of the matrix A are given by the roots of the characteristic equation. To find an

30. Vectors $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -66 \\ 99 \end{pmatrix}$ **are linearly _____**

Independent

Construct a matrix whose rows or columns are the given vectors and calculate its determinant. If that determinant is zero then the vectors are linearly dependent.

31. Vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ are linearly _____.

Dependent

Construct a matrix whose rows or columns are the given vectors and calculate its determinant. If that determinant is zero then the vectors are linearly dependent.

32. **Operator method or systematic elimination equation would be a linear homogenous or a linear non-homogenous differential equation and can be solved by employing one of the methods discussed earlier or obtain one of the _____ variables.**

Dependent…….confirm

This equation would be a linear homogeneous or a linear non-homogeneous differential equation and can be solved by employing one of the methods discussed earlier to obtain one of the dependent variables.

33. **Operator method or systematic elimination provides us a single differential equation in one of the dependent variables that has not been eliminated.**

True…..confirm

This elimination provides us a single differential equation in one of the dependent variables that has not been eliminated.

34. **Operator method or systematic elimination method of solution of a system of linear homogenous or linear non-homogenous differential equations is based on the process of systematic elimination of the _____ variations.**

Dependent variables……confirm

This method of solution of a system of linear homogeneous or linear non-homogeneous differential equations is based on the process of systematic elimination of the dependent variables.

35. The differential equation $5\frac{dx}{dt}+5x-\frac{dy}{dx}=e^{2t}$ in terms of differential operator is _____

$$(D+1)5x - Dy = e^{2t}$$

36. The eigenvalues of matrix $A=\begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$ are_____.

$$\lambda_1 = 5, \lambda_2 = 5$$

_____.....confirm

or $$(3-\lambda)(7-\lambda)+4=0 \Rightarrow (\lambda-5)^2=0$$

Therefore, the characteristic equation has repeated real roots. Thus the matrix has an eigenvalue of multiplicity two.

$$\lambda_1 = \lambda_2 = 5$$

37. A differential equation that can be written in the form dy/dx + p(x)y = q(x)y'' is called _____ equation.

Bernouli equation……confirm

A differential equation that can be written in the form

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

is called Bernoulli equation.

38. A second order linear differential equation of the form $(1-x^2)y'' - 2xy' + n(n+1)y=0$ is called _____ differential equation and any of its solution is called _____ function.

Legendre's , Legendre's…….confirm

A second order linear differential equation of the form

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

is called Legendre's differential equation and any of its solution is called Legendre's

39. A second order differential equation of the form

$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - v^2)\,y = 0$ **is called _____ differential equation**

Bessel…..confirm

A second order linear differential equation of the form

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - v^2)y = 0$$

is called Bessel's differential equation.

40. An immediate consequence of the principle of superposition is that a constant multiple of any situation vector of a homogenous system of first order differential equation is also a solution of the system.

True……confirm

An immediate consequence of the principle of superposition is that a constant multiple of any solution vector of a homogenous system of first order differential equation is also a solution of the system.

41. If A= $A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 & 1 \\ 3 & 5 & 6 \\ 8 & 7 & 5 \end{bmatrix}$ **then the order of matrix A*B is**

3*3…..confirm

42.Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} u & v \\ x & y \end{bmatrix}$ then A+B is

$$A+B = \begin{bmatrix} a+u & b+v \\ c+x & d+y \end{bmatrix}$$ ……confirm

43.If K = $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector of the given matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then AK is

1k

44.The Eigenvalues of a matrix are given by the roots of the characteristics equation.

True……confirm

This equation is called the characteristic equation of the matrix A. Thus the Eigenvalues of the matrix A are given by the roots of the characteristic equation. To find an

45.The Eigenvalues of the matrix are not given by the roots of the characteristics equation.

False…….confirm

This equation is called the characteristic equation of the matrix A. Thus the Eigenvalues of the matrix A are given by the roots of the characteristic equation. To find an

46. The linear normal form of $4\frac{d^3x}{dt^3}+y=e^t$ by using y = x_1, y'=x_1, y''=x_2 and y'''=x_3 is _____

$x_1' = x_2, x_2' = x_3, x_3' = \frac{-1}{4}x_1 + \frac{1}{4}e^t$ **.....confirm**

$$x_1' = x_2$$
$$x_2' = x_3$$
$$x_3' = -\frac{1}{4}x_1 + \frac{1}{4}e^t$$

47.The linear normal form of $2\frac{d^2y}{dx^2}+4\frac{dy}{dx}-5y=0$ by using y=x_1, y'=x_1' and y''=x_2' is _____ .

$x_1' = x_2, x_2' = -2x_2 + \frac{5}{2}x_1$ **.....confirm**

$$x_1' = x_2$$
$$x_2' = -2x_2 + \frac{5}{2}x_1$$

48. The simultaneous ordinary differential equations involve two or more equations that certain derivatives of two or more unknown functions of _____ independent variable.

1……confirm

The simultaneous ordinary differential equations involve two or more equations that contain derivatives of two or more unknown functions of a single independent variable.

49. By converting the equation $(x^2 + 4x + 4)y'' + (x - 2)y' + y = 0$ into general form $y'' + P(x)y' + Q(x)y = 0$ Q(x) is

$$\frac{1}{(x-2)(x+2)}$$

50. It is not always possible to solve the given system for the highest-order derivative of each dependent variable.

False……confirm

to the canonical form. To accomplish this we need to solve the system for the highest order derivative of each dependent variable.

51. It is always possible to solve the given system for the highest-order derivative of each dependent variable.

True…..confirm

to the canonical form. To accomplish this we need to solve the system for the highest order derivative of each dependent variable.

52.The multiplicative identity of 3*3 matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

…..confirm

Multiplicative Identity

For a given positive integer n, the $n \times n$ matrix

$$I = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

53.The vertical lines in a matrix are called the _____ of the matrix

columns……confirm

The horizontal and vertical lines in a matrix are, respectively, called the rows and columns of the matrix.

54.The horizontal lines in a matrix are called the _____ of the matrix

Rows……confirm

The horizontal and vertical lines in a matrix are, respectively, called the rows and columns of the matrix.

55.A singular point $x=x_0$ of the differential equation $a_2(x)y''+a_1(x)y'+a_0(x)y=0$ is said to be a _____ point if both $(x-x_0)P(x)$ and $(x-x_0)^2Q(x)$ are analytical at x_0.

Regular singular point…..confirm

A Singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to be a ***regular singular point*** if both $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at x_0. A

56. A singular point that is not regular is said to be an _____ of the equation.

Irregular singular point……confirm

a ***regular singular point*** if both $(x-x_0)P(x)$ and $(x-x_0)^2Q(x)$ are analytic at x_0. A singular point that is not regular is said to be an ***irregular singular point*** of the equation.



Lecture 22

Applications of Second Order Differential Equation

- A single differential equation can serve as mathematical model for many different phenomena in science and engineering.
- Different forms of the 2nd order linear differential equation

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=f(x)$$

appear in the analysis of problems in physics, chemistry and biology.

- In the present and next lecture we shall focus on one application**;** the motion of a mass attached to a spring.
- We shall see, what the individual terms $a\frac{d^2y}{dx^2}$, $b\frac{dy}{dx}$, cy and $f(x)$ means in the context of vibrational system.
- Except for the terminology and physical interpretation of the terms

$$a\frac{d^2y}{dx^2},\ f(x)$$

b cy, dx

the mathematics of a series circuit is identical to that of a vibrating spring-mass system. Therefore we will discuss an *LRC* circuit in lecture.

Simple Harmonic Motion

When the Newton's 2nd law is combined with the Hook's Law, we can derive a differential equation governing the motion of a mass attached to spring–the simple harmonic motion.

Hook's Law

Suppose that

- A mass is attached to a flexible spring suspended from a rigid support, then
- The spring stretches by an amount '*s*'.
- The spring exerts a restoring *F* opposite to the direction of elongation or stretch.

The Hook's law states that the force F is proportional to the elongation *s*. i.e

$$F=ks$$

Where k is constant of proportionality, and is called spring constant.

Note That

- Different masses stretch a spring by different amount i.e s is different for different m.
- The spring is characterized by the spring constant k.
- For example if $W=10$ lbs and $s=\frac{1}{2}ft$

Then $F=ks$

or $10=\left(\frac{1}{2}\right)k$

ORANGE MONKEY TEAM

or $k = 20$ lbs/ft

If $W = 8$ lbs then $8 = 20(s) \Rightarrow s = 2/5$ ft

Newton's Second Law

When a force F acts upon a body, the acceleration a is produced in the direction of the force whose magnitude is proportional to the magnitude of force. i.e

$$F = ma$$

Where m is constant of proportionality and it represents mass of the body.

Weight

- The gravitational force exerted by the earth on a body of mass m is called weight of the body, denoted by W
- In the absence of air resistance, the only force acting on a freely falling body is its weight. Thus from Newton's 2nd law of motion

  $$W = mg$$

  Where m is measured in slugs, kilograms or grams and $g = 32\text{ft/s}^2$, $9.8m/s^2$ or 980 cm/s^2.

Differential Equation

- When a body of mass m is attached to a spring
- The spring stretches by an amount s and attains an equilibrium position.
- At the equilibrium position, the weight is balanced by the restoring force ks. Thus, the condition of equilibrium is

  $$mg = ks \Rightarrow mg - ks = 0$$

- If the mass is displaced by an amount x from its equilibrium position and then released. The restoring force becomes $k(s + x)$. So that the resultant of weight and the restoring force acting on the body is given by

  $$Resultant = -k(s + x) + mg.$$

  By Newton's 2nd Law of motion, we can written

  $$m\frac{d^2x}{dt^2} = -k(s + x) + mg$$

  or $$m\frac{d^2x}{dt^2} = -kx - ks + mg$$

  Since $mg - ks = 0$

  Therefore $$m\frac{d^2x}{dt^2} = -kx$$

- The negative indicates that the restoring force of the spring acts opposite to the direction of motion.

ORANGE MONKEY TEAM

- The displacements measured below the equilibrium position are positive.
- By dividing with m, the last equation can be written as:

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

or

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

Where $\omega^2 = \frac{k}{m}$. This equation is known as the equation of simple harmonic motion or as the free un-damped motion.

Initial Conditions

Associated with the differential equation

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

are the obvious initial conditions

$$x(0) = \alpha, \quad x'(0) = \beta$$

These initial conditions represent the initial displacement and the initial velocity. For example

- If $\alpha > 0, \ \beta < 0$ then the body starts from a point below the equilibrium position with an imparted upward velocity.
- If $\alpha < 0, \ \beta = 0$ then the body starts from rest $|\alpha|$ units above the equilibrium position.

Solution and Equation of Motion

Consider the equation of simple harmonic motion

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

Put

$$x = e^{mx}, \ \frac{d^2x}{dt^2} = m^2 e^{mx}$$

Then the auxiliary equation is

$$m^2 + \omega^2 = 0 \quad \Rightarrow \ m = \pm i\omega$$

Thus the auxiliary equation has complex roots.

$$m_1 = \omega i, \ m_2 = -\omega i$$

Hence, the general solution of the equation of simple harmonic motion is

$$x(t) = c_1 \cos\omega t + c_2 \sin\omega t$$

ORANGE MONKEY TEAM

Alternative form of Solution

It is often convenient to write the above solution in a alternative simpler form. Consider

$$x(t) = c_1 \cos\omega t + c_2 \sin\omega t$$

and suppose that $A,\ \phi \in R$ such that

$$c_1 = A\sin\phi,\ c_2 = A\cos\phi$$

Then $$A = \sqrt{{c_1}^2 + {c_2}^2}\ ,\ \tan\phi = \frac{c_1}{c_2}$$

So that

$$x(t) = A\sin\omega t\ \cos\phi + B\cos\ \omega t\ \sin\phi$$

or $$x(t) = A\sin(\omega t + \phi)$$

The number ϕ is called the phase angle;

Note that

This form of the solution of the equation of simple harmonic motion is very useful because

- Amplitude of free vibrations becomes very obvious
- The times when the body crosses equilibrium position are given by

  $$x = 0 \Rightarrow \sin(\omega t + \phi) = 0$$

  or $$\omega t + \phi = n\pi$$

  Where n is a non-negative integer.

The Nature of Simple Harmonic Motion

Amplitude

- We know that the solution of the equation of simple harmonic motion can be written as

  $$x(t) = A\sin(\omega t + \phi)$$

- Clearly, the maximum distance that the suspended body can travel on either side of the equilibrium position is A.
- This maximum distance called the amplitude of motion and is given by

$$Amplitude = A = \sqrt{{c_1}^2 + {c_2}^2}$$

A Vibration or a Cycle

In travelling from $x = A$ to $x = -A$ and then back to A, the vibrating body completes one vibration or one cycle.

ORANGE MONKEY TEAM

Period of Vibration

The simple harmonic motion of the suspended body is periodic and it repeats its position after a specific time period T. We know that the distance of the mass at any time t is given by

$$x = A\sin(\omega t + \phi)$$

Since

$$A\sin\left[\omega\left(t + \frac{2\pi}{\omega}\right) + \phi\right]$$

$$= A\sin\left[\left(\omega t + \phi + 2\pi\right)\right]$$

$$= A\sin\left[\left(\omega t + \phi\right)\right]$$

Therefore, the distances of the suspended body from the equilibrium position at the times t and $t + \frac{2\pi}{\omega}$ are same

Further, velocity of the body at any time t is given by

$$\frac{dx}{dt} = A\omega\cos(\omega t + \phi)$$

$$A\omega\cos\left[\omega\left(t + \frac{2\pi}{\omega}\right) + \phi\right]$$

$$= A\omega\cos\left[\omega t + \phi + 2\pi\right]$$

$$= A\omega\cos(\omega t + \phi)$$

Therefore the velocity of the body remains unaltered if t is increased by $2\pi/\omega$. Hence the time period of free vibrations described by the 2nd order differential equation

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

is given by

$$T = \frac{2\pi}{\omega}$$

Frequency

The number of vibration /cycle completed in a unit of time is known as frequency of the free vibrations, denoted by f. Since the cycles completed in time T is 1. Therefore, the number of cycles completed in a unit of time is $1/T$

Hence

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

Example 1

Solve and interpret the initial value problem

$$\frac{d^2x}{dt^2} + 16x = 0$$

$$x(0) = 10,\ \ x'(0) = 0\,.$$

Interpretation

Comparing the initial conditions

$$x(0) = 10,\ \ x'(0) = 0\,.$$

With

$$x(0) = \alpha,\ \ x'(0) = \beta$$

We see that

$$\alpha = 10\ ,\ \beta = 0$$

Thus the problem is equivalent to

- Pulling the mass on a spring 10 units below the equilibrium position.
- Holding it there until time $t = 0$ and then releasing the mass from rest.

Solution

Consider the differential equation

$$\frac{d^2x}{dt^2} + 16x = 0$$

Put $$x = e^{mt}\ ,\ \ \frac{d^2x}{dt^2} = m^2e^{mt}$$

Then, the auxiliary equation is

$$m^2 + 16 = 0$$

$$\Rightarrow m = 0 \pm 4i$$

Therefore, the general solution is:

$$x(t) = c_1 \cos 4t + c_2 \sin 4t$$

Now we apply the initial conditions.

$$x(0) = 10 \Rightarrow c_1.1 + c_2.0 = 10$$

Thus $$c_1 = 10$$

So that $$x(t) = 10\cos 4t + c_2 \sin 4t$$

$$\frac{dx}{dt} = -40 \sin 4t + 4c_{\ 2} \cos 4t$$

ORANGE MONKEY TEAM

Therefore $x'(0) = 0 \quad \Rightarrow -40(0) + 4c_2.1 = 0$

Thus $c_2 = 0$

Hence, the solution of the initial value problem is

$$x(t) = 10\cos 4t$$

Note that

- Clearly, the solution shows that once the system is set into motion, it stays in motion with mass bouncing back and forth with amplitude being10 *units*.
- Since $\omega = 4$. Therefore, the period of oscillation is

$$T = \frac{2\pi}{4} = \frac{\pi}{2} \text{ seconds}$$

Example 2

A mass weighing 2lbs stretches a spring 6 inches. At *t = 0* the mass is released from a point 8 inches below the equilibrium position with an upward velocity of $\frac{4}{3}$ *ft / s* .

Determine the function *x (t)* that describes the subsequent free motion.

Solution

For consistency of units with the engineering system, we make the following conversions

$$6 \text{ inches} = \frac{1}{2} \text{ foot}$$

$$8 \text{ inches} = \frac{2}{3} \text{ foot}.$$

Further weight of the body is given to be

$$W = 2 \text{ lbs}$$

But $W = mg$

Therefore $m = \frac{W}{g} = \frac{2}{32}$

or $m = \frac{1}{16} \text{ slugs.}$

Since $Stretch = s = \frac{1}{2} \text{ foot}$

Therefore by Hook's Law, we can write

$$2 = k\left(\frac{1}{2}\right) \Rightarrow k = 4 \text{ lbs/ft}$$

ORANGE MONKEY TEAM

Hence the equation of simple harmonic motion

$$m\frac{d^2x}{dt^2} = -kx$$

becomes

$$\frac{1}{16}\frac{d^2x}{dt^2} = -4x$$

or $$\frac{d^2x}{dt^2} + 64x = 0.$$

Since the initial displacement is 8 *inches* $= \frac{2}{3}$ f*t* and the initial velocity is $\frac{-4}{3}$ ft/s, the initial conditions are:

$$x(0) = \frac{2}{3}, \quad x'(0) = -\frac{4}{3}$$

The negative sign indicates that the initial velocity is given in the upward i.e negative direction. Thus, we need to solve the initial value problem.

Solve $$\frac{d^2x}{dt^2} + 64x = 0$$

Subject to $$x(0) = \frac{2}{3}, \quad x'(0) = -\frac{4}{3}$$

Putting $$x = e^{mt}, \ \frac{d^2x}{dt^2} = m^2e^{mt}$$

We obtain the auxiliary equation

$$m^2 + 64 = 0$$

or $$m = \pm 8i$$

The general solution of the equation is

$$x(t) = c_1 \cos 8t + c_2 \sin 8t$$

Now, we apply the initial conditions.

$$x(0) = \frac{2}{3} \quad \Rightarrow c_1.1 + c_2.0 = \frac{2}{3}$$

Thus $$c_1 = \frac{2}{3}$$

So that $$x(t) = \frac{2}{3}\cos 8t + c_2 \sin 8t$$

ORANGE MONKEY TEAM

Since

$$x'(t) = -\frac{16}{3}\sin 8t + 8c_2 \cos 8t\,.$$

Therefore

$$x'(0) = -\frac{4}{3} \Rightarrow -\frac{16}{3}.0 + 8c_2.1 = -\frac{4}{3}$$

Thus

$$c_2 = -\frac{1}{6}.$$

Hence, solution of the initial value problem is

$$x(t) = \frac{2}{3}\cos 8t - \frac{1}{6}\sin 8t.$$

Example 3

Write the solution of the initial value problem discussed in the previous example in the form

$$x(t) = A\sin(\omega t + \phi\ )\,.$$

Solution

The initial value discussed in the previous example is:

Solve $$\frac{d^2x}{dt^2} + 64x = 0$$

Subject to $$x(0) = \frac{2}{3}, \quad x'(0) = -\frac{4}{3}$$

Solution of the problem is

$$x(t) = \frac{2}{3}\cos 8t - \frac{1}{6}\sin 8t$$

Thus amplitude of motion is given by

$$A = \sqrt{\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{6}\right)^2} = \frac{\sqrt{17}}{6} \approx 0.69 \text{ ft}$$

and the phase angle is defined by

$$\sin\phi = \frac{2/3}{\sqrt{17}/6} = \frac{4}{\sqrt{17}} > 0$$

$$\cos\phi = \frac{-1/6}{\sqrt{17}/6} = -\frac{1}{\sqrt{17}} < 0$$

ORANGE MONKEY TEAM

Therefore

$$\tan\phi = -4$$

or

$$\tan^{-1}(-4) = -1.326 \text{ radians}$$

Since $\sin\phi > 0,\ \cos\phi < 0,$ the phase angle ϕ must be in 2nd quadrant.

Thus

$$\phi = \pi - 1.326 = 1.816 \text{ radians}$$

Hence the required form of the solution is

$$x(t) = \frac{\sqrt{17}}{6}\sin(8t + 1.816)$$

Example 4

For the motion described by the initial value problem

Solve

$$\frac{d^2x}{dt^2} + 64x = 0$$

Subject to

$$x(0) = \frac{2}{3},\quad x'(0) = -\frac{4}{3}$$

Find the first value of time for which the mass passes through the equilibrium position heading downward.

Solution

We know that the solution of initial value problem is

$$x(t) = \frac{2}{3}\cos 8t - \frac{1}{6}\sin 8t.$$

This solution can be written in the form

$$x(t) = \frac{\sqrt{17}}{6}\sin(8t + 1.816)$$

The values of t for which the mass passes through the equilibrium position i.e for which $x = 0$ are given by

$$wt + \phi = n\pi$$

Where $n = 1,2,...$, therefore, we have

$$8t_1 + 1.816 = \pi,\quad 8t_2 + 1.816 = 2\pi,\quad 8t_3 + 1.816 = 3\pi, ...$$

or

$$t_1 = 0.166,\quad t_2 = 0.558,\quad t_3 = 0.951, ...$$

Hence, the mass passes through the equilibrium position

$$x = 0$$

heading downward first time at $t_2 = 0.558$ seconds.

ORANGE MONKEY TEAM

Exercise

State in words a possible physical interpretation of the given initial-value problems.

1. $\frac{4}{32}x' + 3x = 0, \quad x(0) = -3, \quad x'(0) = -2$
2. $\frac{1}{16}x' + 4x = 0, \quad x(0) = 0.7, \quad x'(0) = 0$

Write the solution of the given initial-value problem in the form $x(t) = A\sin(\varpi t + \phi)$

3. $x' + 25x = 0, \quad x(0) = -2, \quad x'(0) = 10$
4. $\frac{1}{2}x' + 8x = 0, \quad x(0) = 1, \quad x'(0) = -2$
5. $x'' + 2x = 0, \quad x(0) = -1, \quad x'(0) = -2\sqrt{2}$
6. $\frac{1}{4}x' + 16x = 0, \quad x(0) = 4, \quad x'(0) = 16$
7. $0.1x' + 10x = 0, \quad x(0) = 1, \quad x'(0) = 1$
8. $x' + x = 0, \quad x(0) = -4, \quad x'(0) = 3$
9. The period of free undamped oscillations of a mass on a spring is $\pi/4$ seconds. If the spring constant is 16 lb/ft, what is the numerical value of the weight?
10. A 4-lb weight is attached to a spring, whose spring constant is 16 lb/ft . What is period of simple harmonic motion?
11. A 24-lb weight, attached to the spring, stretches it 4 inches. Find the equation of the motion if the weight is released from rest from a point 3 inches above the equilibrium position.
12. A 20-lb weight stretches a spring 6 inches. The weight is released from rest 6 inches below the equilibrium position.
    a) Find the position of the weight at $t = \frac{\pi}{12}, \frac{\pi}{8}, \frac{\pi}{6}, \frac{\pi}{4}, \frac{9\pi}{32}$ seconds.
    b) What is the velocity of the weight when $t = 3\pi/16$ seconds? In which direction is the weight heading at this instant?
    c) At what times does the weight pass through the equilibrium position?

ORANGE MONKEY TEAM

Lecture 23

Damped Motion

In the previous lecture, we discussed the free harmonic motion that assumes no retarding forces acting on the moving mass. However

- No retarding forces acting on the moving body is not realistic, because
- There always exists at least a resisting force due to surrounding medium.

For example a mass can be suspended in a viscous medium. Hence, the damping forces need to be included in a realistic analysis.

Damping Force

In the study of mechanics, the damping forces acting on a body are considered to be proportional to a power of the instantaneous velocity $\frac{dx}{dt}$. In the hydro dynamical problems, the damping force is proportional to $(dx/dt)^2$. So that in these problems

$$Damping\ force = -\beta\left(\frac{dx}{dt}\right)^2$$

Where β is a positive damping constant and negative sign indicates that the damping force acts in a direction opposite to the direction of motion.

In the present discussion, we shall assume that the damping force is proportional to the instantaneous velocity $\frac{dx}{dt}$. Thus for us

$$Damping\ force = -\beta\left(\frac{dx}{dt}\right)$$

The Differential Equation

Suppose That

- A body of mass m is attached to a spring.
- The spring stretches by an amount s to attain the equilibrium position.
- The mass is further displaced by an amount x and then released.
- No external forces are impressed on the system.

Therefore, there are three forces acting on the mass, namely:

a) Weight mg of the body

b) Restoring force $-k(s+x)$

c) Damping force $-\beta\left(\frac{dx}{dt}\right)$

ORANGE MONKEY TEAM

Therefore, total force acting on the mass m is

$$mg - k(s+x) - \beta\left(\frac{dx}{dt}\right)$$

So that by Newton's second law of motion, we have

$$m\frac{d^2x}{dt^2} = mg - k(s+x) - \beta\left(\frac{dx}{dt}\right)$$

Since in the equilibrium position

$$mg - ks = 0$$

Therefore

$$m\frac{d^2x}{dt^2} = -kx - \beta\left(\frac{dx}{dt}\right)$$

Dividing with m, we obtain the differential equation of free damped motion

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = 0$$

For algebraic convenience, we suppose that

$$2\lambda = \frac{\beta}{m}, \quad \omega^2 = \frac{k}{m}$$

Then the equation becomes:

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = 0$$

Solution of the Differential Equation

Consider the equation of the free damped motion

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = 0$$

Put

$$x = e^{mt}, \frac{dx}{dt} = me^{mt}, \frac{d^2x}{dt^2} = m^2e^{mt}$$

Then the auxiliary equation is:

$$m^2 + 2\lambda m + \omega^2 = 0$$

Solving by use of quadratic formula, we obtain

$$m = -\lambda \pm \sqrt{\lambda^2 - \omega^2}$$

Thus the roots of the auxiliary equation are

$$m_1 = -\lambda + \sqrt{\lambda^2 - \omega^2}, \quad m_2 = -\lambda - \sqrt{\lambda^2 - \omega^2}$$

ORANGE MONKEY TEAM

Depending upon the sign of the quantity $\lambda^2 - \omega^2$, we can now distinguish three possible cases of the roots of the auxiliary equation.

Case 1 *Real and distinct roots*

If $\lambda^2 - \omega^2 > 0$ then $\beta > k$ and the system is said to be over-damped. The solution of the equation of free damped motion is

$$x(t) = c_1 e^{m_1 t} + c_2 e^{m_2 t}$$

or

$$x(t) = e^{-\lambda t}\left[c_1 e^{\sqrt{\lambda^2-\omega^2}\,t} + c_2 e^{-\sqrt{\lambda^2-\omega^2}\,t}\right]$$

This equation represents smooth and non oscillatory motion.

Case 2 *Real and equal roots*

If $\lambda^2 - \omega^2 = 0$, then $\beta = k$ and the system is said to be critically damped, because any slight decrease in the damping force would result in oscillatory motion. The general solution of the differential equation of free damped force is

$$x(t) = c_1 e^{m_1 t} + c_2 te^{m_1 t}$$

or

$$x(t) = e^{-\lambda t}(c_1 + c_2 t)$$

Case 3 *Complex roots*

If $\lambda^2 - w^2 < 0$, then $\beta < k$ and the system is said to be under-damped. We need to rewrite the roots of the auxiliary equation as:

$$m_1 = -\lambda + \sqrt{\omega^2 - \lambda^2}\,i, \quad m_2 = -\lambda - \sqrt{\omega^2 - \lambda^2}\,i$$

Thus, the general solution of the equation of free damped motion is

$$x(t) = e^{-\lambda t}\left[c_1 \cos\sqrt{\omega^2 - \lambda^2}\,t + c_2 \sin\sqrt{\omega^2 - \lambda^2}\,t\right]$$

This represents an oscillatory motion; but amplitude of vibration $\to 0$ as $t \to \infty$ because of the coefficient $e^{-\lambda t}$.

Note that

Each of the three solutions contain the damping factor $e^{-\lambda t}, \lambda > 0$, the displacements of the mass become negligible for larger times.

ORANGE MONKEY TEAM

Alternative form of the Solution

When $\lambda^2 - \omega^2 < 0$, the solution of the differential equation of free damped motion

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = 0$$

is $$x(t) = e^{-\lambda t}\left[c_1 \cos\sqrt{\omega^2 - \lambda^2}t + c_2 \sin\sqrt{\omega^2 - \lambda^2}t\right]$$

Suppose that A and ϕ are two real numbers such that

$$\sin\phi = \frac{c_1}{A}, \quad \cos\phi = \frac{c_2}{A}$$

So that $$A = \sqrt{c_1^2 + c_2^2}, \quad \tan\phi = \frac{c_1}{c_2}$$

The number ϕ is known as the ***phase angle***. Then the solution of the equation becomes:

$$x(t) = Ae^{-\lambda t}\left[\sin\sqrt{\omega^2 - \lambda^2}t\cos\phi + \cos\sqrt{\omega^2 - \lambda^2}t\sin\phi\right]$$

or $$x(t) = Ae^{-\lambda t}\sin\left(\sqrt{\omega^2 - \lambda^2}t + \phi\right)$$

Note that

- The coefficient $Ae^{-\lambda t}$ is called the damped amplitude of vibrations.
- The time interval between two successive maxima of $x(t)$ is called ***quasi period***, and is given by the number

$$\frac{2\pi}{\sqrt{\omega^2 - \lambda^2}}$$

- The following number is known as the ***quasi frequency***.

$$\frac{\sqrt{\omega^2 - \lambda^2}}{2\pi}$$

- The graph of the solution

$$x(t) = Ae^{-\lambda t}\sin\left(\sqrt{\omega^2 - \lambda^2}t + \phi\right)$$

crosses positive *t-axis*, i.e the line $x = 0$, at times that are given by

$$\sqrt{\omega^2 - \lambda^2}t + \phi = n\pi$$

Where $n = 1,2,3,\ldots$.

For example, if we have

$$x(t) = e^{-0.5t}\sin\left(2t - \frac{\pi}{3}\right)$$

© Copyright Virtual University of Pakistan

Then $2t - \frac{\pi}{3} = n\pi$

or $2t_1 - \frac{\pi}{3} = 0, 2t_2 - \frac{\pi}{3} = \pi, 2t_3 - \frac{\pi}{3} = 2\pi, \ldots$

or $t_1 = \frac{\pi}{6}, t_2 = \frac{4\pi}{6}, t_3 = \frac{7\pi}{6}, \ldots$

We notice that difference between two successive roots is

$$t_k - t_{k-1} = \frac{\pi}{2} = \frac{1}{2} \text{ quasi period}$$

Since quasi period $= \frac{2\pi}{2} = \pi$. Therefore

$$t_k - t_{k-1} = \frac{\pi}{2} = \frac{1}{2} \text{ quasi period}$$

- Since $|x(t)| \le Ae^{-\lambda t}$ when $\left|\sin\sqrt{\omega^2 - \lambda^2}t + \phi\right| \le 1$, the graph of the solution

$$x(t) = Ae^{-\lambda t}\sin\left(\sqrt{\omega^2 - \lambda^2}t + \phi\right)$$

touches the graphs of the exponential functions

$$\pm Ae^{-\lambda t}$$

at the values of t for which

$$\sin\left(\sqrt{\omega^2 - \lambda^2}t + \phi\right) = \pm 1$$

This means those values of t for which

$$\sqrt{\omega^2 - \lambda^2}t + \phi = (2n+1)\frac{\pi}{2}$$

or $t = \frac{(2n+1)(\pi/2) - \phi}{\sqrt{\omega^2 - \lambda^2}}$ where $n = 0, 1, 2, 3, \ldots$

Again, if we consider

$$x(t) = e^{-0.5t}\sin\left(2t - \frac{\pi}{3}\right)$$

Then $2t_1^* - \frac{\pi}{3} = \frac{\pi}{2}, 2t_2^* - \frac{\pi}{3} = \frac{3\pi}{2}, 2t_3^* - \frac{\pi}{3} = \frac{5\pi}{2}, \ldots$

Or $t_1^* = \frac{5\pi}{12}, t_2^* = \frac{11\pi}{12}, t_3^* = \frac{17\pi}{12}, \ldots$

Again, we notice that the difference between successive values is

$$t_k^* - t_{k-1}^* = \frac{\pi}{2}$$

- The values of t for which the graph of the solution

$$x(t) = Ae^{-\lambda t}\sin\left(\sqrt{\omega^2 - \lambda^2}t + \phi\right)$$

touches the exponential graph are not the values for which the function attains its relative extremum.

Example 1

© Copyright Virtual University of Pakistan

Interpret and solve the initial value problem

$$\frac{d^2x}{dt^2}+5\frac{dx}{dt}+4x=0$$

$$x(0)=1, \quad x'(0)=1$$

Find extreme values of the solution and check whether the graph crosses the equilibrium position.

Interpretation

Comparing the given differential equation

$$\frac{d^2x}{dt^2}+5\frac{dx}{dt}+4x=0$$

with the general equation of the free damped motion

$$\frac{d^2x}{dt^2}+2\lambda\frac{dx}{dt}+\omega^2x=0$$

we see that

$$\lambda=\frac{5}{2}, \quad \omega^2=4$$

so that $\lambda^2-\omega^2>0$

Therefore, the problem represents the over-damped motion of a mass on a spring.

Inspection of the boundary conditions

$$x(0)=1, \quad x'(0)=1$$

reveals that the mass starts 1 unit below the equilibrium position with a downward velocity of 1 ft/sec.

Solution

To solve the differential equation

$$\frac{d^2x}{dt^2}+5\frac{dx}{dt}+4x=0$$

© Copyright Virtual University of Pakistan

We put $x = e^{mt}, \quad \frac{dx}{dt} = me^{mt}, \quad \frac{d^2x}{dt^2} = m^2e^{mt}$

Then the auxiliary equation is

$$m^2 + 5m + 4 = 0$$

$$\Rightarrow (m+4)(m+1) = 0$$

$$\Rightarrow m = -1, \quad m = -4,$$

Therefore, the auxiliary equation has distinct real roots

$$m = -1, \quad m = -4$$

Thus the solution of the differential equation is:

$$x(t) = c_1e^{-t} + c_2e^{-4t}$$

So that $x'(t) = -c_1e^{-t} - 4c_2e^{-4t}$

Now, we apply the boundary conditions

$$x(0) = 1 \Rightarrow c_1.1 + c_2.1 = 1$$

$$x'(0) = 1 \Rightarrow -c_1 - 4c_2 = 1$$

Thus

$$c_1 + c_2 = 1$$

$$-c_1 - 4c_2 = 1$$

Solving these two equations, we have.

$$c_1 = \frac{5}{3}, \quad c_2 = -\frac{2}{3}$$

Therefore, solution of the initial value problem is

$$x(t) = \frac{5}{3}e^{-t} - \frac{2}{3}e^{-4t}$$

Extremum

Since $x(t) = \frac{5}{3}e^{-t} - \frac{2}{3}e^{-4t}$

Therefore $\frac{dx}{dt} = -\frac{5}{3}e^{-t} + \frac{8}{3}e^{-4t}$

So that $x'(t) = 0 \Rightarrow -\frac{5}{3}e^{-t} + \frac{8}{3}e^{-4t} = 0$

or $e^{3t} = \frac{8}{5} \Rightarrow t = \frac{1}{3}\ln\frac{8}{5}$

or $t = 0.157$

© Copyright Virtual University of Pakistan

Since $$\frac{d^2x}{dt^2} = \frac{5}{3}e^{-t} - \frac{32}{3}e^{-4t}$$

Therefore at $t = 0.157,$ we have

$$\frac{d^2x}{dt^2} = \frac{5}{3}e^{-0.157} - \frac{32}{3}e^{-0.628}$$

$$= 1.425 - 5.692 = -4.267 < 0$$

So that the solution $x(t)$ has a maximum at $t = 0.157$ and maximum value of x is:

$$x(0.157) = 1.069$$

Hence the mass attains an extreme displacement of 1.069 ft below the equilibrium position.

Check

Suppose that the graph of $x(t)$ does cross the $t-axis$, that is, the mass passes through the equilibrium position. Then a value of t exists for which

$$x(t) = 0$$

i.e $$\frac{5}{3}e^{-t} - \frac{2}{3}e^{-4t} = 0$$

$$\Rightarrow e^{3t} = \frac{2}{5}$$

or $$t = \frac{1}{3}\ln\frac{2}{5} = -0.305$$

This value of t is physically irrelevant because time can never be negative. Hence, the mass never passes through the equilibrium position.

Example 2

An 8-lb weight stretches a spring 2ft. Assuming that a damping force numerically equals to two times the instantaneous velocity acts on the system. Determine the equation of motion if the weight is released from the equilibrium position with an upward velocity of 3 ft / sec.

Solution

Since

$$\text{Weight} = 8\,\text{lbs}, \quad \text{Stretch} = s = 2\,\text{ft}$$

Therefore, by Hook's law

$$8 = 2k$$

$$\Rightarrow k = 4\ \text{lb/ft}$$

© Copyright Virtual University of Pakistan

Since $Damping \quad force = 2\left(\frac{dx}{dt}\right)$

Therefore $\beta = 2$

Also $mass = \frac{\text{Weight}}{\text{g}} \Rightarrow m = \frac{8}{32} = \frac{1}{4}$ slugs

Thus, the differential equation of motion of the free damped motion is given by

$$m\frac{d^2x}{dt^2} = -kx - \beta\left(\frac{dx}{dt}\right)$$

or $$\frac{1}{4}\frac{d^2x}{dt^2} = -4x - 2\left(\frac{dx}{dt}\right)$$

or $$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x = 0$$

Since the mass is released from equilibrium position with an upward velocity $3\,\text{ft/s}$. Therefore the initial conditions are:

$$x(0) = 0, \qquad x'(0) = -3$$

Thus we need to solve the initial value problem

Solve $$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x = 0$$

Subject to $$x(0) = 0, \quad x'(0) = -3$$

Put $$x = e^{mt}, \quad \frac{dx}{dt} = me^{mt}, \quad \frac{d^2x}{dt^2} = m^2e^{mt}$$

Thus the auxiliary equation is

$$m^2 + 8m + 16 = 0$$

or $$(m+4)^2 = 0 \Rightarrow m = -4, -4$$

So that roots of the auxiliary equation are real and equal.

$$m_1 = -4 = m_2$$

Hence the system is critically damped and the solution of the governing differential equation is

$$x(t) = c_1e^{-4t} + c_2te^{-4t}$$

Moreover, the system is critically damped.

© Copyright Virtual University of Pakistan

We now apply the boundary conditions.

$$x(0) = 0 \Rightarrow c_1.1 + c_2.0 = 0$$

$$\Rightarrow c_1 = 0$$

Thus $x(t) = c_2 te^{-4t}$

$$\Rightarrow \quad \frac{dx}{dt} = c_2 e^{-4t} - 4c_2 te^{-4t}$$

So that $x'(0) = -3 \Rightarrow c_2.1 - 0 = -3$

$$\Rightarrow c_2 = -3$$

Thus solution of the initial value problem is

$$x(t) = -3te^{-4t}$$

Extremum

Since $x(t) = -3te^{-4t}$

Therefore $$\frac{dx}{dt} = -3e^{-4t} + 12te^{-4t}$$

$$= -3e^{-4t}(1 - 4t)$$

Thus $$\frac{dx}{dt} = 0 \Rightarrow t = \frac{1}{4}$$

The corresponding extreme displacement is

$$x\left(\frac{1}{4}\right) = -3\left(\frac{1}{4}\right)e^{-1} = -0.276 \text{ ft}$$

Thus the weight reaches a maximum height of 0.276 ft above the equilibrium position.

Example 3

A 16-lb weight is attached to a 5-ft long spring. At equilibrium the spring measures ft .If the weight is pushed up and released from rest at a point 2-ft above the equilibrium position. Find the displacement $x(t)$ if it is further known that the surrounding medium offers a resistance numerically equal to the instantaneous velocity.

Solution

Length of un - stretched spring = 5 ft

Length of spring at equilibrium = 8.2 ft

Thus Elongation of spring = s = 3.2 ft

By Hook's law, we have

© Copyright Virtual University of Pakistan

$$16 = k(3.2) \Rightarrow k = 5 \text{ lb/ft}$$

Further $$mass = \frac{\text{Weight}}{\text{g}} \quad \Rightarrow m = \frac{16}{32} = \frac{1}{2} \text{ slugs}$$

Since $$Damping \;\; force = \frac{dx}{dt}$$

Therefore $$\beta = 1$$

Thus the differential equation of the free damped motion is given by

$$m\frac{d^2x}{dt^2} = -kx - \beta\frac{dx}{dt}$$

or $$\frac{1}{2}\frac{d^2x}{dt^2} = -5x - \frac{dx}{dt}$$

or $$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 10x = 0$$

Since the spring is released from rest at a point 2 ft above the equilibrium position.

The initial conditions are:

$$x(0) = -2, \quad x'(0) = 0$$

Hence we need to solve the initial value problem

$$\frac{d^2x}{dt^2} + \frac{1}{x}\frac{d}{dt} + 10x = 0$$

$$x(0) = -2, \quad x'(0) = 0$$

To solve the differential equation, we put

$$x = e^{mt}, \quad \frac{dx}{dt} = me^{mt}, \quad \frac{d^2x}{dt^2} = m^2e^{mt}.$$

Then the auxiliary equation is

$$m^2 + 2m + 10 = 0$$

or $$m = -1 \pm 3i$$

So that the auxiliary equation has complex roots

$$m_1 = -1 + 3i, \quad m_2 = -1 - 3i$$

The system is under-damped and the solution of the differential equation is:

$$x(t) = e^{-t}(c_1 \cos 3t + c_2 \sin 3t)$$

© Copyright Virtual University of Pakistan

Now we apply the boundary conditions

$$x(0) = -2 \Rightarrow c_1.1 + c_2.0 = -2$$

$$\Rightarrow c_1 = -2$$

Thus $x(t) = e^{-t}\left(-2\cos 3t + c_2 \sin 3t\right)$

$$\frac{dx}{dt} = e^{-t}\left(6\sin 3t + 3c_2 \cos 3t\right) - e^{-t}\left(-2\cos 3t + c_2 \sin 3t\right)$$

Therefore $x'(0) = 0 \Rightarrow 3c_2 + 2 = 0$

$$c_2 = \frac{-2}{3}$$

Hence, solution of the initial value problem is

$$x(t) = e^{-t}\left(-2\cos 3t - \frac{2}{3}\sin 3t\right)$$

Example 4

Write the solution of the initial value problem

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 10x = 0$$

$$x(0) = -2, \quad x'(0) = 0$$

in the alternative form

$$x(t) = Ae^{-t}\sin(3t + \phi)$$

Solution

We know from previous example that the solution of the initial value problem is

$$x(t) = e^{-t}\left(-2\cos 3t - \frac{2}{3}\sin 3t\right)$$

Suppose that A and ϕ are real numbers such that

$$\sin\phi = -\frac{2}{A}, \quad \cos\phi = \frac{-2/3}{A}$$

Then $A = \sqrt{4 + \frac{4}{9}} = \frac{2}{3}\sqrt{10}$

Also $\tan\phi = \frac{-2}{-2/3} = 3$

Therefore $\tan^{-1}(3) = 1.249$ radian

Since $\sin\phi < 0, \cos\phi < 0,$ the phase angle ϕ must be in 3rd quadrant.

Therefore

$$\phi = \pi + 1.249 = 4.391 \text{ radians}$$

Hence

© Copyright Virtual University of Pakistan

$$x(t) = \frac{2}{3}\sqrt{10}e^{-t}\sin(3t + 4.391)$$

The values of $t = t_\gamma$ where the graph of the solution crosses positive t - axis and the values $t = t_\gamma{}^*$ where the graph of the solution touches the graphs of $\pm\left(\frac{2}{3}\right)\sqrt{10}e^{-t}$ are given in the following table.

γ	t_γ	t_γ^*	$x(t_\gamma^*)$
1	.631	1.154	0.665
2	1.678	2.202	-0.233
3	2.725	3.249	0.082
4	3.772	4.296	-0.029

Quasi Period

Since $$x(t) = \frac{2}{3}\sqrt{10}e^{-t}\sin(3t + 4.391)$$

Therefore $$\sqrt{\lambda^2 - \omega^2} = 3$$

So that the quasi period is given by

$$\frac{2\pi}{\sqrt{\lambda^2 - \omega^2}} = \frac{2\pi}{3} \text{ seconds}$$

Hence, difference between the successive t_γ and t_γ^* is $\frac{\pi}{3}$ units.

© Copyright Virtual University of Pakistan

Exercise

Give a possible interpretation of the given initial value problems.

1. $\frac{1}{6}x'' + 2x' + x = 0, \quad x(0) = 0, \; x'(0) = -1.5$
2. $\frac{16}{32}x'' + x' + 2x = 0, \quad x(0) = -2, \; x'(0) = 1$
3. A 4-lb weight is attached to a spring whose constant is 2 lb /ft. The medium offers a resistance to the motion of the weight numerically equal to the instantaneous velocity. If the weight is released from a point 1 ft above the equilibrium position with a downward velocity of 8 ft / s, determine the time that the weight passes through the equilibrium position. Find the time for which the weight attains its extreme displacement from the equilibrium position. What is the position of the weight at this instant?
4. A 4-ft spring measures 8 ft long after an 8-lb weight is attached to it. The medium through which the weight moves offers a resistance numerically equal to $\sqrt{2}$ times the instantaneous velocity. Find the equation of motion if the weight is released from the equilibrium position with a downward velocity of 5 ft / s. Find the time for which the weight attains its extreme displacement from the equilibrium position. What is the position of the weight at this instant?
5. A 1-kg mass is attached to a spring whose constant is 16 N / m and the entire system is then submerged in to a liquid that imparts a damping force numerically equal to 10 times the instantaneous velocity. Determine the equations of motion if
   a. The weight is released from rest 1m below the equilibrium position; and
   b. The weight is released 1m below the equilibrium position with and upward velocity of 12 m/s.
6. A force of 2-lb stretches a spring 1 ft. A 3.2-lb weight is attached to the spring and the system is then immersed in a medium that imparts damping force numerically equal to 0.4 times the instantaneous velocity.
   a. Find the equation of motion if the weight is released from rest 1 ft above the equilibrium position.
   b. Express the equation of motion in the form $x(t) = Ae^{-\lambda t}\sin\left(\sqrt{\omega^2 - \lambda^2}\, t + \phi\right)$
   c. Find the first times for which the weight passes through the equilibrium position heading upward.

© Copyright Virtual University of Pakistan

7. After a 10-lb weight is attached to a 5-ft spring, the spring measures 7-ft long. The 10-lb weight is removed and replaced with an 8-lb weight and the entire system is placed in a medium offering a resistance numerically equal to the instantaneous velocity.
   a. Find the equation of motion if the weight is released 1/ 2 ft below the equilibrium position with a downward velocity of 1ft / s.
   b. Express the equation of motion in the form $x(t) = Ae^{-\lambda t}\sin\left(\sqrt{\omega^2 - \lambda^2}\ t + \phi\right)$
   c. Find the time for which the weight passes through the equilibrium position heading downward.
8. A 10-lb weight attached to a spring stretches it 2 ft. The weight is attached to a dashpot-damping device that offers a resistance numerically equal to $\beta(\beta > 0)$ times the instantaneous velocity. Determine the values of the damping constant β so that the subsequent motion is
   a. Over-damped
   b. Critically damped
   c. Under-damped
9. A mass of 40 g. stretches a spring 10cm. A damping device imparts a resistance to motion numerically equal to 560 (measured in dynes /(cm / s)) times the instantaneous velocity. Find the equation of motion if the mass is released from the equilibrium position with downward velocity of 2 cm / s.
10. The quasi period of an under-damped, vibrating 1-slugs mass of a spring is $\pi / 2$ seconds. If the spring constant is 25 lb / ft, find the damping constant β .

© Copyright Virtual University of Pakistan

Lecture 24
Forced Motion

In this last lecture on the applications of second order linear differential equations, we consider

- A vibrational system consisting of a body of mass m attached to a spring. The motion of the body is being driven by an external force $f(t)$ i.e. forced motion.
- Flow of current in an electrical circuit that consists of an inductor, resistor and a capacitor connected in series, because of its similarity with the forced motion.

Forced motion with damping

Suppose that we now take into consideration an external force $f(t)$. Then, the forces acting on the system are:

a) Weight of the body $= mg$

b) The restoring force $= -k(s+x)$

c) The damping effect $= -\beta\,(dx/dt)$

d) The external force $= f(t)$.

Hence x denotes the distance of the mass m from the equilibrium position. Thus the total force acting on the mass m is given by

$$Force = mg - k(s+x) - \beta\left(\frac{dx}{dt}\right) + f(t)$$

By the Newton's 2nd law of motion, we have

$$Force = ma = m\frac{d^2x}{dt^2}$$

Therefore
$$m\frac{d^2x}{dt^2} = mg - ks - kx - \beta\left(\frac{dx}{dt}\right) + f(t)$$

But
$$mg - ks = 0$$

So that
$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = \frac{f(t)}{m}$$

or
$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = F(t)$$

where $F(t) = \dfrac{f(t)}{m}$, $2\lambda = \dfrac{\beta}{m}$ and $\omega^2 = \dfrac{k}{m}$.

Note that

- The last equation is a non-homogeneous differential equation governing the forced motion with damping.
- To solve this equation, we use either the method of undetermined coefficients or the variation of parameters.

© Copyright Virtual University of Pakistan

Example 1

Interpret and solve the initial value problem

$$\frac{1}{5}\frac{d^2x}{dt^2}+1.2\frac{dx}{dt}+2x=5\cos 4t$$

$$x(0)=\frac{1}{2},\quad x'(0)=0$$

Interpretation

The problem represents a vibrational system consisting of

- A mass $m=\frac{1}{5}$ slugs or kilograms
- The mass is attached to a spring having spring constant $k=2\,\text{lb/ft}$ or N/m
- The mass is released from rest $\frac{1}{2}$ ft or meter below the equilibrium position
- The motion is damped with damping constant $\beta=1.2$.
- The motion is being driven by an external periodic force $f(t)=5\cos 4t$ that has period $T=\frac{\pi}{2}$.

Solution

Given the differential equation

$$\frac{1}{5}\frac{d^2x}{dt^2}+1.2\frac{dx}{dt}+2x=5\cos 4t$$

or

$$\frac{d^2x}{dt^2}+6\frac{dx}{dt}+10x=25\cos 4t$$

First consider the associated homogeneous differential equation.

$$\frac{d^2x}{dt^2}+6\frac{dx}{dt}+10x=0$$

Put $x=e^{mt},\quad \frac{dx}{dt}=me^{mt},\quad \frac{d^2x}{dt^2}=m^2e^{mt}$

Then the auxiliary equation is:

$$m^2+6m+10=0$$

$$\Rightarrow \quad m=-3\pm i$$

Thus the auxiliary equation has complex roots

$$m_1=-3+i,\ m_2=-3-i$$

So that the complementary function of the equation is

$$x_c=e^{-3t}\left(c_1\cos t+c_2\sin t\right)$$

To find a particular integral of non-homogeneous differential equation we use the undetermined coefficients, we assume that

$$x_p(t)=A\cos 4t+B\sin 4t$$

© Copyright Virtual University of Pakistan

Then
$$x'_p(t) = -4A\sin 4t + 4B\cos 4t$$
$$x''_p(t) = -16A\cos 4t - 16B\sin 4t$$

So that

$$\begin{aligned} x''_p + 6x'_p + 10x_p &= -16A\cos 4t - 16B\sin 4t - 24A\sin 4t \\ &\quad + 24B\cos 4t + 10A\cos 4t + 10B\sin 4t \\ &= (-6A + 24B)\cos 4t + (-24A - 6B)\sin 4t \end{aligned}$$

Substituting in the given non-homogeneous differential equation, we obtain

$$(-6A + 24B)\cos 4t + (-24A - 6B)\sin 4t = 25\cos 4t$$

Equating coefficients, we have

$$-6A + 24B = 25$$
$$-24A - 6B = 0$$

Solving these equations, we obtain

$$A = -\frac{25}{102},\ B = \frac{50}{51}$$

Thus
$$x_p(t) = -\frac{25}{102}\cos 4t + \frac{50}{51}\sin 4t$$

Hence the general solution of the differential equation is:

$$x(t) = e^{-3t}\left[c_1\cos t + c_2\sin t\right] - \frac{25}{102}\cos 4t + \frac{50}{51}\sin 4t$$

$$x'(t) = -3e^{-3t}\left[c_1\cos t + c_2\sin t\right] + e^{-3t}\left(-c_1\sin t + c_2\cos t\right) + \frac{50}{51}\sin 4t + \frac{200}{51}\cos 4t$$

Now
$$x(0) = \frac{1}{2} \text{ gives}$$
$$c_1.1 - \frac{25}{102} = \frac{1}{2}$$

or
$$c_1 = \frac{1}{2} + \frac{25}{102} = \frac{51+25}{102}$$

or
$$c_1 = \frac{38}{51}$$

Also
$$x'(0) = 0 \text{ gives}$$
$$-3c_1 + c_2 + \frac{200}{51} = 0$$

or
$$c_2 = -\frac{200}{51} + \frac{114}{51} = -\frac{86}{51}$$

Hence the solution of the initial value problem is:

$$x(t) = e^{-3t}\left(\frac{38}{51}\cos t - \frac{86}{51}\sin t\right) - \frac{25}{102}\cos 4t + \frac{50}{51}\sin 4t$$

© Copyright Virtual University of Pakistan

Transient and Steady-State Terms

Due to the presence of the factor e^{-3t} we notice that the complementary function

$$x_c(t) = e^{-3t}\left(\frac{38}{\sqrt{51}}\cos t - \frac{86}{51}\sin t\right)$$

possesses the property that

$$\lim_{x\to\infty} x_c(t) = 0$$

Thus for large time, the displacements of the weight are closely approximated by the particular solution

$$x_p(t) = -\frac{25}{102}\cos 4t + \frac{50}{51}\sin 4t$$

Since $x_c(t) \to 0$ as $t \to \infty$, it is said to be transient term or transient solution. The particular solution $x_p(t)$ is called the steady-state solution

Hence, when F is a periodic function, such as

$$F(t) = F_0 \sin \gamma t \quad or \quad F(t) = F_0 \cos \gamma t$$

The general solution of the equation

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = F(t)$$

consists of

$$x(t) = Transient\ solution + Steady\ State\ Solution$$

Example 2

Solve the initial value problem

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 2x = 4\cos t + 2\sin t$$

$$x(0) = 0, \quad x'(0) = 3$$

Solution

First consider the associated homogeneous linear differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 2x = 0$$

Put $x = e^{mx},\ x' = me^{mx},\ x' = m^2e^{mx}$

Then the auxiliary equation is

$$m^2 + 2m + 2 = 0$$

or

$$m = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$$

Thus the complementary function is

$$x_c = e^{-t}(c_1 \cos t + c_2 \sin t)$$

© Copyright Virtual University of Pakistan

For the particular integral we assume that

$$x_p = A\cos t + B\sin t$$
$$x'_p = -A\sin t + B\cos t$$
$$x''_p = -A\cos t - B\sin t$$

So that

$$\frac{d^2x_p}{dt^2} + 2\frac{dx_p}{dt} + 2x_p = -A\cos t - B\sin t - 2A\sin t + 2B\cos t + 2A\cos t + 2B\sin t$$

or $$\frac{d^2x_p}{dt^2} + 2\frac{dx_p}{dt} + 2x_p = (A+2B)\cos t + (-2A+B)\sin t$$

Substituting in the given differential equation, we have

$$(A+2B)\cos t + (-2A+B)\sin t = 4\cos t + 2\sin t$$

Equating coefficients, we obtain

$$A + 2B = 4$$
$$-2A + B = 2$$

Solving these two equations, we have:

$$A = 0, \quad B = 2$$

Thus $$x_p = 2\sin t$$

Hence general solution of the differential equation is

$$x = x_c + x_p$$

or $$x(t) = e^{-t}(c_1\cos t + c_2\sin t) + 2\sin t$$

Thus $$x'(t) = -e^{-t}(c_1\cos t + c_2\sin t) + e^{-t}(-c_1\sin t + c_2\cos t) + 2\cos t$$

Now we apply the boundary conditions

$$x(0) = 0 \Rightarrow c_1.1 + c_2.0 + 0 = 0$$
$$\Rightarrow c_1 = 0$$
$$x'(0) = 3 \Rightarrow -c_1.1 + c_2.1 + 2 = 3$$
$$\Rightarrow c_2 = 1$$

Thus solution of the initial value problem is

$$x = e^{-t}\sin t + 2\sin t$$

Since $$e^{-t}\sin t \to 0 \quad \text{as} \quad t \to 0$$

Therefore

$$e^{-t}\sin t = \textit{Transient Term}, \quad 2\sin t = \textit{Steady State}$$

Hence $$x = \underbrace{e^{-t}\sin t}_{\textit{Transient}} + \underbrace{2\sin t}_{\textit{Steady–state}}$$

We notice that the effect of the transient term becomes negligible for about

$$t > 2\pi$$

© Copyright Virtual University of Pakistan

Motion without Damping

If the system is impressed upon by a periodic force and there is no damping force then there is no transient term in the solution.

Example 3

Solve the initial value problem

$$\frac{d^2x}{dt^2} + \omega^2 x = F_o \sin \gamma t$$

$$x(0) = 0, \quad x'(0) = 0$$

Where F_o is a constant

Solution

For complementary function, consider the associated homogeneous differential equation

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

Put $x = e^{mt}, \; x' = m^2 e^{mt}$

Then the auxiliary equation is

$$m^2 + \omega^2 = 0 \Rightarrow m = \pm \omega i$$

Thus the complementary function is

$$x_c(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

To find a particular solution, we assume that

$$x_p(t) = A\cos \gamma t + B \sin \gamma t$$

Then

$$x'_p(t) = -A\gamma \sin \gamma t + B\gamma \cos \gamma t$$

$$x''_p(t) = -A\gamma^2 \cos \gamma t - B\gamma^2 \sin \gamma t$$

Therefore,

$$x''_p + \omega^2 x_p = -A\gamma^2 \cos \gamma t - B\gamma^2 \sin \gamma t + A\omega^2 \cos \gamma t + B\omega^2 \sin \gamma t$$

$$x''_p + \omega^2 x_p = A(\omega^2 - \gamma^2)\cos \gamma t + B(\omega^2 - \gamma^2)\sin \gamma t$$

Substituting in the given differential equation, we have

$$(\quad^2 \quad) \qquad (\quad^2 \quad) A\,\omega \; -\gamma \quad {}_o \sin \gamma t$$

Equating coefficients, we have

$$A(\omega^2 - \gamma^2) = 0, \quad B(\omega^2 - \gamma^2) = F_o$$

Solving these two equations, we obtain

$$A = 0, \qquad B = \frac{F_o}{\omega^2 - \gamma^2} \qquad (\gamma \neq \omega)$$

$$(\quad F_o \quad)$$

Therefore

$$x_p(t) = \left| \frac{\quad}{\omega^2 - \gamma^2} \right| \sin \gamma t$$

© Copyright Virtual University of Pakistan

Hence, the general solution of the differential equation is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t + \left(\frac{F_o}{\omega^2 - \gamma^2} \right) \sin \gamma t$$

Then $$x'(t) = -c_1 \omega \sin \omega t + c_2 \omega \cos \omega t + \frac{F_o \gamma}{\omega^2 - \gamma^2} \cos \gamma t$$

Now we apply the boundary conditions

$$x(0) = 0 \Rightarrow c_1 .1 + c_2 .0 + 0 = 0$$

$$\Rightarrow c_1 = 0$$

$$x'(0) = 0 \Rightarrow c_1 .0 + c_2 \omega .1 + \frac{F_o \gamma}{\omega^2 - \gamma^2} = 0$$

$$\Rightarrow c_2 = -\frac{\gamma F_o}{\omega\left(\omega^2 - \gamma^2\right)}$$

Thus solution of the initial value problem is

$$x(t) = \frac{F_o}{\omega\left(\omega^2 - \gamma^2\right)} \left(-\gamma \sin \omega t + \omega \sin \gamma t\right), \quad (\gamma \neq \omega)$$

Note that the solution is not defined for = , However $\lim_{\gamma \to \omega} x(t)$ can be obtained using the L'Hôpital's rule

$$x(t) = \lim_{\gamma \to \omega} F_o \frac{-\gamma \sin \omega t + \omega \sin \gamma t}{\omega\left(\omega^2 - \gamma^2\right)}$$

$$= F_o \lim_{\gamma \to \omega} \omega \frac{\frac{d}{d\gamma}\left(-\gamma \sin \omega t + \omega \sin \gamma t\right)}{\frac{d}{d\gamma}\omega\left(\omega^2 - \gamma^2\right)}$$

$$= F_o \lim_{\gamma \to \omega} \omega \frac{-\sin \omega t + \omega t \cos \gamma t}{-2\omega\gamma}$$

$$= F_o \left(\frac{-\sin \omega t + \omega t \cos \omega t}{-2\omega^2} \right)$$

$$= \frac{F_o}{2\omega^2} \sin \omega t - \frac{F_o}{2\omega} t \cos \omega t$$

Clearly $|x(t)| \to \infty$ as $t \to \infty$ .Therefore there is no transient term when there is no damping force in the presence of a periodic impressed force.

© Copyright Virtual University of Pakistan

Electric Circuits

Many different physical systems can be described by a second order linear differential equation similar to the differential equation of the forced motion:

$$m\frac{d^2x}{dt^2} + \beta\frac{dx}{dt} + kx = f(t)$$

One such analogous case is that of an *LRC*-Series circuit. Because of the similarity in mathematics that governs these two systems, it might be possible to use our intuitive understanding of one to help understand the other.

The LRC Series Circuits

The LRC series circuit consist of an inductor, resistor and capacitor connected in series with a time varying source voltage $E(t)$,

Resistor

A resistor is an electrical component that limits or regulates the flow of electrical current in an electrical circuit.

The measure of the extent to which a resistor impedes or resists with the flow of current through it is called resistance, denoted by R.
Clearly higher the resistance, lower the flow of current. Lower the resistance, higher the flow of current. Therefore, we conclude that the flow of current is inversely proportional to the ***resistance***, i.e

$$I = V.\frac{1}{R} \Rightarrow V = IR$$

Where V is constant of proportionality and it represents the voltage. The above equation is mathematical statement of the well known as Ohm's Law.

Inductor

An inductor is a passive electronic component that stores energy in the form of magnetic field. In its simplest form the conductor consists of a wire loop or coil wound on some suitable material.

Whenever current through an inductor changes, i.e increases or decreases, a counter *emf* is induced in it, which tends to oppose this change. This property of the coil due to which it opposes any change of current through it is called the ***inductance***.
Suppose that I denotes the current then the rate of change of current is given by

$$\frac{dI}{dt}$$

This produces a counter *emf* voltage V. Then V is directly proportional to $\frac{dI}{dt}$

$$V\alpha\frac{dI}{dt} \quad \Rightarrow \quad V = L\frac{dI}{dt}$$

Where L is constant of proportionality, which represents inductance of the inductor. The standard unit for measurement of inductance is Henry, denoted by H.

© Copyright Virtual University of Pakistan

Capacitor

A capacitor is a passive electronic component of an electronic circuit that has the ability to store charge and opposes any change of voltage in the circuit. The ability of a capacitor to store charge is called capacitance of the capacitor denoted by C. If $+q$ coulomb of a charge to the capacitor and the potential difference of V volts is established between 2 plates of the capacitor then

$$q \propto C \Rightarrow q = CV$$

or

$$V = \frac{q}{C}$$

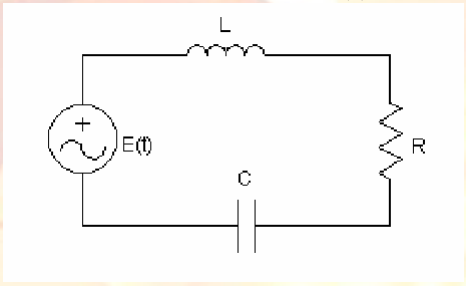
Where C is called constant of proportionality, which represent capacitance. The standard unit to measure capacitance is farad, denoted by F.

Kirchhoff's Voltage Law

The Kirchhoff's 2nd law states that the sum of the voltage drops around any closed loop equals the sum of the voltage rises around that loop. In other words the algebraic sum of voltages around the close loop is zero.

The Differential Equation

Now we consider the following circuit consisting of an inductor, a resistor and a capacitor in series with a time varying voltage source $E(t)$.



If V_L, V_R and V_c denote the voltage drop across the inductor, resistor and capacitor respectively. Then

$$V_L = L\frac{dI}{dt},\ V_R = RI,\ \ V_c = \frac{q}{C}$$

Now by Kirchhoff's law, the sum of V_L, V_R and V_c must equal the source voltage $E(t)$ i.e

$$V_L + V_R + V_c = E(t)$$

or

$$L\frac{dI}{dt} + RI + \frac{q}{C} = E(t)$$

Since the electric current I represents the rate of flow of charge $\frac{dq}{dt}$. Therefore, we can write

$$I = \frac{dq}{dt}$$

Substituting in the last equation, we have:

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = E(t)$$

© Copyright Virtual University of Pakistan

Note that

- We have seen this equation before! It is mathematically exactly the same as the equation for a driven, damped harmonic oscillator.
- If $E(t)=0, R\neq 0$ the electric vibration of the circuit are said to be free damped oscillation.
- If $E(t)=0, R=0$ then the electric vibration can be called free un-damped oscillations.

Solution of the differential equation

The differential equation that governs the flow of charge in an LRC-Series circuit is

$$L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{C}=E(t)$$

This is a non-homogeneous linear differential equation of order-2. Therefore, the general solution of this equation consists of a complementary function and particular integral.

For the complementary function we find general solution of the associated homogeneous differential equation

$$L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{C}=0$$

We put $$q=e^{mt},\quad \frac{dq}{dt}=me^{mt},\quad \frac{d^2q}{dt^2}=m^2e^{mt}$$

Then the auxiliary equation of the associated homogeneous differential equation is:

$$Lm^2+Rm+\frac{1}{C}=0$$

If $R\neq 0$ then, depending on the discriminant, the auxiliary equation may have

- Real and distinct roots
- Real and equal roots
- Complex roots

Case 1 *Real and distinct roots*

If

$$Disc=R^2-\frac{4L}{C}>0$$

Then the auxiliary equation has real and distinct roots. In this case, the circuit is said to be over damped.

Case 2 *Real and equal*
If

$$Disc = R^2 - \frac{4L}{c} = 0$$

Then the auxiliary equation has real and equal roots. In this case, the circuit is said to be critically damped.

Case 3 *Complex roots*
If

$$Disc = R^2 - 4\frac{L}{c} < 0$$

Then the auxiliary equation has complex roots. In this case, the circuit is said to be under damped.

Note that

- Since by the quadratic formula, we know that

$$m = \frac{-R \pm \sqrt{R^2 - 4L/c}}{2L}$$

  In each of the above mentioned three cases, the general solution of the non-homogeneous governing equation contains the factor $e^{-Rt/2L}$. Therefore

$$q(t) \to 0 \text{ as } t \to \infty$$

- In the under damped case when $q(0) = q_o$ the charge on the capacitor oscillates as it decays. This means that the capacitor is charging and discharging as $t \to \infty$
- In the under damped case, i.e. when $E(0) = 0$, and $R = 0$, the electrical vibration do not approach zero as $t \to \infty$. This means that the response of the circuit is Simple Harmonic.

Lecture 25
Forced Motion – Examples

Example 1

Consider an LC series circuit in which .

$$E(t) = 0$$

Determine the charge $q(t)$ on the capacitor for $t > 0$ if its initial charge is q_o and if initially there is no current flowing in the circuit.

Solution

Since in an LC series circuit, there is no resistor. Therefore,

$$R\frac{dq}{dt} = 0$$

So that, the governing differential equation becomes

$$L\frac{d^2q}{dt^2} + \frac{1}{c}q = 0$$

The initial conditions for the circuit are

$$q(0) = q_o, \quad I(0) = 0$$

Since $$\frac{dq}{dt} = I(t)$$

Therefore the initial conditions are equivalent to

$$q(0) = q_o, \quad q'(0) = 0$$

Thus, we have to solve the initial value problem.

$$L\frac{d^2q}{dt^2} + \frac{1}{c}q = 0$$

$$q(0) = q_o, \quad q'(0) = 0$$

To solve the governing differential equation, we put

$$q = e^{mt}, \quad \frac{d^2q}{dt^2} = m^2e^{mt}$$

So that the auxiliary equation is:

$$Lm^2 + \frac{1}{c} = 0$$

$$\Rightarrow \quad m^2 = -\frac{1}{Lc}$$

$$\Rightarrow \quad m = \pm\left(\frac{1}{\sqrt{Lc}}\right)i$$

Therefore, the solution of the differential equation is :

$$q(t) = c_1 \cos\left(\frac{1}{\sqrt{Lc}}t\right) + c_2 \sin\left(\frac{1}{\sqrt{Lc}}t\right)$$

Now, we apply the boundary conditions

$$q(0) = q_o \Rightarrow q_o = c_1.1 + c_2.0$$

$$\Rightarrow c_1 = q_o$$

Thus $$q(t) = q_o \cos\left(\frac{1}{\sqrt{Lc}}t\right) + c_2 \sin\left(\frac{1}{\sqrt{Lc}}t\right)$$

Differentiating *w.r to* t , we have:

$$\frac{dq}{dt} = -\frac{q_o}{\sqrt{Lc}} \sin\left(\frac{1}{\sqrt{Lc}}t\right) + \frac{c_2}{\sqrt{Lc}} \cos\left(\frac{1}{\sqrt{Lc}}t\right)$$

Now $$q'(0) = 0 \Rightarrow 0 + \frac{c_2}{\sqrt{Lc}}.1 = 0$$

$$\Rightarrow c_2 = 0$$

Hence

$$q(t) = q_o \cos\frac{1}{\sqrt{Lc}}t$$

Since $$I(t) = \frac{dq}{dt}$$

Therefore, current in the circuit is given by

$$I(t) = -\frac{q_o}{\sqrt{Lc}} \sin\left(\frac{1}{\sqrt{LC}}t\right)$$

Example 2

Find the charge $q(t)$ on the capacitor in an LRC series circuit when L=0.25 Henry, R=10 Ohms, C=0.001 farad, $E(t) = 0$, $q(0) = q_o$ and $I(0)$=0.

Solution

We know that for an LRC circuit, the governing differential equation is

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = E(t)$$

Since $L = 0.25 = \frac{1}{4}$, $R = 10$,$C = 0.001 = \frac{1}{1000}$

Therefore, the equation becomes:

$$\frac{1}{4}\frac{d^2q}{dt^2} + 10\frac{dq}{dt} + 1000q = 0$$

or $$\frac{d^2q}{dt^2} + 40\frac{dq}{dt} + 4000q = 0$$

The initial conditions are

$$q(0) = q_o,\ I(0) = 0$$

or $$q(0) = q_o,\ q'(0) = 0$$

To solve the differential equation, we put

$$q = e^{mt}, \quad \frac{dq}{dt} = me^{mt}, \quad \frac{d^2q}{dt^2} = m^2e^{mt}$$

Therefore, the auxiliary equation is

$$m^2 + 40m + 4000 = 0$$

$$\Rightarrow m = \frac{-40 \pm \sqrt{1600 - 16000}}{2}$$

$$\Rightarrow m = -20 \pm 60i$$

Thus, the solution of the differential equation is

$$q(t) = e^{-20t}\left(c_1 \cos 60t + c_2 \sin 60t\right)$$

Now, we apply the initial conditions

$$q(0) = q_o \Rightarrow c_1.1 + c_2.0 = q_o$$

$$\Rightarrow c_1 = q_o$$

Therefore $$q(t) = e^{-20t}\left(q_o \cos 60t + c_2 \sin 60t\right)$$

Now $$q'(t) = -20e^{-20t}\left(q_o \cos 60t + c_2 \cos 60t\right) + e^{-20t}\left(-60q_o \sin 60\,t + 60\,c_2 \cos 60t\right)$$

Thus $q'(0) = 0 \Rightarrow -20q_o - 20c_2 + 60c_2.1 = 0$

$$\Rightarrow c_2 = \frac{q_o}{2}$$

Hence the solution of the initial value problem is

$$q(t) = q_o e^{-20t}\left(\cos 60t + \frac{1}{2}\sin 60t\right)$$

As discussed in the previous lectures, a single sine function

$$q(t) = \frac{q_o\sqrt{10}}{3}e^{-20t}\sin(60t + 1.249)$$

Since $R \neq 0$ and $\lim_{t\to\infty} q(t) = 0$

Therefore the solution of the given differential equation is transient solution.

Note that

The electric vibrations in this case are free damped oscillations as there is no impressed voltage $E(t)$ on the circuit.

Example 3

Find the steady state of solution $q_p(t)$ and the steady state current in an *LRC* series circuit when the impressed voltage is

$$E(t) = E_o \sin\gamma t$$

Solution

The steady state solution $q_p(t)$ is a particular solution of the differential equation

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E_o \sin\gamma t$$

We use the method of undetermined coefficients, for finding $q_p(t)$. Therefore, we assume

$$q(t) = A\sin\gamma t + B\cos\gamma t$$

Then

$$q'(t) = A\gamma\cos\gamma t - B\gamma\sin\gamma t$$

$$q'(t) = -A\gamma^2\sin\gamma t - B\gamma^2\cos\gamma t$$

Therefore

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = -AL\gamma^2\sin\gamma t - BL\gamma^2\cos\gamma t + AR\gamma\cos\gamma t$$

$$- BR\gamma\sin\gamma t + \frac{A}{C}\sin\gamma t + \frac{B}{C}\cos\gamma t$$

$$= \left[\frac{A}{C} - AL\gamma^2 - BR\gamma\right]\sin\gamma t + \left[\frac{B}{C} - BL\gamma^2 + AR\gamma\right]\cos\gamma t$$

Substituting in the given differential equation, we obtain

$$\left[\frac{A}{C} - AL\gamma^2 - BR\gamma\right]\sin\gamma t + \left[\frac{B}{C} - BL\gamma^2 + AR\gamma\right]\cos\gamma t = E_o \sin\gamma t$$

Equating coefficients of $\sin\gamma t$ and $\cos\gamma t$, we obtain

$$\frac{A}{C} - AL\gamma^2 - BR\gamma = E_o$$

$$\frac{B}{C} - BL\gamma^2 + AR\gamma = 0$$

or

$$\left(\frac{1}{C} - L\gamma^2\right)A - BR\gamma = E_o$$

$$AR\gamma + \left(\frac{1}{C} - L\gamma^2\right)B = 0$$

To solve these equations, we have from second equation

$$B = \frac{-AR\gamma}{\frac{1}{C} - L\gamma^2}$$

Substituting in the first equation and simplifying, we obtain

$$A = \frac{E_o\left(L\gamma - \frac{1}{C\gamma}\right)}{-\gamma\left[L^2\gamma^2 - \frac{2L}{C} + \frac{1}{C^2\gamma^2} + R^2\right]}$$

Using this value of A and simplifying yields

$$B = \frac{E_o R}{-\gamma\left[L^2\gamma^2 - \frac{2L}{C} + \frac{1}{C^2\gamma^2} + R^2\right]}$$

If we use the notations

$$X = L\gamma - \frac{1}{C\gamma} \text{ then } X^2 = L^2\gamma^2 - \frac{2L}{C} + \frac{1}{C^2\gamma^2}$$

$$Z = \sqrt{X^2 + R^2} \text{ then } Z^2 = L^2\gamma^2 - \frac{2L}{C} + \frac{1}{C^2\gamma^2} + R^2$$

Then $$A = \frac{E_o X}{-\gamma Z^2}, \qquad B = \frac{E_o R}{-\gamma Z^2}$$

Therefore, the steady-state charge is given by

$$q_p(t) = -\frac{E_o X}{\gamma Z^2}\sin\gamma t - \frac{E_o R}{\gamma Z^2}\cos\gamma t$$

So that the steady-state current is given by

$$I_p(t) = \frac{E_o}{Z}\left(\frac{R}{Z}\sin\gamma t - \frac{X}{Z}\cos\gamma t\right)$$

Note that

- The quantity $X = L\gamma - \frac{1}{C\gamma}$ is called the ***reactance*** of the circuit.
- The quantity $Z = \sqrt{X^2 + R^2}$ is called ***impedance*** of the circuit.
- Both the reactance and the impedance are measured in ohms.

Exercise

1. A 16-lb weight stretches a spring 8/3 ft. Initially the weight starts from rest 2-ft below the equilibrium position and the subsequent motion takes place in a medium that offers a damping force numerically equal to ½ the instantaneous velocity. Find the equation of motion, if the weight is driven by an external force equal to $f(t) = 10\cos 3t$.
2. A mass 1-slug, when attached to a spring, stretches it 2-ft and then comes to rest in the equilibrium position. Starting at $t = 0$, an external force equal to $f(t) = 8\sin 4t$ is applied to the system. Find the equation of motion if the surrounding medium offers a damping force numerically equal to 8 times the instantaneous velocity.
3. In problem 2 determine the equation of motion if the external force is $f(t) = e^{-t}\sin 4t$. Analyze the displacements for $t \to \infty$.
4. When a mass of 2 kilograms is attached to a spring whose constant is 32 N/m, it comes to rest in the equilibrium position. Starting at $t = 0$, a force equal to $f(t) = 68e^{-2t}\cos 4t$ is applied to the system. Find the equation of motion in the absence of damping.
5. In problem 4 write the equation of motion in the form
$$x(t) = A\sin(\omega t + \varphi) + Be^{-2t}\sin(4t + \theta).$$
What is the amplitude of vibrations after a very long time?
6. Find the charge on the capacitor and the current in an LC series circuit. Where $L = 1$ Henry, $C = \frac{1}{16}$ farad, $E(t) = 60$ volts. Assuming that $q(0) = 0$ and $i(0) = 0$.
7. Determine whether an LRC series circuit, where $L = 3$ Henrys, $R = 10$ ohms, $C = 0.1$farad is over-damped, critically damped or under-damped.
8. Find the charge on the capacitor in an LRC series circuit when $L = 1/4$ Henry, $R = 20$ ohms, $C = 1/300$ farad, $E(t) = 0$ volts, $q(0) = 4$ coulombsand $i(0) = 0$ amperes
Is the charge on the capacitor ever equal to zero?

Find the charge on the capacitor and the current in the given LRC series circuit. Find the maximum charge on the capacitor.

9. $L = 5/3$ henrys, $R = 10$ ohms, $C = 1/30$ farad, $E(t) = 300$ volts, $q(0) = 0$ coulombs, $i(0) = 0$ amperes
10. $L = 1$ henry, $R = 100$ ohms, $C = 0.0004$ farad, $E(t) = 30$ volts, $q(0) = 0$ coulombs, $i(0) = 2$ amperes

Virtual
University
ORANGE
MONKEY TEACH

Lecture 26

Differential Equations with Variable Coefficients

So far we have been solving Linear Differential Equations with constant coefficients.

We will now discuss the Differential Equations with non-constant (variable) coefficients.

These equations normally arise in applications such as temperature or potential u in the region bounded between two concentric spheres. Then under some circumstances we have to solve the differential equation:

$$r\frac{d^2u}{dr^2}+2\frac{du}{dr}=0$$

where the variable $r>0$ represents the radial distance measured outward from the center of the spheres.

Differential equations with variable coefficients such as

$$x^2y'+xy'+(x^2-v^2)y=0$$

$$(1-x^2)y'-2xy'+n(n+1)y=0$$

and $y'-2xy'+2ny=0$

occur in applications ranging from potential problems, temperature distributions and vibration phenomena to quantum mechanics.

The differential equations with variable coefficients cannot be solved so easily.

Cauchy- Euler Equation:

Any linear differential equation of the form

$$a_n x^n\frac{d^n y}{dx^n}+a_{n-1}x^{n-1}\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1x\frac{dy}{dx}+a_0y=g(x)$$

where $a_n, a_{n-1},\cdots,a_0$ are constants, is said to be a *Cauchy-Euler* equation or equi-dimensional equation. The degree of each monomial coefficient matches the order of differentiation i.e x^n is the coefficient of nth derivative of y, x^{n-1} of (n-1)th derivative of y, etc.

For convenience we consider a homogeneous second-order differential equation

$$ax^2\frac{d^2y}{dx^2}+bx\frac{dy}{dx}+cy=0,\qquad x\neq 0$$

The solution of higher-order equations follows analogously.

Also, we can solve the non-homogeneous equation

$$ax^2\frac{d^2y}{dx^2}+bx\frac{dy}{dx}+cy=g(x),\qquad x\neq 0$$

by variation of parameters after finding the complementary function $y_c(x)$.

We find the general solution on the interval $(0,\infty)$ and the solution on $(0,-\infty)$ can be obtained by substituting $t=-x$ in the differential equation.

Method of Solution:

We try a solution of the form $y=x^m$, where m is to be determined. The first and second derivatives are, respectively,

$$\frac{dy}{dx}=mx^{m-1} \quad \text{and} \quad \frac{d^2y}{dx^2}=m(m-1)x^{m-2}$$

Consequently the differential equation becomes

$$ax^2\frac{d^2y}{dx^2}+bx\frac{dy}{dx}+cy=ax^2\cdot m(m-1)x^{m-2}+bx\cdot mx^{m-1}+cx^m$$
$$=am(m-1)x^m+bmx^m+cx^m$$
$$=x^m(am(m-1)+bm+c)$$

Thus $y=x^m$ is a solution of the differential equation whenever m is a solution of the ***auxiliary equation***

$$(am(m-1)+bm+c)=0 \text{ or } am^2+(b-a)m+c=0$$

The solution of the differential equation depends on the roots of the AE.

Case-I: Distinct Real Roots

Let m_1 and m_2 denote the real roots of the auxiliary equation such that $m_1\neq m_2$. Then

$y=x^{m_1}$ and $y=x^{m_2}$ form a fundamental set of solutions.

Hence the general solution is

$$y=c_1x^{m_1}+c_2x^{m_2}.$$

Example 1

Solve $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = 0$

Solution:

Suppose that $y = x^m$, then

$$\frac{dy}{dx} = mx^{m-1} \quad , \quad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Now substituting in the differential equation, we get:

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = x^2 \cdot m(m-1)x^{m-2} - 2x \cdot mx^{m-1} - 4x^m$$

$$= x^m(m(m-1) - 2m - 4)$$

$$x^m(m^2 - 3m - 4) = 0 \quad \text{if } m^2 - 3m - 4 = 0$$

This implies $m_1 = -1, m_2 = 4$; roots are real and distinct.

So the solution is $y = c_1x^{-1} + c_2x^4$.

Case II: Repeated Real Roots

If the roots of the auxiliary equation are repeated, that is, then we obtain only one solution $y = x^{m_1}$.

To construct a second solution y_2, we first write the *Cauchy-Euler equation* in the form

$$\frac{d^2y}{dx^2} + \frac{b}{ax}\frac{dy}{dx} + \frac{c}{ax^2}y = 0$$

Comparing with

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

We make the identification $P(x) = \frac{b}{ax}$. Thus

$$y_2 = x^{m_1}\int \frac{e^{\int \frac{b}{ax}dx}}{(x^{m_1})^2}dx$$

$$= x^{m_1}\int \frac{e^{-(\frac{b}{a})\ln x}}{x^{2m_1}}dx$$

$$= x^{m1}\int x^{-\frac{b}{a}}.x^{-2m1}dx$$

Since roots of the AE $am^2 + (b-a)m + c = 0$ are equal, therefore discriminant is zero

$$\text{i.e } m_1 = -\frac{(b-a)}{2a} \quad \text{or} \quad -2m_1 = +\frac{(b-a)}{a}$$

$$y_2 = x^{m_1}\int x^{\frac{-b}{a}}.x^{\frac{b-a}{a}}\,dx$$

$$y_2 = x^{m_1}\int \frac{dx}{x} = x^{m_1}\ln x.$$

The general solution is then

$$y = c_1 x^{m_1} + c_2 x^{m_1}\ln x$$

Example 2

Solve $4x^2\dfrac{d^2y}{dx^2} + 8x\dfrac{dy}{dx} + y = 0.$

Solution:

Suppose that $y = x^m$, then

$$\frac{dy}{dx} = mx^{m-1}, \quad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}.$$

Substituting in the differential equation, we get:

$$4x^2\frac{d^2y}{dx^2} + 8x\frac{dy}{dx} + y = x^m(4m(m-1) + 8m + 1) = x^m(4m^2 + 4m + 1) = 0$$

if $4m^2 + 4m + 1 = 0$ or $(2m+1)^2 = 0$.

Since $m_1 = -\dfrac{1}{2}$, the general solution is

$$y = c_1 x^{-\frac{1}{2}} + c_2 x^{-\frac{1}{2}}\ln x.$$

For higher order equations, if m_1 is a root of multiplicity k, then it can be shown that: $x^{m_1}, x^{m_1}\ln x, x^{m_1}(\ln x)^2, \cdots, x^{m_1}(\ln x)^{k-1}$ are k linearly independent solutions.

Correspondingly, the general solution of the differential equation must then contain a linear combination of these k solutions.

Case III Conjugate Complex Roots

If the roots of the auxiliary equation are the conjugate pair

$$m_1 = \alpha + i\beta, \quad m_2 = \alpha - i\beta$$

where and $\beta > 0$ are real, then the solution is

$$y = c_1 x^{\alpha + i\beta} + c_2 x^{\alpha - i\beta}.$$

But, as in the case of equations with constant coefficients, when the roots of the auxiliary equation are complex, we wish to write the solution in terms of real functions only. We note the identity

$$x^{i\beta} = (e^{\ln x})^{i\beta} = e^{i\beta \ln x},$$

which, by *Euler's formula*, is the same as

$$x^{i\beta} = \cos(\beta \ln x) + i \sin(\beta \ln x)$$

Similarly we have

$$x^{-i\beta} = \cos(\beta \ln x) - i \sin(\beta \ln x)$$

Adding and subtracting last two results yields, respectively,

$$x^{i\beta} + x^{-i\beta} = 2\cos(\beta \ln x)$$

and $$x^{i\beta} - x^{-i\beta} = 2i \sin(\beta \ln x)$$

From the fact that $y = c_1 x^{\alpha + i\beta} + c_2 x^{\alpha - i\beta}$ is the solution of $ax^2 y'' + bxy' + cy = 0$, for any values of constants c_1 and c_2, we see that

$$y_1 = x^{\alpha}(x^{i\beta} + x^{-i\beta}), \quad (c_1 = c_2 = 1)$$

$$y_2 = x^{\alpha}(x^{i\beta} - x^{-i\beta}), \quad (c_1 = 1, c_2 = -1)$$

or $$y_1 = 2x^{\alpha}(\cos(\beta \ln x))$$

$$y_2 = 2x^{\alpha}(\sin(\beta \ln x))$$ are also solutions.

Since $W(x^{\alpha} \cos(\beta \ln x), x^{\alpha} \sin(\beta \ln x)) = \beta x^{2\alpha - 1} \neq 0; \beta > 0$, on the interval $(0, \infty)$, we conclude that

$$y_1 = x^{\alpha} \cos(\beta \ln x) \text{ and } y_2 = x^{\alpha} \sin(\beta \ln x)$$

constitute a fundamental set of real solutions of the differential equation.

Hence the general solution is

$$y_1 = x^{\alpha}[c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$$

Example 3

Solve the initial value problem

$$x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + 3y = 0, \qquad y(1) = 1, y'(1) = -5$$

Solution:

Let us suppose that: $y = x^m$, then $\frac{dy}{dx} = mx^{m-1}$ and $\frac{d^2y}{dx^2} = m(m-1)x^{m-2}$.

$$x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + 3y = x^m(m(m-1) + 3m + 3) = x^m(m^2 + 2m + 3) = 0$$

if $m^2 + 2m + 3 = 0$.

From the quadratic formula we find that $m_1 = -1 + \sqrt{2}i$ and $m_1 = -1 - \sqrt{2}i$. If we make the identifications $\alpha = -1$ and $\beta = \sqrt{2}$, so the general solution of the differential equation is

$$y_1 = x^{-1}[c_1 \cos(\sqrt{2} \ln x) + c_2 \sin(\sqrt{2} \ln x)].$$

By applying the conditions $y(1) = 1, y'(1) = -5$, we find that

$$c_1 = 1 \quad \text{and} \quad c_2 = -2\sqrt{2}.$$

Thus the solution to the initial value problem is

$$y_1 = x^{-1}[\cos(\sqrt{2} \ln x) - 2\sqrt{2} \sin(\sqrt{2} \ln x)]$$

Example 4

Solve the *third-order Cauchy-Euler differential equation*

$$x^3 \frac{d^3y}{dx^3} + 5x^2 \frac{d^2y}{dx^2} + 7x \frac{dy}{dx} + 8y = 0,$$

Solution

The first three derivative of $y = x^m$ are

$$\frac{dy}{dx} = mx^{m-1}, \quad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}, \quad \frac{d^3y}{dx^3} = m(m-1)(m-2)x^{m-3},$$

so the given differential equation becomes

$$x^3 \frac{d^3y}{dx^3} + 5x^2 \frac{d^2y}{dx^2} + 7x \frac{dy}{dx} + 8y = x^3 m(m-1)(m-2)x^{m-3} + 5x^2 m(m-1)x^{m-2} + 7xmx^{m-1} + 8x^m,$$

$$= x^m (m(m-1)(m-2) + 5m(m-1) + 7m + 8)$$
$$= x^m (m^3 + 2m^2 + 4m + 8)$$

In this case we see that $y = x^m$ is a solution of the differential equation, provided m is a root of the cubic equation

$$m^3 + 2m^2 + 4m + 8 = 0$$
or $(m+2)(m^2+4) = 0$

The roots are: $m_1 = -2, m_2 = 2i, m_3 = -2i$.

Hence the general solution is

$$y_1 = c_1 x^{-2} + c_2 \cos(2 \ln x) + c_3 \sin(2 \ln x)$$

Example 5

Solve the non-homogeneous equation

$$x^2y'' - 3xy' + 3y = 2x^4e^x$$

Solution

Put $y = x^m$

$$\frac{dy}{dx} = mx^{m-1}, \quad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Therefore we get the auxiliary equation,

$$m(m-1) - 3m + 3 = 0 \text{ or } (m-1)(m-3) = 0 \text{ or } m = 1,3$$

Thus $y_c = c_1x + c_2x^3$

Before using variation of parameters to find the particular solution $y_p = u_1y_1 + u_2y_2$, recall that the formulas $u'_1 = \frac{W_1}{W}$ and $u'_2 = \frac{W_2}{W}$, where $W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}$, $W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$, and W is the Wronskian of y_1 and y_2, were derived under the assumption that the differential equation has been put into special form . $y' + P(x)y' + Q(x)y = f(x)$

Therefore we divide the given equation by x^2, and form $y' - \frac{3}{x}y' + \frac{3}{x^2}y = 2x^2e^x$

we make the identification $f(x) = 2x^2e^x$. Now with $y_1 = x$, $y_2 = x^2$, and

$$W = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix} = 2x^3, \quad W_1 = \begin{vmatrix} 0 & x^3 \\ 2x^2e^x & 3x^2 \end{vmatrix} = -2x^5e^x, \quad W_2 = \begin{vmatrix} x & x \\ 1 & 2x^2e^x \end{vmatrix} = 2x^3e^x$$

we find

$$u'_1 = \frac{2x^5e^x}{2x^3} = -x^2e^x \text{ and } \quad u'_2 = \frac{2x^3e^x}{2x^3} = e^x$$

$$u_1 = -x^2e^x + 2xe^x - 2e^x \text{ and } u_2 = e^x .$$

Hence $y_p = u_1y_1 + u_2y_2$

$$= (-x^2e^x + 2xe^x - 2e^x)x + e^x x^3 = 2x^2e^x - 2xe^x$$

Finally we have $y = y_c + y_p = c_1x + c_2x^3 + 2x^2e^x - 2xe^x$

Exercises

1. $4x^2y'' + y = 0$
2. $xy'' - y' = 0$

3. $x^2y'' + 5xy' + 3y = 0$
4. $4x^2y'' + 4xy' - y = 0$
5. $x^2y'' - 7xy' + 41y = 0$
6. $x^3\frac{d^3y}{dx^3} - 2x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} - 4y = 0$
7. $x^4\frac{d^4y}{dx^4} + 6x^3\frac{d^3y}{dx^3} + 9x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0$
8. $x^2y'' - 5xy' + 8y = 0; y(1) = 0, y'(1) = 4$
9. $x^2y'' - 2xy' + 2y = x^3 \ln x$
10. $x^3\frac{d^3y}{dx^3} - 3x^2\frac{d^2y}{dx^2} + 6x\frac{dy}{dx} - 6y = 3 + \ln x^3$

Lecture 27

Cauchy-Euler Equation: Alternative Method of Solution

We reduce any Cauchy-Euler differential equation to a differential equation with constant coefficients through the substitution

$$x = e^t \quad \text{or} \quad t = \ln x$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx} = \frac{1}{x}\cdot\frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\cdot\frac{dy}{dt}\right) = \frac{1}{x}\cdot\frac{d}{dx}\left(\frac{dy}{dt}\right) - \frac{1}{x^2}\cdot\frac{dy}{dt}$$

or $$\frac{d^2y}{dx^2} = \frac{1}{x}\cdot\frac{d}{dt}\left(\frac{dy}{dt}\right)\frac{dt}{dx} - \frac{1}{x^2}\cdot\frac{dy}{dt}$$

or $$\frac{d^2y}{dx^2} = \frac{1}{x^2}\cdot\frac{d^2y}{dt^2} - \frac{1}{x^2}\cdot\frac{dy}{dt}$$

Therefore $$x\frac{dy}{dx} = \frac{dy}{dt}, \; x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}$$

Now introduce the notation

$$D = \frac{d}{dx}, D^2 = \frac{d^2}{dx^2}, \text{ etc.}$$

and $$\Delta = \frac{d}{dt}, \Delta^2 = \frac{d^2}{dt^2}, \quad \text{etc.}$$

Therefore, we have

$$xD = \Delta$$

$$x^2D^2 = \Delta^2 - \Delta = \Delta(\Delta - 1)$$

Similarly

$$x^3D^3 = \Delta(\Delta - 1)(\Delta - 2)$$

$$x^4D^4 = \Delta(\Delta - 1)(\Delta - 2)(\Delta - 3) \quad \text{so on so forth.}$$

This substitution in a given *Cauchy-Euler* differential equation will reduce it into a differential equation with constant coefficients.

At this stage we suppose $y = e^{mt}$ to obtain an auxiliary equation and write the solution in terms of y and t. We then go back to x through $x = e^t$.

Example 1

Solve $x^2 \frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = 0$

Solution

The given differential equation can be written as

$$(x^2D^2 - 2xD - 4)y = 0$$

With the substitution $x = e^t$ or $t = \ln x$, we obtain

$$xD = \Delta, \quad x^2D^2 = \Delta(\Delta - 1)$$

Therefore the equation becomes:

$$[\Delta(\Delta - 1) - 2\Delta - 4]y = 0$$

or $$(\Delta^2 - 3\Delta - 4)y = 0$$

or $$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} - 4y = 0$$

Now substitute: $y = e^{mt}$ then $\frac{dy}{dt} = me^{mt}$, $\frac{d^2y}{dt^2} = m^2e^{mt}$

Thus $(m^2 - 3m - 4)e^{mt} = 0$ or $m^2 - 3m - 4 = 0$, which is the auxiliary equation.

$$(m+1)(m-4) = 0 \qquad m = -1, 4$$

The roots of the auxiliary equation are distinct and real, so the solution is

$$y = c_1e^{-t} + c_2e^{4t}$$

But $x = e^t$, therefore the answer will be

$$y = c_1x^{-1} + c_2x^4$$

Example 2

Solve $4x^2\frac{d^2y}{dx^2} + 8x\frac{dy}{dx} + y = 0$

Solution

The differential equation can be written as:

$$(4x^2D^2 + 8xD + 1)y = 0$$

Where $D = \frac{d}{dx}, D^2 = \frac{d^2}{dx^2}$

Now with the substitution $x = e^t$ or $t = \ln x$, $xD = \Delta$, $x^2D^2 = \Delta(\Delta - 1)$ where $\Delta = \frac{d}{dt}$

The equation becomes:

$$(4\Delta(\Delta - 1) + 8\Delta + 1)y = 0 \qquad \text{or } (4\Delta^2 + 4\Delta + 1)y = 0$$

$$4\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + y = 0$$

© Copyright Virtual University of Pakistan

Now substituting $y = e^{mt}$ then $\frac{dy}{dt} = me^{mt}$, $\frac{d^2y}{dt^2} = m^2e^{mt}$, we get

$$(4m^2 + 4m + 1)e^{mt} = 0$$

or $4m^2 + 4m + 1 = 0$ or $(2m+1)^2 = 0$

or $m = -\frac{1}{2}, -\frac{1}{2}$; the roots are real but repeated.

Therefore the solution is

$$y = (c_1 + c_2 t)e^{-\frac{1}{2}t}$$

or $y = (c_1 + c_2 \ln x)x^{-\frac{1}{2}}$

i-e $y = c_1 x^{-\frac{1}{2}} + c_2 x^{-\frac{1}{2}} \ln x$

Example 3

Solve the initial value problem

$x^2 \frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + 3y = 0$, $y(1) = -5$, $y'(1) = -$

Solution

The given differential can be written as:

$$(x^2D^2 + 3xD + 3)y = 0$$

Now with the substitution $x = e^t$ or $t = \ln x$ we have:

$$xD = \Delta \ , x^2D^2 = \Delta(\Delta - 1)$$

Thus the equation becomes:

$$(\Delta(\Delta - 1) + 3\Delta + 3)y = 0 \quad \text{or} \quad (\Delta^2 + 2\Delta + 3)y = 0$$

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 3y = 0$$

Put $y = e^{mt}$ then the A.E. equation is:

or $m^2 + 2m + 3 = 0$

or $m = \frac{-2 \pm \sqrt{4-12}}{2} = -1 \pm i\sqrt{2}$

So that solution is:

$$y = e^{-t}(c_1 \cos\sqrt{2}t + c_2 \sin\sqrt{2}t)$$

or $y = x^{-1}(c_1 \cos\sqrt{2}\ln x + c_2 \sin\sqrt{2}\ln x)$

© Copyright Virtual University of Pakistan

Now $y(1)=1$ gives, $1=(c_1\cos 0+c_2\sin 0) \Rightarrow c_1=1$

$y'=-x^{-2}(c_1\cos\sqrt{2}\ln x+c_2\sin\sqrt{2}\ln x)+x^{-2}(-\sqrt{2}c_1\sin\sqrt{2}\ln x+\sqrt{2}c_2\cos\sqrt{2}\ln x)$

$\therefore y'(1)=-5$ gives: $-5=-[c_1+0]+[\sqrt{2}c_2]$ or $\sqrt{2}c_2=c_1-5=-4$, $c_2=\dfrac{-4}{\sqrt{2}}=-2\sqrt{2}$

Hence solution of the IVP is:

$$y=x^{-1}[\cos(\sqrt{2}\ln x)-2\sqrt{2}\sin(\sqrt{2}\ln x)].$$

Example 4

Solve $x^3\dfrac{d^3y}{dx^3}+5x^2\dfrac{d^2y}{dx^2}+7x\dfrac{dy}{dx}+8y=0$

Solution

The given differential equation can be written as:

$$(x^3D^3+5x^2D^2+7xD+8)y=0$$

Now with the substitution $x=e^t$ or $t=\ln x$ we have:

$$xD=\Delta,\ x^2D^2=\Delta(\Delta-1),\ x^3D^3=\Delta(\Delta-1)(\Delta-2)$$

So the equation becomes:

$$(\Delta(\Delta-1)(\Delta-2)+5\Delta(\Delta-1)+7\Delta+8)y=0$$

or $(\Delta^3-3\Delta^2+2\Delta+5\Delta^2-5\Delta+7\Delta+8)y=0$

or $(\Delta^3+2\Delta^2+4\Delta+8)y=0$

or $\dfrac{d^3y}{dt^3}+2\dfrac{d^2y}{dt^2}+4\dfrac{dy}{dt}+8y=0$

Put $y=e^{mt}$, then the auxiliary equation is:

$$m^3+2m^2+4m+8=0$$

or $(m^2+4)(m+2)=0$

$m=-2,\ \text{or}\pm 2i$

So the solution is:

$$y=c_1e^{-2t}+c_2\cos 2t+c_3\sin 2t$$

or $$y=c_1x^{-2}+c_2\cos(2\ln x)+c_3\sin(2\ln x)$$

© Copyright Virtual University of Pakistan

Example 5

Solve the non-homogeneous differential equation

$$x^2y'' - 3xy' + 3y = 2x^4e^x$$

Solution

First consider the associated homogeneous differential equation.

$$x^2y'' - 3xy' + 3y = 0$$

With the notation $\frac{d}{dx} = D$, $\frac{d^2}{dx^2} = D^2$, the differential equation becomes:

$$(x^2D^2 - 3xD + 3)y = 0$$

With the substitution $x = e^t$ or $t = \ln x$, we have:

$$xD = \Delta, \quad x^2D^2 = \Delta(\Delta - 1)$$

So the homogeneous differential equation becomes:

$$[\Delta(\Delta - 1) - 3\Delta + 3]y = 0$$

$$(\Delta^2 - 4\Delta + 3)y = 0$$

or $$\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 3y = 0$$

Put $y = e^{mt}$ then the AE is:

$$m^2 - 4m + 3 = 0 \text{ or } (m-3)(m-1) = 0, \text{ or } m = 1,3$$

$$\therefore y_c = c_1e^t + c_2e^{3t}, \text{ as } x = e^t$$

$$y_c = c_1x + c_2x^3$$

For y_p we write the differential equation as:

$$y'' - \frac{3}{x}y' + \frac{3}{x^2}y = 2x^2e^x$$

$y_p = u_1x + u_2x^3$, where u_1 and u_2 are functions given by

$$u_1' = \frac{W_1}{W}, \quad u_2' = \frac{W_2}{W},$$

with

$$W = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix} = 2x^3, \; W_1 = \begin{vmatrix} 0 & x^3 \\ 2x^2e^x & 3x^2 \end{vmatrix} = -2x^5e^x \text{ and}$$

$$W_2 = \begin{vmatrix} x & 0 \\ 1 & 2x^2e^x \end{vmatrix} = 2x^3e^x$$

© Copyright Virtual University of Pakistan

So that $u'_1 = \frac{2x^5 e^x}{2x^3} = -x^2 e^x$ and $u'_2 = \frac{2x^3 e^x}{2x^3} = e^x$

$$\therefore u_1 = -\int x^2 e^x dx = -[x^2 e^x - 2\int xe^x dx]$$
$$= -x^2 e^x + 2[xe^x - \int e^x dx]$$
$$= -x^2 e^x + 2xe^x - 2e^x$$

and $u_2 = \int e^x dx = e^x$.

Therefore

$$y_p = x(-x^2 e^x + 2xe^x - 2e^x) + x^3 e^x = 2x^2 e^x - 2xe^x$$

Hence the general solution is:

$$y = y_c + y_p$$
$$y = c_1 x + c_2 x^3 + 2x^2 e^x - 2xe^x$$

Example 6

Solve $x^2 \frac{d^2 y}{dx^2} - x\frac{dy}{dx} + y = \ln x$

Solution

Consider the associated homogeneous differential equation.

$$x^2 \frac{d^2 y}{dx^2} - x\frac{dy}{dx} + y = 0$$

or $(x^2 D^2 - xD + 1)y = 0$

With the substitution $x = e^t$, we have:

$$xD = \Delta, \quad x^2 D^2 = \Delta(\Delta - 1)$$

So the homogeneous differential equation becomes:

$$[\Delta(\Delta - 1) - \Delta + 1]y = 0$$
$$(\Delta^2 - 2\Delta + 1)y = 0$$

or $\frac{d^2 y}{dt^2} - 2\frac{dy}{dt} + y = 0$

Putting $y = e^{mt}$, we get the auxiliary equation as:

$m^2 - 2m + 1 = 0$ or $(m-1)^2 = 0$ or $m = 1,1$

© Copyright Virtual University of Pakistan

$$\therefore y_c = c_1 e^t + c_2 te^t$$

or $y_c = c_1 x + c_2 x \ln x$.

Now the non-homogeneous differential equation becomes:

$$\frac{d^2y}{dt^2} - 2\frac{dy}{dt} + y = t$$

By the method of undetermined coefficients we try a particular solution of the form $y_p = A + Bt$. This assumption leads to

$-2B + A + Bt = t$ so that $A=2$ and $B=1$.

Using $y = y_c + y_p$, we get

$$y_c = c_1 e^t + c_2 te^t + 2 + t ;$$

So the general solution of the original differential equation on the interval $(0, \infty)$ is

$$y_c = c_1 x + c_2 x \ln x + 2 + \ln x$$



© Copyright Virtual University of Pakistan

Exercises

Solve using $x = e^t$

1. $x \dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = 0$
2. $x^2 \dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + 4y = 0$
3. $x^2 \dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} - 2y = 0$
4. $25x^2 \dfrac{d^2y}{dx^2} + 25x\dfrac{dy}{dx} + y = 0$
5. $3x^2 \dfrac{d^2y}{dx^2} + 6x\dfrac{dy}{dx} + y = 0$
6. $x\dfrac{d^4y}{dx^4} + 6\dfrac{d^3y}{dx^3} = 0$
7. $x^2 \dfrac{d^2y}{dx^2} + 3x\dfrac{dy}{dx} = 0, y(1) = 0, y'(0) = 4$
8. $x^2 \dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0, y(1) = 1, y'(1) = 2$
9. $x^2 \dfrac{d^2y}{dx^2} + 10x\dfrac{dy}{dx} + 8y = x^2$
10. $x^2 \dfrac{d^2y}{dx^2} + 9x\dfrac{dy}{dx} - 20y = \dfrac{5}{x^3}$

© Copyright Virtual University of Pakistan

Lecture 28

Power Series: An Introduction

- A standard technique for solving linear differential equations with variable coefficients is to find a solution as an infinite series. Often this solution can be found in the form of a power series.

- Therefore, in this lecture we discuss some of the more important facts about power series.

- However, for an in-depth review of the infinite series concept one should consult a standard calculus text.

Power Series

A power series in $(x-a)$ is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \cdots.$$

The coefficients c_0, c_1, c_2,... and a are constants and x represents a variable. In this discussion we will only be concerned with the cases where the coefficients, x and a are real numbers. The number a is known as the centre of the power series.

Example 1

The infinite series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} x^n = x - \frac{x^2}{2^2} + \frac{x^3}{3^2} - \cdots$$

is a power series in x. This series is centered at zero.

Convergence and Divergence

- If we choose a specified value of the variable x then the power series becomes an infinite series of constants. If, for the given x, the sum of terms of the power series equals a finite real number, then the series is said to be convergent at x.

- A power series that is not convergent is said to be a divergent series. This means that the sum of terms of a divergent power series is not equal to a finite real number.

© Copyright Virtual University of Pakistan

Example 2

(a) Consider the power series

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Since for $x = 1$ the series become

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots = e$$

Therefore, the power series converges $x = 1$ to the number e

(b) Consider the power series

$$\sum_{n=0}^{\infty} n!(x+2)^n = 1 + (x+2) + 2!(x+2)^2 + 3!(x+2)^3 + \cdots$$

The series diverges $\forall\ x$, except at $x = -2$. For instance, if we take $x = 1$ then the series becomes

$$\sum_{n=0}^{\infty} n!(x+2)^n = 1 + 3 + 18 + \cdots$$

Clearly the sum of all terms on right hand side is not a finite number. Therefore, the series is divergent at $x = 1$. Similarly, we can see its divergence at all other values of $x \neq -2$

The Ratio Test

To determine for which values of x a power series is convergent, one can often use the Ratio Test. The Ratio test states that if

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} c_n (x-a)^n$$

is a power series and

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| |x - a| = L$$

Then:

- The power series converges absolutely for those values of x for which $L < 1$.
- The power series diverges for those values of x for which $L > 1$ or $L = \infty$.
- The test is inconclusive for those values of x for which $L = 1$.

© Copyright Virtual University of Pakistan

Interval of Convergence

The set of all real values of x for which a power series

$$\sum_{n=0}^{\infty} c_n \left(x-a \right)^n$$

converges is known as the interval of convergence of the power series.

Radius of Convergence

Consider a power series

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

Then exactly one of the following three possibilities is true:

- The series converges only at its center $x = a$.
- The series converges for all values of x.
- There is a number $R > 0$ such that the series converges absolutely $\forall x$ satisfying $|x-a| < R$ and diverges for $|x-a| > R$. This means that the series converges for $x \in (a-R, a+R)$ and diverges out side this interval.

The number R is called the radius of convergence of the power series. If first possibility holds then $R = 0$ and in case of 2nd possibility we write $R = \infty$.

From the Ratio test we can clearly see that the radius of convergence is given by

$$R = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+1}} \right|$$

provided the limit exists.

Convergence at an Endpoint

If the radius of convergence of a power series is $R > 0$, then the interval of convergence of the series is one of the following

$$(a-R, a+R),\ (a-R, a+R],\ [a-R, a+R),\ \left[a-R, a+R \right]$$

To determine which of these intervals is the interval of convergence, we must conduct separate investigations for the numbers $x = a - R \, and \, x = a + R$.

Example 3

Consider the power series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} x^n$$

Then

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{x^n} \right|$$

© Copyright Virtual University of Pakistan

or $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\sqrt{\frac{n}{n+1}}\cdot x\right| = \lim_{n\to\infty}\left|\sqrt{\frac{n}{n+1}}\right| |x| = |x|$$

Therefore, it follows from the Ratio Test that the power series converges absolutely for those values of x which satisfy

$$|x| < 1$$

This means that the power series converges if x belongs to the interval

$$(-1,1)$$

The series diverges outside this interval i.e. when $x > 1$ or $x < -1$. The convergence of the power series at the numbers 1 and -1 must be investigated separately by substituting into the power series.

a) When we substitute $x = 1$, we obtain

$$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}(1)^n = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} + \cdots$$

which is a divergent p-series, with $p = \frac{1}{2}$.

b) When we substitute $x = -1$, we obtain

$$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}(-1)^n = -1 + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} + \cdots + \frac{(-1)^n}{\sqrt{n}} + \cdots$$

which converges, by alternating series test.

Hence, the interval of convergence of the power series is $[-1,1)$. This means that the series is convergent for those vales of x which satisfy

$$-1 \le x < 1$$

Example 4

Find the interval of convergence of the power series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty}\frac{(x-3)^n}{2^n \cdot n}$$

Solution

The power series is centered at 3 and the radius of convergence of the series is

$$R = \lim_{n\to\infty}\frac{2^{n+1}(n+1)}{2^n \cdot n} = 2$$

Hence, the series converges absolutely for those values of x which satisfy the inequality

$$|x-3| < 2 \Rightarrow 1 < x < 5$$

(a) At the left endpoint we substitute $x = 1$ in the given power series to obtain the series of constants:

© Copyright Virtual University of Pakistan

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

This series is convergent by the alternating series test.

(b) At the right endpoint we substitute $x = 5$ in the given series and obtain the following harmonic series of constants

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

Since a harmonic series is always divergent, the above power series is divergent.

Hence, the series the interval of convergence of the given power series is a half open and half closed interval $[1, 5)$.

Absolute Convergence

Within its interval of convergence a power series converges absolutely. In other words, the series of absolute values

$$\sum_{n=o}^{\infty} |c_n \| (x-a)^n|$$

converges for all values x in the interval of convergence.

A Power Series Represent Functions

A power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ determines a function f whose domain is the interval of convergence of the power series. Thus for all x in the interval of convergence, we write

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots$$

If a function is f is defined in this way, we say that $\sum_{n=0}^{\infty} c_n (x-a)^n$ is a power series representation for $f(x)$. We also say that f is represented by the power series

Theorem

Suppose that a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ has a radius of convergence $R > 0$ and for every x in the interval of convergence a function f is defined by

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots$$

Then

© Copyright Virtual University of Pakistan

- The function f is continuous, differentiable, and integrable on the interval $(a-R,\ a+R)$.
- Moreover, $f'(x)$ and $\int f(x)dx$ can be found from term-by-term differentiation and ntegration.

Therefore

$$f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \cdots = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$$

$$\int f(x)\,dx = C + c_0(x-a) + c_1\frac{(x-a)^2}{2} + c_2\frac{(x-a)^3}{3} + \cdots$$

$$= C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

The series obtained by differentiation and integration have same radius of convergence. However, the convergence at the end points $x = a-R$ and $x = a+R$ of the interval may change. This means that the interval of convergence may be different from the interval of convergence of the original series.

Example 5

Find a function f that is represented by the power series

$$1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots$$

Solution

The given power series is a geometric series whose common ratio is $r = -x$. Therefore, if $|x| < 1$ then the series converges and its sum is

$$S = \frac{a}{1-r} = \frac{1}{1+x}$$

Hence we can write

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots$$

This last expression is the power series representation for the function $f(x) = \dfrac{1}{1+x}$.

Series that are Identically Zero

If for all real numbers x in the interval of convergence, a power series is identically zero i.e.

$$\sum_{n=o}^{\infty} c_n (x-a)^n = 0, \qquad R > 0$$

Then all the coefficients in the power series are zero. Thus we can write

$$c_n = 0, \qquad \forall\ n = 0, 1, 2, \ldots$$

© Copyright Virtual University of Pakistan

Analytic at a Point

A function f is said to be analytic at point a if the function can be represented by power series in $(x-a)$ with a positive radius of convergence. The notion of analyticity at a point will be important in finding power series solution of a differential equation.

Example 6

Since the functions e^x, $\cos x$, and $\ln(1+x)$ can be represented by the power series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

Therefore, these functions are analytic at the point $x = 0$.

Arithmetic of Power Series

- Power series can be combined through the operations of addition, multiplication, and division.
- The procedure for addition, multiplication and division of power series is similar to the way in which polynomials are added, multiplied, and divided.
- Thus we add coefficients of like powers of x, use the distributive law and collect like terms, and perform long division.

Example 7

If both of the following power series converge for $|x| < R$

$$f(x) = \sum_{n=0}^{\infty} c_n x^n, \quad g(x) = \sum_{n=0}^{\infty} b_n x^n$$

Then

$$f(x) + g(x) = \sum_{n=0}^{\infty} (c_n + b_n) x^n$$

and

$$f(x) \cdot g(x) = c_0 b_0 + (c_0 b_1 + c_1 b_0) x + (c_0 b_2 + c_1 b_1 + c_2 b_0) x^2 + \cdots$$

© Copyright Virtual University of Pakistan

Lecture 29

Power Series: An Introduction

Example 8

Find the first four terms of a power series in x for the product $e^x \cos x$.

Solution

From calculus the Maclaurin series for e^x and $\cos x$ are, respectively,

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots.$$

Multiplying the two series and collecting the like terms yields

$$e^x \cos x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots\right)\left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots\right)$$

$$= 1 + (1)x + \left(-\frac{1}{2} + \frac{1}{2}\right)x^2 + \left(-\frac{1}{2} + \frac{1}{6}\right)x^3 + \left(\frac{1}{24} = \frac{1}{4} + \frac{1}{24}\right)x^4 + \cdots$$

$$= 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \cdots$$

The interval of convergence of the power series for both the functions e^x and $\cos x$ is $(-\infty, \infty)$. Consequently the interval of convergence of the power series for their product $e^x \cos x$ is also $(-\infty, \infty)$.

Example 9

Find the first four terms of a power series in x for the function $\sec x$.

Solution

We know that

$$\sec x = \frac{1}{\cos x}, \quad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots$$

© Copyright Virtual University of Pakistan

Therefore using long division, we have

$$1-\frac{x^2}{2}+\frac{x^4}{24}-\frac{x^6}{720}+\cdots \;\Big)\; 1$$

$$1+\frac{x^2}{2}+\frac{5x^4}{24}+\frac{61x^6}{720}+\cdots$$

$$1-\frac{x^2}{2}+\frac{x^4}{24}-\frac{x^6}{720}+\cdots$$

$$\frac{x^2}{2}-\frac{x^4}{24}+\frac{x^6}{720}-\cdots$$

$$\frac{x^2}{2}-\frac{x^4}{4}+\frac{x^6}{48}-\cdots$$

$$\frac{5x^4}{24}-\frac{7x^6}{360}+\cdots$$

$$\frac{5x^4}{24}-\frac{5x^6}{48}+\cdots$$

$$\frac{61x^6}{720}-\cdots$$

Hence, the power series for the function $f(x)=\sec x$ is

$$\sec x = 1+\frac{x^2}{2}+\frac{5x^4}{24}+\frac{61x^6}{720}+\cdots$$

The interval of convergence of this series is $(-\pi/2, \pi/2)$.

Note that

- The procedures illustrated in examples 2 and 3 are obviously tedious to do by hand.
- Therefore, problems of this sort can be done using a computer algebra system (CAS) such as Mathematica.
- When we type the command: Series$[Sec[x], \{x, 0, 8\}]$ and enter, the Mathematica immediately gives the result obtained in the above example.
- For finding power series solutions it is important that we become adept at simplifying the sum of two or more power series, each series expressed in summation (sigma) notation, to an expression with a single $\sum$. This often requires a shift of the summation indices.

© Copyright Virtual University of Pakistan

- In order to add any two power series, we must ensure that:

  (a) That summation indices in both series start with the same number.
  (b) That the powers of x in each of the power series be "in phase".

  Therefore, if one series starts with a multiple of, say, x to the first power, then the other series must also start with the same power of the same power of x.

Example 10

Write the following sum of two series as one power series

$$\sum_{n=1}^{\infty} 2nc_n x^{n-1} + \sum_{n=0}^{\infty} 6nc_n x^{n+1}$$

Solution

To write the given sum power series as one series, we write it as follows:

$$\sum_{n=1}^{\infty} 2nc_n x^{n-1} + \sum_{n=0}^{\infty} 6nc_n x^{n+1} = 2\cdot 1c_1 x^0 + \sum_{n=2}^{\infty} 2nc_n x^{n-1} + \sum_{n=0}^{\infty} 6nc_n x^{n+1}$$

The first series on right hand side starts with x^1 for $n=2$ and the second series also starts with x^1 for $n=0$. Both the series on the right side start with x^1.

To get the same summation index we are inspired by the exponents of x which is $n-1$ in the first series and $n+1$ in the second series. Therefore, we let

$$k = n-1, \quad k = n+1$$

in the first series and second series, respectively. So that the right side becomes:

$$2c_1 + \sum_{k=1}^{\infty} 2(k+1)c_{k+1} x^k + \sum_{k=1}^{\infty} 6(k-1)c_{k-1} x^k .$$

Recall that the summation index is a "dummy" variable. The fact that $k = n-1$ in one case and $k = n+1$ in the other should cause no confusion if you keep in mind that it is the *value* of the summation index that is important. In both cases k takes on the same successive values $1,2,3,\ldots$ for $n = 2,3,4,\ldots$ (for $k = n-1$) and $n = 0,1,2,\ldots$ (for $k = n+1$)

We are now in a position to add the two series in the given sum term by term:

$$\sum_{n=1}^{\infty} 2nc_n x^{n-1} + \sum_{n=0}^{\infty} 6nc_n x^{n+1} = 2c_1 + \sum_{k=1}^{\infty} \left[2(k+1)c_{k+1} + 6(k-1)c_{k-1}\right] x^k$$

If you are not convinced, then write out a few terms on both series of the last equation.

© Copyright Virtual University of Pakistan

Lecture 29

Power Series Solution of a Differential Equation

We know that the explicit solution of the linear first-order differential equation

$$\frac{dy}{dx} - 2xy = 0$$

is $y = e^{x^2}$

Also $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$

If we replace x by x^2 in the series representation of e^x, we can write the solution of the differential equation as

$$y = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

This last series converges for all real values of x. In other words, knowing the solution in advance, we were able to find an infinite series solution of the differential equation.

We now propose to obtain a **power series solution** of the differential equation directly; the method of attack is similar to the technique of undetermined coefficients.

Example 11

Find a solution of the differential equation

$$\frac{dy}{dx} - 2xy = 0$$

in the form of power series in x.

Solution

If we assume that a solution of the given equation exists in the form

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + \sum_{n=1}^{\infty} c_n x^n$$

The question is that: Can we determine coefficients c_n for which the power series converges to a function satisfying the differential equation? Now term-by-term differentiation of the proposed series solution gives

$$\frac{dy}{dx} = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

Using the last result and the assumed solution, we have

© Copyright Virtual University of Pakistan

$$\frac{dy}{dx} - 2xy = \sum_{n=1}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} 2c_n x^{n+1}$$

We would like to add the two series in this equation. To this end we write

$$\frac{dy}{dx} - 2xy = 1 \cdot c_1 x^0 + \sum_{n=2}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} 2c_n x^{n+1}$$

and then proceed as in the previous example by letting

$$k = n-1, \quad k = n+1$$

in the first and second series, respectively. Therefore, last equation becomes

$$\frac{dy}{dx} - 2xy = c_1 + \sum_{k=1}^{\infty} (k+1) c_{k+1} x^k - \sum_{k=1}^{\infty} 2c_{k-1} x^k$$

After we add the series term wise, it follows that

$$\frac{dy}{dx} - 2xy = c_1 + \sum_{k=1}^{\infty} \left[(k+1)c_{k+1} - 2c_{k-1} \right] x^k$$

Substituting in the given differential equation, we obtain

$$c_1 + \sum_{k=1}^{\infty} \left[(k+1) c_{k+1} - 2c_{k-1} \right] x^k = 0$$

In order to have this true, it is necessary that all the coefficients must be zero. This means that

$$c_1 = 0, \quad (k+1) c_{k+1} - 2c_{k-1} = 0, \qquad k = 1, 2, 3, \ldots$$

This equation provides a recurrence relation that determines the coefficient c_k. Since $k+1 \neq 0$ for all the indicated values of k, we can write as

$$c_{k+1} = \frac{2c_{k-1}}{k+1}$$

Iteration of this last formula then gives

$$k = 1, \qquad c_2 = \frac{2}{2} c_0 = c_0$$

$$k = 2, \qquad c_3 = \frac{2}{3} c_1 = 0$$

© Copyright Virtual University of Pakistan

$$k=3, \quad c_4 = \frac{2}{4}c_2 = \frac{1}{2}c_0 = \frac{1}{2!}c_0$$

$$k=4, \quad c_5 = \frac{2}{5}c_3 = 0$$

$$k=5, \quad c_6 = \frac{2}{6}c_4 = \frac{1}{3\cdot 2!}c_0 = \frac{1}{3!}c_0$$

$$k=6, \quad c_7 = \frac{2}{7}c_5 = 0$$

$$k=7, \quad c_8 = \frac{2}{8}c_6 = \frac{1}{4\cdot 3!}c_0 = \frac{1}{4!}c_0$$

and so on. Thus from the original assumption (7), we find

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + \cdots$$

$$= c_0 + 0 + c_0 x^2 + 0 + \frac{1}{2!}c_0 x^4 + 0 + \frac{1}{3!}c_0 x^6 + 0 + \cdots$$

$$= c_0\left[1 + x^2 + \frac{1}{2!}x^4 + \frac{1}{3!}x^6 + \cdots\right] = c_0 \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

Since the coefficient c_0 remains completely undetermined, we have in fact found the general solution of the differential equation.

Note that

The differential equation in this example and the differential equation in the following example can be easily solved by the other methods. The point of these two examples is to prepare ourselves for finding the power series solution of the differential equations with variable coefficients.

Example 12

Find solution of the differential equation

$$4y' + y = 0$$

in the form of a powers series in x.

Solution

We assume that a solution of the given differential equation exists in the form of

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + \sum_{n=1}^{\infty} c_n x^n$$

Then term by term differentiation of the proposed series solution yields

© Copyright Virtual University of Pakistan

$$y' = \sum_{n=1}^{\infty} nc_n x^{n-1} = c_1 + \sum_{n=2}^{\infty} nc_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}$$

Substituting the expression for y' and y, we obtain

$$4y'' + y = \sum_{n=2}^{\infty} 4n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^n$$

Notice that both series start with x^0. If we, respectively, substitute

$$k = n-2, \quad k = n, \quad k = 0,1,2,...$$

in the first series and second series on the right hand side of the last equation. Then we after using, in turn, $n = k+2$ and $n = k$, we get

$$4y'' + y = \sum_{k=0}^{\infty} 4(k+2)(k+1)c_{k+2}x^k + \sum_{k=0}^{\infty} c_k x^k$$

or

$$4y'' + y = \sum_{k=0}^{\infty} \left[4(k+2)(k+1)c_{k+2} + c_k\right] x^k$$

Substituting in the given differential equation, we obtain

$$\sum_{k=0}^{\infty} \left[4(k+2)(k+1)c_{k+2} + c_k\right] x^k = 0$$

From this last identity we conclude that

$$4(k+2)(k+1)c_{k+2} + c_k = 0$$

or

$$c_{k+2} = \frac{-c_k}{4(k+2)(k+1)}, \quad k = 0,1,2,...$$

From iteration of this recurrence relation it follows that

© Copyright Virtual University of Pakistan

$$c_2 = \frac{-c_0}{4.2.1} = -\frac{c_0}{2^2.2!}$$
$$c_3 = \frac{-c_1}{4.3.2} = -\frac{c_1}{2^2.3!}$$
$$c_4 = \frac{-c_2}{4.4.3} = +\frac{c_0}{2^4.4!}$$
$$c_5 = \frac{-c_3}{4.5.4} = +\frac{c_1}{2^4.5!}$$
$$c_6 = \frac{-c_4}{4.6.5} = -\frac{c_0}{2^6.6!}$$
$$c_7 = \frac{-c_5}{4.7.6} = -\frac{c_1}{2^6.7!}$$

and so forth. This iteration leaves both c_0 and c_1 arbitrary. From the original assumption we have

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + c_7 x^7 + \cdots$$
$$= c_0 + c_1 x - \frac{c_0}{2^2.2!}x^2 - \frac{c_1}{2^2.3!}x^3 + \frac{c_0}{2^4.4!}x^4 + \frac{c_1}{2^4.5!}x^5 - \frac{c_0}{2^6.6!}x^6 - \frac{c_1}{2^6.7!}x^7 + \cdots$$

or

$$y = c_0\left[1 - \frac{1}{2^2.2!}x^2 + \frac{1}{2^4.4!}x^4 - \frac{1}{2^6.6!}x^6 + \cdots\right] + c_1\left[x - \frac{1}{2^2.3!}x^3 + \frac{1}{2^4.5!}x^5 - \frac{1}{2^6.7!}x^7 + \cdots\right]$$

is a general solution. When the series are written in summation notation,

$$y_1(x) = c_0\sum_{k=0}^{\infty}\frac{(-1)^k (x)^{2k}}{(2k)!\left(\frac{1}{2}\right)} \quad \text{and} \quad y_2(x) = 2c_1\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}\left(\frac{x}{2}\right)^{2k+1}$$

the ratio test can be applied to show that both series converges for all x. You might also recognize the Maclaurin series as $y_2(x) = c_0\cos(x/2)$ and $y_2(x) = 2c_1\sin(x/2)$.

© Copyright Virtual University of Pakistan

Exercise

Find the interval of convergence of the given power series.

1. $\sum_{k=1}^{\infty} \frac{2^k}{k} x^k$
2. $\sum_{n=1}^{\infty} \frac{(x+7)^n}{\sqrt{n}}$
3. $\sum_{k=0}^{\infty} k!2^k x^k$
4. $\sum_{k=0}^{\infty} \frac{k-1}{k^{2k}} x^k$

Find the first four terms of a power series in x for the given function.

5. $e^x \sin x$
6. $e^x \ln(1-x)$
7. $\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right)^2$

Solve each differential equation in the manner of the previous chapters and then compare the results with the solutions obtained by assuming a power series solution

$$y = \sum_{n=0}^{\infty} c_n x^n$$

8. $y' - x^2 y = 0$
9. $y'' + y = 0$
10. $2y'' + y' = 0$

© Copyright Virtual University of Pakistan

Lecture 30
Solution about Ordinary Points

Analytic Function: A function *f* is said to be analytic at a point *a* if it can be represented by a power series in *(x-a)* with a positive radius of convergence.

Suppose the linear second-order differential equation

$$a_2(x)\,y'' + a_1(x)\,y' + a_0(x)\,y = 0 \qquad (1)$$

is put into the form

$$y'' + P(x)\,y' + Q(x)\,y = 0 \qquad (2)$$

by dividing by the leading coefficient $a_2(x)$.

Ordinary and singular points: A point x_0 is said to be a ***ordinary point*** of a differential equation (1) if both *P(x)* and *Q(x)* are analytic at x_0. A point that is not an ordinary point is said to be ***singular point*** of the equation.

Polynomial Coefficients:

If $a_2(x)$, $a_1(x)$ and $a_0(x)$ are polynomials with no common factors, then $x = x_0$ is

(i) an ordinary point if $a_2(x) \neq 0$ or

(ii) a singular point if $a_2(x) = 0$.

Example

(a) The singular points of the equation $(x^2 - 1)y'' + 2xy' + 6y = 0$ are the solutions of $x^2 - 1 = 0$ or $x = \pm 1$. All other finite values of *x* are the ordinary points.

(b) The singular points need not be real numbers.

The equation $(x^2 + 1)y'' + 2xy' + 6y = 0$ has the singular points at the solutions of $x^2 + 1 = 0$, namely, $x = \pm i$.

All other finite values, real or complex, are ordinary points.

Example

The Cauchy-Euler equation $ax^2y'' + bxy' + cy = 0$, where *a*, *b* and *c* are constants, has singular point at $x = 0$.

All other finite values of *x*, real or complex, are ordinary points.

© Copyright Virtual University of Pakistan

THEOREM (Existence of Power Series Solution)

If $x = x_0$ is an ordinary point of the differential equation $y' + P(x)y' + Q(x)y = 0$, we can always find two linearly independent solutions in the form of power series centered at x_0 :

$$y = \sum_{n=0}^{\infty} c_n (x - x_0)^n .$$

A series solution converges at least for $|x - x_0| < R$, where R is the distance from x_0 to the closest singular point (real or complex).

Example

Solve $y' - 2xy = 0$.

Solution

We see that $x = 0$ is an ordinary point of the equation. Since there are no finite singular points, there exist two solutions of the form $y = \sum_{n=0}^{\infty} c_n x^n$ convergent for $|x| < \infty$.

Proceeding, we write

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$y'' - 2xy = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} 2c_n x^{n+1}$$

$$= 2 \cdot 1 c_2 x^0 + \sum_{n=3}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} 2c_n x^{n+1}$$

both series start with x

Letting $k = n - 2$ in the first series and $k = n + 1$ in the second, we have

$$y'' - 2xy = 2c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} 2c_{k-1} x^k$$

$$= 2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1) c_{k+2} - 2c_{k-1}] x^k = 0$$

$$2c_2 = 0 \quad \text{and} \quad (k+2)(k+1) c_{k+2} - 2c_{k-1} = 0$$

© Copyright Virtual University of Pakistan

The last expression is same as

$$c_{k+2} = \frac{2c_{k-1}}{(k+2)(k+1)}, \quad k = 1,2,3,\cdots$$

Iteration gives $c_3 = \frac{2c_0}{3\cdot 2}$

$$c_4 = \frac{2c_1}{4\cdot 3}$$

$$c_5 = \frac{2c_2}{5\cdot 4} = 0 \qquad \text{because } c_2 = 0$$

$$c_6 = \frac{2c_3}{6\cdot 5} = \frac{2^2}{6\cdot 5\cdot 3\cdot 2}c_0$$

$$c_7 = \frac{2c_4}{7\cdot 6} = \frac{2^2}{7\cdot 6\cdot 4\cdot 3}c_1$$

$$c_8 = \frac{2c_5}{8\cdot 7} = 0$$

$$c_9 = \frac{2c_6}{9\cdot 8} = \frac{2^3}{9\cdot 8\cdot 6\cdot 5\cdot 3\cdot 2}c_0$$

$$c_{10} = \frac{2c_7}{10\cdot 9} = \frac{2^3}{10\cdot 9\cdot 7\cdot 6\cdot 4\cdot 3}c_1$$

$$c_{11} = \frac{2c_8}{11\cdot 10} = 0 \qquad \text{, and so on.}$$

It is obvious that both c_0 and c_1 are arbitrary. Now

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + c_7 x^7 + c_8 x^8 + c_9 x^9 + c_{10} x^{10} + c_{11} x^{11} + \cdots$$

$$y = c_0 + c_1 x + 0 + \frac{2}{3\cdot 2}c_0 x^3 + \frac{2}{4\cdot 3}c_1 x^4 + 0 + \frac{2^2}{6\cdot 5\cdot 3\cdot 2}c_0 x^6 + \frac{2^2}{7\cdot 6\cdot 4\cdot 3}c_1 x^7 + 0$$

$$+ \frac{2^3}{9\cdot 8\cdot 6\cdot 5\cdot 3\cdot 2}c_0 x^9 + \frac{2^3}{10\cdot 9\cdot 7\cdot 6\cdot 4\cdot 3}c_1 x^{10} + 0 + \cdots$$

$$y = c_0\left[1 + \frac{2}{3\cdot 2}x^3 + \frac{2^2}{6\cdot 5\cdot 3\cdot 2}x^6 + \frac{2^3}{9\cdot 8\cdot 6\cdot 5\cdot 3\cdot 2}x^9 + \cdots\right]$$

$$+ c_1\left[x + \frac{2}{4\cdot 3}x^4 + \frac{2^2}{7\cdot 6\cdot 4\cdot 3}x^7 + \frac{2^3}{10\cdot 9\cdot 7\cdot 6\cdot 4\cdot 3}x^{10} + \cdots\right].$$

© Copyright Virtual University of Pakistan

Example

Solve $(x^2+1)y'' + xy' - y = 0.$

Solution

Since the singular points are $x = \pm i$, $x = 0$ is the ordinary point, a power series will converge at least for $|x| < 1$. The assumption $y = \sum_{n=0}^{\infty} c_n x^n$ leads to

$$(x^2+1)\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x\sum_{n=1}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n$$

$$= \sum_{n=2}^{\infty} n(n-1)c_n x^n + \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=1}^{\infty} nc_n x^n - \sum_{n=0}^{\infty} c_n x^n$$

$$= 2c_2 x^0 - c_0 x^0 + 6c_3 x + c_1 x - c_1 x + \sum_{n=2}^{\infty} n(n-1)c_n x^n + \sum_{n=4}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=2}^{\infty} nc_n x^n - \sum_{n=2}^{\infty} c_n x^n$$

$k=n$ $\quad$ $k=n-2$ $\quad$ $k=n$ $\quad$ $k=n$

$$= 2c_2 - c_0 + 6c_3 x + \sum_{k=2}^{\infty} [k(k-1)c_k + (k+2)(k+1)c_{k+2} + kc_k - c_k]x^k = 0$$

or $$2c_2 - c_0 + 6c_3 x + \sum_{k=2}^{\infty} [(k+1)(k-1)c_k + (k+2)(k+1)c_{k+2}]x^k = 0.$$

Thus $$2c_2 - c_0 = 0$$

$$c_3 = 0$$

$$(k+1)(k-1)c_k + (k+2)(k+1)c_{k+2} = 0$$

This implies

$$c_2 = \frac{1}{2}c_0$$

$$c_3 = 0$$

$$c_{k+2} = \frac{-(k-1)}{(k+2)}c_k, \quad k = 2, 3, \cdots$$

© Copyright Virtual University of Pakistan

Iteration of the last formula gives

$$c_4 = -\frac{1}{4}c_2 = -\frac{1}{2\cdot 4}c_0 = -\frac{1}{2^2 2!}c_0$$

$$c_5 = -\frac{2}{5}c_3 = 0$$

$$c_6 = -\frac{3}{6}c_4 = \frac{3}{2\cdot 4\cdot 6}c_0 = \frac{1\cdot 3}{2^3 3!}c_0$$

$$c_7 = -\frac{4}{7}c_5 = 0$$

$$c_8 = -\frac{5}{8}c_6 = -\frac{3\cdot 5}{2\cdot 4\cdot 6\cdot 8}c_0 = -\frac{1\cdot 3\cdot 5}{2^4 4!}c_0$$

$$c_9 = -\frac{6}{9}c_7 = 0$$

$$c_{10} = -\frac{7}{10}c_8 = -\frac{3\cdot 5\cdot 7}{2\cdot 4\cdot 6\cdot 8\cdot 10}c_0 = \frac{1\cdot 3\cdot 5\cdot 7}{2^5 5!}c_0 \text{ and so on.}$$

Therefore

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + c_7 x^7 + c_8 x^8 + \cdots$$

$$y = c_1 x + c_0\left[1 + \frac{1}{2}x^2 - \frac{1}{2^2 2!}x^4 + \frac{1\cdot 3}{2^3 3!}x^6 - \frac{1\cdot 3\cdot 5}{2^4 4!}x^8 + \frac{1\cdot 3\cdot 5\cdot 7}{2^5 5!}x^{10} - \cdots\right]$$

The solutions are

$$y_1(x) = c_0\left[1 + \frac{1}{2}x^2 + \sum_{n=2}^{\infty}(-1)^{n-1}\frac{1\cdot 3\cdot 5\cdots(2n-3)}{2^n n!}x^{2n}\right], \quad |x| < 1$$

$$y_2(x) = c_1 x.$$

Example

If we seek a solution $y = \sum_{n=0}^{\infty} c_n x^n$ for the equation

$$y'' - (1+x)y = 0,$$

we obtain $c_2 = c_0/2$ and the three-term recurrence relation

$$c_{k+2} = \frac{c_k + c_{k-1}}{(k+1)(k+2)}, \quad k = 1,2,3,\cdots$$

To simplify the iteration we can first choose $c_0 \neq 0$, $c_1 = 0$; this yields one solution. The other solution follows from next choosing $c_0 = 0$, $c_1 \neq 0$. With the first assumption we find

© Copyright Virtual University of Pakistan

$$c_2 = \frac{1}{2}c_0$$

$$c_3 = \frac{c_1 + c_0}{2\cdot 3} = \frac{c_0}{2\cdot 3} = \frac{1}{6}c_0$$

$$c_4 = \frac{c_2 + c_1}{3\cdot 4} = \frac{c_0}{2\cdot 3\cdot 4} = \frac{1}{24}c_0$$

$$c_5 = \frac{c_3 + c_2}{4\cdot 5} = \frac{c_0}{4\cdot 5}[\frac{1}{2\cdot 3} + \frac{1}{2}] = \frac{1}{30}c_0 \text{ and so on.}$$

Thus one solution is

$$y_1(x) = c_0[1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{30}x^5 + \cdots].$$

Similarly if we choose $c_0 = 0$, then

$$c_2 = 0$$

$$c_3 = \frac{c_1 + c_0}{2\cdot 3} = \frac{c_1}{2\cdot 3} = \frac{1}{6}c_1$$

$$c_4 = \frac{c_2 + c_1}{3\cdot 4} = \frac{c_1}{3\cdot 4} = \frac{1}{12}c_1$$

$$c_5 = \frac{c_3 + c_2}{4\cdot 5} = \frac{c_1}{2\cdot 3\cdot 4\cdot 5} = \frac{1}{120}c_1 \text{ and so on.}$$

Hence another solution is

$$y_2(x) = c_1[x + \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{120}x^5 + \cdots].$$

Each series converges for all finite values of *x*.

Non-polynomial Coefficients

The next example illustrates how to find a power series solution about an ordinary point of a differential equation when its coefficients are not polynomials. In this example we see an application of multiplication of two power series that we discussed earlier.

© Copyright Virtual University of Pakistan

Example

Solve $y' + (\cos x)y = 0$

Solution:

The equation has no singular point.

Since $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$, it is seen that $x = 0$ is an ordinary point.

Thus the assumption $y = \sum_{n=0}^{\infty} c_n x^n$ leads to

$$y'' + (\cos x)y = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-} - (1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots)\sum_{n=0}^{\infty} c_n x^n$$

$$= (2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \cdots) + (1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots)(c_0 + c_1 x + c_2 x^2 + \cdots)$$

$$= 2c_2 + c_0 + (6c_3 + c_1)x + (12c_4 + c_2 - \frac{1}{2}c_0)x^2 + (20c_5 + c_3 - \frac{1}{2}c_1)x^3 + \cdots$$

If the last line be identically zero, we must have

$$2c_2 + c_0 = 0 \Rightarrow c_2 = -\frac{c_0}{2}$$

$$6c_3 + c_1 = 0 \Rightarrow c_3 = -\frac{c_1}{6}$$

$$12c_4 + c_2 - \frac{1}{2}c_0 = 0 \Rightarrow c_4 = \frac{c_0}{12}$$

$20c_5 + c_3 - \frac{1}{2}c_1 = 0 \Rightarrow c_5 = \frac{c_1}{30}$ and so on. c_0 and c_1 are arbitrary.

Now

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + \cdots$$

or $$y = c_0 + c_1 x - \frac{c_0}{2}x^2 - \frac{c_1}{6}x^3 + \frac{c_0}{12}x^4 + \frac{c_1}{30}x^5 - \cdots$$

$$y = c_0(1 - \frac{1}{2}x^2 + \frac{1}{12}x^4 - \cdots) + c_1(x - \frac{1}{6}x^3 + \frac{1}{30}x^5 - \cdots)$$

$y_1(x) = c_0[1 - \frac{1}{2}x^2 + \frac{1}{12}x^4 - \cdots]$ and $y_2(x) = c_1[x - \frac{1}{6}x^3 + \frac{1}{30}x^5 - \cdots]$

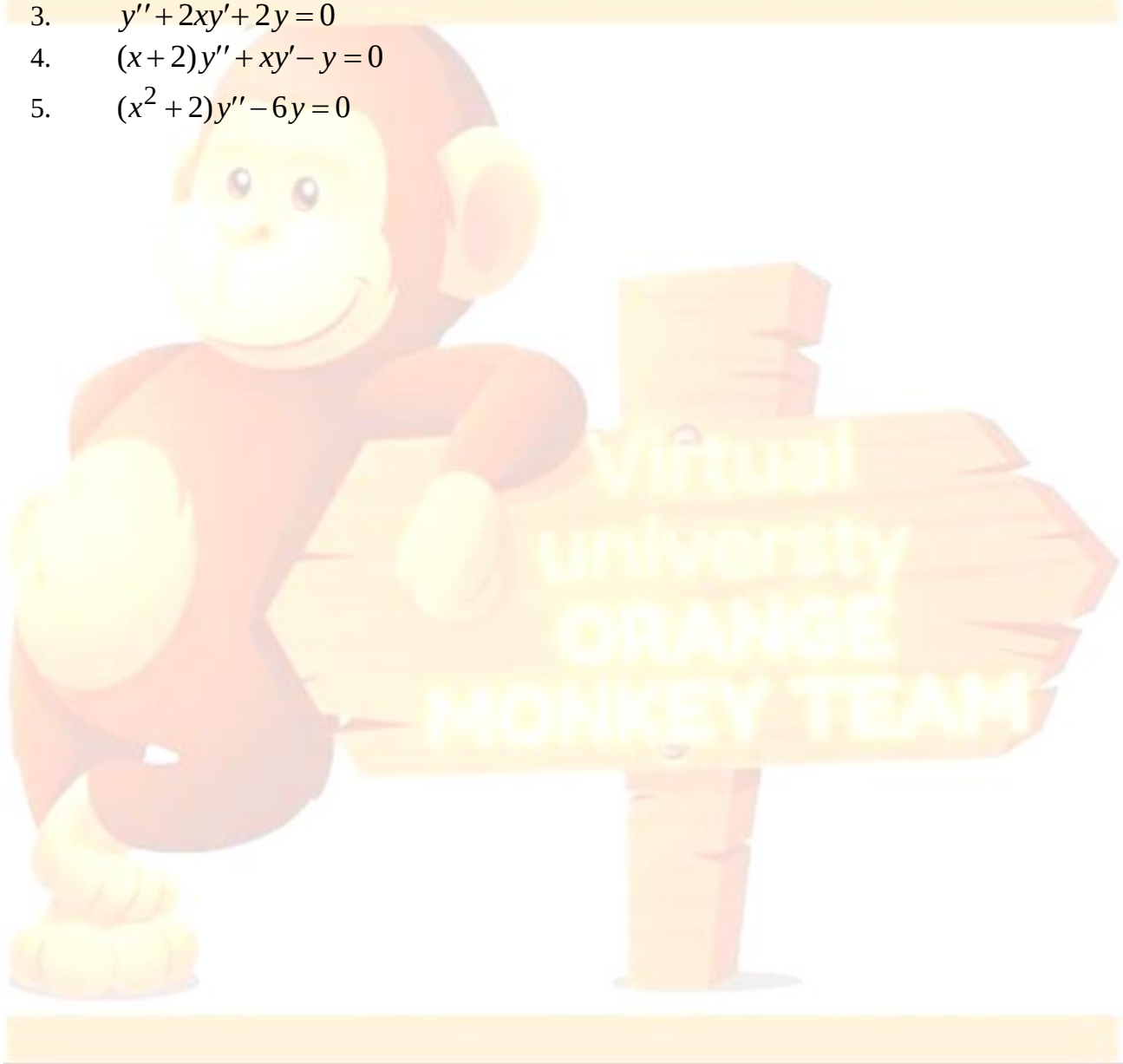
Since the differential equation has no singular point, both series converge for all finite values of x.

© Copyright Virtual University of Pakistan

Exercise

In each of the following problems find two linearly independent power series solutions about the ordinary point $x = 0$.

1. $y'' + x^2 y = 0$
2. $y'' - xy' + 2y = 0$
3. $y'' + 2xy' + 2y = 0$
4. $(x+2)y'' + xy' - y = 0$
5. $(x^2+2)y'' - 6y = 0$



© Copyright Virtual University of Pakistan

Lecture 31
Solution about Singular Points

If $x = x_0$ is singular point, it is not always possible to find a solution of the form

$$y = \sum_{n=0}^{\infty} c_n (x - x_0)^n \quad \text{for the equation } a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

However, we may be able to find a solution of the form

$$y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$$, where r is constant to be determined.

To define regular/irregular singular points, we put the given equation into the standard form

$$y'' + P(x)y' + Q(x)y = 0$$

Definition: Regular and Irregular Singular Points

A Singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to be a ***regular singular point*** if both $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at x_0. A singular point that is not regular is said to be an ***irregular singular point*** of the equation.

Polynomial Coefficients

If the coefficients in the given differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ are polynomials with no common factors, above definition is equivalent to the following:

Let $a_2(x_0) = 0$ *Form* $P(x)$ *and* $Q(x)$ *by reducing* $\frac{a_1(x)}{a_2(x)}$ *and* $\frac{a_0(x)}{a_2(x)}$ *to lowest terms, respectively. If the factor* $(x - x_0)$ *appears at most to the first powers in the denominator of* $P(x)$ *and at most to the second power in the denominator of* $Q(x)$, *then* $x = x_0$ *is a* ***regular singular point****.*

Example 1

$x = \pm 2$ are singular points of the equation

$$(x^2 - 4)^2 y'' + (x - 2)y' + y = 0$$

Dividing the equation by $(x^2 - 4)^2 = (x - 2)^2 (x + 2)^2$, we find that

© Copyright Virtual University of Pakistan

$$P(x) = \frac{1}{(x-2)(x+2)^2} \text{ and } Q(x) = \frac{1}{(x-2)^2(x+2)^2}$$

1. $x = 2$ is a regular singular point because power of $x-2$ in $P(x)$ is 1 and in $Q(x)$ is 2.
2. $x = -2$ is an irregular singular point because power of $x+2$ in $P(x)$ is 2.

The 1st condition is violated.

Example 2

Both $x = 0$ and $x = -1$are singular points of the differential equation

$$x^2(x+1)^2 y'' + (x^2-1)y' + 2y = 0$$

Because $x^2(x+1)^2 = 0$ or $x = 0, -1$

Now write the equation in the form

$$y'' + \frac{x^2-1}{x^2(x+1)^2} y' + \frac{2}{x^2(x+1)^2} y = 0$$

or $$y'' + \frac{x-1}{x^2(x+1)} y' + \frac{2}{x^2(x+1)^2} y = 0$$

So $$P(x) = \frac{x-1}{x^2(x+1)} \text{ and } Q(x) = \frac{2}{x^2(x+1)^2}$$

Shows that $x = 0$ is a irregular singular point since $(x-0)$ appears to the second powers in the denominator of $P(x)$.

Note, however, $x = -1$ is a regular singular point.

Example 3

a) $x = 1$ and $x = -1$ are singular points of the differential equation

$$(1-x^2)y'' + -2xy' + 30y = 0$$

Because $1 - x^2 = 0$ or $x = \pm 1$.

Now write the equation in the form

$$y'' - \frac{2x}{(1-x^2)} y' + \frac{30}{1-x^2} y = 0$$

or $$y'' - \frac{2x}{(1-x)(1+x)} y' + \frac{30}{(1-x)(1+x)} y = 0$$

© Copyright Virtual University of Pakistan

$$P(x) = \frac{-2x}{(1-x)(1+x)} \quad \text{and} \quad Q(x) = \frac{30}{(1-x\ )(1+x)}$$

Clearly $x = \pm 1$ are regular singular points.

(b) $x = 0$ is an irregular singular points of the differential equation

$$x^3 y'' - 2xy' + 5y = 0$$

or $y'' - \frac{2}{x^2} y' + \frac{5}{x^3} y = 0$ giving $Q(x) = \frac{5}{x^3}$.

(c) $x = 0$ is a regular singular points of the differential equation

$$x\, y'' - 2xy' + 5y = 0$$

Because the equation can be written as $y'' - 2y' + \frac{5}{x} y = 0$ giving $P(x) = -2$ and

$Q(x) = \frac{5}{x}$

In part (c) of Example 3 we noticed that $(x-0)$ and $(x-0)^2$ do not even appear in the denominators of $P(x)$ and $Q(x)$ respectively. Remember, these factors can appear at most in this fashion. For a singular point $x = x_0$, any nonnegative power of $(x - x_0)$ less than one (namely, zero) and nonnegative power less than two (namely, zero and one) in the denominators of $P(x)$ and $Q(x)$, respectively, imply x_0 is a ***regular singular point.***

Please note that the singular points can also be complex numbers.
For example, $x = \pm 3i$ are regular singular points of the equation

$$(x^2 + 9)y'' + -3xy' + (1-x)y = 0$$

Because the equation can be written as

$$y' - \frac{3x}{x^2+9} y' + \frac{1-x}{x^2+9} y = 0.$$

$$\therefore P(x) = \frac{-3x}{(x-3i)(x+3i)}. \quad Q(x) = \frac{1-x}{(x-3i\ )(x+3i)}.$$

Method of Frobenius

To solve a differential equation $a_2(x)y' + a_1(x)y' + a_0(x)y = 0$ about a regular singular point we employ the Frobenius' Theorem.

Frobenius' Theorem

If $x = x_0$ is a regular singular point of equation $a_2(x)y' + a_1(x)y' + a_0(x)y = 0$, then there exists at least one series solution of the form

© Copyright Virtual University of Pakistan

$$y = (x - x_0)^r \sum_{n=0}^{\infty} c_n (x - x_0)^n = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$$

where the number r is a constant that must be determined. The series will converge at least on some interval $0 < x - x_0 < R$.

Note that the solutions of the form $y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$ are not guaranteed.

Method of Frobenius

1. Identify regular singular point x_0,
2. Substitute $y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$ in the given differential equation,
3. Determine the unknown exponent r and the coefficients c_n.
4. For simplicity assume that $x_0 = 0$.

Example 4

As $x = 0$ is regular singular points of the differential equation
$3xy' + y' - y = 0.$

We try a solution of the form $y = \sum_{n=0}^{\infty} c_n x^{n+r}$.

Therefore $y' = \sum_{n=0}^{\infty} (n + r) c_n x^{n+r-1}.$

And $y' = \sum_{n=0}^{\infty} (n + r)(n + r - 1) c_n x^{n+r-2}.$

$$3xy' + y' - y = 3\sum_{n=0}^{\infty} (n + r)(n + r - 1) c_n x^{n+r-1} + \sum_{n=0}^{\infty} (n + r) c_n x^{n+r-1} - \sum_{n=0}^{\infty} c_n x^{n+r}.$$

$$= \sum_{n=0}^{\infty} (n + r)(3n + 3r - 2) c_n x^{n+r-1} - \sum_{n=0}^{\infty} c_n x^{n+r}.$$

$$= x^r \left[r(3r - 2) c_0 x^{-1} + \sum_{n=1}^{\infty} (n + r)(3n + 3r - 2) c_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n \right].$$

$$= x^{r-1} \left[r(3r - 2) c_0 + \sum_{k=0}^{\infty} [(k + r + 1)(3k + 3r + 1) c_{k+1} - c_k] x^{k} \right] = 0$$

(where $k = n - 1$ in the first sum and $k = n$ in the second)

which implies $r(3r - 2)c_0 = 0$

© Copyright Virtual University of Pakistan

$(k+r+1)(3k+3r+1)c_{k+1} - c_k = 0$, $k = 0,1,2,...$

Since nothing is gained by taking $c_0 = 0$, we must then have

$r(3r-2) = 0$ [called the indicial equation and its roots $r = \frac{2}{3}, 0$ are called indicial roots or exponents of the singularity.]

and $c_{k+1} = \frac{c_k}{(k+r+1)(3k+3r+1)}, k = 0,1,2,...$

Substitute $r_1 = \frac{2}{3}$ and $r_2 = 0$ in the above equation and these values will give two different recurrence relations:

For $r_1 = \frac{2}{3}$, $c_{k+1} = \frac{c_k}{(3k+5)(k+1)}$, $k = 0,1,2,...$ (1)

For $r_2 = 0$ $c_{k+1} = \frac{c_k}{(k+1)(3k+1)}$, $k = 0,1,2,...$ (2)

Iteration of (1) gives

$$c_1 = \frac{c_0}{5.1}$$

$$c_2 = \frac{c_1}{8.2} = \frac{c_0}{2!5.8}$$

$$c_3 = \frac{c_2}{11.3} = \frac{c_0}{3!5.8.11}$$

$$c_4 = \frac{c_3}{14.4} = \frac{c_0}{4!5.8.11.14}$$

.

In general $c_n = \frac{c_0}{n!5.8.11.14...(3n+2)}$, $n = 1,2,...$

Iteration of (2) gives

$$c_1 = \frac{c_0}{1.1}$$

$$c_2 = \frac{c_1}{2.4} = \frac{c_0}{2!1.4}$$

$$c_3 = \frac{c_2}{3.7} = \frac{c_0}{3!1.4.7}$$

$$c_4 = \frac{c_3}{4.10} = \frac{c_0}{4!1.4.7.10}$$

In general $c_n = \frac{c_0}{n!1.4.7...(3n-2)}$, $n = 1,2,...$

© Copyright Virtual University of Pakistan

Thus we obtain two series solutions

$$y_1 = c_0 x^{\frac{2}{3}}\left[1+\sum_{n=1}^{\infty}\frac{1}{n!5.8.11.14...(3n+2)}x^n\right] \qquad (1)$$

$$y_2 = c_0 x^0\left[1+\sum_{n=1}^{\infty}\frac{1}{n!1.4.7...(3n-2)}x^n\right]. \qquad (4)$$

By the ratio test it can be demonstrated that both (3) and (4) converge for all finite values of x. Also it should be clear from the form of (3) and (4) that neither series is a constant multiple of the other and therefore, $y_1(x)$ and $y_2(x)$ are linearly independent on the x-axis. Hence by the superposition principle

$$y = C_1 y_1(x) + C_2 y_2(x) = C_1\left[x^{\frac{2}{3}}+\sum_{n=1}^{\infty}\frac{1}{n!5.8.11.14...(3n+2)}x^{n+\frac{2}{3}}\right]$$

$$+C_2\left[1+\sum_{n=1}^{\infty}\frac{1}{n!1.4.7...(3n-2)}x^n\right], \quad |x|<\infty$$

is an other solution of the differential equation. On any interval not containing the origin, this combination represents the general solution of the differential equation

Remark: The method of Frobenius may not always provide 2 solutions.

Example 5

The differential equation

$xy'' + 3y' - y = 0$ has regular singular point at $x = 0$

We try a solution of the form $y = \sum_{n=0}^{\infty} c_n x^{n+r}$

Therefore $y' = \sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1}$ and $y' = \sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-2}$.

so that

$$xy' + 3y' - y = x^r\left[r(r+2)c_0 x^{-1} + \sum_{k=0}^{\infty}[(k+r+1)(k+r+3)c_{k+1} - c_k]x^k\right] = 0$$

so that the indicial equation and exponent are $r(r+2) = 0$ and $r_1 = 0$, $r_2 = -2$, respectively.

Since $(k+r+1)(k+r+3)c_{k+1} - c_k = 0$, $k = 0,1,2,...$ (1)

© Copyright Virtual University of Pakistan

it follows that when $r_1 = 0$,

$$c_{k+1} = \frac{c_k}{(k+1)(k+3)},$$

$$c_1 = \frac{c_0}{1.3}$$

$$c_2 = \frac{c_1}{2.4} = \frac{2c_0}{2!4!}$$

$$c_3 = \frac{c_2}{3.5} = \frac{2c_0}{3!5!}$$

$$c_4 = \frac{c_3}{4.6} = \frac{2c_0}{4!6!}$$

$$\vdots$$

$$c_n = \frac{2c_0}{n!(n+2)!}, \quad n = 1, 2, \ldots$$

Thus one series solution is

$$y_1 = c_0 x^0 \left[1 + \sum_{n=1}^{\infty} \frac{2}{n!(n+2)!} x^n\right] = c_0 \sum_{n=0}^{\infty} \frac{2}{n!(n+2)!} x^n, \ |x| < \infty.$$

Now when $r_2 = -2$, (1) becomes

$$(k-1)(k+1)c_{k+1} - c_k = 0 \qquad (2)$$

but note here that we do not divide by $(k-1)(k+1)$ immediately since this term is zero for $k = 1$. However, we use the recurrence relation (2) for the cases $k = 0$ and $k = 1$:

$$-1.1c_1 - c_0 = 0 \quad \text{and} \quad 0.2c_2 - c_1 = 0$$

The latter equation implies that $c_1 = 0$ and so the former equation implies that $c_0 = 0$. Continuing, we find

$$c_{k+1} = \frac{c_k}{(k-1)(k+1)} \qquad k = 2, 3, \ldots$$

$$c_3 = \frac{c_2}{1.3}$$

$$c_4 = \frac{c_3}{2.4} = \frac{2c_2}{2!.4!}$$

$$c_5 = \frac{c_4}{3.5} = \frac{2c_2}{3!.5!}, \ldots$$

$$\vdots$$

In general $\quad c_n = \dfrac{2c_2}{(n-2)!n!}, \quad n = 3, 4, 5, \ldots$

© Copyright Virtual University of Pakistan

Thus $$y_2 = c_2 x^{-2}\left[x^2 + \sum_{n=3}^{\infty} \frac{2}{(n-2)!n!} x^n\right]. \qquad (3)$$

However, close inspection of (3) reveals that y_2 is simply constant multiple of y_1. To see this, let $k = n - 2$ in (3). We conclude that the method of Frobenius gives only one series solution of the given differential equation.

Cases of Indicial Roots

When using the method of Frobenius, we usually distinguish three cases corresponding to the nature of the indicial roots. For the sake of discussion let us suppose that r_1 and r_2 are the real solutions of the indicial equation and that, when appropriate, r_1 denotes the largest root.

Case I: Roots not Differing by an Integer

If r_1 and r_2 are distinct and do not differ by an integer, then their exist two linearly independent solutions of the differential equation of the form

$$y_1 = \sum_{n=0}^{\infty} c_n x^{n+r_1} .. c_0 \neq 0, \text{ and } y_2 = \sum_{n=0}^{\infty} b_n x^{n+r_2}, \quad b_0 \neq 0.$$

Example 6

Solve $2xy' + (1+x)y' + y = 0.$

Solution

If $y = \sum_{n=0}^{\infty} c_n x^{n+r}$, then

$$2xy' + (1+x)y' + y = 2\sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1} +$$
$$\sum_{n=0}^{\infty}(n+r)c_n x^{n+r} + \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$= \sum_{n=0}^{\infty}(n+r)(2n+2r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r+1)c_n x^{n+r}$$

$$= x^r\left[r(2r-1)c_0 x^{-1} + \sum_{n=1}^{\infty}(n+r)(2n+2r-1)c_n x^{n-1} + \sum_{k=0}^{\infty}(n+r+1)c_n x^n\right]$$

$n = k+1$ $\qquad$ $k = n$

© Copyright Virtual University of Pakistan

$$= x^r \left[r(2r-1)c_0 x^{-1} + \sum_{k=0}^{\infty} [(k+r+1)(2k+2r+1)c_{k+1} + (k+r+1)c_k]x^k \right] = 0$$

which implies $r(2r-1)=0$

$$(k+r+1)(2k+2r+1)c_{k+1} + (k+r+1)c_k = 0,\ k = 0,1,2,... \qquad (1)$$

For $r_1 = \frac{1}{2}$, we can divide by $k+\frac{3}{2}$ in the above equation to obtain

$$c_{k+1} = \frac{-c_k}{2(k+1)},$$

$$c_1 = \frac{-c_0}{2.1}$$

$$c_2 = \frac{-c_1}{2.2} = \frac{c_0}{2^2.2!}$$

$$c_3 = \frac{-c_2}{2.3} = \frac{-c_0}{2^3.3!}$$

$$\vdots$$

In general $c_n = \frac{(-1)^n c_0}{2^n n!},\ n = 1,2,3,...$

Thus we have

$$y_1 = c_0 x^{\frac{1}{2}} \left[1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{2^n n!} x^n\right],$$ which converges for $x \geq 0$.

As given, the series is not meaningful for $x < 0$ because of the presence of $x^{\frac{1}{2}}$.

Now for $r_2 = 0$, (1) becomes

$$c_{k+1} = \frac{-c_k}{2k+1}$$

$$c_1 = \frac{-c_0}{1}$$

$$c_2 = \frac{-c_1}{3} = \frac{c_0}{1.3}$$

$$c_3 = \frac{-c_2}{5} = \frac{-c_0}{1.3.5}$$

$$c_4 = \frac{-c_3}{7} = \frac{c_0}{1.3.5.7}$$

$$\vdots$$

In general $c_n = \frac{(-1)^n c_0}{1.3.5.7...(2n-1)},\ n = 1, 2, 3, ...$

© Copyright Virtual University of Pakistan

Thus second solution is

$$y_2 = c_0\left[1+\sum_{n=1}\frac{(-1)^n}{1.3.5.7...(2n-1)}x^n\right]. \quad |x|<\infty.$$

On the interval $(0,\infty)$, the general solution is

$$y = C_1 y_1(x) + C_2 y_2(x).$$

© Copyright Virtual University of Pakistan

Lecture 32
Solutions about Singular Points

Method of Frobenius-Cases II and III

When the roots of the indicial equation differ by a positive integer, we may or may not be able to find two solutions of

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \qquad (1)$$

having form

$$y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r} \qquad (2)$$

If not, then one solution corresponding to the smaller root contains a logarithmic term. When the exponents are equal, a second solution always contains a logarithm. This latter situation is similar to the solution of the Cauchy-Euler differencial equation when the roots of the auxiliary equation are equal. We have the next two cases.

Case II: Roots Differing by a Positive Integer

If $r_1 - r_2 = N$, where N is a positive integer, then there exist two linearly independent solutions of the form

$$y_1 = \sum_{n=0}^{\infty} c_n x^{n+r_1}, c_0 \neq 0 \qquad (3a)$$

$$y_2 = C y_1(x) \ln x + \sum_{n=0}^{\infty} b_n x^{n+r_2}, b_0 \neq 0 \qquad (3b)$$

Where C is a constant that could be zero.

Case III: Equal Indicial Roots:

If $r_1 = r_2$, there always exist two linearly independent solutions of (1) of the form

$$y_1 = \sum_{n=0}^{\infty} c_n x^{n+r_1}, c_0 \neq 0 \qquad (4a)$$

$$y_2 = y_1(x)\ln x + \sum_{n=1}^{\infty} b_n x^{n+r_1} \qquad \because r_1 = r_2 \qquad (4b)$$

© Copyright Virtual University of Pakistan

Example 7: Solve $xy' + (x-6)y' - 3y = 0$ (1)

Solution: The assumption $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ leads to

$$xy' + (x-6)y' - 3y$$

$$= \sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-1} - 6\sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r)c_n x^{n+r} - 3\sum_{n=0}^{\infty} c_n x^{n+r}$$

$$= x^r\left[r(r-7)c_0 x^{-1} + \sum_{n=1}^{\infty}(n+r)(n+r-7)c_n x^{n-1} + \sum_{n=0}^{\infty}(n+r-3)c_n x^n\right]$$

$$x^r\left[r(r-7)c_0 x^{-1} + \sum_{k=0}^{\infty}\left[(k+r+1)(k+r-6)c_{k+1} + (k+r-3)c_k\right]x^k\right] = 0$$

Thus $r(r-7) = 0$ so that $r_1 = 7, r_2 = 0, r_1 - r_2 = 7,$ *and*

$$(k+r+1)(k+r-6)\,c_{k+1} + (k+r-3)c_k = 0, \qquad k = 0,1,2,3,... \qquad (2)$$

For smaller root $r_2 = 0, (2)$ *becomes*

$$(k+1)(k-6)c_{k+1} + (k-3)c_k = 0 \qquad (3)$$

recurrence relation becomes

$$c_{k+1} = -\frac{(k-3)}{(k+1)(k-6)}c_k$$

Since k-6=0, when, k=6, we do not divide by this term until k>6.we find

$$1.(-6)c_1 + (-3)c_0 = 0$$
$$2.(-5)c_2 + (-2)c_1 = 0$$
$$3.(-4)c_3 + (-1).c_2 = 0$$
$$4.(-3)c_4 + 0.c_3 = 0$$
$$5.\ (-2)c_5 + 1.c_4 = 0$$
$$6.\ (-1)c_6 + 2.c_5 = 0$$
$$7.0.c_7 + 3.c_6 = 0$$

This implies that

$c_4 = c_5 = c_6 = 0,$ But c_0 and c_7 can be chosen arbitrarily.

Hence $c_1 = -\frac{1}{2}c_0$

© Copyright Virtual University of Pakistan

$$c_2 = -\frac{1}{5}c_1 = \frac{1}{10}c_0$$

$$c_3 = -\frac{1}{12}c_2 = -\frac{1}{120}c_0 \qquad (4)$$

and for $k \geq 7$

$$c_{k+1} = \frac{-(k-3)}{(k+1)(k-6)}c_k$$

$$c_8 = \frac{-4}{8.1}c_7$$

$$c_9 = -\frac{5}{9.2}c_8 = \frac{4.5}{2!8.9}c_7$$

$$c_{10} = \frac{-6}{10.3}c_9 = \frac{-4.5.6}{3!8.9.10}c_7$$

.

.

.

$$c_n = \frac{(-1)^{n+1}4\cdot 5\cdot 6\cdots(n-4)}{(n-7)!8\cdot 9\cdot 10\cdots(n)}c_7, \qquad n = 8,9,10,\cdots \qquad (5)$$

If we choose $c_7 = 0 and c_0 \neq 0$ It follows that we obtain the polynomial solution

$$y_1 = c_0[1 - \frac{1}{2}x + \frac{1}{10}x^2 - \frac{1}{120}x^3],$$

But when $c_7 \neq = 0 and c_0 = 0$, It follows that a second, though infinite series solution is

$$y_2 = c_7[x^7 + \sum_{n=8}^{\infty}\frac{(-1)^{n+1}4\cdot 5\cdot 6\cdots(n-4)}{(n-7)!\,8\cdot 9\cdot 10\cdots n}x^n]$$

$$= c_7[x^7 + \sum_{k=1}^{\infty}\frac{(-1)^k 4\cdot 5\cdot 6\cdots(k+3)}{k!8\cdot 9\cdot 10\cdots(k+7)}x^{k+3}], \qquad |x| < \infty \qquad (6)$$

Finally the general solution of equation (1) on the interval $(0,\infty)$ is

$$Y = c_1 y_1(x) + c_2 y_2(x)$$

$$= c_1[1 - \frac{1}{2}x + \frac{1}{10}x^2 - \frac{1}{120}x^3] + c_2[x^7 + \sum_{n=1}^{\infty}\frac{(-1)^k 4\cdot 5\cdot 6\cdots(k+3)}{k!\cdot 8\cdot 9\cdot 10\cdots(k+7)}x^{k+7}]$$

It is interesting to observe that in example 9 the larger root $r_1 = 7$ were not used. Had we done so, we would have obtained a series solution of the form*

$$y = \sum_{n=0}^{\infty} c_n x^{n+7} \qquad (7)$$

Where c_n are given by equation (2) with $r_1 = 7$

© Copyright Virtual University of Pakistan

$$c_{k+1} = \frac{-(k+4)}{(k+8)(k+1)} c_k, \qquad k = 0,1,2...$$

Iteration of this latter recurrence relation then would yield only one solution, namely the solution given by (6) with c_0 playing the role of c_7)

When the roots of indicial equation differ by a positive integer, the second solution may contain a logarithm.

On the other hand if we fail to find second series type solution, we can always use the fact that

$$y_2 = y_1(x)\int \frac{e^{-\int p(x)dx}}{y_1^{\,2}(x)}dx \qquad (8)$$

is a solution of the equation $y' + P(x)y' + Q(x)y = 0$,whenever y_1 is a known solution.

Note: In case 2 it is always a good idea to work with smaller roots first.

Example8:

Find the general solution of $xy' + 3y' - y = 0$

Solution The method of frobenius provide only one solution to this equation, namely,

$$y_1 = \sum_{n=0}^{\infty} \frac{2}{n!(n+2)!} x^n = 1 + \frac{1}{3}x + \frac{1}{24}x^2 + \frac{1}{360}x^3 + \cdots \qquad (9)$$

From (8) we obtain a second solution

$$y_2 = y_1(x)\int \frac{e^{-\int p(x)dx}}{y_1^{\,2}(x)}dx = y_1(x)\int \frac{dx}{x^3[1+\frac{1}{3}x+\frac{1}{24}x^2+\frac{1}{360}x^3+\cdots]^2}$$

$$= y_1(x)\int \frac{dx}{x^3[1+\frac{2}{3}x+\frac{7}{36}x^2+\frac{1}{90}x^3+\cdots]}$$

$$= y_1(x)\int \frac{1}{x^3}[1-\frac{2}{3}x+\frac{13}{36}x^2-\frac{19}{270}x^3+\cdots]dx$$

$$= y_1(x)\left[-\frac{1}{2x^2}+\frac{2}{3x}+\frac{1}{4}\ln x-\frac{19}{270}x+\ldots\right]$$

$$= \frac{1}{4}y_1(x)\ln x + y_1(x)\left[-\frac{1}{2x^2}+\frac{2}{3x}-\frac{19}{270}x+\ldots\right] \qquad (*)$$

$$\therefore y = c_1 y_1(x) + c_2\left[\frac{1}{4}y_1(x)\ln x + y_1(x)\left(-\frac{1}{2x^2}+\frac{2}{3x}-\frac{19}{270}x+\ldots\right)\right] \qquad (**)$$

© Copyright Virtual University of Pakistan

Example 9:
Find the general solution of

$$xy'' + 3y' - y = 0$$

Solution :

$$y_2 = y_1 \ln x + \sum_{n=0}^{\infty} b_n x^{n-2} \qquad (10)$$

$$y_1 = \sum_{n=0}^{\infty} \frac{2}{n!(n+2)!} x^n \qquad (11)$$

differentiate (10) gives

$$y_2' = \frac{y_1}{x} + y_1' \ln x + \sum_{n=0}^{\infty} (n-2) b_n x^{n-3}$$

$$y_2'' = -\frac{y_1}{x^2} + \frac{2y_1'}{x} + y_1'' \ln x + \sum_{n=0}^{\infty} (n-2)(n-3) b_n x^{n-4}$$

so that

$$xy_2'' + 3y_2' - y_2 = \ln x \left[xy_1'' + 3y_1' - y_1 \right] + 2y_1' + \frac{2y_1}{x} + \sum_{n=0}^{\infty} (n-2)(n-3) b_n x^{n-3}$$

$$+ 3\sum_{n=0}^{\infty} (n-2) b_n x^{n-3} - \sum_{n=0}^{\infty} b_n x^{n-2}$$

$$= 2y_1' + \frac{2y_1}{x} + \sum_{n=0}^{\infty} (n-2) n b_n x^{n-3} - \sum_{n=0}^{\infty} b_n x^{n-2} \qquad (12)$$

where we have combined the 1st two summations and used the fact that

$$xy_1'' + 3y_1' - y_1 = 0$$

Differentiate (11) we can write (12) as

$$\sum_{n=0}^{\infty} \frac{4n}{n!(n+2)!} x^{n-1} + \sum_{n=0}^{\infty} \frac{4}{n!(n+2)!} x^{n-1} + \sum_{n=0}^{\infty} (n-2) n b_n x^{n-3} - \sum_{n=0}^{\infty} b_n x^{n-2}$$

$$= 0(-2) b_0 x^{-3} + (-b_0 - b_1) x^{-2} + \sum_{n=0}^{\infty} \frac{4(n+1)}{n!(n+2)!} x^{n-1} + \sum_{n=2}^{\infty} (n-2) n b_n x^{n-3} - \sum_{n=1}^{\infty} b_n x^{n-2}$$

$$-(b_0 + b_1) x^{-2} + \sum_{k=0}^{\infty} \left[\frac{4(k+1)}{k!(k+2)!} + k(k+2) b_{k+2} - b_{k+1} \right] x^{k-1}. \qquad \textbf{(13)}$$

Setting (13) equal to zero then gives $b_1 = -b_0$ **and**

$$\frac{4(k+1)}{k!(k+1)!} + k(k+2) b_{k+2} - b_{k+1} = 0, \qquad \textbf{For k=0, 1, 2, …} \qquad \textbf{(14)}$$

© Copyright Virtual University of Pakistan

When k=0 in equation (14) we have $2+0\cdot 2b_2 - b_1 = 0$ so that but $b_1 = 2$, $b_0 = -2$, but b_2 is arbitrary

Rewriting equation (14) as

$$b_{k+2} = \frac{b_{k+1}}{k(k+2)} - \frac{4(k+1)}{k!(k+2)!k(k+2)} \qquad (15)$$

and evaluating for k=1,2,… gives

$$b_3 = \frac{b_2}{3} - \frac{4}{9}$$

$$b_4 = \frac{1}{8}b_3 - \frac{1}{32} = \frac{1}{24}b_2 - \frac{25}{288}$$

and so on. Thus we can finally write

$$y_2 = y_1 \ln x + b_0 x^{-2} + b_1 x^{-1} + b_2 + b_3 x + \cdots$$
$$= y_1 \ln x - 2x^{-2} + 2x^{-1} + b_2 + \left(\frac{b_2}{3} - \frac{4}{9}\right)x + \cdots \qquad (16)$$

Where b_2 is arbitrary.

Equivalent Solution

At this point you may be wondering whether (*) and (16) are really equivalent. If we choose $c_2 = 4$ in equation (**), then

$$y_2 = y_1 \ln x + \left(-\frac{2}{x^2} + \frac{8}{3x} - \frac{38}{135}\right)\left(1 + x + \frac{1}{24}x^2 + \frac{1}{360}x^3 + \cdots\right)$$

$$y_2 = y_1 \ln x + \left(\frac{2}{3} - \frac{8}{24} + \frac{38}{360}\right)\left(-\frac{2}{x^2} + \frac{8}{3x} - \frac{38}{135}x + \cdots\right) \qquad (17)$$

$$= y_1 \ln x - 2x^{-2} + 2x^{-1} + \frac{29}{36} - \frac{19}{108}x + \ldots$$

Which is precisely obtained what we obtained from (16). If b_2 is chosen as $\frac{29}{36}$

The next example illustrates the case when the indicial roots are equal.

© Copyright Virtual University of Pakistan

Example:10

Find the general solution of $xy'' + y' - 4y = 0$ (18)

Solution:*The* assumption $y=\sum_{n=0}^{\infty} c_n x^{n+r}$ leads to

$$xy'' + y' - 4y = \sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1} - 4\sum_{n=0}^{\infty} c_n x^{n+r}$$

$$=\sum_{n=0}^{\infty}(n+r)^2 c_n x^{n+r-1} - 4\sum_{n=0}^{\infty} c_n x^{n+r}$$

$$= x^{r}\left[r^2 c_0 x^{-1} + \sum_{n=1}^{\infty}(n+r)^2 c_n x^{n-1} - 4\sum_{n=0}^{\infty} c_n x^{n}\right]$$

$$= x^{r}\left[r^2 c_0 x^{-1} + \sum_{k=0}^{\infty}(k+r+1)^2 c_{k+1} - 4c_k\right] x^{k} = 0$$

Therefore r^2 =0, and so the indicial roots are equal: $r_1 = r_2 = 0$. Moreover we have

$$(k+r+1)^2 c_{k+1} - 4c_k = 0, \text{k=0,1,2,}\ldots \quad (19)$$

Clearly the roots $r_1 = 0$ will yield one solution corresponding to the coefficients defined by the iteration of

$$c_{k+1} = \frac{4c_k}{(k+1)^2} \qquad \text{k=0,1,2,}\ldots$$

The result is

$$y_1 = c_0 \sum_{n=0}^{\infty} \frac{4^n}{(n!)^2} x^n, |x| < \infty \quad (20)$$

$$y_2 = y_1(x)\int \frac{e^{-\int \frac{1}{x})dx}\,dx}{y_1^2(x)} = y_1(x)\int \frac{dx}{x\left[1+4x+4x^2+\frac{16}{9}x^3+\cdots\right]^2}$$

$$= y_1(x)\int \frac{1}{x}\left[1-8x+40x^2-\frac{1472}{9}x^3+\cdots\right]dx$$

$$= y_1(x)\int \left[\frac{1}{x}-8+40x-\frac{1472}{9}x^2+\ldots\right]dx$$

© Copyright Virtual University of Pakistan

$$= y_1(x)\left[\ln x - 8x + 20x^2 - \frac{1472}{27}x^3 + \cdots\right]$$

Thus on the interval $(0,\infty)$ the general solution of (18) is

$$y = c_1 y_1(x) + c_2\left[y_1(x)\ln x + y_1(x)\left(-8x + 20x^2 - \frac{1472}{27}x^3 + \ldots\right)\right]$$

where $y_1(x)$ is defined by (20)

In caseII we can also determine $y_2(x)$ of example9 directly from assumption (4b)

Exercises

In problem 1-10 determine the singular points of each differential equation. Classify each the singular point as regular or irregular.

1 $x^3y' + 4x^2y' + 3y = 0$

2 $xy' - (x+3)^{-2}y = 0$

3 $(x^2 - 9)y' + (x+3) + 2y = 0$

4 $y' - \frac{1}{x}y' + \frac{1}{(x-1)^3}y = 0$

5 $(x^3 + 4x)y' - 2xy' + 6y = 0$)

6 $x^2(x-5)^2y' + 4xy' + (x-2)y = 0$

7 $(x^2 + x - 6)^2y' + (x+3)y' + (x-2)y = 0$

8 $x(x^2+1)^2y' + y = 0$

9 $x^3(x^2 - 25)(x-2)^2y' + 3x(x-2)y' + 7(x+5)y = 0$

10 $(x^3 - 2x^2 - 3x)^2y' + x(x+3)^2y' + (x+1)y = 0$

In problem 11-22 show that the indicial roots do not differ by an integer. Use the method of frobenius to obtain two linearly independent series solutions about the regular singular point $x_0 = 0$ Form the general solution on $(0,\infty)$

11. $2xy' - y' + 2y = 0$

12. $2xy' + 5y' + xy = 0$

13. $4xy' + \frac{1}{2}y' + y = 0$

14. $2x^2y' - xy' + (x^2+1)y = 0$

15. $3xy' + (2-x)y' - 2y = 0$

16. $x^2y' - \left(x - \frac{2}{9}\right)y' + xy = 0$

© Copyright Virtual University of Pakistan

17. $2xy' + (3+2x)y' + y = 0$

18. $x^2y' + xy' + \left(x^2 - \frac{4}{9}\right)y = 0$

19. $9x^2y' + 9x^2y' + 2y = 0$

20. $2x^2y' + 3xy' + (2x-1)y = 0$

21. $2x^2y' - x(x-1)y' - y = 0$

22. $x(x-2)y' - y' - 2y = 0$

In problem 23-34 show that the indicial roots differ by an integer. Use the method of frobenius to obtain two linearly independent series solutions about the regular singular point $x_0 = 0$ Form the general solution on $(0, \infty)$

23. $xy' + 2y' - xy = 0$

24. $x^2y' + xy' + \left(x^2 - \frac{1}{4}\right)y = 0$

25. $x(x-1)y' + 3y' - 2y = 0$

26. $y' + \frac{3}{x}y' - 2y = 0$

27. $xy' + (1-x)y' - y = 0$

28. $xy' + y = 0$

29. $xy' + y' + y = 0$

30. $xy' - y' + y = 0$

31. $x^2y' + x(x-1)y' + y = 0$

32. $xy' + y' - 4xy = 0$

33. $x^2y' + (x-1)y' - 2y = 0$

34. $xy' - y' + x^3y = 0$

© Copyright Virtual University of Pakistan

Lecture 33

Bessel's Differential Equation

A second order linear differential equation of the form

$$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+\left(x^2-v^2\right)y=0$$

is called Bessel's differential equation.

Solution of this equation is usually denoted by $J_v(x)$ and is known as Bessel's function. This equation occurs frequently in advanced studies in applied mathematics, physics and engineering.

Series Solution of Bessel's Differential Equation

Bessel's differential equation is

$$x^2y''+xy'+\left(x^2-v^2\right)y=0 \qquad (1)$$

If we assume that

$$y=\sum_{n=0}^{\infty}C_nx^{n+r}$$

Then

$$y'=\sum_{n=0}^{\infty}C_n(n+r)x^{n+r-1}$$

$$y'=\sum_{n=0}^{\infty}C_n(n+r)(n+r-1)x^{n+r-2}$$

So that

$$x^2y'+xy'+\left(x^2-v^2\right)y=\sum_{n=0}^{\infty}C_n(n+r)(n+r-1)x^{n+r}+\sum_{n=0}^{\infty}C_n(n+r)x^{n+r}$$

$$+\sum_{n=0}^{\infty}C_nx^{n+r+2}-v^2\sum_{n=0}^{\infty}C_nx^{n+r}=0$$

$$C_0\left(r^2-v^2\right)x^r+x^r\sum_{n=1}^{\infty}C_n\left[(n+r)(n+r-1)+(n+r)-v^2\right]x^n+x^r\sum_{n=0}^{\infty}C_nx^{n+2}=0 \quad \ldots (2)$$

From (2) we see that the indicial equation is $r^2-v^2=0$, so the indicial roots are $r_1=v$, $r_2=-v$. When $r_1=v$ then (2) becomes

© Copyright Virtual University of Pakistan

$$x^v \sum_{n=1}^{\infty} C_n n(n+2v)x^n + x^v \sum_{n=0}^{\infty} C_n x^{n+2} =$$

$$x^v \left[(1+2v)C_1 x + \sum_{n=2}^{\infty} C_n n(n+2v)x^n + \sum_{n=0}^{\infty} C_n x^{n+2} \right] = 0$$

$$\underset{k=n-2}{\qquad} \qquad \underset{k=n}{\qquad}$$

$$x^v \left[(1+2v)C_1 x + \sum_{k=0}^{\infty} \left[(k+2)(k+2+2v)C_{k+2} + C_k \right] x^{k+2} \right] = 0$$

We can write

$$(1+2v)C_1 = 0$$

$$(k+2)(k+2+2v)C_{k+2} + C_k = 0$$

$$C_{k+2} = \frac{-C_k}{(k+2)(k+2+2v)} \qquad (3)$$

$$k = 0,1,2,...$$

The choice $C_1 = 0$ in (3) implies

$$C_1 = C_3 = C_5 = ... = 0$$

so for $k = 0,2,4,...$ we find, after letting $k+2 = 2n$, $n = 1,2,3,...$ that

$$C_{2n} = \frac{-C_{2n-2}}{2^2 n(n+v)} \qquad \textbf{(2)}\ (4)$$

Thus

$$C_2 = -\frac{C_0}{2^2 \cdot 1 \cdot (1+v)}$$

$$C_4 = -\frac{C_2}{2^2 \cdot 2 \cdot (2+v)} = \frac{C_0}{2^4 \cdot 1 \cdot 2 \cdot (1+v)(2+v)}$$

$$C_6 = -\frac{C_4}{2^2 \cdot 3 \cdot (3+v)} = -\frac{C_0}{2^6 \cdot 1 \cdot 2 \cdot 3 \cdot (1+v)(2+v)(3+v)} \qquad \textbf{(3)}$$

… … …

$$C_{2n} = \frac{(-1)^n C_0}{2^{2n} \cdot n!(1+v)(2+v)\cdots(n+v)} \qquad n = 1,2,3,...$$

© Copyright Virtual University of Pakistan

It is standard practice to choose C_0 to be a specific value namely

$$C_0 = \frac{1}{2^v \Gamma(1+v)}$$

where $\Gamma(1+v)$ the Gamma function. Also

$$\Gamma(1+\alpha) = \alpha\Gamma(\alpha).$$

Using this property, we can reduce the indicated product in the denominator of (5) to one term. For example

$$\Gamma(1+v+1) = (1+v)\Gamma(1+v)$$

$$\Gamma(1+v+2) = (2+v)\Gamma(2+v)$$

$$= (2+v)(1+v)\Gamma(1+v)$$

Hence we can write (5) as

$$C_{2n} = \frac{(-1)^n}{2^{2n+v}\, n!(1+v)(2+v)\cdots(n+v)\Gamma(1+v)}$$

$$= \frac{(-1)^n}{2^{2n+v}\, n!\Gamma(1+v+n)}, \qquad n = 0,1,2,...$$

So the solution is

$$y = \sum_{n=0}^{\infty} C_{2n} x^{2n+v} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v}$$

If $v \geq 0$, the series converges at least on the interval $[0 \quad \infty)$.

Bessel's Function of the First Kind

As for $r_1 = v$, we have

$$J_v(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v} \qquad \textbf{(4)}$$

Also for the second exponent $r_2 = -v$, we have

$$J_{-v}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)\Gamma(1-v+n)} \left(\frac{x}{2}\right)^{2n-v} \qquad \textbf{(5)}$$

The function $J_v(x)$ and $J_{-v}(x)$ are called Bessel function of the first kind of order v and $-v$ respectively.

© Copyright Virtual University of Pakistan

Now some care must be taken in writing the general solution of (1). When $v = 0$, it is clear that (6) and (7) are the same. If $v > 0$ and $r_1 - r_2 = v - (-v) = 2v$ is not a positive integer, then $J_v(x)$ and $J_{-v}(x)$ are linearly independent solutions of (1) on $(0, \infty)$ and so the general solution of the interval would be

$$y = C_1 J_v(x) + C_2 J_{-v}(x)$$

If $r_1 - r_2 = 2v$ is a positive integer, a second series solution of (1) may exists.

Example 1

Find the general solution of the equation

$$x^2 y' + xy' + \left(x^2 - \frac{1}{4}\right) y = 0 \text{ on } (0, \infty)$$

Solution

The Bessel differential equation is

$$x^2 y' + xy' + (x^2 - v^2) y = 0 \quad (1)$$

$$x^2 y' + xy' + \left(x^2 - \frac{1}{4}\right) y = 0 \quad (2)$$

Comparing (1) and (2), we get $v^2 = \frac{1}{4}$, therefore $v = \pm\frac{1}{2}$

So general solution of (1) is $y = C_1 J_{1/2}(x) + C_2 J_{-1/2}(x)$

Example 2

Find the general solution of the equation

$$x^2 y' + xy' + \left(x^2 - \frac{1}{9}\right) y = 0$$

Solution

We identify $v^2 = \frac{1}{9}$, therefore $v = \pm\frac{1}{3}$

So general solution is $y = C_1 J_{1/3}(x) + C_2 J_{-1/3}(x)$

© Copyright Virtual University of Pakistan

Example 3

Derive the formula $xJ'_v(x) = vJ_v(x) - xJ_{v+1}(x)$

Solution

As $J_v(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v}$

$$xJ'_v(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2n+v)}{n!\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v}$$

$$= v \cdot \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v} + 2 \cdot \sum_{n=0}^{\infty} \frac{(-1)^n n}{n!\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v}$$

$$= vJ_v(x) + x \cdot \sum_{n=0}^{\infty} \frac{(-1)^n}{(n-1)!\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v-1}$$

$$= vJ_v(x) - x \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\Gamma(2+v+k)} \left(\frac{x}{2}\right)^{2k+v+1}$$

$$= vJ_v(x) - xJ_{v+1}(x)$$

So $xJ'_v(x) = vJ_v(x) - xJ_{v+1}(x)$

Example 4

Derive the recurrence relation $2J'_n(x) = J_{n-1}(x) - J_{n+1}(x)$

Solution:

As $J_n(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!} \left(\frac{x}{2}\right)^{n+2s}$

$$J'_n(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!} (n+2s) \left(\frac{x}{2}\right)^{n+2s-1} \left(\frac{1}{2}\right)$$

$$= \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!} (n+s+s) \left(\frac{x}{2}\right)^{n+2s-1} \left(\frac{1}{2}\right)$$

© Copyright Virtual University of Pakistan

$$= \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!}\,(n+s)\left(\frac{x}{2}\right)^{n+2s-1}\left(\frac{1}{2}\right) + \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!}\,s\cdot\left(\frac{x}{2}\right)^{n+2s-1}\left(\frac{1}{2}\right)$$

$$= \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)(n+s-1)!}\,(n+s)\left(\frac{x}{2}\right)^{n+2s-1}\left(\frac{1}{2}\right) + \sum_{s=0}^{\infty} \frac{(-1)^s s}{s(s-1)!(n+s)!}\cdot\left(\frac{x}{2}\right)^{n+2s-1}\left(\frac{1}{2}\right)$$

$$= \frac{1}{2}\sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n-1+s)!}\left(\frac{x}{2}\right)^{n-1+2s} + \frac{1}{2}\sum_{s=1}^{\infty} \frac{(-1)^s}{(s-1)!(n+s)!}\cdot\left(\frac{x}{2}\right)^{n+2s-1}$$

$$= \frac{1}{2}J_{n-1}(x) + \frac{1}{2}\sum_{s=1}^{\infty} \frac{(-1)^s}{(s-1)!(n+s)!}\left(\frac{x}{2}\right)^{n+2s-1}$$

Put $s-1=p$ in 2nd term $\Rightarrow s=p+1$

$$= \frac{1}{2}J_{n-1}(x) + \frac{1}{2}\sum_{p=0}^{\infty} \frac{(-1)^{p+1}}{p!(n+p+1)!}\left(\frac{x}{2}\right)^{n+2(p+1)-1}$$

$$= \frac{1}{2}J_{n-1}(x) + \frac{1}{2}\sum_{p=0}^{\infty} \frac{-1(-1)^p}{p!(n+1+p)!}\left(\frac{x}{2}\right)^{n+1+2p}$$

$$J_n'(x) = \frac{1}{2}J_{n-1}(x) - \frac{1}{2}J_{n+1}(x)$$

$$\Rightarrow 2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$$

Example 5

Derive the expression of $J_n(x)$ for $n=\pm\frac{1}{2}$

Solution:

Consider $$J_n(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!}\left(\frac{x}{2}\right)^{n+2s}$$

As $n! = \Gamma(n+1)$

$$\Rightarrow J_n(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{\Gamma(s+1)\Gamma(n+s+1)}\left(\frac{x}{2}\right)^{n+2s}$$

put $n = 1/2$

$$J_{1/2}(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{\Gamma(s+1)\Gamma(1/2+s+1)}\left(\frac{x}{2}\right)^{\frac{1}{2}+2s}$$

© Copyright Virtual University of Pakistan

$$= \sum_{s=0}^{\infty} \frac{(-1)^s}{\Gamma(s+1)\Gamma(s+3/2)} \left(\frac{x}{2}\right)^{\frac{1}{2}+2s}$$

Expanding R.H.S of above

$$J_{1/2}(x) = \frac{(-1)^0}{\Gamma(0+1)\Gamma(0+3/2)} \left(\frac{x}{2}\right)^{\frac{1}{2}+0} + \frac{(-1)^1}{\Gamma(1+1)\Gamma(1+3/2)} \left(\frac{x}{2}\right)^{\frac{1}{2}+2(1)}$$

$$+ \frac{(-1)^2}{\Gamma(2+1)\Gamma(2+3/2)} \left(\frac{x}{2}\right)^{\frac{1}{2}+2(2)} + \frac{(-1)^3}{\Gamma(3+1)\Gamma(3+3/2)} \left(\frac{x}{2}\right)^{\frac{1}{2}+2(3)} + \cdots$$

$$= \frac{2}{\sqrt{\pi}} \left(\frac{x}{2}\right)^{\frac{1}{2}} - \frac{2 \cdot 2}{3\sqrt{\pi}} \left(\frac{x}{2}\right)^{\frac{5}{2}} + \frac{2 \cdot 2 \cdot 2}{2 \cdot 5 \cdot 3\sqrt{\pi}} \left(\frac{x}{2}\right)^{\frac{9}{2}} - \cdots$$

$$= \frac{1}{\sqrt{\pi}} \left[2 \frac{\sqrt{x}}{\sqrt{2}} - \frac{4}{3} \frac{\sqrt{x} \cdot x^2}{2^{5/2}} + \frac{4}{15} \frac{\sqrt{x} \cdot x^4}{2^{9/2}} - \cdots \right]$$

$$= \frac{\sqrt{x}}{\sqrt{\pi}} \left[\frac{2}{\sqrt{2}} - \frac{4x^2}{3 \cdot 2^{5/2}} + \frac{4}{15} \frac{x^4}{2^{9/2}} - \cdots \right]$$

$$= \frac{\sqrt{2} \cdot \sqrt{x}}{\sqrt{\pi}} \left[\frac{2}{\sqrt{2}\sqrt{2}} - \frac{4x^2}{3 \cdot \sqrt{2} \cdot 2^{5/2}} + \frac{4x^4}{15 \cdot \sqrt{2} \cdot 2^{9/2}} - \cdots \right]$$

$$= \frac{\sqrt{2} \cdot \sqrt{x}}{\sqrt{\pi}} \left[1 - \frac{x^2}{6} + \frac{x^4}{120} - \cdots \right]$$

$$= \frac{\sqrt{2} \cdot \sqrt{x}}{\sqrt{\pi}} \cdot \frac{1}{x} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \right)$$

$$= \frac{\sqrt{2} \cdot \sqrt{x}}{\sqrt{\pi}} \sin x$$

$$\Rightarrow J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

© Copyright Virtual University of Pakistan

Similarly for $n=-1/2$, we proceed further as before,

$$J_n(x)=\sum_{s=0}^{\infty}\frac{(-1)^s}{s!(n+s)!}\left(\frac{x}{2}\right)^{n+2s} \quad \text{where } n!=\Gamma(n+1)$$

$$\Rightarrow J_n(x)=\sum_{s=0}^{\infty}\frac{(-1)^s}{\Gamma(s+1)\Gamma(n+s+1)}\left(\frac{x}{2}\right)^{n+2s}$$

put $n=-\frac{1}{2}$

$$J_{-1/2}(x)=\sum_{s=0}^{\infty}\frac{(-1)^s}{\Gamma(s+1)\Gamma(-1/2+s+1)}\left(\frac{x}{2}\right)^{-\frac{1}{2}+2s}$$

$$J_{-1/2}(x)=\sum_{s=0}^{\infty}\frac{(-1)^s}{\Gamma(s+1)\Gamma(s+1/2)}\left(\frac{x}{2}\right)^{-\frac{1}{2}+2s}$$

Expanding the R.H.S of above we get

$$J_{-1/2}(x)=\frac{(-1)^0}{\Gamma(0+1)\Gamma(0+1/2)}\left(\frac{x}{2}\right)^{-\frac{1}{2}}+\frac{(-1)^1}{\Gamma(1+1)\Gamma(1+1/2)}\left(\frac{x}{2}\right)^{-\frac{1}{2}+2(1)}$$
$$+\frac{(-1)^2}{\Gamma(2+1)\Gamma(2+1/2)}\left(\frac{x}{2}\right)^{-\frac{1}{2}+2(2)}+\cdots$$

$$J_{-1/2}(x)=\frac{1}{\Gamma(1)\Gamma(1/2)}\sqrt{\frac{2}{x}}-\frac{1}{\Gamma(2)\Gamma(3/2)}\left(\frac{x}{2}\right)^{\frac{3}{2}}+\frac{1}{\Gamma(3)\Gamma(5/2)}\left(\frac{x}{2}\right)^{\frac{7}{2}}-\cdots$$

$$=\frac{1}{(1)\Gamma(1/2)}\sqrt{\frac{2}{x}}-\frac{1}{1\cdot\frac{1}{2}\cdot\Gamma(1/2)}\left(\frac{x}{2}\right)^{\frac{3}{2}}+\frac{1}{2\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2)}\left(\frac{x}{2}\right)^{\frac{7}{2}}-\cdots$$

$$=\frac{1}{\Gamma(1/2)}\left[\frac{\sqrt{2}}{\sqrt{x}}-\frac{2x^{3/2}}{2^{3/2}}+\frac{2\cdot 2}{2\cdot 3}\frac{x^{7/2}}{2^{7/2}}-\cdots\right]$$

$$=\frac{1}{\sqrt{\pi}}\left[\frac{\sqrt{2}}{\sqrt{x}}-\frac{2x^{3/2}}{2^{3/2}}+\frac{2}{3}\frac{x^{7/2}}{2^{7/2}}-\cdots\right]$$

$$=\frac{\sqrt{2}}{\sqrt{\pi}}\left[\frac{\sqrt{2}}{\sqrt{x}\sqrt{2}}-\frac{2x^{3/2}}{4}+\frac{2}{3}\frac{x^{7/2}}{16}-\cdots\right]$$

© Copyright Virtual University of Pakistan

$$= \frac{\sqrt{2}}{\sqrt{\pi}}\left[\frac{\sqrt{2}}{\sqrt{x}\sqrt{2}} - \frac{x^{3/2}}{2} + \frac{1}{3}\frac{x^{7/2}}{8} - \cdots\right]$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}}\left[\frac{1}{\sqrt{x}} - \frac{x^{3/2}}{2} + \frac{x^{7/2}}{24} - \cdots\right]$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}\sqrt{x}}\left[\frac{\sqrt{x}}{\sqrt{x}} - \frac{x^{3/2}}{2} + \frac{x^{7/2}}{24} - \cdots\right]$$

$$= \frac{\sqrt{2}}{\sqrt{\pi x}}\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right]$$

$$= \sqrt{\frac{2}{\pi x}}\cos x \qquad \because \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\Rightarrow J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\cos x$$

Remarks:

Bessel functions of index half an odd integer are called Spherical Bessel functions. Like other Bessel functions spherical Bessel functions are used in many physical problems.

© Copyright Virtual University of Pakistan

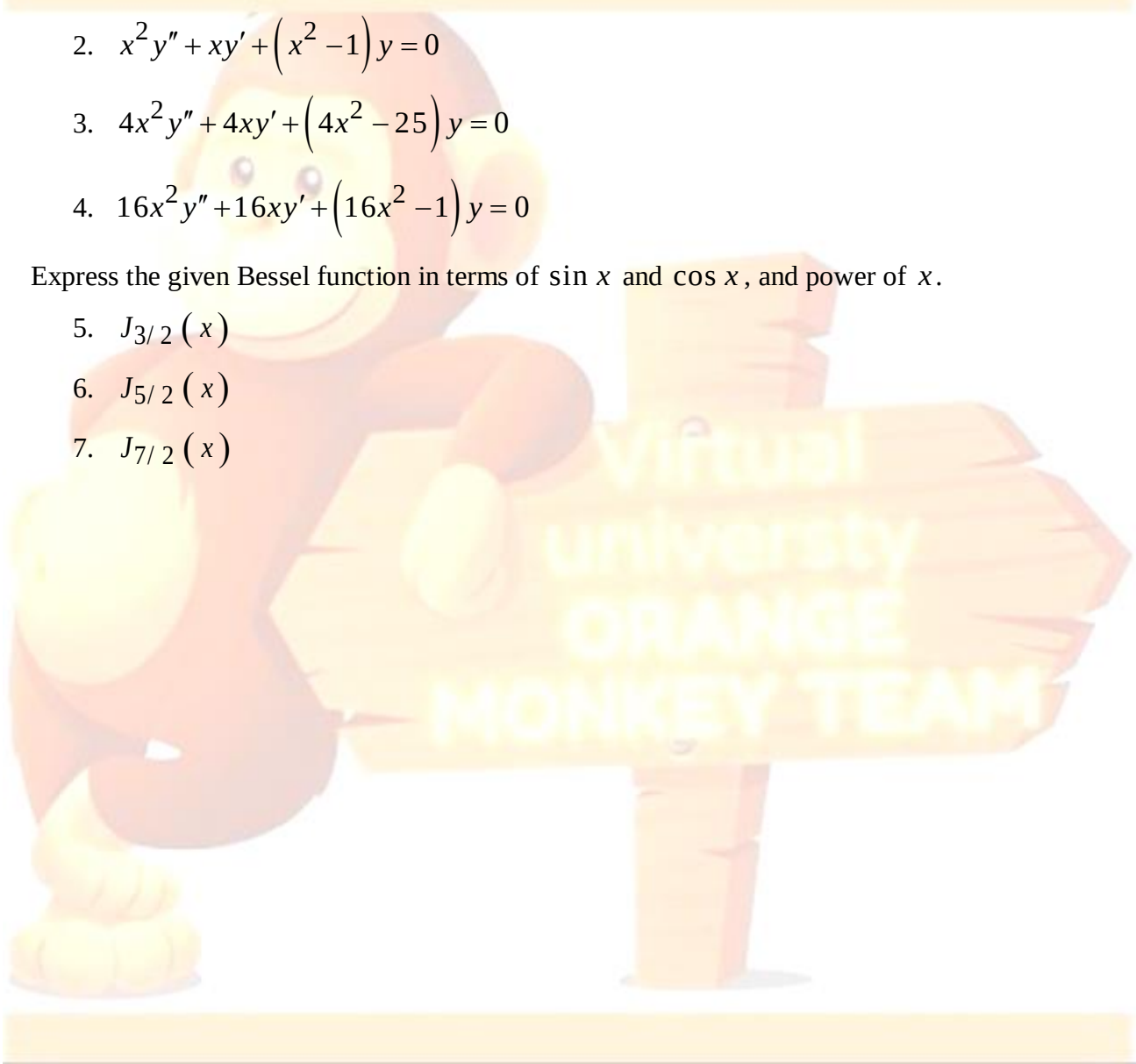
Exercise

Find the general solution of the given differential equation on $(0, \infty)$.

1. $x^2 y'' + xy' + \left(x^2 - \frac{1}{9}\right) y = 0$
2. $x^2 y'' + xy' + \left(x^2 - 1\right) y = 0$
3. $4x^2 y'' + 4xy' + \left(4x^2 - 25\right) y = 0$
4. $16x^2 y'' + 16xy' + \left(16x^2 - 1\right) y = 0$

Express the given Bessel function in terms of $\sin x$ and $\cos x$, and power of x.

5. $J_{3/2}(x)$
6. $J_{5/2}(x)$
7. $J_{7/2}(x)$



© Copyright Virtual University of Pakistan

Lecture 34

Legendre's Differential Equation

A second order linear differential equation of the form

$$\left(1-x^2\right)y' - 2xy' + n(n+1)y = 0$$

is called Legendre's differential equation and any of its solution is called Legendre's function. If n is positive integer then the solution of Legendre's differential equation is called a Legendere's polynomial of degree n and is denoted by $P_n(x)$.

We assume a solution of the form $y = \sum_{k=0}^{\infty} C_k x_k$

$$\therefore \left(1-x^2\right)y'' - 2xy' + n(n+1)y =$$

$$\left(1-x^2\right)\sum_{k=2}^{\infty} C_k k(k-1)x^{k-2} - 2\sum_{k=1}^{\infty} C_k k x^k + n(n+1)\sum_{k=0}^{\infty} C_k x^k$$

$$= \sum_{k=2}^{\infty} C_k k(k-1)x^{k-2} - \sum_{k=2}^{\infty} C_k k(k-1)x^k - 2\sum_{k=1}^{\infty} C_k k x^k + n(n+1)\sum_{k=0}^{\infty} C_k x^k$$

$$= \left[n(n+1)C_0 + 2C_2\right]x^0 + \left[n(n+1)C_1 - 2C_1 + 6C_3\right]x + \sum_{k=4}^{\infty} C_k k(k-1)x^{k-2}$$

$j=k-2$

$$- \sum_{k=2}^{\infty} C_k k(k-1)x^k - 2\sum_{k=2}^{\infty} C_k k x^k + n(n+1)\sum_{k=2}^{\infty} C_k x^k$$

$j=k$ $\quad$ $j=k$ $\quad$ $j=k$

$$= \left[n(n+1)C_0 + 2C_2\right] + \left[(n-1)(n+2)C_1 + 6C_3\right]x$$

$$+ \sum_{j=2}^{\infty} \left[(j+2)(j+1)C_{j+2} + (n-j)(n+j+1)C_j\right]x^j = 0$$

$$\Rightarrow \quad n(n+1)C_0 + 2C_2 = 0$$

$$(n-1)(n+2)C_1 + 6C_3 = 0$$

$$(j+2)(j+1)C_{j+2} + (n-j)(n+j+1)C_j = 0, \quad j = 2,3,4,\ldots$$

or $$C_2 = -\frac{n(n+1)}{2!}C_0$$

© Copyright Virtual University of Pakistan

$$C_3 = -\frac{(n-1)(n+2)}{3!}C_1$$

$$C_{j+2} = -\frac{(n-j)(n+j+1)}{(j+2)(j+1)}C_j; \quad j = 2,3,\ldots \quad (1)$$

From Iteration formula (1)

$$C_4 = -\frac{(n-2)(n+3)}{4\cdot 3}C_2 = \frac{(n-2)(n)(n+1)(n+3)}{4!}C_0$$

$$C_5 = -\frac{(n-3)(n+4)}{5\cdot 4}C_3 = \frac{(n-3)(n-1)(n+2)(n+4)}{5!}C_1$$

$$C_6 = -\frac{(n-4)(n+5)}{5\cdot 6}C_4 = -\frac{(n-4)(n-2)\ n(n+1)(n+3)(n+5)}{6!}C_0$$

$$C_7 = -\frac{(n-5)(n+6)}{7\cdot 6}C_5 = -\frac{(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!}C_1$$

and so on. Thus at least $|x|<1$, we obtain two linearly independent power series solutions.

$$y_1(x) = C_0\left[1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 - \frac{(n-4)(n-2)n(n+1)(n+3)(n+5)}{6!}x^6 + \cdots\right]$$

$$y_2(x) = C_1\left[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - \frac{(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!}x^7 + \cdots\right]$$

Note that if n is even integer, the first series terminates, where $y_2(x)$ is an infinite series. For example if $n=4$, then

$$y_1(x) = C_0\left[1 - \frac{4\cdot 5}{2!}x^2 + \frac{2\cdot 4\cdot 5\cdot 7}{4!}x^4\right] = C_0\left[1 - 10x^2 + \frac{35}{3}x^4\right]$$

Similarly, when n is an odd integer, the series for $y_2(x)$ terminates with x^n .i.e when n is a non-negative integer, we obtain an nth-degree polynomial solution of Legendre's equation. Since we know that a constant multiple of a solution of Legendre's equation is

© Copyright Virtual University of Pakistan

also a solution, it is traditional to choose specific values for C_0 and C_1 depending on whether n is even or odd positive integer, respectively.

For $n=0$, we choose $C_0=1$ and for $n=2,4,6,...$

$$C_0=(-1)^{n/2}\frac{1\cdot 3\cdot ...(n-1)}{2\cdot 4\cdot ...(n)}$$

Whereas for $n=1$, we choose $C_1=1$and for $n=3,5,7,...$

$$C_1=(-1)^{(n-1)/2}\frac{1\cdot 3\cdot ...n}{2\cdot 4...(n)}$$

For example, when $n=4$, we have

$$y_1(x)=(-1)^{4/2}\frac{1\cdot 3}{2\cdot 4}\left[1-10x^2+\frac{35}{3}x^4\right]$$

$$=\frac{3}{8}-\frac{30}{8}x^2+\frac{35}{8}x^4$$

$$y_1(x)=\frac{1}{8}\left(35x^4-30x^2+3\right)$$

Legendre's Polynomials are specific n[th] degree polynomials and are denoted by $P_n(x)$. From the series for $y_1(x)$and $y_2(x)$and from the above choices of C_0 and C_1, we find that the first several Legendre's polynomials are

$$P_0(x)=1$$

$$P_1(x)=x$$

$$P_2(x)=\frac{1}{2}\left(3x^2-1\right)$$

$$P_3(x)=\frac{1}{2}\left(5x^3-3x\right)$$

$$P_4(x)=\frac{1}{8}\left(35x^4-30x^2+3\right)$$

$$P_5(x)=\frac{1}{8}\left(63x^5-70x^3+15x\right)$$

Note that $P_0(x),P_1(x),P_2(x),P_3(x),...$are, in turn particular solution of the differential equations

© Copyright Virtual University of Pakistan

$n=0$	$\left(1-x^2\right)y'-2xy'=0$
$n=1$	$\left(1-x^2\right)y'-2xy'-2y=0$
$n=2$	$\left(1-x^2\right)y'-2xy'+6y=0$
$n=3$	$\left(1-x^2\right)y'-2xy'+12y=0$
…	… … … …

Rodrigues Formula for Legendre's Polynomials

The Legendre Polynomials are also generated by Rodrigues formula

$$P_n(x)=\frac{1}{2^n n!}\frac{d^n}{dx^n}\left(x^2-1\right)^n$$

Generating Function For Legendre's Polynomials

The Legendre's polynomials are the coefficient of z^n in the expansion of

$$\phi=\left(1-2xz+z^2\right)^{-\frac{1}{2}}$$

in ascending powers of z.

Now $$\phi=\left(1-2xz+z^2\right)^{-\frac{1}{2}}=\left\{1-z\left(2x-z\right)\right\}^{-\frac{1}{2}}$$

Therefore by Binomial Series

$$\phi=1+\frac{1}{2}z\left(2x-z\right)+\frac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2!}\left\{-z\left(2x-z\right)\right\}^2+\frac{-\frac{1}{2}\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}\left\{-z\left(2x-z\right)\right\}^3+\cdots$$

$$=1+\frac{1}{2}z\left(2x-z\right)+\frac{3}{8}z^2\left(4x^2+z^2-4xz\right)+\frac{5}{16}z^3\left(8x^3-z^3-12x^2z+6xz^2\right)+\cdots$$

$$=1+zx-\frac{1}{2}z^2+\frac{3}{2}x^2z^2+\frac{3}{8}z^4-\frac{3}{2}xz^3-\frac{5}{2}x^3z^3-\frac{5}{16}z^6-\frac{15}{4}x^2z^4+\frac{15}{8}xz^5+\cdots$$

$$=1+xz+\frac{1}{2}\left(3x^2-1\right)z^2+\frac{1}{2}\left(5x^3-3x\right)z^3+\frac{1}{8}\left(35x^4-30x^2+3\right)z^4+\cdots \qquad (1)$$

Also

$$\sum_{n=0}^{\infty}P_n(x)z^n=P_0(x)+P_1(x)z+P_2(x)z^2+P_3(x)z^3+\cdots$$

Equating Coefficients of (1) and (2)

© Copyright Virtual University of Pakistan

$$P_0(x) = 1$$
$$P_1(x) = x$$
$$P_2(x) = \frac{1}{2}\left(3x^2 - 1\right)$$
$$P_3(x) = \frac{1}{2}\left(5x^3 - 3x\right)$$
$$P_4(x) = \frac{1}{8}\left(35x^4 - 30x^2 + 3\right)$$

Which are Legendre's Polynomials

Recurrence Relation

Recurrence relations that relate Legendre's polynomials of different degrees are also very important in some aspects of their application. We shall derive one such relation using the formula

$$\left(1 - 2xt + t^2\right)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(x) \cdot t^n \qquad (1)$$

Differentiating both sides of (1) with respect to t gives

$$\left(1 - 2xt + t^2\right)^{-\frac{3}{2}}(x - t) = \sum_{n=0}^{\infty} nP_n(x)t^{n-1} = \sum_{n=1}^{\infty} nP_n(x)t^{n-1}$$

so that after multiplying by $1 - 2xt + t^2$, we have

$$(x - t)\left(1 - 2xt + t^2\right)^{-\frac{1}{2}} = \left(1 - 2xt + t^2\right)\sum_{n=1}^{\infty} nP_n(x)t^{n-1}$$

$$(x - t)\sum_{n=0}^{\infty} P_n(x)t^n = \left(1 - 2xt + t^2\right)\sum_{n=1}^{\infty} nP_n(x)t^{n-1}$$

$$\sum_{n=0}^{\infty} xP_n(x)t^n - \sum_{n=0}^{\infty} P_n(x)t^{n+1} - \sum_{n=1}^{\infty} nP_n(x)t^{n-1} + 2x\sum_{n=1}^{\infty} nP_n(x)t^n$$
$$-\sum_{n=1}^{\infty} nP_n(x)t^{n+1} = 0$$

$$x + x2t + \sum_{n=2}^{\infty} xP_n(x)t^n - t - \sum_{n=1}^{\infty} P_n(x)t^{n+1} - x - 2\left(\frac{3x^2 - 1}{2}\right)t$$
$$-\sum_{n=3}^{\infty} nP_n(x)t^{n-1} + 2x^2t + 2x\sum_{n=2}^{\infty} nP_n(x)t^n - \sum_{n=1}^{\infty} nP_n(x)t^{n+1} = 0$$

© Copyright Virtual University of Pakistan

Observing the appropriate cancellations, simplifying and changing the summation indices

$$\sum_{k=2}^{\infty}\left[-(k+1)P_{k+1}(x)+(2k+1)xP_k(x)-kP_{k-1}(x)\right]t^k=0$$

Equating the total coefficient of t^k to zero gives the three-term recurrence relation

$$(k+1)P_{k+1}(x)-(2k+1)xP_k(x)+kP_{k-1}(x)=0, \qquad k=2,3,4,...$$

Legendre's Polynomials are orthogonal

Proof:

Legendre's Differential Equation is $\left(1-x^2\right)y''-2xy'+n(n+1)y=0$

Let $P_n(x)$ and $P_m(x)$ are two solutions of Legendre's differential equation then

$$\left(1-x^2\right)P_n''(x)-2xP_n'(x)+n(n+1)P_n(x)=0\text{, and}$$

$$\left(1-x^2\right)P_m''(x)-2xP_m'(x)+m(m+1)P_m(x)=0$$

which we can write

$$\left[\left(1-x^2\right)P_n'(x)\right]'+n(n+1)P_n(x)=0 \qquad (1)$$

$$\left[\left(1-x^2\right)P_m'(x)\right]'+m(m+1)P_m(x)=0 \qquad (2)$$

Multiplying (1) by $P_m(x)$ and (2) by $P_n(x)$ and subtracting, we get

$$P_m(x)\left\{\left(1-x^2\right)P_n'\right\}'-P_n(x)\left\{\left(1-x^2\right)P_m'(x)\right\}' \qquad (3)$$
$$+\left\{n(n+1)-m(m+1)\right\}P_m(x)P_n(x)=0$$

Now

Add and subtract $(1-x^2)P'_mP'_n$ *to formulizethe above*

$$P_m(x)\left\{(1-x^2)P'_n\right\}'-P_n(x)\left\{(1-x^2)P'_m\right\}'$$

© Copyright Virtual University of Pakistan

$$= \left(1-x^2\right) P_m'(x) P_n'(x) + P_m(x) \left[\left(1-x^2\right) P_n'(x)\right]'$$
$$- \left(1-x^2\right) P_m'(x) P_n'(x) + P_n(x) \left[\left(1-x^2\right) P_m'(x)\right]',$$

$$= \left(1-x^2\right)\left[P_n(x)P_n'(x) - P_m'(x)P_n(x)\right]'$$

Which shows that (3) can be written as

$$\left[\left(1-x^2\right)\left\{P_m(x)P_n'(x) - P_m'(x)P_n(x)\right\}\right]'$$
$$+\left[n(n+1) - m(m+1)\right] P_m(x) P_n(x) = 0$$

$$\left(\left(1-x^2\right)\left\{P_m(x)P_n'(x) - P_m'(x)P_n\right\}\right)' + (n-m)(n+m+1) P_m(x) P_n(x) = 0$$

$$(n-m)(m+n+1) P_m(x) P_n(x) = \left(\left(1-x^2\right)\left\{P_m'(x)P_n(x) - P_m(x)P_n'(x)\right\}\right)'$$

$$(n-m)(m+n+1)\int_a^b P_m(x) P_n(x)dx = \int_a^b \left(\left(1-x^2\right)\left\{P_m(x)P_n'(x) - P_m(x)P_n'(x)\right\}\right)' dx$$

$$(n-m)(m+n+1)\int_a^b P_m(x) P_n(x)dx = \left(1-x^2\right)\left\{P_m'(x)P_n(x) - P_m(x)P_n'(x)\right\}\Big|_a^b$$

As $1-x^2 = 0$ for $x = \pm 1$ so

$$(n-m)(n+m+1)\int_{-1}^{1} P_m(x) P_n(x)\,dx = 0 \qquad \text{for } x = \pm 1$$

Since m & n are non-negative

$$\Rightarrow \int_{-1}^{1} P_m(x) P_n(x)\,dx = 0 \quad \text{for } m \neq n$$

which shows that Legendre's Polynomials are orthogonal w.r.to the weight function $w(x) = 1$ over the interval $[-1 \quad 1]$

© Copyright Virtual University of Pakistan

Normality condition for Legendre' Polynomials

Consider the generating function

$$\left(1-2xt+t^2\right)^{-\frac{1}{2}} = \sum_{m=0}^{\infty} P_m(x)\,t^m \qquad (1)$$

Also

$$\left(1-2xt+t^2\right)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(x)\,t^n \qquad (2)$$

Multiplying (1) and (2)

$$\left(1-2xt+t^2\right)^{-1} = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} P_m(x)\,P_n(x)\,t^{m+n}$$

Integrating from -1 to 1

$$\int_{-1}^{1} \frac{1}{\left(1-2xt+t^2\right)}dx = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\int_{-1}^{1} P_m(x)\,P_n(x)\,t^{m+n}\,dx$$

$$-\frac{1}{2t}\int_{-1}^{1} \frac{-2t}{\left(1-2xt+t^2\right)}dx = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\int_{-1}^{1} P_m(x)\,P_n(x)\,t^{m+n}\,dx$$

$$-\frac{1}{2t}\ln\left(1-2xt+t^2\right)\Big|_{-1}^{1} = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\int_{-1}^{1} P_m(x)\,P_n(x)\,t^{m+n}\,dx$$

$$\Rightarrow \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\int_{-1}^{1} P_m(x)\,P_n(x)\,t^{m+n}\,dx = -\frac{1}{2t}\left[\ln\left(1-2t+t^2\right)-\ln\left(1+2t+t^2\right)\right]$$

$$= -\frac{1}{2t}\left[\ln(1-t)^2-\ln\left(1+t^2\right)\right]$$

$$= -\frac{1}{2t}\left\{\ln(1+t)^2-\ln\left(1-t^2\right)\right\}$$

© Copyright Virtual University of Pakistan

$$= -\frac{1}{t}\left[\ln(1+t) - \ln(1-t)\right]$$

$$= \frac{1}{t}\left[\left(t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \cdots\right) - \left(-t - \frac{t^2}{2} - \frac{t^3}{3} - \frac{t^4}{4} - \cdots\right)\right]$$

$$= \frac{1}{t}\left\{2t + \frac{2t^3}{3} + \frac{2t^5}{5} + \cdots\right\}$$

$$= \frac{2}{t}\left\{t + \frac{t^3}{3} + \frac{t^5}{5} + \cdots\right\}$$

$$= 2\left\{1 + \frac{t^2}{3} + \frac{t^4}{5} + \cdots\right\}$$

$$\Rightarrow \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\int_{-1}^{1} P_m(x)\,P_n(x)\,t^{m+n}\,dx = 2\left\{1 + \frac{t^2}{3} + \frac{t^4}{5} + \cdots\right\}$$

for $m = n$

$$\Rightarrow \sum_{n=0}^{\infty}\int_{-1}^{1} P_n(x)\,P_n(x)\,t^{n+n}\,dx = 2\left\{1 + \frac{t^2}{3} + \frac{t^4}{5} + \cdots\right\}$$

$$\Rightarrow \sum_{n=0}^{\infty}\int_{-1}^{1}\left[P_n(x)\right]^2 t^{2n}\,dx = 2\left\{1 + \frac{t^{2(1)}}{2(1)+1} + \frac{t^{2(2)}}{2(2)+1} + \cdots + \frac{t^{2n}}{2(n)+1}\right\}$$

Equating coefficient of t^{2n} on both sides

$$\Rightarrow \int_{-1}^{1}\left[P_n(x)\right]^2 dx = \frac{2}{2n+1}$$

$$\Rightarrow \int_{-1}^{1} P_n(x)\,P_n(x)\,dx = \frac{2}{2n+1}$$

$$\Rightarrow \int_{-1}^{1} P_n(x)\,P_n(x)\,\frac{2n+1}{2}\,dx = 1$$

© Copyright Virtual University of Pakistan

which shows that Legender polynomials are normal with respect to the weight function $w(x) = \frac{2n+1}{2}$ over the interval $-1 < x < 1$.

Remark:

Orthognality condition for $P_n(x)$ can also be written as

$$\int_{-1}^{1} P_n(x) P_n(x) dx = \left(\frac{2}{2n+1}\right)\delta_{m,n}$$

where $\delta_{m,n} = \begin{cases} 0, & \text{if } m \neq n \\ 1, & \text{otherwise} \end{cases}$

Exercise

1. Show that the Legendre's equation has an alternative form
$$\frac{d}{dx}\left[\left(1-x^2\right)\frac{dy}{dx}\right] + n(n+1)y = 0$$

2. Show that the equation
$\sin\theta \frac{d^2y}{d\theta^2} + \cos\theta \frac{dy}{d\theta} + n(n+1)(\sin\theta)y = 0$ can be transformed into Legendre's equation by means of the substitution $x = \cos\theta$

3. Use the explicit Legendre's polynomials $P_1(x)$, $P_2(x)$, $P_2(x)$, and $P_3(x)$ to evaluate $\int_{-1}^{1} P_n^2 dx$ for $n = 0,1,2,3$. Generalize the results.

4. Use the explicit Legendre polynomials $P_1(x)$, $P_2(x)$, $P_2(x)$, and $P_3(x)$ to evaluate $\int_{-1}^{1} P_n(x) P_m(x) dx$ for $n \neq m$. Generalize the results.

5. The Legendre's polynomials are also generated by **Rodrigues' formula**
$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}\left(x^2-1\right)^n$$
verify the results for $n = 0,1,2,3$.

© Copyright Virtual University of Pakistan

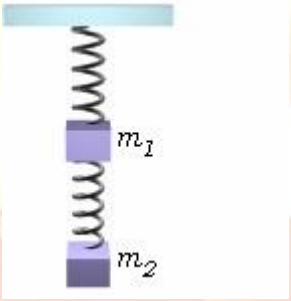
Lecture 35

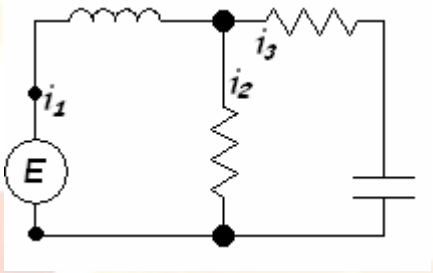
Systems of Linear Differential Equations

- Recall that the mathematical model for the motion of a mass attached to a spring or for the response of a series electrical circuit is a differential equation.

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=f(x)$$

- However, we can attach two or more springs together to hold two masses m_1 and m_2 . Similarly a network of parallel circuits can be formed.





- To model these latter situations, we would need two or more coupled or simultaneous equations to describe the motion of the masses or the response of the network.
- Therefore, in this lecture we will discuss the theory and solution of the systems of simultaneous linear differential equations with constant coefficients.

Note that

An nth order linear differential equation with constant coefficients a_0, a_1, ..., a_n is an equation of the form

$$a_n\frac{d^ny}{dx^n}+a_{n-1}\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1\frac{dy}{dx}+a_0y=g(x)$$

If we write $D=\frac{d}{dx}, D^2=\frac{d^2}{dx^2},\cdots, D^n=\frac{d^n}{dx^n}$ then this equation can be written as follows

$$\left(a_nD^n+a_{n-1}D^{(n-1)}+\cdots+a_1D+a_0\right)y=g(t)$$

Simultaneous Differential Equations

The simultaneous ordinary differential equations involve two or more equations that contain derivatives of two or more unknown functions of a single independent variable.

© Copyright Virtual University of Pakistan

Example 1

If x, y and z are functions of the variable t, then

$$4\frac{d^2x}{dt^2} = -5x + y$$

$$2\frac{d^2y}{dt^2} = 3x - y$$

and

$$x' - 3x + y' + z' = 5$$

$$x + y' - 6z' = t - 1$$

are systems of simultaneous differential equations.

Solution of a System

A solution of a system of differential equations is a set of differentiable functions

$$x = f(t),\ y = g(t),\ x = h(t),...$$

those satisfy each equation of the system on some interval I.

Systematic Elimination: Operator Method

- This method of solution of a system of linear homogeneous or linear non-homogeneous differential equations is based on the process of systematic elimination of the dependent variables.
- This elimination provides us a single differential equation in one of the dependent variables that has not been eliminated.
- This equation would be a linear homogeneous or a linear non-homogeneous differential equation and can be solved by employing one of the methods discussed earlier to obtain one of the dependent variables.

Notice that the analogue of multiplying an algebraic equation by a constant is operating on a differential equation with some combination of derivatives.

The Method

Step 1 First write the differential equations of the system in a form that involves the differential operator D.

Step 2 We retain first of the dependent variables and eliminate the rest from the differential equations of the system.

Step 3 The result of this elimination is to be a single linear differential equation with constant coefficients in the retained variable. We solve this equation to obtain the value of this variable.

Step 4 Next, we retain second of the dependent variables and eliminate all others variables

Step 5 The result of the elimination performed in step 4 is to be again a single linear differential equation with constant coefficients in the retained 2nd variable. We again solve this equation and obtain the value of the second dependent variable. This process of elimination is continued untill all the variables are taken care of.

© Copyright Virtual University of Pakistan

Step 6 The computed values of the dependent variables don't satisfy the given system for every choice of all the arbitrary constants. By substituting the values of the dependent variables computed in step 5 into an equation of the original system, we can reduce the number of constant from the solution set.

Step 7 We use the work done in step number 6 to write the solution set of the given system of linear differential equations.

Example 1

Solve the system of differential equations

$$\frac{dy}{dt} = 2x, \qquad \frac{dx}{dt} = 3y$$

Solution:

Step 1 The given system of linear differential equations can be written in the differential operator form as

$$Dy = 2x, \qquad Dx = 3y$$

or

$$2x - Dy = 0, \qquad Dx - 3y = 0$$

Step 2 Next we eliminate one of the two variables, say x , from the two differential equations. Operating on the first equation by D while multiplying the second by 2 and then subtracting eliminates x from the system. It follows that

$$-D^2y + 6y = 0 \quad \text{or} \quad D^2y - 6y = 0.$$

Step 3 Clearly, the result is a single linear differential equation with constant coefficients in the retained variable y . The roots of the auxiliary equation are real and distinct

$$m_1 = \sqrt{6} \quad \text{and} \quad m_2 = -\sqrt{6},$$

Therefore,

$$y(t) = c_1 e^{\sqrt{6}\,t} + c_2 e^{-\sqrt{6}\,t}$$

Step 4 We now eliminate the variable y that was retained in the previous step. Multiplying the first equation by -3 , while operating on the second by D and then adding gives the differential equation for x,

$$D^2x - 6x = 0.$$

Step 5 Again, the result is a single linear differential equation with constant coefficients in the retained variable x . We now solve this equation and obtain the value of the second dependent variable. The roots of the auxiliary equation are $m = \pm\sqrt{6}$. It follows that

$$x(t) = c_3 e^{\sqrt{6}\,t} + c_4 e^{-\sqrt{6}\,t}$$

Hence the values of the dependent variables $x(t)$, $y(t)$ are.

$$x(t) = c_3 e^{\sqrt{6}\,t} + c_4 e^{-\sqrt{6}\,t}$$

$$y(t) = c_1 e^{\sqrt{6}\,t} + c_2 e^{-\sqrt{6}\,t}$$

© Copyright Virtual University of Pakistan

Step 6 Substituting the values of $x(t)$ and $y(t)$ from step 5 into first equation of the given system, we have

$$\left(\sqrt{6}c_1 - 2c_3\right)e^{\sqrt{6}t} + \left(-\sqrt{6}c_2 - 2c_4\right)e^{-\sqrt{6}t} = 0.$$

Since this expression is to be zero for all values of t, we must have

$$\sqrt{6}c_1 - 2c_3 = 0, \quad -\sqrt{6}c_2 - 2c_4 = 0$$

or $$c_3 = \frac{\sqrt{6}}{2}c_1, \qquad c_4 = -\frac{\sqrt{6}}{2}c_2$$

Notice that if we substitute the computed values of $x(t)$ and $y(t)$ into the second equation of the system, we shall find that the same relationship holds between the constants.

Step 7 Hence, by using the above values of c_1 and c_2, we write the solution of the given system as

$$x(t) = \frac{\sqrt{6}}{2}c_1 e^{\sqrt{6}t} - \frac{\sqrt{6}}{2}c_2 e^{-\sqrt{6}t}$$

$$y(t) = c_1 e^{\sqrt{6}t} + c_2 e^{-\sqrt{6}t}$$

Example 2
Solve the following system of differential equations

$$Dx + (D+2)y = 0$$
$$(D-3)x - 2y = 0$$

Solution:

Step 1 The differential equations of the given system are already in the operator form.
Step 2 We eliminate the variable x from the two equations of the system. Thus operating on the first equation by $D-3$ and on the second by D and then subtracting eliminates x from the system. The resulting differential equation for the retained variable y is

$$\left[(D-3)(D+2) + 2D\right]y = 0$$

$$\left(D^2 + D - 6\right)y = 0$$

Step 3 The auxiliary equation of the differential equation for y obtained in the last step is

$$m^2 + m - 6 = 0 \Rightarrow (m-2)(m+3) = 0$$

Since the roots of the auxiliary equation are

$$m_1 = 2, \; m_2 = -3$$

© Copyright Virtual University of Pakistan

Therefore, the solution for the dependent variable y is

$$y(t) = c_1 e^{2t} + c_2 e^{-3t}$$

Step 4 Multiplying the first equation by 2 while operating on the second by $(D+2)$ and then adding yields the differential equation for x

$$(D^2 + D - 6)x = 0,$$

Step 5 The auxiliary equation for this equation for x is

$$m^2 + m - 6 = 0 = (m-2)(m+3)$$

The roots of this auxiliary equation are

$$m_1 = 2,\ m_2 = -3$$

Thus, the solution for the retained variable x is

$$x(t) = c_3 e^{2t} + c_4 e^{-3t}$$

Writing two solutions together, we have

$$x(t) = c_3 e^{2t} + c_4 e^{-3t}$$
$$y(t) = c_1 e^{2t} + c_2 e^{-3t}$$

Step 6 To reduce the number of constants, we substitute the last two equations into the first equation of the given system to obtain

$$(4c_1 + 2c_3)e^{2t} + (-c_2 - 3c_4)e^{-3t} = 0$$

Since this relation is to hold for all values of the independent variable t. Therefore, we must have

$$4c_1 + 2c_3 = 0, \qquad -c_2 - 3c_4 = 0.$$

or $$c_3 = -2c_1, \qquad c_4 = -\frac{1}{3}c_2$$

Step 7 Hence, a solution of the given system of differential equations is

$$x(t) = -2c_1 e^{2t} - \frac{1}{3}c_2 e^{-3t}$$
$$y(t) = c_1 e^{2t} + c_2 e^{-3t}$$

Example 3
Solve the system

$$\frac{dx}{dt} - 4x + \frac{d^2y}{dt^2} = t^2$$
$$\frac{dx}{dt} + x + \frac{dy}{dt} = 0$$

Solution:
Step 1 First we write the differential equations of the system in the differential operator form:

© Copyright Virtual University of Pakistan

$$(D-4)x + D^2 y = t^2$$
$$(D+1)x + Dy = 0$$

Step 2 Then we eliminate one of the dependent variables, say x . Operating on the first equation with the operator $D + 1$, on the second equation with the operator $D - 4$ and then subtracting, we obtain

$$[(D+1)D^2 - (D-4)D]\, y = (D+1)t^2$$

or $$\left(D^3 + 4D\right) y = t^2 + 2t.$$

Step 3 The auxiliary equation of the differential equation found in the previous step is

$$m^3 + 4m = 0 = m\,(m^2 + 4)$$

Therefore, roots of the auxiliary equation are

$$m_1 = 0, \quad m_2 = 2\,i, \quad m_3 = -2\,i$$

So that the complementary function for the retained variable y is

$$y_C = c_1 + c_2 \cos 2t + c_3 \sin 2t.$$

To determine the particular solution y_p we use undetermined coefficients. Therefore, we assume

$$y_p = At^3 + Bt^2 + Ct.$$

So that $$y_p' = 3At^2 + 2Bt + C,$$
$$y_p'' = 6\,At + 2\,B, \quad y_p'' = 6A$$

Thus $$y_p''' + 4y_p' = 12At^2 + 8Bt + 6A + 4C$$

Substituting in the differential equation found in step, we obtain

$$12At^2 + 8Bt + 6A + 4C = t^2 + 2t$$

Equating coefficients of t^2, t and constant terms yields

$$12A = 1, \quad 8B = 2, \quad 6A + 4C = 0,$$

Solving these equations give

$$A = 1/12,\ B = 1/4,\ C = -1/8.$$

Hence, the solution for the variable y is given by

$$y = y_c + y_p$$

or $$y = c_1 + c_2 \cos 2t + c_3 \sin 2t + \frac{1}{12}t^3 + \frac{1}{4}t^2 - \frac{1}{8}t.$$

Step 4 Next we eliminate the variable y from the given system. For this purpose we multiply first equation with 1 while operate on the second equation with the operator D and then subtracting, we obtain

$$[(D-4) - D(D+1)]x = \quad t^2$$

© Copyright Virtual University of Pakistan

or $(D^2 + 3)x = -t^2$

Step 5 The auxiliary equation of the differential equation for x is

$$m^2 + 4 = 0 \Rightarrow m = \pm 2i$$

The roots of the auxiliary equation are complex. Therefore, the complementary function for x

$$x_c = c_4 \cos 2t + c_5 \sin 2t$$

The method of undetermined coefficients can be applied to obtain a particular solution. We assume that

$$x_p = At^2 + Bt + C.$$

Then $x'_p = 2At + B,\ x''_p = 2A$

Therefore $x''_p + 4x_p = 2A + 4At^2 + 4Bt + 4C$

Substituting in the differential equation for x, we obtain

$$4At^2 + 4Bt + 2A + 4C = -t^2$$

Equating the coefficients of t^2, t and constant terms, we have

$$4A = -1,\quad 4B = 0,\quad 2A + 4C = 0$$

Solving these equations we obtain

$$A = -1/4,\ B = 0,\ C = 1/8$$

Thus $x_p = -\frac{1}{4}t^2 + \frac{1}{8}$

So that $x = x_c + x_p = c_4 \cos 2t + c_5 \sin 2t - \frac{1}{4}t^2 + \frac{1}{8}$

Hence, we have

$$x = x_c + x_p = c_4 \cos 2t + c_5 \sin 2t - \frac{1}{4}t^2 + \frac{1}{8}$$

$$y = c_1 + c_2 \cos 2t + c_3 \sin 2t + \frac{1}{12}t^3 + \frac{1}{4}t^2 - \frac{1}{8}t.$$

Step 6 Now c_4 and c_5 can be expressed in terms of c_2 and c_3 by substituting these values of x and y into the second equation of the given system and we find, after combining the terms,

$$(c_5 - 2c_4 - 2c_2)\sin 2t + (2c_5 + c_4 + 2c_3)\cos 2t = 0$$

So that $c_5 - 2c_4 - 2c_2 = 0,\quad 2c_5 + c_4 + 2c_3 = 0$

© Copyright Virtual University of Pakistan

Solving the last two equations for c_4 and c_5 in terms of c_2 and c_3 gives

$$c_4 = -\frac{1}{5}(4c_2 + 2c_3), \qquad c_5 = \frac{1}{5}(2c_2 - 4c_3).$$

Step 7 Finally, a solution of the given system is found to be

$$x(t) = -\frac{1}{5}(4c_2 + 2c_3)\cos 2t + \frac{1}{5}(2c_2 - 4c_3)\sin 2t - \frac{1}{4}t^2 + \frac{1}{8}t$$

$$y(t) = c_1 + c_2\cos 2t + c_3 \sin 2t + \frac{1}{12}t^3 + \frac{1}{4}t^2 - \frac{1}{8}t.$$

Exercise

Solve, if possible, the given system of differential equations by either systematic elimination.

1. $\frac{dx}{dt} = x + 7y, \qquad \frac{dy}{dt} = x - 2y$
2. $\frac{dx}{dt} - 4y = 1, \qquad x + \frac{dy}{dt} = 2$
3. $(D+1)x + (D-1)y = 2, \qquad 3x + (D+2)y = -1$
4. $\frac{d^2x}{dt^2} + \frac{dy}{dt} = -5x, \qquad \frac{dx}{dt} + \frac{dy}{dt} = -x + 4y$
5. $D^2x - Dy = t, \qquad (D+3)x + (D+3)y = 2$
6. $\frac{dx}{dt} + \frac{dy}{dt} = e^t, \qquad -\frac{d^2x}{dt^2} + \frac{dx}{dt} + x + y = 0$
7. $(D-1)x + (D^2+1)y = 1, \qquad (D^2-1)x + (D+1)y = 2$
8. $Dx = y, \qquad Dy = z, \qquad Dz = x$
9. $\frac{dx}{dt} = -x + z, \qquad \frac{dy}{dt} = -y + z, \qquad \frac{dz}{dt} = -x + y$
10. $Dx - 2Dy = t^2, \qquad (D+1)x - 2(D+1)y = 1$

© Copyright Virtual University of Pakistan

Lecture 36
Systems of Linear Differential Equations

Solution of Using Determinants

If L_1, L_2, L_3 and L_4 denote linear differential operators with constant coefficients, then a system of linear differential equations in two variables x and y can be written as

$$L_1 x + L_2 y = g_1(t)$$
$$L_3 x + L_4 y = g_2(t)$$

To eliminate y, we operate on the first equation with L_4 and on the second equation with L_2 and then subtracting, we obtain

$$(L_1 L_4 - L_2 L_3)x = L_4 g_1 - L_2 g_2$$

Similarly, operating on the first equation with L_3 and second equation with L_1 and then subtracting, we obtain

$$(L_1 L_4 - L_2 L_3)y = L_1 g_2 - L_3 g_1$$

Since $$L_1 L_4 - L_2 L_3 = \begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}$$

Therefore $$L_4 g_1 - L_2 g_2 = \begin{vmatrix} g_1 & L_2 \\ g_2 & L_4 \end{vmatrix}$$

and $$L_1 g_2 - L_3 g_1 = \begin{vmatrix} L_1 & g_1 \\ L_3 & g_2 \end{vmatrix}$$

Hence, the given system of differential equations can be decoupled into *nth* order differential equations. These equations use determinants similar to those used in Cramer's rule:

$$\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix} x = \begin{vmatrix} g_1 & L_2 \\ g_2 & L_4 \end{vmatrix} \quad \text{and} \quad \begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix} y = \begin{vmatrix} L_1 & g_1 \\ L_3 & g_2 \end{vmatrix}$$

The uncoupled differential equations can be solved in the usual manner.

© Copyright Virtual University of Pakistan

Note that

- The determinant on left hand side in each of these equations can be expanded in the usual algebraic sense. This means that the symbol D occurring in L_i is to be treated as an algebraic quantity. The result of this expansion is a differential operator of order n, which is operated on x and y.

- However, some care should be exercised in the expansion of the determinant on the right hand side. We must expand these determinants in the sense of the internal differential operators actually operating on the functions g_1 and g_2. Therefore, the symbol D occurring in L_i is to be treated as an algebraic quantity.

The Method

The steps involved in application of the method of detailed above can be summarized as follows:

Step 1 First we have to write the differential equations of the given system in the differential operator form

$$L_1 x + L_2 y = g_1(t)$$
$$L_3 x + L_4 y = g_2(t)$$

Step 2 We find the determinants

$$\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}, \begin{vmatrix} g_1 & L_2 \\ g_2 & L_4 \end{vmatrix}, \begin{vmatrix} L_1 & g_1 \\ L_3 & g_2 \end{vmatrix}$$

Step 3 If the first determinant is non-zero, then it represents an nth order differential operator and we decoupled the given system by writing the differential equations

$$\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix} \cdot x = \begin{vmatrix} g_1 & L_2 \\ g_2 & L_4 \end{vmatrix}$$
$$\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix} \cdot y = \begin{vmatrix} L_1 & g_1 \\ L_3 & g_2 \end{vmatrix}$$

Step 4 Find the complementary functions for the two equations. Remember that the auxiliary equation and hence the complementary function of each of these differential equations is the same.

Step 5 Find the particular integrals x_p and y_p using method of undetermined coefficients or the method of variation of parameters.

Step 6 Finally, we write the general solutions for both the dependent variables x and y

$$x = x_c + x_p, \; y = y_c + y_p.$$

© Copyright Virtual University of Pakistan

Step 7 Reduce the number of constants by substituting in one of the differential equations of the given system

Note that

If the determinant found in step 2 is zero, then the system may have a solution containing any number of independent constants or the system may have no solution at all. Similar remarks hold for systems larger than system indicated in the previous discussion.

Example 1

Solve the following homogeneous system of differential equations

$$2\frac{dx}{dt} - 5x + \frac{dy}{dt} = e^t$$
$$\frac{dx}{dt} - x + \frac{dy}{dt} = 5e^t$$

Solution:

Step 1 First we write the differential equations of the system in terms of the differential operator D

$$(2D-5)x + Dy = e^t$$
$$(D-1)x + Dy = 5e^t$$

Step 2 We form the determinant

$$\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix}, \quad \begin{vmatrix} e^t & D \\ 5e^t & D \end{vmatrix}, \quad \begin{vmatrix} 2D-5 & e^t \\ D-1 & 5e^t \end{vmatrix}$$

Step 3 Since the 1st determinant is non-zero

$$\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix} = (2D-5)D - (D-1)D$$

or

$$\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix} = D^2 - 4D \neq 0$$

Therefore, we write the decoupled equations

$$\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix} x = \begin{vmatrix} e^t & D \\ 5e^t & D \end{vmatrix}$$

$$\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix} y = \begin{vmatrix} 2D-5 & e^t \\ D-1 & 5e^t \end{vmatrix}$$

© Copyright Virtual University of Pakistan

After expanding we find that

$$\left(D^2 - 4D\right)x = De^t - D(5\ e^t) = -4e^t$$

$$\left(D^2 - 4D\right)y = (2D-5)(5e^t) - (D-1)e^t = -15e^t$$

Step 4 We find the complementary function for the two equations. The auxiliary equation for both of the differential equations is:

$$m^2 - 4m = 0 \Rightarrow m = 0, 4$$

The auxiliary equation has real and distinct roots

$$x_C = c_1 + c_2 e^{4t}$$

$$y_C = c_3 + c_4 e^{4t}$$

Step 5 We now use the method of undetermined coefficients to find the particular integrals x_p and y_p.

Since $g_1(t) = -4\ e^t, \quad g_2(t) = -15\ e^t$

We assume that

$$x_p = Ae^t, \quad y_p = Be^t$$

Then $D\ x_p = Ae^t, \quad D^2 x_p = Ae^t$

And $D\ y_p = Be^t, \quad D^2 y_p = Be^t$

Substituting in the differential equations, we have

$$Ae^t - 4Ae^t = -4e^t$$

$$Be^t - 4Be^t = -15e^t$$

or $-3Ae^t = -4e^t, \quad -3Be^t = -15e^t$

Equating coefficients of e^t and constant terms, we obtain

$$A = \frac{4}{3}, \quad B = 5$$

So that $x_p = \frac{4}{3}e^t, \quad y_p = 5e^t$

Step 6 Hence, the general solution of the two decoupled equations

$$x = x_C + x_p = c_1 + c_2 e^{4t} + \frac{4}{3}e^t$$

$$y = y_C + y_p = c_3 + c_4 e^{4t} + 5e^t$$

© Copyright Virtual University of Pakistan

Step 7 Substituting these solutions for x and y into the second equation of the given system, we obtain

$$-c_1 + \left(3c_2 + 4c_4\right)e^{4t} = 0$$

or $$c_1 = 0, \quad c_4 = -\frac{3}{4}c_2.$$

Hence, the general solution of the given system of differential equations is

$$x(t) = c_2 e^{4t} + \frac{4}{3}e^{t}$$

$$y(t) = c_3 - \frac{3}{4}c_2 e^{4t} + 5e^{t}$$

If we re-notate the constants c_2 and c_3 as c_1 and c_2, respectively. Then the solution of the system can be written as:

$$x(t) = c_1 e^{4t} + \frac{4}{3}e^{t}$$

$$y(t) = -\frac{3}{4}c_1 e^{4t} + c_2 + 5e^{t}$$

Example 2

Solve

$$x' = 3x - y - 1$$

$$y' = x + y + 4e^{t}$$

Solution:

Step 1 First we write the differential equations of the system in terms of the differential operator D

$$(D-3)x + y = -1$$

$$-x + (D-1)y = 4e^{t}$$

Step 2 We form the determinant

$$\begin{vmatrix} D-3 & 1 \\ -1 & D-1 \end{vmatrix}, \begin{vmatrix} -1 & 1 \\ 4e^{t} & D-1 \end{vmatrix}, \begin{vmatrix} D-3 & 1 \\ -1 & 4e^{t} \end{vmatrix}$$

Step 3 Since the 1st determinant is non-zero

$$\begin{vmatrix} D-3 & 1 \\ -1 & D-1 \end{vmatrix} = D^2 - 4D + 4 \neq 0$$

© Copyright Virtual University of Pakistan

Therefore, we write the decoupled equations

$$\begin{vmatrix} D-3 & 1 \\ -1 & D-1 \end{vmatrix} x = \begin{vmatrix} -1 & 1 \\ 4e^t & D-1 \end{vmatrix}$$

$$\begin{vmatrix} D-3 & 1 \\ -1 & D-1 \end{vmatrix} y = \begin{vmatrix} D-3 & -1 \\ -1 & 4e^t \end{vmatrix}$$

After expanding we find that

$$(D-2)^2 x = 1 - 4e^t$$

$$(D-2)^2 y = -1 - 8e^t.$$

Step 4 We find the complementary function for the two equations. The auxiliary equation for both of the differential equations is:

$$(m-2)^2 = 0 \Rightarrow m = 2,2$$

The auxiliary equation has real and equal roots

$$x_C = c_1 e^{2t} + c_2 te^{2t}$$

$$y_C = c_3 e^{2t} + c_4 te^{2t}$$

Step 5 We now use the method of undetermined coefficients to find the particular integrals x_p and y_p.

Since $g_1(t) = 1 - 4e^t, \quad g_2(t) = -1 - 8e^t$

We assume that

$$x_p = A + Be^t, \quad y_p = C + Ee^t$$

Then $Dx_p = Be^t, \quad D^2 x_p = Be^t$

And $Dy_p = Ee^t, \quad D^2 y_p = Ee^t$

Substituting in the differential equations

$$(D-2)^2 x_p = D^2 x_p - 4Dx_p + 4x_p = 1 - 4e^t$$

$$(D-2)^2 y_p = D^2 y_p - 4Dy_p + 4y_p = -1 - 8e^t$$

Therefore, we have

$$Be^t - 4Be^t + 4A + 4Be^t = 1 - 4e^t$$

$$Ee^t - 4Ee^t + 4C + 4Ee^t = -1 - 8e^t$$

or $Be^t + 4A = 1 - 4e^t, \quad Ee^t + 4C = -1 - 8e^t$

© Copyright Virtual University of Pakistan

Equating coefficients of e^t and constant terms, we obtain

$$B = -4, \quad A = \frac{1}{4}$$

$$C = -\frac{1}{4}, \quad E = -8$$

So that $$x_p = \frac{1}{4} - 4e^t, \quad y_p = -\frac{1}{4} - 8e^t$$

Step 6 Hence, the general solution of the two decoupled equations

$$x = x_c + x_p = c_1 e^{2t} + c_2 te^{2t} + \frac{1}{4} - 4e^t$$

$$y = y_c + y_p = c_3 e^{2t} + c_4 te^{2t} - \frac{1}{4} - 8e^t$$

Step 7 Substituting these solutions for x and y into the second equation of the given system, we obtain

$$(c_3 - c_1 + c_4)e^{2t} + (c_4 - c_2)te^{2t} = 0$$

or $$c_4 = c_2, \quad c_3 = c_1 - c_4 = c_1 - c_2.$$

Hence, a solution of the given system of differential equations is

$$x(t) = c_1 e^{2t} + c_2 te^{2t} + \frac{1}{4} - 4e^t$$

$$y(t) = (c_1 - c_2)e^{2t} + c_2 te^{2t} - \frac{1}{4} - 8e^t$$

Example 3

Given the system

$$\begin{aligned} Dx + Dz &= t^2 \\ 2x + D^2 y &= e^t \\ -2Dx - 2y + (D+1)z &= 0 \end{aligned}$$

Find the differential equation for the dependent variables x, y and z.

Solution:

Step1 The differential equations of the system are already written in the differential operator form.

Step 2 We form the determinant

© Copyright Virtual University of Pakistan

$$\begin{vmatrix} D & 0 & D \\ 2 & D^2 & 0 \\ -2D & -2 & D+1 \end{vmatrix}, \begin{vmatrix} t^2 & 0 & D \\ e^t & D^2 & 0 \\ 0 & -2 & D+1 \end{vmatrix}, \begin{vmatrix} D & t^2 & D \\ 2 & e^t & 0 \\ -2D & 0 & D+1 \end{vmatrix}, \begin{vmatrix} D & 0 & t^2 \\ 2 & D^2 & e^t \\ -2D & -2 & 0 \end{vmatrix}$$

Step 3 Since the first determinant is non-zero.

$$\begin{vmatrix} D & 0 & D \\ 2 & D^2 & 0 \\ -2D & -2 & D+1 \end{vmatrix} = D\begin{vmatrix} D^2 & 0 \\ -2 & D+1 \end{vmatrix} + D\begin{vmatrix} 2 & D^2 \\ -2D & -2 \end{vmatrix}$$

or

$$\begin{vmatrix} D & 0 & D \\ 2 & D^2 & 0 \\ -2D & -2 & D+1 \end{vmatrix} = D\left(3D^3 + D^2 - 4\right) \neq 0$$

Therefore, we can write the decoupled equations

$$\begin{vmatrix} D & 0 & D \\ 2 & D^2 & 0 \\ -2D & -2 & D+1 \end{vmatrix} \cdot x = \begin{vmatrix} t^2 & 0 & D \\ e^t & D^2 & 0 \\ 0 & -2 & D+1 \end{vmatrix}$$

$$\begin{vmatrix} D & 0 & D \\ 2 & D^2 & 0 \\ -2D & -2 & D+1 \end{vmatrix} \cdot y = \begin{vmatrix} D & t^2 & D \\ 2 & e^t & 0 \\ -2D & 0 & D+1 \end{vmatrix}$$

$$\begin{vmatrix} D & 0 & D \\ 2 & D^2 & 0 \\ -2D & -2 & D+1 \end{vmatrix} \cdot z = \begin{vmatrix} D & 0 & t^2 \\ 2 & D^2 & e^t \\ -2D & -2 & 0 \end{vmatrix}$$

The determinant on the left hand side in these equations has already been expanded. Now we expand the determinants on the right hand side by the cofactors of an appropriate row.

$$\begin{vmatrix} t^2 & 0 & D \\ e^t & D^2 & 0 \\ 0 & -2 & D+1 \end{vmatrix} = \begin{vmatrix} D^2 & 0 \\ -2 & D+1 \end{vmatrix} t^2 + D\begin{vmatrix} e^t & D^2 \\ 0 & -2 \end{vmatrix}$$

$$= D^2(D+1)t^2 + D(-2e^t) = (D^3 + D^2)t^2 - 2e^t$$

$$= 2 - 2e^t$$

© Copyright Virtual University of Pakistan

$$\begin{vmatrix} D & t^2 & D \\ 2 & e^t & 0 \\ -2D & 0 & D+1 \end{vmatrix} = D\begin{vmatrix} e^t & 0 \\ 0 & D+1 \end{vmatrix} - \begin{vmatrix} 2 & 0 \\ -2D & D+1 \end{vmatrix} t^2 + D\begin{vmatrix} 2 & e^t \\ -2D & 0 \end{vmatrix}$$

$$= D[(D+1)e^t] - [(D+1)(2t^2)] + D[2De^t]$$

$$= 2e^t - 4t - 2t^2 + 2e^t = 4e^t - 2t^2 - 4t.$$

$$\begin{vmatrix} D & 0 & t^2 \\ 2 & D^2 & e^t \\ -2D & -2 & 0 \end{vmatrix} = D\begin{vmatrix} D^2 & e^t \\ -2 & 0 \end{vmatrix} + \begin{vmatrix} 2 & D^2 \\ -2D & -2 \end{vmatrix} t^2$$

$$= D(2e^t) + (-4 + 2D^3)t^2 = 2e^t - 4t^2 + 0$$

$$= 2e^t - 4t^2$$

Hence the differential equations for the dependent variables x, y and z can be written as

$$D\left(3D^3 + D^2 - 4y\right)x = 2 - 2e^t$$

or

$$D\left(3D^3 + D^2 - 4y\right)y = 4e^t - 2t^2 - 4t.$$

$$D\left(3D^3 + D^2 - 4y\right)z = 2e^t - 4t^2$$

Again we remind that the D symbol on the left-hand side is to be treated as an algebraic quantity, but this is not the case on the right-hand side.

© Copyright Virtual University of Pakistan

Exercise

Solve, if possible, the given system of differential equations by use of determinants.

1. $\frac{dx}{dt} = 2x - y, \qquad \frac{dy}{dt} = x - 2y$
2. $\frac{dx}{dt} = -y + t, \qquad \frac{dy}{dt} = x - t$
3. $\left(D^2 + 5\right)x - 2y = 0, \qquad -2x + \left(D^2 + 2\right)y = 0$
4. $\frac{d^2x}{dt^2} = 4y + e^t, \qquad \frac{d^2y}{dt^2} = 4x - e^t$
5. $\frac{d^2x}{dt^2} + \frac{dy}{dt} = -5x, \qquad \frac{dx}{dt} + \frac{dy}{dt} = -x + 4y$
6. $Dx + D^2y = e^{3t}, \quad (D+1)x + (D-1)y = 4e^{3t}$
7. $\left(D^2 - 1\right)x - y = 0, \qquad \left(D - 1\right)x + Dy = 0$
8. $(2D^2 - D - 1)x - (2D + 1)y = 1, \qquad (D-1)x + Dy = -1$
9. $\frac{dx}{dt} + \frac{dy}{dt} = e^t, \qquad -\frac{d^2x}{dt^2} + \frac{dx}{dt} + x + y = 0$
10. $2Dx + (D-1)y = t, \qquad Dx + Dy = t^2$

© Copyright Virtual University of Pakistan

Lecture 37
Systems of Linear First-Order Equation

In Previous Lecture

In the preceding lectures we dealt with linear systems of the form

$$\begin{aligned} P_{11}(D)\,x_1 + P_{12}(D)\,x_2 + \cdots + P_{1n}(D)\,x_n &= b_1(t) \\ P_{21}(D)\,x_1 + P_{22}(D)\,x_2 + \cdots + P_{2n}(D)\,x_n &= b_2(t) \\ \vdots \qquad\quad \vdots \qquad\quad \vdots \qquad\quad \vdots \\ P_{n1}(D)\,x_1 + P_{n2}(D)\,x_2 + \cdots + P_{nn}(D)\,x_n &= b_n(t) \end{aligned}$$

where the P_{ij} were polynomials in the differential operator D.

The *nth* Order System

1. The study of systems of first-order differential equations

$$\begin{aligned} \frac{dx_1}{dt} &= g_1(t, x_1, x_2, \ldots, x_n) \\ \frac{dx_2}{dt} &= g_2(t, x_1, x_2, \ldots, x_n) \\ &\vdots \\ \frac{dx_n}{dt} &= g_n(t, x_1, x_2, \ldots, x_n) \end{aligned}$$

is also particularly important in advanced mathematics. This system of n first-order equations is called and ***n*th-order system.**

2. Every *n*th-order differential equation

$$y^{(n)} = F\left(t, y, y', \ldots, y^{(n-1)}\right)$$

as well as ***most systems*** of differential equations, can be reduced to the ***nth-order system***.

© Copyright Virtual University of Pakistan

Linear Normal Form

A particularly, but important, case of the *nth-order system* is of those systems having the linear normal or canonical form:

$$\begin{aligned}
\frac{dx_1}{dt} &= a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n + f_1(t) \\
\frac{dx_2}{dt} &= a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n + f_2(t) \\
&\vdots \qquad\qquad \vdots \qquad\qquad \vdots \\
\frac{dx_n}{dt} &= a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n + f_n(t)
\end{aligned}$$

where the coefficients a_{ij} and the f_i are the continuous functions on a common interval I. When $f_i(t) = 0, i = 1, 2, ..., n,$ the system is said to be **homogeneous**; otherwise it is called **non-homogeneous**.

Reduction of Equation to a System

Suppose a linear *n*th-order differential equation is first written as

$$\frac{d^n y}{dt^n} = -\frac{a_0}{a_n}y - \frac{a_1}{a_n}y' - \cdots - \frac{a_{n-1}}{a_n}y^{(n-1)} + f(t).$$

If we then introduce the variables

$$y = x_1, \quad y' = x_2, \quad y' = x_3, ..., y^{(n-1)} = x_n$$

it follows that

$$y' = x_1' = x_2, \quad y' = x_2' = x_3, ..., y^{(n-1)} = x_{n-1}' = x_n, \quad y^{(n)} = x_n'$$

Hence the given *n*th-order differential equation can be expressed as an *n*th-order system:

$$\begin{aligned}
x_1' &= x_2 \\
x_2' &= x_3 \\
x_3' &= x_4 \\
&\vdots \\
x_{n-1}' &= x_n \\
x_n' &= -\frac{a_0}{a_n}x_1 - \frac{a_1}{a_n}x_2 - \cdots - \frac{a_{n-1}}{a_n}x_n + f(t).
\end{aligned}$$

Inspection of this system reveals that it is in the form of an *n*th-order system.

© Copyright Virtual University of Pakistan

Example 1

Reduce the third-order equation

$$2y'' = -y - 4y' + 6y' + \sin t$$

or

$$2y'' - 6y' + 4y' + y = \sin t$$

to the normal form.

Solution: Write the differential equation as

$$y''' = -\frac{1}{2}y - 2y' + 3y' + \frac{1}{2}\sin t$$

Now introduce the variables

$$y = x_1, y' = x_2, y' = x_3.$$

Then

$$x_1' = y' = x_2$$
$$x_2' = y' = x_3$$
$$x_3' = y''$$

Hence, we can write the given differential equation in the linear normal form

$$x_1' = x_2$$
$$x_2' = x_3$$
$$x_3' = -\frac{1}{2}x_1 - 2x_2 + 3x_3 + \frac{1}{2}\sin t$$

Example 2

Rewrite the given second order differential equation as a system in the normal form

$$2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 5y = 0$$

Solution:

We write the given the differential equation as

$$\frac{d^2y}{dx^2} = -2\frac{dy}{dx} + \frac{5}{2}y$$

Now introduce the variables

$$y = x_1, \quad y' = x_2$$

Then

$$y' = x_1' = x_2$$
$$y'' = x_2'$$

So that the given differential equation can be written in the form of a system

$$x_1' = x_2$$
$$x_2' = -2x_2 + \frac{5}{2}x_1$$

This is the linear normal or canonical form.

© Copyright Virtual University of Pakistan

Example 3

Write the following differential equation as an equivalent system in the Canonical form.

$$4\frac{d^3y}{dt^3} + y = e^t$$

Solution:

First write the given differential equation as

$$4\frac{d^3y}{dt^3} = -y + e^t$$

dividing by 4 on both sides

or $$\frac{d^3y}{dt^3} = -\frac{1}{4}y + \frac{1}{4}e^t$$

Now introduce the variables

$$y = x_1,\ y' = x_2,\ y' = x_3$$

Then

$$y' = x_1' = x_2$$
$$y'' = x_2' = x_3$$
$$y''' = x_3'$$

Hence, the given differential equation can be written as an equivalent system.

$$x_1' = x_2$$
$$x_2' = x_3$$
$$x_3' = -\frac{1}{4}x_1 + \frac{1}{4}e^t$$

Clearly, this system is in the linear normal or the Canonical form.

Example 4

Rewrite the differential equation in the linear normal form

$$t^2y'' + ty' + (t^2 - 4)y = 0$$

Solution:

First we write the equation in the form

$$t^2y' = -ty' - (t^2 - 4)y$$

or $$y' = -\frac{1}{t}y' - \frac{(t^2 - 4)}{t^2}y, \quad t \neq 0$$

or $$y'' = -\frac{1}{t}y' - \frac{t^2 - 4}{t^2}y$$

Then introduce the variables

$$y = x_1,\ y' = x_2$$

© Copyright Virtual University of Pakistan

Then

$$y' = x_1' = x_2$$
$$y'' = x_2'$$

Hence, the given equation is equivalent to the following system.

$$x_1' = x_2$$
$$x_2' = -\frac{1}{t}x_2 - \frac{t^2 - 4}{t^2}x_1$$

The system is in the required linear normal or the cnonical form.

Systems Reduced to Normal Form

Using Procedure similar to that used for a single equation, we can reduce most systems of the linear form

$$P_{11}(D)x_1 + P_{12}(D)x_2 + \cdots + P_{1n}(D)x_n = b_1(t)$$
$$P_{21}(D)x_1 + P_{22}(D)x_2 + \cdots + P_{2n}(D)x_n = b_2(t)$$
$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$
$$P_{n1}(D)x_1 + P_{n2}(D)x_2 + \cdots + P_{nn}(D)x_n = b_n(t)$$

to the canonical form. To accomplish this we need to solve the system for the highest order derivative of each dependent variable.

Note:
It is not always possible to solve the given system for the highest-order derivative of each dependent variable.

Example 5
Reduce the following system to the normal form.

$$\left(D^2 - D + 5\right)x + 2D^2 y = e^t$$
$$-2x + \left(D^2 + 2\right)y = 3t^2$$

Solution:

First write the given system in the differential operator form

$$D^2 x + 2D^2 y = e^t - 5x + Dx$$
$$D^2 y = 3t^2 + 2x - 2y$$

Then eliminate $D^2 y$ by multiplying the second equation by 2 and subtracting from first equation to have

$$D^2 x = e^t - 6t^2 - 9x + 4y + Dx.$$

© Copyright Virtual University of Pakistan

Also $D^2 y = 3t^2 + 2x - 2y$

We are now in a position to introduce the new variables. Therefore, we suppose that

$$Dx = u, \quad Dy = v$$

Thus, the expressions for $D^2 x$ and $D^2 y$, respectively, become

$$Du = e^t - 6t^2 - 9x + 4y + u$$
$$Dv = 3t^2 + 2x - 2y.$$

Thus the original system can be written as

$$Dx = u$$
$$Dy = v$$
$$Du = -9x + 4y + u + e^t - 6t^2$$
$$Dv = 2x - 2y + 3t^2$$

Clearly, this system is in the canonical form.

Example 6

If possible, re-write the given system in the canonical form

$$x' + 4x - y' = 7t$$
$$x' + y' - 2y = 3t$$

Solution:

First we write the differential equations of the system in the differential operator form

$$Dx + 4x - Dy = 7t$$
$$Dx + Dy - 2y = 3t$$

To eliminate Dy we add the two equations of the system, to obtain

$$2Dx = 10t - 4x + 2y$$

or $$Dx = -2x + y + 5t$$

Next to solve for the Dy, we eliminate Dx. For this purpose we simply subtract the first equation from second equation of the system, to have

$$-4x + 2Dy - 2y = -4t$$
$$2Dy = 4x + 2y - 4t$$

or $$Dy = 2x + y - 2t$$

Hence the original system is equivalent to the following system

$$Dx = -2x + y + 5t$$
$$Dy = 2x + y - 2t$$

Clearly the system is in the normal form.

© Copyright Virtual University of Pakistan

Example 7

If possible, re-write the given system in the linear normal form

$$\frac{d^3x}{dt^3} = 4x - 3\frac{d^2x}{dt^2} + 4\frac{dy}{dt}$$

$$\frac{d^2y}{dt^2} = 10t^2 - 4\frac{dx}{dt} + 3\frac{dy}{dt}$$

Solution:

First write the given system in the differential operator form

$$D^3x = 4x - 3D^2x + 4Dy$$

$$D^2y = 10t^2 - 4Dx + 3Dy$$

No need to eliminate anything as the equations are already expressing the highest-order derivatives of x and y in terms of the remaining functions and derivatives. Therefore, we are now in a position to introduce new variables. Suppose that

$$Dx = u, \quad Dy = v$$

$$D^2x = Du = w$$

$$D^2y = Dv, \quad D^3x = Dw$$

Then the expressions for D^3x and for D^2y can be written as

$$Dw = 4x + 4v - 3w$$

$$Dv = 10t^2 - 4u + 3v$$

Hence, the given system of differential equations is equivalent to the following system

$$Dx = u$$

$$Dy = v$$

$$Du = w$$

$$Dv = 10t^2 - 4u + 3v$$

$$Dw = 4x + 4v - 3w$$

This new system is clearly in the linear normal form.

Degenerate Systems

The systems of differential equations of the form

$$P_{11}(D)x_1 + P_{12}(D)x_2 + \cdots + P_{1n}(D)x_n = b_1(t)$$

$$P_{21}(D)x_1 + P_{22}(D)x_2 + \cdots + P_{2n}(D)x_n = b_2(t)$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$P_{n1}(D)x_1 + P_{n2}(D)x_2 + \cdots + P_{nn}(D)x_n = b_n(t)$$

those cannot be reduced to a linear system in normal form is said to be a degenerate system.

© Copyright Virtual University of Pakistan

Example 8

If possible, re-write the following system in a linear normal form

$$(D+1)x+(D+1)y=0$$
$$2Dx+(2D+1)y=0$$

Solution:

The given system is already written in the differential operator form. The system can be written in the form

$$Dx+x+Dy+y=0$$
$$2Dx+2Dy+y=0$$

We eliminate Dx to solve for the highest derivative Dy by multiplying the first equation with 2 and then subtracting second equation from the first one. Thus we have

$$\begin{array}{r} 2Dx+2x+2Dy+2y=0 \\ \underline{\pm 2Dx \pm 2Dy \pm y=0} \\ 2x+y=0 \end{array}$$

Therefore, it is impossible to solve the system for the highest derivative of each dependent variable; the system cannot be reduced to the canonical form. Hence the system is a degenerate.

Example 9

If possible, re-write the following system of differential equations in the canonical form

$$x''+y'=1$$
$$x''+y'=-1$$

Solution:

We write the system in the operator form

$$D^2x+Dy=1$$
$$D^2x+Dy=-1$$

To solve for a highest order derivative of y in terms of the remaining functions and derivatives, we subtract the second equation from the first and we obtain

$$\begin{array}{r} D^2x+Dy=1 \\ \underline{\pm D^2x \pm Dy=-1} \\ 0=2 \end{array}$$

This is absurd. Thus the given system cannot be reduced to a canonical form. Hence the system is a degenerate system.

© Copyright Virtual University of Pakistan

Example 10

If possible, re-write the given system

$$(2D+1)x - 2Dy = 4$$

$$Dx - Dy = e^t$$

Solution:
The given system is already in the operator form and can be written as

$$2Dx + x - 2Dy = 4$$

$$Dx - Dy = e^t$$

To solve for the highest derivative Dy , we eliminate the highest derivative Dx . Therefore, multiply the second equation with 2 and then subtract from the first equation to have

$$2Dx + x - 2Dy = 4$$

$$\pm 2Dx \mp 2\mathrm{Dy} = \pm 2\mathrm{e}^{\mathrm{t}}$$

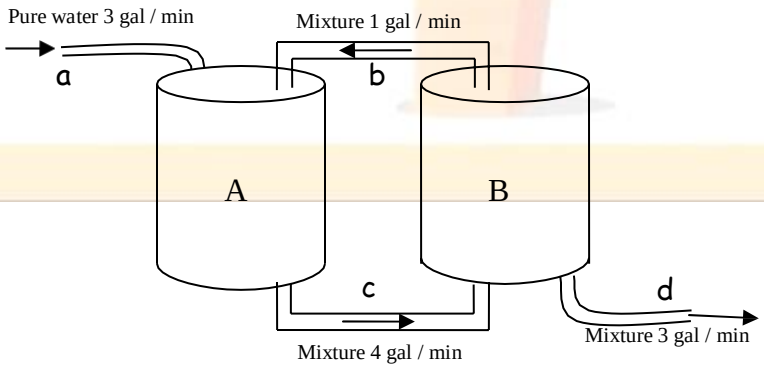
$$x = 4 - 2e^t$$

Therefore, it is impossible to solve the system for the highest derivatives of each variable. Thus the system cannot be reduced to the linear normal form. Hence, the system is a degenerate system.

Applications

The systems having the linear normal form arise naturally in some physical applications. The following example provides an application of a homogeneous linear normal system in two dependent variables.

Example 11
Tank *A* contains 50 gallons of water in which 25 pounds of salt are dissolved. A second tank *B* contains 50 gallons of pure water. Liquid is pumped in and out of the tank at rates shown in Figure. Derive the differential equations that describe the number of pounds $x_1(t)$ and $x_2(t)$ of salt at any time in tanks *A* and *B*, respectively.



© Copyright Virtual University of Pakistan

Solution:

Tank A

Input through pipe $a = (3 \text{ gal/min}) \cdot (0 \text{ lb/gal}) = 0$

Input through pipe $b = (1 \text{ gal/min}) \cdot \left(\frac{x_2}{50} \text{ lb/gal}\right) = \frac{x_2}{50} \text{ lb/min}$

Thus, total input for the tank $A = 0 + \frac{x_2}{50} = \frac{x_2}{50}$

Output through pipe $c = (4 \text{ gal/min}) \cdot \left(\frac{x_1}{50} \text{lb/gal}\right) = \frac{4x_1}{50} \text{lb/min}$

Hence, the net rate of change of $x_1(t)$ in lb / min is given by

$$\frac{dx_1}{dt} = input - output$$

or
$$\frac{dx_1}{dt} = \frac{x_2}{50} - \frac{4x_1}{50}$$

or
$$\frac{dx_1}{dt} = \frac{-2}{25}x_1 + \frac{x_2}{50}$$

Tank B

Input through pipe c is $4 \text{ gal/min} = \frac{4x_1}{50} \text{ lb/min}$

Output through pipe b is $1 \text{ gal/min} = \frac{x_2}{50} \text{lb/min}$

Similarly output through pipe d is $3 \text{ gal/min} = \frac{3x_2}{50} \text{ lb/min}$

Total output for the tank $b = \frac{x_2}{50} + \frac{3x_2}{50} = \frac{4x_2}{50}$

Hence, the net rate of change of $x_2(t)$ in $lb/\min$

$$\frac{dx_2}{dt} = input - output$$

or
$$\frac{dx_2}{dt} = \frac{4x_1}{50} - \frac{4x_2}{50}$$

or
$$\frac{dx_2}{dt} = \frac{2x_1}{25} - \frac{2x_2}{25}$$

© Copyright Virtual University of Pakistan

Thus we obtain the first order system

$$\frac{dx_1}{dt} = \frac{-2}{25}x_1 + \frac{x_2}{50}$$

$$\frac{dx_2}{dt} = \frac{2x_1}{25} - \frac{2x_2}{25}$$

We observe that the foregoing system is accompanied the initial conditions

$$x_1(0) = 25, \quad x_2(0) = 0.$$

Exercise

Rewrite the given differential equation as a system in linear normal form.

1. $\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 4y = \sin 3t$
2. $y''' - 3y'' + 6y' - 10y = t^2 + 1$
3. $\frac{d^4y}{dt^4} - 2\frac{d^2y}{dt^2} + 4\frac{dy}{dx} + y = t$
4. $2\frac{d^4y}{dt^4} + \frac{d^3y}{dt^3} - 8y = 10$

Rewrite, if possible, the given system in the linear normal form.

5. $(D-1)x - Dy = t^2, \quad x + Dy = 5t - 2$
6. $x'' - 2y'' = \sin t, \quad x'' + y'' = \cos t$
7. $m_1x_1'' = -k_1x_1 + k_2(x_2 - x_1), \qquad m_2x_2'' = -k_2(x_2 - x_1)$
8. $D^2x + Dy = 4t, \qquad -D^2x + (D+1)y = 6t^2 + 10$

© Copyright Virtual University of Pakistan

Lecture 38
Introduction to Matrices

Matrix

A rectangular array of numbers or functions subject to certain rules and conditions is called a matrix. Matrices are denoted by capital letters $A, B,...,Y, Z$. The numbers or functions are called elements or entries of the matrix. The elements of a matrix are denoted by small letters $a, b,..., y, z$.

Rows and Columns

The horizontal and vertical lines in a matrix are, respectively, called the rows and columns of the matrix.

Order of a Matrix

If a matrix has m rows and n columns then we say that the size or order of the matrix is $m \times n$. If A is a matrix having m rows and n columns then the matrix can be written as

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Square Matrix

A matrix having n rows and n columns is said to be a $n \times n$ square matrix or a square matrix of order n. The element, or entry, in the *ith* row and *jth* column of a $m \times n$ matrix A is written as a_{ij}. Therefore a 1 x 1 matrix is simply a constant or a function.

Equality of matrix

Any two matrices A and B are said to be equal if and only if they have the same orders and the corresponding elements of the two matrices are equal. Thus if $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ then

$$A = B \Leftrightarrow a_{ij} = b_{ij}, \quad \forall\, i, j$$

Column Matrix

A column matrix X is any matrix having n rows and only one column. Thus the column matrix X can be written as

$$X = \begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \\ \vdots \\ b_{n1} \end{pmatrix} = [b_{i1}]_{n \times 1}$$

A column matrix is also called a column vector or simply a vector.

© Copyright Virtual University of Pakistan

Multiple of matrices

A multiple of a matrix A is defined to be

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ ka_{21} & ka_{22} & \cdots & ka_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ ka_{m1} & ka_{m2} & \cdots & ka_{mn} \end{bmatrix} = [ka_{ij}]_{m\times n}$$

Where k is a constant or it is a function. Notice that the product kA is same as the product Ak. Therefore, we can write

$$kA = Ak$$

Example 1

(a)
$$5 \cdot \begin{bmatrix} 2 & -3 \\ 4 & -1 \\ 1/5 & 6 \end{bmatrix} = \begin{bmatrix} 10 & -15 \\ 20 & -5 \\ 1 & 30 \end{bmatrix}$$

(b)
$$e^{t} \cdot \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} e^{t} \\ -2e^{t} \\ 4e^{t} \end{bmatrix}$$

Since we know that $kA = Ak$. Therefore, we can write

$$e^{-3t} \cdot \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2e^{-3t} \\ 5e^{-3t} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} e^{-3t}$$

Addition of Matrices

Any two matrices can be added only when they have same orders and the resulting matrix is obtained by adding the corresponding entries. Therefore, if $A = [a_{ij}]$ and $B = [b_{ij}]$ are two $m \times n$ matrices then their sum is defined to be the matrix $A + B$ defined by

$$A + B = [a_{ij} + b_{ij}]$$

© Copyright Virtual University of Pakistan

Example 2

Consider the following two matrices of order 3×3

$$A=\begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 6 \\ -6 & 10 & -5 \end{pmatrix},\quad B=\begin{pmatrix} 4 & 7 & -8 \\ 9 & 3 & 5 \\ 1 & -1 & 2 \end{pmatrix}$$

Since the given matrices have same orders. Therefore, these matrices can be added and their sum is given by

$$A+B=\begin{pmatrix} 2+4 & -1+7 & 3+(-8) \\ 0+9 & 4+3 & 6+5 \\ -6+1 & 10+(-1) & -5+2 \end{pmatrix}=\begin{pmatrix} 6 & 6 & -5 \\ 9 & 7 & 11 \\ -5 & 9 & -3 \end{pmatrix}$$

Example 3

Write the following single column matrix as the sum of three column vectors

$$\begin{pmatrix} 3t^2-2e^t \\ t^2+7t \\ 5t \end{pmatrix}$$

Solution

$$\begin{pmatrix} 3t^2-2e^t \\ t^2+7t \\ 5t \end{pmatrix}=\begin{pmatrix} 3t^2 \\ t^2 \\ 0 \end{pmatrix}+\begin{pmatrix} 0 \\ 7t \\ 5t \end{pmatrix}+\begin{pmatrix} -2e^t \\ 0 \\ 0 \end{pmatrix}=\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}t^2+\begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix}t+\begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}e^t$$

Difference of Matrices

The difference of two matrices A and B of same order $m\times n$ is defined to be the matrix

$$A-B=A+(-B)$$

The matrix $-B$ is obtained by multiplying the matrix B with -1. So that

$$-B=(-1)B$$

Multiplication of Matrices

Any two matrices A and B are conformable for the product AB, if the number of columns in the first matrix A is equal to the number of rows in the second matrix B. Thus if the order of the matrix A is $m\times n$ then to make the product AB possible order of the matrix B must be $n\times p$. Then the order of the product matrix AB is $m\times p$. Thus

$$A_{m\times n}\cdot B_{n\times p}=C_{m\times p}$$

© Copyright Virtual University of Pakistan

If the matrices A and B are given by

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix},\quad B=\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

Then

$$AB=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

$$=\begin{pmatrix} a_{11}b_{11}+a_{12}b_{21}+\cdots+a_{1n}b_{n1} & \cdots & a_{11}b_{1p}+a_{12}b_{2p}+\cdots+a_{1n}b_{np} \\ a_{21}b_{11}+a_{22}b_{21}+\cdots+a_{2n}b_{n1} & \cdots & a_{21}b_{1p}+a_{22}b_{2p}+\cdots+a_{2n}b_{np} \\ \vdots & & \vdots \\ a_{m1}b_{11}+a_{m2}b_{21}+\cdots+a_{mn}b_{n1} & \cdots & a_{m1}b_{1p}+a_{m2}b_{2p}+\cdots+a_{mn}b_{np} \end{pmatrix}$$

$$=\left(\sum_{k=1}^{n} a_{ik}b_{kj}\right)_{n\times p}$$

Example 4

If possible, find the products AB and BA, when

(a) $$A=\begin{pmatrix} 4 & 7 \\ 3 & 5 \end{pmatrix},\quad B=\begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix}$$

(b) $$A=\begin{pmatrix} 5 & 8 \\ 1 & 0 \\ 2 & 7 \end{pmatrix},\quad B=\begin{pmatrix} -4 & -3 \\ 2 & 0 \end{pmatrix}$$

Solution

© Copyright Virtual University of Pakistan

(a) The matrices A and B are square matrices of order 2. Therefore, both of the products AB and BA are possible.

$$AB = \begin{pmatrix} 4 & 7 \\ 3 & 5 \end{pmatrix}\begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix} = \begin{pmatrix} 4\cdot 9+7\cdot 6 & 4\cdot(-2)+7\cdot 8 \\ 3\cdot 9+5\cdot 6 & 3\cdot(-2)+5\cdot 8 \end{pmatrix} = \begin{pmatrix} 78 & 48 \\ 57 & 34 \end{pmatrix}$$

Similarly
$$BA = \begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix}\begin{pmatrix} 4 & 7 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 9\cdot 4+(-2)\cdot 3 & 9\cdot 7+(-2)\cdot 5 \\ 6\cdot 4+8\cdot 3 & 6\cdot 7+8\cdot 5 \end{pmatrix} = \begin{pmatrix} 30 & 53 \\ 48 & 82 \end{pmatrix}$$

(b) The product AB is possible as the number of columns in the matrix A and the number of rows in B is 2. However, the product BA is not possible because the number of rows in the matrix B and the number of rows in A is not same.

$$AB = \begin{pmatrix} 5\cdot(-4)+8\cdot 2 & 5\cdot(-3)+8\cdot 0 \\ 1\cdot(-4)+0\cdot 2 & 1\cdot(-3)+0\cdot 0 \\ 2\cdot(-4)+7\cdot 2 & 2\cdot(-3)+7\cdot 0 \end{pmatrix} = \begin{pmatrix} -4 & -15 \\ -4 & -3 \\ 6 & -6 \end{pmatrix}$$

Note that

In general, matrix multiplication is not commutative. This means that $AB \neq BA$. For example, we observe in part (a) of the previous example

$$AB = \begin{pmatrix} 78 & 48 \\ 57 & 34 \end{pmatrix}, \quad BA = \begin{pmatrix} 30 & 53 \\ 48 & 82 \end{pmatrix}$$

Clearly $AB \neq BA$.. Similarly in part (b) of the example, we have

$$AB = \begin{pmatrix} -4 & -15 \\ -4 & -3 \\ 6 & -6 \end{pmatrix}$$

However**,** the product BA is not possible.

© Copyright Virtual University of Pakistan

Example 5

(a)
$$\begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ 1 & -7 & 9 \end{pmatrix}\begin{pmatrix} -3 \\ 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 2\cdot(-3)+(-1)\cdot 6+3\cdot 4 \\ 0\cdot(-3)+4\cdot 6+5\cdot 6 \\ 1\cdot(-3)+(-7)\cdot 6+9\cdot 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 44 \\ -9 \end{pmatrix}$$

(b)
$$\begin{pmatrix} -4 & 2 \\ 3 & 8 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4x+2y \\ 3x+8y \end{pmatrix}$$

Multiplicative Identity

For a given positive integer n, the $n \times n$ matrix

$$I = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

is called the multiplicative identity matrix. If A is a matrix of order $n \times n$, then it can be verified that

$$I \cdot A = A \cdot I = A$$

Also, it is readily verified that if X is any $n \times 1$ column matrix, then $I \cdot X = X$

Zero Matrix

A matrix consisting of all zero entries is called a zero matrix or null matrix and is denoted by O. For example

$$O = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

and so on. If A and O are $m \times n$ matrices, then

$$A + O = O + A = A$$

© Copyright Virtual University of Pakistan

Associative Law

The matrix multiplication is associative. This means that if A, B and C are $m \times p$, $p \times r$ and $r \times n$ matrices, then

$$A(BC) = (AB)C$$

The result is a $m \times n$ matrix.

Distributive Law

If B and C are matrices of order $r \times n$ and A is a matrix of order $m \times r$, then the distributive law states that

$$A(B + C) = AB + AC$$

Furthermore, if the product $(A + B)C$ is defined, then

$$(A + B)C = AC + BC$$

Determinant of a Matrix

Associated with every square matrix A of constants, there is a number called the determinant of the matrix, which is denoted by $\det(A)$ or $|A|$

Example 6

Find the determinant of the following matrix

$$A = \begin{pmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{pmatrix}$$

Solution

The determinant of the matrix A is given by

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

We expand the $\det(A)$ by cofactors of the first row, we obtain

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix} = 3\begin{vmatrix} 5 & 1 \\ 2 & 4 \end{vmatrix} - 6\begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} + 2\begin{vmatrix} 2 & 5 \\ -1 & 2 \end{vmatrix}$$

or $$\det(A) = 3(20 - 2) - 6(8 + 1) + 2(4 + 5) = 18$$

© Copyright Virtual University of Pakistan

Transpose Of a Matrix

The transpose of a $m \times n$ matrix A is obtained by interchanging rows and columns of the matrix and is denoted by A^{tr}. In other words, rows of A become the columns of A^{tr}. If

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \cdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Then

$$A^{tr} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$$

Since order of the matrix A is $m \times n$, the order of the transpose matrix A^{tr} is $n \times m$.

Example 7

(a) The transpose of matrix

$$A = \begin{pmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{pmatrix}$$

is $$A^{tr} = \begin{pmatrix} 3 & 2 & -1 \\ 6 & 5 & 2 \\ 2 & 1 & 4 \end{pmatrix}$$

(b) If X denotes the matrix

$$X = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix}$$

Then $$X^{tr} = \begin{bmatrix} 5 & 0 & 3 \end{bmatrix}$$

© Copyright Virtual University of Pakistan

Multiplicative Inverse of a Matrix

Suppose that A is a square matrix of order $n \times n$. If there exists an $n \times n$ matrix B such that

$$AB = BA = I$$

Then B is said to be the multiplicative inverse of the matrix A and is denoted by $B = A^{-1}$.

Non-Singular Matrices

A square matrix A of order $n \times n$ is said to be a non-singular matrix if

$$\det(A) \neq 0$$

Otherwise the square matrix A is said to be singular. Thus for a singular A we must have

$$\det(A) = 0$$

Theorem

If A is a square matrix of order $n \times n$ then the matrix has a multiplicative inverse A^{-1} if and only if the matrix A is non-singular.

.

Theorem

Let A be a non singular matrix of order $n \times n$ and let C_{ij} denote the cofactor (signed minor) of the corresponding entry a_{ij} in the matrix A i.e.

$$C_{ij} = (-1)^{i+j} M_{ij}$$

M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting the *ith* row and *jth* column from A. Then inverse of the matrix A is given by

$$A^{-1} = \frac{1}{\det(A)} (C_{ij})^{tr}$$

Further Explanation

1. For further reference we take $n = 2$ so that A is a 2×2 non-singular matrix given by

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Therefore $C_{11} = a_{22}$, $C_{12} = -a_{21}$, $C_{21} = -a_{12}$ and $C_{22} = a_{11}$. So that

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{21} \\ -a_{12} & a_{11} \end{pmatrix}^{tr} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

© Copyright Virtual University of Pakistan

2. For a 3×3 non-singular matrix

$$A=\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$C_{11}=\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, C_{12}=-\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \quad C_{13}=\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

and so on. Therefore, inverse of the matrix A is given by

$$A^{-1}=\frac{1}{\det A}\begin{pmatrix} C_{11} & C_{21} & a_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}.$$

Example 8

Find, if possible, the multiplicative inverse for the matrix

$$A=\begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}.$$

Solution:

The matrix A is non-singular because

$$\det(A)=\begin{vmatrix} 1 & 4 \\ 2 & 10 \end{vmatrix}=10-8=2$$

Therefore, A^{-1}exists and is given by

.

$$A^{-1}=\frac{1}{2}\begin{pmatrix} 10 & -4 \\ -2 & 1 \end{pmatrix}=\begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix}$$

Check

$$AA^{-1}=\begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}\begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix}=\begin{pmatrix} 5-4 & -2+2 \\ 10-10 & -4+5 \end{pmatrix}=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}=I$$

$$AA^{-1}=\begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}=\begin{pmatrix} 5-4 & 20-20 \\ -1+1 & -4+5 \end{pmatrix}=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}=I$$

© Copyright Virtual University of Pakistan

Example 9

Find, if possible, the multiplicative inverse of the following matrix

$$A = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}$$

Solution:

The matrix is singular because

$$\det(A) = \begin{vmatrix} 2 & 2 \\ 3 & 3 \end{vmatrix} = 2 \cdot 3 - 2 \cdot 3 = 0$$

Therefore, the multiplicative inverse A^{-1} of the matrix does not exist.

Example 10

Find the multiplicative inverse for the following matrix

$$A = \begin{pmatrix} 2 & 2 & 0 \\ -2 & 1 & 1 \\ 3 & 0 & 1 \end{pmatrix}.$$

Solution:

Since $$\det(A) = \begin{vmatrix} 2 & 2 & 0 \\ -2 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix} = 2(1-0) - 2(-2-3) + 0(0-3) = 12 \neq 0$$

Therefore, the given matrix is non singular. So that, the multiplicative inverse A^{-1} of the matrix A exists. The cofactors corresponding to the entries in each row are

$$C_{11} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1, \qquad C_{12} = -\begin{vmatrix} -2 & 1 \\ 3 & 1 \end{vmatrix} = 5, \qquad C_{13} = \begin{vmatrix} -2 & 1 \\ 3 & 0 \end{vmatrix} = -3$$

$$C_{21} = -\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = -2, \qquad C_{22} = \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix} = 2, \qquad C_{23} = -\begin{vmatrix} 2 & 2 \\ 3 & 0 \end{vmatrix} = 6$$

$$C_{31} = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2, \qquad C_{32} = -\begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix} = -2, \qquad C_{33} = \begin{vmatrix} 2 & 2 \\ -2 & 1 \end{vmatrix} = 6$$

Hence $$A^{-1} = \frac{1}{12}\begin{pmatrix} 1 & -2 & 2 \\ 5 & 2 & -2 \\ -3 & 6 & 6 \end{pmatrix} = \begin{pmatrix} 1/12 & -1/6 & 1/6 \\ 5/12 & 1/6 & -1/6 \\ -1/4 & 1/2 & 1/2 \end{pmatrix}$$

Please verify that $A \cdot A^{-1} = A^{-1} \cdot A = I$

© Copyright Virtual University of Pakistan

Derivative of a Matrix of functions

Suppose that

$$A(t) = \left[a_{ij}(t) \right]_{m\times n}$$

is a matrix whose entries are functions those are differentiable on a common interval, then derivative of the matrix $A(t)$ is a matrix whose entries are derivatives of the corresponding entries of the matrix $A(t)$. Thus

$$\frac{dA}{dt} = \left[\frac{da_{ij}}{dt} \right]_{m\times n}$$

The derivative of a matrix is also denoted by $A'(t)$.

Integral of a Matrix of Functions

Suppose that $A(t) = \left(a_{ij}(t)\right)_{m\times n}$ is a matrix whose entries are functions those are continuous on a common interval containing t, then integral of the matrix $A(t)$ is a matrix whose entries are integrals of the corresponding entries of the matrix $A(t)$. Thus

$$\int_{t_0}^{t} A(s)ds = \left(\int_{t_0}^{t} a_{ij}(s)ds \right)_{m\times n}$$

Example 11

Find the derivative and the integral of the following matrix

$$X(t) = \begin{pmatrix} \sin 2t \\ e^{3t} \\ 8t-1 \end{pmatrix}$$

Solution:

The derivative and integral of the given matrix are, respectively, given by

$$X'(t) = \begin{pmatrix} \frac{d}{dt}(\sin 2t) \\ \frac{d}{dt}(e^{3t}) \\ \frac{d}{dt}(8t-1) \end{pmatrix} = \begin{pmatrix} 2\cos 2t \\ 3e^{3t} \\ 8 \end{pmatrix}$$

$$\int_0^t X(s)ds = \begin{pmatrix} \int_0^t \sin 2t ds \\ \int_{t0}^{3t} e^{3t} ds \\ \int_0^t 8t-1 ds \end{pmatrix} = \begin{pmatrix} -1/2\cos 2t + 1/2 \\ 1/3e^{3t} - 1/3 \\ 4t^2 - t \end{pmatrix}$$

© Copyright Virtual University of Pakistan

Augmented Matrix

Consider an algebraic system of n linear equations in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \quad & \quad \vdots \quad \quad \vdots \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Suppose that A denotes the coefficient matrix in the above algebraic system, then

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

It is well known that Cramer's rule can be used to solve the system, whenever $\det(A) \neq 0$. However, it is also well known that a Herculean effort is required to solve the system if $n > 3$. Thus for larger systems the Gaussian and Gauss-Jordon elimination methods are preferred and in these methods we apply elementary row operations on augmented matrix. The augmented matrix of the system of linear equations is the following $n \times (n+1)$ matrix

$$A_b = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{211} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{bmatrix}$$

If B denotes the column matrix of the b_i, $\forall i = 1, 2, ..., n$ then the augmented matrix of the above mentioned system of linear algebraic equations can be written as $(A \mid B)$.

Elementary Row Operations

The elementary row operations consist of the following three operations

- Multiply a row by a non-zero constant.
- Interchange any row with another row.
- Add a non-zero constant multiple of one row to another row.

These row operations on the augmented matrix of a system are equivalent to, multiplying an equation by a non-zero constant, interchanging position of any two equations of the system and adding a constant multiple of an equation to another equation.

© Copyright Virtual University of Pakistan

The Gaussian and Gauss-Jordon Methods

In the Gaussian Elimination method we carry out a succession of elementary row operations on the augmented matrix of the system of linear equations to be solved until it is transformed into row-echelon form, a matrix that has the following structure:

- The first non-zero entry in a non-zero row is 1.
- In consecutive nonzero rows the first entry 1 in the lower row appears to the right of the first 1 in the higher row.
- Rows consisting of all 0's are at the bottom of the matrix.

In the Gauss-Jordan method the row operations are continued until the augmented matrix is transformed into the reduced row-echelon form**.** A reduced row-echelon matrix has the structure similar to row-echelon, but with an additional property.

- The first non-zero entry in a non-zero row is 1.
- In consecutive nonzero rows the first entry 1 in the lower row appears to the right of the first 1 in the higher row.
- Rows consisting of all 0's are at the bottom of the matrix.
- A column containing a first entry 1 has 0's everywhere else.

Example 1

(a) The following two matrices are in row-echelon form.

$$\left(\begin{array}{ccc|c} 1 & 5 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccccc|c} 0 & 0 & 1 & -6 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array}\right)$$

Please verify that the three conditions of the structure of the echelon form are satisfied.

(b) The following two matrices are in reduced row-echelon form**.**

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccccc|c} 0 & 0 & 1 & -6 & 0 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array}\right)$$

Please notice that all remaining entries in the columns containing a leading entry 1 are 0.

Notation

© Copyright Virtual University of Pakistan

To keep track of the row operations on an augmented matrix, we utilized the following notation:

Symbol	Meaning
R_{ij}	Interchange the rows i and j.
cR_i	Multiply the ith row by a nonzero constant c.
$cR_i + R_j$	Multiply the ith row by c and then add to the jth row.

Example 2

Solve the following system of linear algebraic equations by the (a) Gaussian elimination and (b) Gauss-Jordan elimination

$$\begin{aligned} 2x_1 + 6x_2 + x_3 &= 7 \\ x_1 + 2x_1 - x_3 &= -1 \\ 5x_1 + 7x_2 - 4x_3 &= 9 \end{aligned}$$

Solution

(a) The augmented matrix of the system is

$$\left(\begin{array}{ccc|c} 2 & 6 & 1 & 7 \\ 1 & 2 & -1 & -1 \\ 5 & 7 & -4 & 9 \end{array}\right)$$

By interchanging first and second row i.e. by R_{12}, we obtain

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 2 & 6 & 1 & 7 \\ 5 & 7 & -4 & 9 \end{array}\right)$$

Multiplying first row with -2 and -5 and then adding to 2nd and 3rd row i.e. by $-R_1 + R_2$ and $-5R_1 + R_3$, we obtain

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 2 & 3 & 9 \\ 0 & -3 & 1 & 14 \end{array}\right)$$

Multiply the second row with $1/2$, i.e. the operation $\frac{1}{2}R_2$, yields

© Copyright Virtual University of Pakistan

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & 3/2 & 9/2 \\ 0 & -3 & 1 & 14 \end{array}\right)$$

Next add three times the second row to the third row, the operation $3R_2 + R_3$ gives

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & 3/2 & 9/2 \\ 0 & 0 & 11/2 & 55/2 \end{array}\right)$$

Finally, multiply the third row with $2/11$. This means the operation $\frac{2}{11}R_1$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & 3/2 & 9/2 \\ 0 & 0 & 1 & 5 \end{array}\right)$$

The last matrix is in row-echelon form and represents the system

$$x_1 + x_2 - x_3 = 1$$
$$x_2 + \frac{3}{2}x_3 = 9/2$$
$$x_3 = 5$$

Now by the backward substitution we obtain the solution set of the given system of linear algebraic equations

$$x_1 = 10, \quad x_2 = -3, \quad x_3 = 5$$

(b) W start with the last matrix in part (a). Since the first in the second and third rows are 1's we must, in turn, making the remaining entries in the second and third columns 0s:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & 3/2 & 9/2 \\ 0 & 0 & 1 & 5 \end{array}\right)$$

Adding -2 times the 2nd row to first row, this means the operation $-2R_2 + R_1$, we have

$$\left(\begin{array}{ccc|c} 1 & 0 & -4 & -10 \\ 0 & 1 & 3/2 & 9/2 \\ 0 & 0 & 1 & 5 \end{array}\right)$$

Finally by 4 times the third row to first and $-1/2$ times the third row to second row, i.e. the operations $4R_3 + R_1$ and $\frac{-1}{2}R_3 + R_2$, yields

© Copyright Virtual University of Pakistan

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -10 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 5 \end{array}\right).$$

The last matrix is now in reduce row-echelon form .Because of what the matrix means in terms of equations, it evident that the solution of the system

$$x_1 = 10, \quad x_2 = -3, \quad x_3 = 5$$

Example 3

Use the Gauss-Jordan elimination to solve the following system of linear algebraic equations.

$$x + 3y - 2z = -7$$
$$4x + y + 3z = 5$$
$$2x - 5y + 7z = 19$$

Solution:

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & -7 \\ 4 & 1 & 3 & 5 \\ 2 & -5 & 7 & 19 \end{array}\right)$$

$-4R_1 + R_2$ and $-2R_1 + R_3$ yields

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & -7 \\ 0 & -11 & 11 & 33 \\ 0 & -11 & 11 & 33 \end{array}\right)$$

$\frac{-1}{11}R_2$ and $\frac{-1}{11}R_3$ produces

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & -7 \\ 0 & 1 & -1 & -3 \\ 0 & 1 & -1 & -3 \end{array}\right)$$

$3R_2 + R_1$ and $-R_2 + R_3$ gives

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

In this case the last matrix in reduced row-echelon form implies that the original system of three equations in three unknowns.

© Copyright Virtual University of Pakistan

$$x + z = 2,\ \ y - z = -3$$

We can assign an arbitrarily value to z. If we let $z = t,\ t \in R$, then we see that the system has infinitely many solutions:

$$x = 2 - t,\ \ y = -3 + t,\ \ z = t$$

Geometrically, these equations are the parametric equations for the line of intersection of the planes

$$x + 0y + 0z = 2,\ \ 0x + y - z = -3$$

Exercise

Write the given sum as a single column matrix

1. $3\begin{pmatrix} 2 \\ t \\ -1 \end{pmatrix} + (t-1)\begin{pmatrix} -1 \\ t \\ -3 \end{pmatrix} - 2\begin{pmatrix} t \\ 4 \\ -5t \end{pmatrix}$

2. $\begin{pmatrix} 1 & -3 & 4 \\ 2 & 5 & -1 \\ 0 & -4 & -2 \end{pmatrix}\begin{pmatrix} t \\ 2t-1 \\ -t \end{pmatrix} + \begin{pmatrix} -t \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 8 \\ -6 \end{pmatrix}$

Determine whether the given matrix is singular or non-singular. If singular, find A^{-1}.

3. $A = \begin{pmatrix} 3 & 2 & 1 \\ 4 & 1 & 0 \\ -2 & 5 & -1 \end{pmatrix}$

4. $A = \begin{pmatrix} 4 & 1 & -1 \\ 6 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$

Find $\dfrac{dX}{dt}$

5. $X = \begin{pmatrix} 1 \\ 2 - \sin 2t - 4\cos 2t \\ -3\sin 2t + 5\cos 2t \end{pmatrix}$

6. If $A(t) = \begin{pmatrix} e^{4t} & \cos \pi t & 2 \\ 2t & 3t & -1 \end{pmatrix}$ then find (a) $\int_0^2 A(t)dt$, (b) $\int_0^t A(s)ds$.

7. Find the integral $\int_1^2 B(t)dt$ if $B(t) = \begin{pmatrix} 6t & 2 \\ 1/t & 4t \end{pmatrix}$

Solve the given system of equations by either Gaussian elimination or by the Gauss-Jordon elimination.

© Copyright Virtual University of Pakistan

8. $5x - 2y + 4z = 10$
   $x + y + z = 9$
   $4x - 3y + 3z = 1$

9. $x_1 + x_2 - x_3 - x_4 = -1$
   $x_1 + x_2 + x_3 + x_4 = 3$
   $x_1 - x_2 + x_3 - x_4 = 3$
   $4x_1 + x_2 - 2x_3 + x_4 = 0$

10. $x_1 + x_2 - x_3 + 3x_4 = 1$
    $x_2 - x_3 - 4x_4 = 0$
    $x_1 + 2x_2 - 2x_3 - x_4 = 6$
    $4x_1 + 7x_2 - 7x_3 = 9$

© Copyright Virtual University of Pakistan

Lecture 39
The Eigenvalue problem

Eigenvalues and Eigenvectors

Let A be a $n \times n$ matrix. A number λ is said to be an eigenvalue of A if there exists a nonzero solution vector K of the system of linear differential equations:

$$AK = \lambda K$$

The solution vector K is said to be an eigenvector corresponding to the eigenvalue λ. Using properties of matrix algebra, we can write the above equation in the following alternative form

$$(A - \lambda I)K = 0$$

where I is the identity matrix.

If we let $K = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ \vdots \\ k_n \end{pmatrix}$

Then the above system is same as the following system of linear algebraic equations

$$\begin{aligned} (a_{11} - \lambda)k_1 + a_{12}k_2 + \cdots + a_{1n}k_n &= 0 \\ a_{21}k_1 + (a_{22} - \lambda)k_2 + \cdots + a_{2n}k_n &= 0 \\ \vdots \qquad\qquad \vdots \qquad\qquad & \vdots \\ a_{n1}k_n + a_{n2}k_2 + \cdots + (a_{nn} - \lambda)k_n &= 0 \end{aligned}$$

Clearly, an obvious solution of this system is the trivial solution

$$k_1 = k_2 = \ldots = k_n = 0$$

However, we are seeking only a non-trivial solution of the system.

The Non-trivial solution

The non-trivial solution of the system exists only when

$$\det(A - \lambda I) = 0$$

This equation is called the characteristic equation of the matrix A. Thus the Eigenvalues of the matrix A are given by the roots of the characteristic equation. To find an eigenvector corresponding to an eigenvalue λ we simply solve the system of linear algebraic equations

$$\det(A - \lambda I)K = 0$$

© Copyright Virtual University of Pakistan

This system of equations can be solved by applying the Gauss-Jordan elimination to the augmented matrix

$$\left(A - \lambda I \quad \middle| 0 \right).$$

Example 4

Verify that the following column vector is an eigenvector

$$K = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

is an eigenvector of the following 3×3 matrix

$$A = \begin{pmatrix} 0 & -1 & -3 \\ 2 & 3 & 3 \\ -2 & 1 & 1 \end{pmatrix}$$

Solution:

By carrying out the multiplication AK, we see that

$$AK = \begin{pmatrix} 0 & -1 & -3 \\ 2 & 3 & 3 \\ -2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = (-2) \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} = (-2) K$$

Hence the number $\lambda = -2$ is an eigenvalue of the given matrix A.

Example 5

Find the eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

Solution:

Eigenvalues

The characteristic equation of the matrix A is

$$\det \left(A - \lambda I \right) = \begin{vmatrix} 1-\lambda & 2 & 1 \\ 6 & -1-\lambda & 0 \\ -1 & -2 & -1-\lambda \end{vmatrix} = 0$$

Expanding the determinant by the cofactors of the second row, we obtain

© Copyright Virtual University of Pakistan

$$-\lambda^3 - \lambda^2 + 12\lambda = 0$$

This is so much easy given below the explanation of the above kindly see it and let me know if you have any more query

L: STAND FOR LEMDA

(1-L)((-1-L) (-1-L) -0)-2(6(-1-L)-0) +1(6(-2) +1(-1-L) =0

(1-L)(1+L^2+2L)-2(-6-6L) +1(-12 -1-L) =0

(1-L)(1+L^2+2L)+12+12L+1(-13-L) =0

1+L^2+2L-L-L^3-2L^2+12+12L-13-L=0

-L^3-L^2+12L=0

$$\lambda(\lambda + 4)(\lambda - 3) = 0$$

Hence the eigenvalues of the matrix are

$$\lambda_1 = 0,\ \lambda_2 = -4,\ \lambda_3 = 3.$$

Eigenvectors

For $\lambda_1 = 0$ we have

$$(A - 0|\ 0) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 6 & -1 & 0 & 0 \\ -1 & -2 & -1 & 0 \end{array}\right)$$

By $-6R_1 + R_2,\ R_1 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -13 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

By $-\frac{1}{13}R_2$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 6/13 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

By $-2R_2 + R_1$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1/13 & 0 \\ 0 & 1 & 6/13 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

© Copyright Virtual University of Pakistan

Thus we have the following equations in k_1, k_2 and k_3. The number k_3 can be chosen arbitrarily

$$k_1 = -(1/13)k_3,\ k_2 = -(6/13)k_3$$

Choosing $k_3 = -13$, we get $k_1 = 1$ and $k_2 = 6$. Hence, the eigenvector corresponding $\lambda_1 = 0$ is

$$K_1 = \begin{pmatrix} 1 \\ 6 \\ -13 \end{pmatrix}$$

For $\lambda_2 = -4$, we have

$$(A+4 \mid 0) = \left(\begin{array}{ccc|c} 5 & 2 & 1 & 0 \\ 6 & 3 & 0 & 0 \\ -1 & -2 & 3 & 0 \end{array}\right)$$

By $(-1)R_3,\ R_{32}$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 6 & 3 & 0 & 0 \\ 5 & 2 & 1 & 0 \end{array}\right)$$

By $-6R_1 + R_2, -5R_1 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & -9 & 18 & 0 \\ 0 & -8 & 16 & 0 \end{array}\right)$$

By $-\frac{1}{9}R_2,\ -\frac{1}{8}R_3$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right)$$

By $-2R_2 + R_1,\ -R_2 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Hence we obtain the following two equations involving k_1, k_2 and k_3.

$$k_1 = -k_3,\ k_2 = 2k_3$$

© Copyright Virtual University of Pakistan

Choosing $k_3 = 1$, we have $k_1 = -1,\ k_2 = 2$. Hence we have an eigenvector corresponding to the eigenvalue $\lambda_2 = -4$

$$K_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Finally, for $\lambda_3 = 3$, we have

$$(A - 3I \mid 0) = \left(\begin{array}{ccc|c} -2 & 2 & 1 & 0 \\ 6 & -4 & 0 & 0 \\ -1 & -2 & -4 & 0 \end{array}\right)$$

By using the Gauss Jordon elimination as used for other values, we obtain (verify!)

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 3/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

So that we obtain the equations

$$k_1 = -k_3, \quad k_2 = (-3/2)k_3$$

The choice $k_3 = -2$ leads to $k_1 = 2, \quad k_2 = 3$. Hence, we have the following eigenvector

$$K_3 = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$$

Note that

The component k_3 could be chosen as any nonzero number. Therefore, a nonzero constant multiple of an eigenvector is also an eigenvector.

Example 6

Find the eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$$

Solution:

From the characteristic equation of the given matrix is

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 4 \\ -1 & 7-\lambda \end{vmatrix} = 0$$

© Copyright Virtual University of Pakistan

or $$(3-\lambda)(7-\lambda)+4=0 \Rightarrow (\lambda-5)^2=0$$

Therefore, the characteristic equation has repeated real roots. Thus the matrix has an eigenvalue of multiplicity two.

$$\lambda_1=\lambda_2=5$$

In the case of a 2×2 matrix there is no need to use Gauss-Jordan elimination. To find the eigenvector(s) corresponding to $\lambda_1=5$ we resort to the system of linear equations

$$(A-5I)K=0$$

or in its equivalent form

$$-2k_1+4k_2=0$$
$$k_1+2k_2=0$$

It is apparent from this system that

$$k_1=2k_2\,.$$

Thus if we choose $k_2=1$, we find the single eigenvector

$$K_1=\begin{pmatrix}2\\1\end{pmatrix}$$

Example 7

Find the eigenvalues and eigenvectors of

$$A=\begin{pmatrix}9 & 1 & 1\\ 1 & 9 & 1\\ 1 & 1 & 9\end{pmatrix}$$

Solution

The characteristic equation of the given matrix is

$$\det(A-\lambda I)=\begin{vmatrix}9-\lambda & 1 & 1\\ 1 & 9-\lambda & 1\\ 1 & 1 & 9-\lambda\end{vmatrix}=0$$

or $$(\lambda-11)(\lambda-8)^2=0 \Rightarrow \lambda=11,\ 8,\ 8$$

Thus the eigenvalues of the matrix are $\lambda_1=11, \lambda_2=\lambda_3=8$

For $\lambda_1=11$, we have

$$(A-11I\,|\,0)=\left(\begin{array}{ccc|c}-2 & 1 & 1 & 0\\ 1 & -2 & 1 & 0\\ 1 & 1 & -2 & 0\end{array}\right)$$

The Gauss-Jordan elimination gives

© Copyright Virtual University of Pakistan

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Hence, $k_1 = k_3$, $k_2 = k_3$. If $k_3 = 1$, then

$$K_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Now for $\lambda_2 = 8$ we have

$$(A - 8I \mid 0) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right)$$

Again the Gauss-Jordon elimination gives

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Therefore, $k_1 + k_2 + k_3 = 0$

We are free to select two of the variables arbitrarily. Choosing, on the one hand, $k_2 = 1, k_3 = 0$ and, on the other, $k_2 = 0, k_3 = 1$, we obtain two linearly independent eigenvectors corresponding to a single eigenvalue

$$K_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \; K_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Note that

Thus we note that when a $n \times n$ matrix A possesses n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, a set of n linearly independent eigenvectors $K_1, K_2, \ldots, K_n$ can be found.

However, when the characteristic equation has repeated roots, it may not be possible to find n linearly independent eigenvectors of the matrix.

Exercise

Find the eigenvalues and eigenvectors of the given matrix.

1. $\begin{pmatrix} -1 & 2 \\ -7 & 8 \end{pmatrix}$
2. $\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$

© Copyright Virtual University of Pakistan

3. $\begin{pmatrix} -8 & -1 \\ 16 & 0 \end{pmatrix}$

4. $\begin{pmatrix} 5 & -1 & 0 \\ 0 & -5 & 9 \\ 5 & -1 & 0 \end{pmatrix}$

5. $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 4 & 0 & 1 \end{pmatrix}$

6. $\begin{pmatrix} 0 & 4 & 0 \\ -1 & -4 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

Show that the given matrix has complex eigenvalues.

7. $\begin{pmatrix} -1 & 2 \\ -5 & 1 \end{pmatrix}$

8. $\begin{pmatrix} 2 & -1 & 0 \\ 5 & 2 & 4 \\ 0 & 1 & 2 \end{pmatrix}$

© Copyright Virtual University of Pakistan

Lecture 40
Matrices and Systems of Linear First-Order Equations

Matrix form of a system

Consider the following system of linear first-order differential equations

$$\frac{dx_1}{dt} = a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n + f_1(t)$$
$$\frac{dx_2}{dt} = a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n + f_2(t)$$
$$\vdots$$
$$\frac{dx_n}{dt} = a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n + f_n(t)$$

Suppose that X, $A(t)$ and $F(t)$, respectively, denote the following matrices

$$X = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad A(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{pmatrix}, \quad F(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

Then the system of differential equations can be written as

$$\frac{d}{dt}\begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{pmatrix}\begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} + \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

or simply

$$\frac{dX}{dt} = A(t)X + F(t)$$

If the system of differential equations is homogenous, then $F(t) = 0$ and we can write

$$\frac{dX}{dt} = A(t)X$$

Both the non-homogeneous and the homogeneous systems can also be written as

$$X' = AX + F, \quad X' = AX$$

© Copyright Virtual University of Pakistan

Example 1

Write the following non-homogeneous system of differential equations in the matrix form

$$\frac{dx}{dt} = -2x + 5y + e^t - 2t$$

$$\frac{dy}{dt} = 4x - 3y + 10t$$

Solution:

If we suppose that

$$X = \begin{pmatrix} x \\ y \end{pmatrix}$$

Then, the given non-homogeneous differential equations can be written as

$$\frac{dX}{dt} = \begin{pmatrix} -2 & 5 \\ 4 & -3 \end{pmatrix} X + \begin{pmatrix} e^t - 2t \\ 10t \end{pmatrix}$$

or

$$X' = \begin{pmatrix} -2 & 5 \\ 4 & -3 \end{pmatrix} X + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + \begin{pmatrix} -2 \\ 10 \end{pmatrix} t$$

Solution Vector

Consider a homogeneous system of differential equations

$$\frac{dX}{dt} = AX$$

A solution vector on an interval I of the homogeneous system is any column matrix

$$X = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

The entries of the solution vector have to be differentiable functions satisfying each equation of the system on the interval I.

Example 2

Verify that

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}, \quad X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} = \begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix}$$

are solution of the following system of the homogeneous differential equations

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

on the interval $(-\infty, \infty)$

© Copyright Virtual University of Pakistan

Solution:

Since

$$X_1 = \begin{pmatrix} e^{-2t} \\ -e_{-2t} \end{pmatrix} \Rightarrow X_1^{/} = \begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$$

Further

$$AX_1 = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix} = \begin{pmatrix} e^{-2t} - 3e^{-2t} \\ 5e^{-2t} - 3e^{-2t} \end{pmatrix}$$

or $$AX_1 = \begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix} = X_1^{/}$$

Similarly

$$X_2 = \begin{pmatrix} 3\ e^{6t} \\ 5\ e_{6t} \end{pmatrix} \Rightarrow X_2^{/} = \begin{pmatrix} 18\ e^{6t} \\ 30\ e6t \end{pmatrix}$$

and $$AX_2 = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3\ e^{6t} \\ 5\ e6t \end{pmatrix} = \begin{pmatrix} 3\ e^{6t} + 15\ e^{6t} \\ 15\ e^{6t} + 15\ e^{6t} \end{pmatrix}$$

or $$AX_2 = \begin{pmatrix} 18e^{6t} \\ 30e_{6t} \end{pmatrix} = X_2^{/}$$

Thus, the vectors X_1 and X_2 satisfy the homogeneous linear system

$$X^{/} = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

Hence, the given vectors are solutions of the given homogeneous system of differential equations.

Note that

Much of the theory of the systems of n linear first-order differential equations is similar to that of the linear nth -order differential equations.

© Copyright Virtual University of Pakistan

Initial –Value Problem

Let t_0 denote any point in some interval denoted by I and

$$X(t_o) = \begin{pmatrix} x_1(t_o) \\ x_2(t_o) \\ \vdots \\ x_n(t_o) \end{pmatrix}, \quad X_o = \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_n \end{pmatrix}$$

$\gamma_i; i = 1, 2, ..., n$ are given constants. Then the problem of solving the system of differential equations

$$\frac{dX}{dt} = A(t)X + F(t)$$

Subject to the initial conditions

$$X(t_0) = X_0$$

is called an initial value problem on the interval I.

Theorem: *Existence of a unique Solution*

Suppose that the entries of the matrices $A(t)$ and $F(t)$ in the system of differential equations

$$\frac{dX}{dt} = A(t)X + F(t)$$

being considered in the above mentioned initial value problem, are continuous functions on a common interval I that contains the point t_0. Then there exist a unique solution of the initial–value problem on the interval I.

Superposition Principle

Suppose that $X_1, X_2, ..., X_n$ be a set of solution vectors of the homogenous system

$$\frac{dX}{dt} = A(t)\,X$$

on an interval I. Then the principle of superposition states that linear combination

$$X = c_1X_1 + c_2X_2 + \cdots + c_kX_k$$

$c_i; i = 1, 2, ..., k$ being arbitrary constants, is also a solution of the system on the same interval I.

Note that

An immediate consequence of the principle of superposition is that a constant multiple of any solution vector of a homogenous system of first order differential equation is also a solution of the system.

© Copyright Virtual University of Pakistan

Example 3

Consider the following homogeneous system of differential equations

$$X' = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} X$$

Also consider a solution vector X_1 of the system that is given by

$$X_1 = \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix}$$

For any constant c_1 the vector $X = c_1 X_1$ is also a solution of the homogeneous system. To verify this we differentiae the vector X with respect to t

$$\frac{dX}{dt} = c_1 \frac{dX}{dt} = c_1 \begin{pmatrix} -\sin t \\ \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t + \sin t \end{pmatrix}$$

Also

$$AX = c_1 \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix}$$

$$AX = c_1 \begin{pmatrix} -\sin t \\ \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t + \sin t \end{pmatrix}$$

Thus, we have verified that**:**

$$\frac{dX}{dt} = AX$$

Hence the vector $c_1 X_1$ is also a solution vector of the homogeneous system of differential equations.

© Copyright Virtual University of Pakistan

Example 4

Consider the following system considered in the previous example 4

$$X' = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} X$$

We know from the previous example that the vector X_1 is a solution of the system

$$X_1 = \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix}$$

If $$X_2 = \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

Then $$X'_2 = \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

and $$AX_2 = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

Therefore

$$AX_2 = X'_2$$

Hence the vector X_2 is a solution vector of the homogeneous system. We can verify that the following vector is also a solution of the homogeneous system.

$$X = c_1X_1 + c_2X_2$$

or

$$X = c_1 \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

© Copyright Virtual University of Pakistan

Linear Dependence of Solution Vectors

Let $X_1, X_2, X_3, ..., X_k$ be a set of solution vectors, on an interval I, of the homogenous system of differential equations

$$\frac{dX}{dt} = AX$$

We say that the set is linearly dependent on I if there exist constants $c_1, c_2, c_3 ..., c_k$ not all zero such that

$$X(t) = c_1 X_1(t) + c_2 X_2(t) + \cdots + c_k X_k(t) = 0, \qquad \forall\ t \in I$$

Note that

- Any two solution vectors X_1 and X_2 are linearly dependent if and only if one of the two vectors is a constant multiple of the other.
- For $k > 2$ if the set of k solution vectors is linearly dependent then we can express at least one of the solution vectors as a linear combination of the remaining vectors.

Linear Independence of Solution Vectors

Suppose that $X_1, X_2, ..., X_k$ is a set of solution vectors, on an interval I, of the homogenous system of differential equations

$$\frac{dX}{dt} = AX$$

Then the set of solution vectors is said to be linearly independent if it is not linearly dependent on the interval I. This means that

$$X(t) = c_1 X_1(t) + c_2 X_2(t) + \cdots + c_k X_k(t) = 0$$

only when each $c_i = 0$.

Example 5

Consider the following two column vectors

$$X_1 = \begin{pmatrix} 3e^t \\ e^t \end{pmatrix}, \quad X_2 = \begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}$$

Since $$\frac{dX_1}{dt} = \begin{pmatrix} 3e^t \\ e^t \end{pmatrix}, \quad \frac{dX_2}{dt} = \begin{pmatrix} -e^{-t} \\ -e^{-t} \end{pmatrix}$$

and $$\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 3e^t \\ e^t \end{pmatrix} = \begin{pmatrix} 6e^t - 3e^t \\ 3e^t - 2e^t \end{pmatrix} = \begin{pmatrix} 3e^t \\ e^t \end{pmatrix} = \frac{dX_1}{dt}$$

© Copyright Virtual University of Pakistan

Similarly

$$\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}=\begin{pmatrix} 2e^{-t}-3e^{-t} \\ e^{-t}-2e^{-t} \end{pmatrix}=\begin{pmatrix} -e^{-t} \\ -e^{-t} \end{pmatrix}=\frac{dX_2}{dt}$$

Hence both the vectors X_1 and X_2 are solutions of the homogeneous system

$$X'=\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}X$$

Now suppose that $c_1,\ c_2$ are any two arbitrary real constants such that

$$c_1X_1+c_2X_2=0$$

or

$$c_1\begin{pmatrix} 3 \\ 1 \end{pmatrix}e^{t}+c_2\begin{pmatrix} 1 \\ 1 \end{pmatrix}e^{-t}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This means that

$$3c_1e^{t}+c_2e^{-t}=0$$

$$c_1e^{t}+c_2e^{-t}=0$$

The only solution of these equations for the arbitrary constants c_1 and c_2 is

$$c_1=c_2=0$$

Hence, the solution vectors X_1 and X_2 are linearly independent on $(-\infty,\infty)$.

Example 6

Again consider the same homogeneous system as considered in the previous example

$$X'=\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}X$$

We have already seen that the vectors $X_1,\ X_2$ i.e.

$$X_1=\begin{pmatrix} 3e^{t} \\ e^{t} \end{pmatrix},\quad X_2=\begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}$$

are solutions of the homogeneous system. We can verify that the following vector X_3

$$X^3=\begin{pmatrix} e^{t}+\cosh t \\ \cosh t \end{pmatrix}$$

is also a solution of the homogeneous system However, the set of solutions that consists of $X_1,\ X_2$ and X_3 is linearly dependent because X_3 is a linear combination of the other two vectors

$$X_3=\frac{1}{2}X_1+\frac{1}{2}X_2$$

© Copyright Virtual University of Pakistan

Exercise

Write the given system in matrix form.

1. $\frac{dx}{dt} = x - y + z + t - 1$

   $\frac{dy}{dt} = 2x + y - z - 3t^2$

   $\frac{dz}{dt} = x + y + z + t^2 - t + 2$

2. $\frac{dx}{dt} = -3x + 4y + e^{-t}\sin 2t$

   $\frac{dx}{dt} = 5x + 9y + 4e^{-t}\cos 2t$

3. $\frac{dx}{dt} = -3x + 4y - 9z$

   $\frac{dy}{dt} = 6x - y$

   $\frac{dz}{dt} = 10x + 4y + 3z$

4. $\frac{dx}{dt} = -3x + 4y + e^{-t}\sin 2t$

   $\frac{dy}{dt} = 5x + 9y + 4e^{-t}\cos 2t$

Write the given system without of use of matrices

5. $X' = \begin{pmatrix} 7 & 5 & -9 \\ 4 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix} X + \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} e^{5t} - \begin{pmatrix} 8 \\ 0 \\ 3 \end{pmatrix} e^{-2t}$

6. $\frac{d}{dt}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix}\sin t + \begin{pmatrix} t-4 \\ 2t+1 \end{pmatrix} e^{4t}$

7. $\frac{d}{dt}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -1 & 2 \\ 3 & -4 & 1 \\ -2 & 5 & 6 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} e^{-t} - \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} t$

Verify that the vector X is the solution of the given system

8. $\frac{dx}{dt} = -2x + 5y$

   $\frac{dx}{dt} = -2x + 4y, \qquad X = \begin{pmatrix} 5\cos t \\ 3\cos t - \sin t \end{pmatrix} e^t$

© Copyright Virtual University of Pakistan

9. $X' = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} X$, $X = \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{t} + \begin{pmatrix} 4 \\ -4 \end{pmatrix} te^{t}$

10. $\frac{dX}{dt} = \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix} X; \quad X = \begin{pmatrix} 1 \\ 6 \\ -13 \end{pmatrix}$

© Copyright Virtual University of Pakistan

Lecture 41
Matrices and Systems of Linear 1st-Order Equations (Continued)

Theorem: A necessary and sufficient condition that the set of solutions, on an interval I., consisting of the vectors

$$X_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}, X_2 = \begin{pmatrix} x_{12} \\ x_{21} \\ \vdots \\ x_{n2} \end{pmatrix}, \ldots, X_n = \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix}$$

of the homogenous system $X' = AX$ to be linearly independent is that the Wronskian of these solutions is non-zero for every $t \in I$. Thus

$$W(X_1, X_2, \ldots, X_n) = \begin{vmatrix} x_{11} & x_{12} & \ldots & x_{1n} \\ x_{21} & x_{22} & \ldots & x_{2n} \\ \vdots & \vdots & \ldots & \vdots \\ x_{n1} & x_{n2} & \ldots & x_{nn} \end{vmatrix} \neq 0, \quad \forall t \in I$$

Note that

- It can be shown that if $X_1, X_2, \ldots, X_n$ are solution vectors of the system, then either

  $$W(X_1, X_2, \ldots, X_n) \neq 0, \ \forall t \in I$$

  or

  $$W(X_1, X_2, \ldots, X_n) = 0, \ \forall t \in I$$

  Thus if we can show that $W \neq 0$ for some $t_0 \in I$, then $W \neq 0, \ \forall t \in I$ and hence the solutions are linearly independent on I

- Unlike our previous definition of the Wronskian, the determinant does not involve any differentiation.

Example 1

As verified earlier that the vectors

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}, \quad X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

are solutions of the following homogeneous system.

© Copyright Virtual University of Pakistan

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

Clearly, X_1 and X_2 are linearly independent on $(-\infty, \infty)$ as neither of the vectors is a constant multiple of the other. We now compute Wronskian of the solution vectors X_1 and X_2.

$$W(X_1, X_2) = \begin{vmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{vmatrix} = 8e^{4t} \neq 0, \quad \forall t \in (-\infty, \infty)$$

Fundamental set of solution

Suppose that $\{X_1, X_2, \ldots, X_n\}$ is a set of n solution vectors, on an interval I, of a homogenous system $X' = AX$. The set is said to be a fundamental set of solutions of the system on the interval I if the solution vectors $X_1, X_2, \ldots, X_n$ are linearly independent.

Theorem: *Existence of a Fundamental Set*

There exist a fundamental set of solution for the homogenous system $X' = AX$ on an interval I

General solution

Suppose that $X_1, X_2, \ldots, X_n$ is a fundamental set of solution of the homogenous system $X' = AX$ on an interval I. Then any linear combination of the solution vectors $X_1, X_2, \ldots, X_n$ of the form

$$X = c_1 X_1 + c_2 X_2 + \cdots + c_n X_n$$

$c_i; i = 1, 2, \ldots, n$ being arbitrary constants is said to be the general solution of the system on the interval I.

Note that

For appropriate choices of the arbitrary constants $c_1, c_2, \ldots, c_n$ any solution, on the interval I, of the homogeneous system $X' = AX$ can be obtained from the general solution.

Example 2

As discussed in the Example 1, the following vectors are linearly independent solutions

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}, \quad X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

of the following homogeneous system of differential equations on $(-\infty, \infty)$

© Copyright Virtual University of Pakistan

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

Hence X_1 and X_2 form a fundamental set of solution of the system on the interval $(-\infty, \infty)$. Hence, the general solution of the system on $(-\infty, \infty)$ is

$$X = c_1 X_1 + c_2 X_2 = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

Example 3

Consider the vectors X_1, X_2 and X_3 these vectors are given by

$$X_1 = \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^t, \quad X_3 = \begin{pmatrix} \sin t \\ -\frac{1}{2}\sin t + \frac{1}{2}\cos t \end{pmatrix}$$

It has been verified in the last lecture that the vectors X_1 and X_2 are solutions of the homogeneous system

$$X' = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} X$$

It can be easily verified that the vector X_3 is also a solution of the system. We now compute the Wronskian of the solution vectors X_1, X_2 and X_3

$$W(X_1, X_2, X_3) = \begin{vmatrix} \cos t & 0 & \sin t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t & e^t & \frac{1}{2}\sin t - \frac{1}{2}\cos t \\ -\cos t - \sin t & 0 & -\sin t + \cos t \end{vmatrix}$$

Expand from 2nd column

or $$W(X_1, X_2, X_3) = e^t \begin{vmatrix} \cos t & \sin t \\ -\cos t - \sin t & \sin t + \cos t \end{vmatrix}$$

or $$W(X_1, X_2, X_3) = e^t \neq 0, \quad \forall t \in R$$

Thus, we conclude that X_1, X_2 and X_3 form a fundamental set of solution on $(-\infty, \infty)$. Hence, the general solution of the system on $(-\infty, \infty)$ is

$$X = c_1 X_1 + c_2 X_2 + c_3 X_3$$

or

© Copyright Virtual University of Pakistan

$$X = c_1\begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \end{pmatrix} + c_2\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^{t} + c_3\begin{pmatrix} \sin t \\ -\frac{1}{2}\sin t - \frac{1}{2}\cos t \end{pmatrix}$$

Non-homogeneous Systems

As stated earlier in this lecture that a system of differential equations such as

$$\frac{dX}{dt} = A(t)X + F(t)$$

is non-homogeneous if $F(t) \neq 0,\ \forall t$. The general solution of such a system consists of a complementary function and a particular integral.

Particular Integral

A particular solution, on an interval I, of a non-homogeneous system is any vector X_p free of arbitrary parameters, whose entries are functions that satisfy each equation of the system.

Example 4

Show that the vector

$$X_p = \begin{pmatrix} 3t-4 \\ -5t+6 \end{pmatrix}$$

is a particular solution of the following non-homogeneous system on the interval $(-\infty, \infty)$

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

Solution:

Differentiating the given vector with respect to t, we obtain

$$X'_p = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

Further

$$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 3t-4 \\ -5t+6 \end{pmatrix} + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

or

$$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} (3t-4)+3(-5t+6) \\ 5(3t-4)+3(-5t+6) \end{pmatrix} + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

or

$$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} -12t+14 \\ -2 \end{pmatrix} + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

© Copyright Virtual University of Pakistan

or $$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ -5 \end{pmatrix} = X^P$$

Thus the given vector X_p satisfies the non-homogeneous system of differential equations. Hence, the given vector X_p is a particular solution of the non-homogeneous system.

Theorem

Let $X_1, X_2, \ldots, X_k$ be a set of solution vectors of the homogenous system $X' = AX$ on an interval I and let X_p be any solution vector of the non-homogenous system $X' = AX + F(t)$ on the same interval I. Then $\exists$ constants $c_1, c_2, \ldots, c_k$ such that

$$X_p = c_1X_1 + c_2X_2 + \ldots + c_kX_k + X_p$$

is also a solution of the non-homogenous system on the interval.

Complementary function

Let $X_1, X_2, \ldots, X_n$ be solution vectors of the homogenous system $X' = AX$ on an interval I, then the general solution

$$X = c_1X_1 + c_2X_2 + \ldots + c_nX_n$$

of the homogeneous system is called the complementary function of the non-homogeneous system $X' = AX + F(t)$ on the same interval I.

General solution-Non homogenous systems

Let X_p be a particular integral and X_c the complementary function, on an interval I, of the non-homogenous system

$$X' = A(t)X + F(t).$$

The general solution of the non-homogenous system on the interval I is defined to be

$$X = X_c + X_p$$

Example 5

In Example 4 it was verified that

$$X_p = \begin{pmatrix} 3t-4 \\ -5t+6 \end{pmatrix}$$

© Copyright Virtual University of Pakistan

is a particular solution, on $(-\infty, \infty)$, of the non-homogenous system

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X + \begin{pmatrix} 12t - 11 \\ -3 \end{pmatrix}$$

As we have seen earlier, the general solution of the associated homogeneous system i.e. the complementary function of the given non-homogeneous system is

$$X_c = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

Hence the general solution, on $(-\infty, \infty)$, of the non-homogeneous system is

$$X = X_c + X_p$$
$$X = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} + \begin{pmatrix} 3t - 4 \\ -5t + 6 \end{pmatrix}$$

Fundamental Matrix

Suppose that the a fundamental set of n solution vectors of a homogeneous system $X' = AX$, on an interval I, consists of the vectors

$$X_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}, X_2 = \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix}, \ldots, X_n = \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix}$$

Then a fundamental matrix of the system on the interval I is given by

$$\phi(t) = \begin{pmatrix} x_{11} & x_{12} & \ldots & x_{1n} \\ x_{21} & x_{22} & \ldots & x_{2n} \\ \vdots & \vdots & \ldots & \vdots \\ x_{n1} & x_{n2} & \ldots & x_{nn} \end{pmatrix}$$

Example 6

As verified earlier, the following vectors

© Copyright Virtual University of Pakistan

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$$

$$X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} = \begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix}$$

form a fundamental set of solutions of the system on $(-\infty, \infty)$

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

So that the general solution of the system is

$$X = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

Hence, a fundamental matrix of the system on the interval is

$$\phi(t) = \begin{pmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{pmatrix}$$

Note that

- The general solution of the system can be written as

$$X = \begin{pmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Or $X = \phi(t)C, \quad C = \begin{pmatrix} c_1 & c_2 \end{pmatrix}^{tr}$

- Since $X = \phi(t)C$ is a solution of the system $X' = A(t)X$. Therefore

$$\phi'(t)C = A(t)\phi(t)C$$

Or $$[\phi'(t) - A(t)\phi(t)]C = 0$$

Since the last equation is to hold for every t in the interval I for every possible column matrix of constants C, we must have

$$\phi'(t) - A(t)\phi(t) = 0$$

Or $$\phi'(t) = A(t)\phi(t)$$

Note that

- The fundamental matrix $\phi(t)$ of a homogenous system $X' = A(t)X$ is non-singular because the determinant $\det(\phi(t))$ coincides with the Wronskian of the solution vectors of the system and linear independence of the solution vectors guarantees that $\det(\phi(t)) \neq 0$.

© Copyright Virtual University of Pakistan

- Let $\phi(t)$ be a fundamental matrix of the homogenous system $X' = A(t)X$ on an interval I. Then, in view of the above mentioned observation, the inverse of the matrix $\phi^{-1}(t)$ exists for every value of t in the interval I.

© Copyright Virtual University of Pakistan

Exercise

The given vectors are the solutions of a system $X' = AX$. Determine whether the vectors form a fundamental set on $(-\infty, \infty)$.

1. $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t, X_2 = \begin{pmatrix} 2 \\ 6 \end{pmatrix} e^t + \begin{pmatrix} 8 \\ -8 \end{pmatrix} te^t$

2. $X_1 = \begin{pmatrix} 1 \\ 6 \\ -13 \end{pmatrix}, X_2 = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} e^{-4t}, \; X_3 = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} e^{3t}$

3. $X' = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} X - \begin{pmatrix} 1 \\ 7 \end{pmatrix} e^t; \quad X_p = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \begin{pmatrix} 1 \\ -1 \end{pmatrix} te^t$

Verify that vector X_p is a particular solution of the given systems

4. $\dfrac{dx}{dt} = x + 4y + 2t - 7, \quad \dfrac{dy}{dt} = 3x + 2y - 4t - 18$

$X_p = \begin{pmatrix} 2 \\ -1 \end{pmatrix} t + \begin{pmatrix} 5 \\ 1 \end{pmatrix}$

5. $X' = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} X + \begin{pmatrix} -5 \\ 2 \end{pmatrix}; \quad X_p = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

6. $X' = \begin{pmatrix} 1 & 2 & 3 \\ -4 & 2 & 0 \\ -6 & 1 & 0 \end{pmatrix} X + \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} \sin 3t; \quad X_p \begin{pmatrix} \sin 3t \\ 0 \\ \cos 3t \end{pmatrix}$

7. $X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t}, \quad X_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-6t}$

8. $X_1 = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 3 \\ -6 \\ 12 \end{pmatrix} + t \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}$

9. Prove that the general solution of the homogeneous system

$$X' = \begin{pmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} X$$

on the interval $(-\infty, \infty)$ is

$$X = c_1 \begin{pmatrix} 6 \\ -1 \\ -5 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} e^{-2t} + c_3 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} e^{3t}$$

© Copyright Virtual University of Pakistan

© Copyright Virtual University of Pakistan

Lecture 42
Homogeneous Linear Systems

Most of the theory developed for a single linear differential equation can be extended to a system of such differential equations. The extension is not entirely obvious. However, using the notation and some ideas of matrix algebra discussed in a previous lecture most effectively carry it out. Therefore, in the present and in the next lecture we will learn to solve the homogeneous linear systems of linear differential equations with real constant coefficients.

Example 1

Consider the homogeneous system of differential equations

$$\frac{dx}{dt} = x + 3y$$

$$\frac{dy}{dt} = 5x + 3y$$

In matrix form the system can be written as

$$\begin{pmatrix} dx/dt \\ dy/dt \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

If we suppose that

$$X = \begin{pmatrix} x \\ y \end{pmatrix}$$

Then the system can again be re-written as

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

Now suppose that X_1 and X_2 denote the vectors

$$X_1 = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}, \quad X_2 = \begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix}$$

Then

$$X_1' = \begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix}, \quad X_2' = \begin{pmatrix} 18\ e^{6t} \\ 30\ e^{6t} \end{pmatrix}$$

Now

$$AX_1 = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix} = \begin{pmatrix} e^{-2t} - 3e^{-2t} \\ 5e^{-2t} - 3e^{-2t} \end{pmatrix}$$

or

$$AX_1 = \begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix} = X_1'$$

© Copyright Virtual University of Pakistan

Similarly

$$AX_2 = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix} = \begin{pmatrix} 3e^{6t} + 15e^{6t} \\ 15e^{6t} + 15e^{6t} \end{pmatrix}$$

or

$$AX_2 = \begin{pmatrix} 18\ e^{6t} \\ 30\ e_{6t} \end{pmatrix} = X_2'$$

Hence, X_1 and X_2 are solutions of the homogeneous system of differential equations $X' = AX$. Further

$$W(X_1, X_2) = \begin{vmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{vmatrix} = 8e^{4t} \neq o, \quad \forall t \in R$$

Thus, the solutions vectors X_1 and X_2 are linearly independent. Hence, these vectors form a fundamental set of solutions on $(-\infty, \infty)$. Therefore, the general solution of the system on $(-\infty, \infty)$ is

$$X = c_1X_1 + c_2X_2$$

$$X = c_1\begin{pmatrix} 1 \\ -1 \end{pmatrix}e^{-2t} + c_2\begin{pmatrix} 3 \\ 5 \end{pmatrix}e^{6t}$$

Note that

- Each of the solution vectors X_1 and X_2 are of the form

$$X = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}e^{\lambda t}$$

  Where k_1 and k_2 are constants.
- The question arises whether we can always find a solution of the homogeneous system $X' = AX$, A is $n \times n$ matrix of constants, of the form

$$X = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix}e^{\lambda t} = Ke^{\lambda t}$$

  for the homogenous linear 1st order system.

© Copyright Virtual University of Pakistan

Eigenvalues and Eigenvectors

Suppose that

$$X = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix} e^{\lambda t} = Ke^{\lambda t}$$

is a solution of the system

$$\frac{dX}{dt} = AX$$

where A is an $n \times n$ matrix of constants then

$$\frac{dX}{dt} = K \ \lambda \ e^{\lambda t}$$

Substituting this last equation in the homogeneous system $X' = AX$, we have

$$K\lambda e^{\lambda t} = AKe^{\lambda t} \Rightarrow AK = \lambda K$$

or $$(A - \lambda \ I \ ) \ K = 0$$

This represents a system of linear algebraic equations. The linear 1st order homogenous system of differential equations

$$\frac{dX}{dt} = AX$$

has a non-trivial solution X if there exist a non-trivial solution K of the system of algebraic equations

$$\det(A - \lambda I) = 0$$

This equation is called characteristic equation of the matrix A and represents an nth degree polynomial in λ.

Case 1 *Distinct real eigenvalues*

Suppose that the coefficient matrix A in the homogeneous system of differential equations

$$\frac{dX}{dt} = AX$$

has n distinct eigenvalues $\lambda_1, \lambda_2, \lambda_3, ..., \lambda_n$ and $K_1, K_2, ..., K_n$ be the corresponding eigenvectors. Then the general solution of the system on $(-\infty, \infty)$ is given by

$$X = c_1 k_1 e^{\lambda_1 t} + c_2 k_2 e^{\lambda_2 t} + c_3 k_3 e^{\lambda_3 t} + ... + c_n k_n e^{\lambda_n t}$$

© Copyright Virtual University of Pakistan

Example 2

Solve the following homogeneous system of differential equations

$$\frac{dx}{dt} = 2x + 3y$$
$$\frac{dy}{dt} = 2x + y$$

Solution

The given system can be written in the matrix form as

$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Therefore, the coefficient matrix

$$A = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$$

Now we find the eigenvalues and eigenvectors of the coefficient A. The characteristics equation is

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{vmatrix}$$

$$\det(A - \lambda I) = \lambda^2 - 3\lambda - 4$$

Therefore, the characteristic equation is

$$\det(A - \lambda I) = 0 = \lambda^2 - 3\lambda - 4$$

or $$(\lambda + 1)(\lambda - 4) = 0 \Rightarrow \lambda = -1, 4$$

Therefore, roots of the characteristic equation are real and distinct and so are the eigenvalues.

For $\lambda = -1$**,** we have

$$(A - \lambda I)K = \begin{pmatrix} 2+1 & 3 \\ 2 & 1+1 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

or $$(A - \lambda I)K = \begin{pmatrix} 3k_1 + 3k_2 \\ 2k_1 + 2k_2 \end{pmatrix}$$

Hence $$(A - \lambda I)K = 0 \Rightarrow \begin{cases} 3k_1 + 3k_2 = 0 \\ 2k_1 + 2k_2 = 0 \end{cases}$$

These two equations are no different and represent the equation

$$k_1 + k_2 = 0 \Rightarrow k_1 = -k_2$$

© Copyright Virtual University of Pakistan

Thus we can choose value of the constant k_2 arbitrarily. If we choose $k_2 = -1$ then $k_1 = 1$. Hence the corresponding eigenvector is

$$K_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

For $\lambda = 4$ we have

$$(A - \lambda I)K = \begin{pmatrix} 2-4 & 3 \\ 2 & 1-4 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

or

$$(A - \lambda I)K = \begin{pmatrix} -2k_1 + 3k_2 \\ 2k_1 - 3k_2 \end{pmatrix}$$

Hence

$$(A - \lambda I)K = 0 \Rightarrow \begin{cases} -2k_1 + 3k_2 = 0 \\ 2k_1 - 3k_2 = 0 \end{cases}$$

Again the above two equations are not different and represent the equation

$$2k_1 - 3k_2 = 0 \Rightarrow k_1 = \frac{3k_2}{2}$$

Again, the constant k_2 can be chosen arbitrarily. Let us choose $k_2 = 2$ then $k_1 = 3$. Thus the corresponding eigenvector is

$$K_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Therefore, we obtain two linearly independent solution vectors of the given homogeneous system.

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}, \quad X_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{4t}$$

Hence the general solution of the system is the following

$$X = c_1 X_1 + c_2 X_2$$

or

$$X = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{4t}$$

or

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} c_1 e^{-t} + 3c_2 e^{4t} \\ -c_1 e^{-t} + 2c_2 e^{4t} \end{pmatrix}$$

This means that the solution of the system is

$$x(t) = c_1 e^{-t} + 3c_2 e^{4t}$$

$$y(t) = -c_1 e^{-t} + 2c_2 e^{4t}$$

© Copyright Virtual University of Pakistan

Example 3

Solve the homogeneous system

$$\frac{dx}{dt} = -4x + y + z$$
$$\frac{dy}{dt} = x + 5y - z$$
$$\frac{dz}{dt} = y - 3z$$

Solution:

The given system can be written as

$$\begin{pmatrix} dx/dt \\ dy/dt \\ dz/dt \end{pmatrix} = \begin{pmatrix} -4 & 1 & 1 \\ 1 & 5 & -1 \\ 0 & 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Therefore the coefficient matrix of the system of differential equations is

$$A = \begin{pmatrix} -4 & 1 & 1 \\ 1 & 5 & -1 \\ 0 & 1 & -3 \end{pmatrix}$$

Therefore $$A - \lambda I = \begin{pmatrix} -4-\lambda & 1 & 1 \\ 1 & 5-\lambda & -1 \\ 0 & 1 & -3-\lambda \end{pmatrix}$$

Thus the characteristic equation is

$$\det(A - \lambda I) = \begin{vmatrix} -4-\lambda & 1 & 1 \\ 1 & 5-\lambda & -1 \\ 0 & 1 & -3-\lambda \end{vmatrix} = 0$$

Expanding the determinant using cofactors of third row, we obtain

$$-(\lambda + 3)(\lambda + 4)(\lambda - 5) = 0$$
$$\lambda = -3,\ -4,\ 5$$

Thus the characteristic equation has real and distinct roots and so are the eigenvalues of the coefficient matrix A. To find the eigenvectors corresponding to these computed eigenvalues, we need to solve the following system of linear algebraic equations for k_1, k_2 and k_3 when $\lambda = -3,\ -4,\ 5$, successively.

$$\det(A - \lambda I)K = 0 \Rightarrow \begin{pmatrix} -4-\lambda & 1 & 1 \\ 1 & 5-\lambda & -1 \\ 0 & 1 & -3-\lambda \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

© Copyright Virtual University of Pakistan

For solving this system we use Gauss-Jordon elimination technique, which consists of reducing the augmented matrix to the reduced echelon form by applying the elementary row operations. The augmented matrix of the system of linear algebraic equations is

$$\left(\begin{array}{cccc} -4-\lambda & 1 & 1 & 0 \\ 1 & 5-\lambda & -1 & 0 \\ 0 & 1 & -3-\lambda & 0 \end{array}\right)$$

For $\lambda = -3$**,** the augmented matrix becomes:

$$\left(\begin{array}{cccc} -1 & 1 & 1 & 0 \\ 1 & 8 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right)$$

Appling the row operation R_{12} , $R_2 + R_1$, R_{23} , $R_3 - 9R_2$, $R_1 - 8R_2$ in succession reduces the augmented matrix in the reduced echelon form.

$$\left(\begin{array}{cccc} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

So that we have the following equivalent system

$$\left(\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} k_1 \\ k_2 \\ k_3 \end{array}\right) = \left(\begin{array}{c} 0 \\ 0 \\ 0 \end{array}\right)$$

or $$k_1 = k_3,\ k_2 = 0$$

Therefore, the constant k_3 can be chosen arbitrarily. If we choose $k_3 = 1$, then $k_1 = 1$, So that the corresponding eigenvector is

$$K_1 = \left(\begin{array}{c} 1 \\ 0 \\ 1 \end{array}\right)$$

For $\lambda_2 = -4$, the augmented matrix becomes

$$((A + 4I) \mid 0) = \left(\begin{array}{cccc} 0 & 1 & 1 & 0 \\ 1 & 9 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right)$$

We apply elementary row operations to transform the matrix to the following reduced echelon form:

$$\left(\begin{array}{cccc} 1 & 0 & -10 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $$k_1 = 10k_3,\ \ k_2 = -k_3$$

© Copyright Virtual University of Pakistan

Again k_3 can be chosen arbitrarily, therefore choosing $k_3 = 1$ we get $k_1 = 10$, $k_2 = -1$
Hence, the second eigenvector is

$$K_2 = \begin{pmatrix} 10 \\ -1 \\ 1 \end{pmatrix}$$

Finally, when $\lambda_3 = 5$ the augmented matrix becomes

$$((A - 5\ I) \mid 0) = \left(\begin{array}{cccc} -9 & 1 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -8 & 0 \end{array}\right)$$

The application of the elementary row operation transforms the augmented matrix to the reduced echelon form

$$\left(\begin{array}{cccc} 1 & 0 & -1 & 0 \\ 0 & 1 & -8 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $k_1 = k_3, \quad k_2 = 8k_3$

If we choose $k_3 = 1$, then $k_1 = 1$ and $k_2 = 8$. Thus the eigenvector corresponding to $\lambda_3 = 5$ is

$$K_3 = \begin{pmatrix} 1 \\ 8 \\ 1 \end{pmatrix}$$

Thus we obtain three linearly independent solution vectors

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{-3t}, \quad X_2 = \begin{pmatrix} 10 \\ -1 \\ 1 \end{pmatrix} e^{-4t}, \quad X_3 = \begin{pmatrix} 1 \\ 8 \\ 1 \end{pmatrix} e^{5t}$$

Hence, the general solution of the given homogeneous system is

$$X = c_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} 10 \\ -1 \\ 1 \end{pmatrix} e^{-4t} + c_3 \begin{pmatrix} 1 \\ 8 \\ 1 \end{pmatrix} e^{5t}$$

© Copyright Virtual University of Pakistan

Case 2 *Complex eigenvalues*

Suppose that the coefficient matrix A in the homogeneous system of differential equations

$$\frac{dX}{dt} = AX$$

has complex eigenvalues. This means that roots of the characteristic equation

$$\det(A - \lambda I) = 0$$

are imaginary.

Theorem: *Solutions corresponding to complex eigenvalues*

Suppose that K is an eigenvector corresponding to the complex eigenvalue

$$\lambda_1 = \alpha + i\beta; \quad \alpha, \beta \in R$$

of the coefficient matrix A with real entries, then the vectors X_1 and X_2 given by

$$X_1 = K_1 e^{\lambda_1 t}, \quad X_2 = \overline{K_1} e^{\overline{\lambda_1} t}$$

are solution of the homogeneous system.

$$\frac{dX}{dt} = AX$$

Example 4

Consider the following homogeneous system of differential equations

$$\frac{dx}{dt} = 6x - y$$

$$\frac{dy}{dt} = 5x + 4y$$

The system can be written as

or
$$\begin{pmatrix} dx/dt \\ dy/dt \end{pmatrix} = \begin{pmatrix} 6 & -1 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Therefore the coefficient matrix of the system is

$$A = \begin{pmatrix} 6 & -1 \\ 5 & 4 \end{pmatrix}$$

So that the characteristic equation is

$$\det(A - \lambda I) = \begin{vmatrix} 6-\lambda & -1 \\ 5 & 4-\lambda \end{vmatrix} = 0$$

or
$$(6-\lambda)(4-\lambda) + 5 = 0 = \lambda^2 - 10\lambda + 29$$

Now using the quadratic formula we have

$$\lambda_1 = 5 + 2i, \quad \lambda_2 = 5 - 2i$$

© Copyright Virtual University of Pakistan

For, $\lambda_1 = 5+2i$, we must solve the system of linear algebraic equations

$$\left.\begin{aligned}(1-2i)k_1 - k_2 &= 0\\ 5k_1 - (1+2i)k_2 &= 0\end{aligned}\right\} \Rightarrow (1-2i)k_1 - k_2 = 0$$

or $k_2 = (1-2i)k_1$

Therefore, it follows that after we choose $k_1 = 1$ then $k_2 = 1-2i.$ So that one eigenvector is given by

$$K_1 = \begin{pmatrix} 1 \\ 1-2i \end{pmatrix}$$

Similarly for $\lambda_2 = 5-2i$ we must solve the system of linear algebraic equations

$$\left.\begin{aligned}(1+2i)k_1 - k_2 &= 0\\ 5k_1 - (1-2i)k_2 &= 0\end{aligned}\right\} \Rightarrow (1+2i)k_1 - k_2 = 0$$

or $k_2 = (1+2i)k_1$

Therefore, it follows that after we choose $k_1 = 1$ then $k_2 = 1+2i.$ So that second eigenvector is given by

$$K_2 = \begin{pmatrix} 1 \\ 1+2i \end{pmatrix}$$

Consequently, two solution of the homogeneous system are

$$X_1 = \begin{pmatrix} 1 \\ 1-2i \end{pmatrix} e^{(5+2i)t}, \quad X_2 = \begin{pmatrix} 1 \\ 1+2i \end{pmatrix} e^{(5-2i)t}$$

By the superposition principle another solution of the system is

$$X = c_1 \begin{pmatrix} 1 \\ 1-2i \end{pmatrix} e^{(5+2i)t} + c_2 \begin{pmatrix} 1 \\ 1+2i \end{pmatrix} e^{(5-2i)t}$$

Note that

The entries in K_2 corresponding to λ_2 are the conjugates of the entries in K_1 corresponding to λ_1. Further, λ_2 is conjugate of λ_1. Therefore, we can write this as

$$\lambda_2 = \overline{\lambda}_1, \quad K_2 = \overline{K}_1$$

Theorem *Real solutions corresponding to a complex eigenvalue*

© Copyright Virtual University of Pakistan

Suppose that

- $\lambda_1 = \alpha + i\beta$ is a complex eigenvalue of the matrix A in the system $\frac{dX}{dt} = AX$
- K_1 is an eigenvector corresponding to the eigen value λ_1
- $B_1 = \frac{1}{2}(K_1 + \overline{K_1}) = \text{Re}(K_1),\ B_2 = \frac{i}{2}(-K_1 + \overline{K_1}) = \text{Im}(K_1)$

Then two linearly independent solutions of the system on $(-\infty, \infty)$ are given by

$$X_1 = (B_1 \cos \beta t - B_2 \sin \beta t)e^{\alpha t}$$

$$X_2 = (B_2 \cos \beta t + B_1 \sin \beta t)e^{\alpha t}$$

Example 5

Solve the system

$$X' = \begin{pmatrix} 2 & 8 \\ -1 & -2 \end{pmatrix} X$$

The coefficient matrix of the system is

$$A = \begin{pmatrix} 2 & 8 \\ -1 & -2 \end{pmatrix}$$

Therefore

$$A - \lambda I = \begin{pmatrix} 2-\lambda & 8 \\ -1 & -2-\lambda \end{pmatrix}$$

Thus, the characteristic equation is

$$\det(A - \lambda I) = 0 = \begin{vmatrix} 2-\lambda & 8 \\ -1 & -2-\lambda \end{vmatrix}$$

$$-(2-\lambda)(2+\lambda) + 8 = 0 = \lambda^2 + 4$$

Thus the Eigenvalues are of the coefficient matrix are $\lambda_1 = 2i$ and $\lambda_2 = \overline{\lambda_1} = -2i$.

For λ_1 we see that the system of linear algebraic equations $(A - \lambda I)K = 0$

$$(2-2i)k_1 + 8k_2 = 0$$

$$-k_1 - (2+2i)k_2 = 0$$

Solving these equations, we obtain

$$k_1 = -(2+2i)k_2$$

© Copyright Virtual University of Pakistan

Choosing $k_2 = -1$ gives $k_1 = (2+2i)k_2$. Thus the corresponding eigenvector is

$$K_1 = \begin{pmatrix} 2+2i \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} + i\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

So that $$B_1 = \mathrm{Re}(K_1) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \; B_2 = \mathrm{Im}(K_1) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

Since $\alpha = 0$, the general solution of the given system of differential equations is

$$X = c_1\left[\begin{pmatrix} 2 \\ -1 \end{pmatrix}\cos 2t - \begin{pmatrix} 2 \\ 0 \end{pmatrix}\sin 2t\right] + c_2\left[\begin{pmatrix} 2 \\ 0 \end{pmatrix}\cos 2t + \begin{pmatrix} 2 \\ -1 \end{pmatrix}\sin 2t\right]$$

$$X = c_1\begin{pmatrix} 2\cos 2t - 2\sin 2t \\ -\cos 2t \end{pmatrix} + c_2\begin{pmatrix} 2\cos 2t + 2\sin 2t \\ -\sin 2t \end{pmatrix}$$

Example 6

Solve the following system of differential equations

$$X' = \begin{pmatrix} 1 & 2 \\ -1/2 & 1 \end{pmatrix} X$$

Solution:

The coefficient matrix of the given system is

$$A = \begin{pmatrix} 1 & 2 \\ -1/2 & 1 \end{pmatrix}$$

Thus $$A - \lambda I = \begin{pmatrix} 1-\lambda & 2 \\ -1/2 & 1-\lambda \end{pmatrix}$$

So that the characteristic equation is

$$\det(A - \lambda I) = 0 = \begin{vmatrix} 1-\lambda & 2 \\ -1/2 & 1-\lambda \end{vmatrix}$$

or $$\lambda^2 - 2\lambda + 2 = 0$$

Therefore, by the quadratic formula we obtain

$$\lambda = \left(2 \pm \sqrt{4-8}\right)/2$$

Thus the eigenvalues of the coefficient matrix are

$$\lambda_1 = 1+i, \;\; \lambda_2 = \bar{\lambda}_1 = 1-i$$

Now an eigenvector associated with the eigenvalue λ_1 is

$$K_1 = \begin{pmatrix} 2 \\ i \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + i\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

From $$B_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

So that we have the following two linearly independent solutions of the system

© Copyright Virtual University of Pakistan

$$X_1 = \left[\begin{pmatrix} 2 \\ 0 \end{pmatrix}\cos t - \begin{pmatrix} 0 \\ 1 \end{pmatrix}\sin t\right]e^t, \quad X_2 = \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\cos t + \begin{pmatrix} 2 \\ 0 \end{pmatrix}\sin t\right]e^t$$

Hence, the general solution of the system is

$$X = c_1\left[\begin{pmatrix} 2 \\ 0 \end{pmatrix}\cos t - \begin{pmatrix} 0 \\ 1 \end{pmatrix}\sin t\right]e^t + c_2\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\cos t + \begin{pmatrix} 2 \\ 0 \end{pmatrix}\sin t\right]e^t$$

or

$$X = c_1\begin{pmatrix} 2\cos t \\ -\sin t \end{pmatrix}e^t + c_2\begin{pmatrix} 2\sin t \\ \cos t \end{pmatrix}e^t$$

Exercise

Find the general solution of the given system

1. $\dfrac{dx}{dt} = x + 2y$

   $\dfrac{dy}{dt} = 4x + 3y$

2. $\dfrac{dx}{dt} = \dfrac{1}{2}x + 9y$

   $\dfrac{dy}{dt} = \dfrac{1}{2}x + 2y$

3. $X' = \begin{pmatrix} -6 & 2 \\ -3 & 1 \end{pmatrix}X$

4. $\dfrac{dx}{dt} = 2y$

   $\dfrac{dy}{dt} = 8x$

5. $X' = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}X$

6. $\dfrac{dx}{dt} = 6x - 9y$

   $\dfrac{dy}{dt} = 5x + 2y$

7. $\dfrac{dx}{dt} = x + y$

   $\dfrac{dy}{dt} = -2x - y$

8. $\dfrac{dx}{dt} = 4x + 5y$

   $\dfrac{dy}{dt} = -2x + 6y$

© Copyright Virtual University of Pakistan

9. $X' = \begin{pmatrix} 4 & -5 \\ 5 & -4 \end{pmatrix} X$

10. $X' = \begin{pmatrix} 1 & -8 \\ 1 & -3 \end{pmatrix} X$

© Copyright Virtual University of Pakistan

Lecture 43

Real and Repeated Eigenvalues

In the previous lecture we tried to learn how to solve a system of linear differential equations having a coefficient matrix that has real distinct and complex eigenvalues. In this lecture, we consider the systems

$$X' = AX$$

in which some of the n eigenvalue $\lambda_1, \lambda_2, \lambda_3, ..., \lambda_n$ of the $n \times n$ coefficient matrix A are repeated.

Eigenvalue of multiplicity m

Suppose that m is a positive integer and $(\lambda - \lambda_1)^m$ is a factor of the characteristic equation

$$\det(A - \lambda I) = 0$$

Further, suppose that $(\lambda - \lambda_1)^{m+1}$ is not a factor of the characteristic equation. Then the number λ_1 is said to be an eigenvalue of the coefficient matrix of multiplicity m.

Method of solution:

Consider the following system of n linear differential equations in n unknowns

$$X' = AX$$

Suppose that the coefficient matrix has an eigenvalue of multiplicity of m. There are two possibilities of the existence of the eigenvectors corresponding to this repeated eigenvalue:

- For the $n \times n$ coefficient matrix A, it may be possible to find m linearly independent eigenvectors $K_1, K_2, ..., K_m$ corresponding to the eigenvalue λ_1 of multiplicity $m \leq n$. In this case the general solution of the system contains the linear combination

$$c_1 K_1 e^{\lambda_1 t} + c_2 K_2 e^{\lambda_1 t} + \cdots + c_n K_n e^{\lambda_1 t}$$

- If there is only one eigenvector corresponding to the eigenvalue λ_1 of multiplicity m, then m linearly independent solutions of the form

$$X_1 = K_{11} e^{\lambda_1 t}$$

$$X_2 = K_{21} e^{\lambda_1 t} + K_{22} e^{\lambda_1 t}$$

$$\vdots$$

$$X_m = K_{m1} \frac{t^{m-1}}{(m-1)!} e^{\lambda_1 t} + K_{m2} \frac{t^{m-2}}{(m-2)!} e^{\lambda_1 t} + \cdots + K_{mm} e^{\lambda_1 t}$$

where the column vectors K_{ij} can always be found.

© Copyright Virtual University of Pakistan

Eigenvalue of Multiplicity Two

We begin by considering the systems of differential equations $X' = AX$ in which the coefficient matrix A has an eigenvalue λ_1 of multiplicity two. Then there are two possibilities;

- Whether we can find two linearly independent eigenvectors corresponding to eigenvalue λ_1 or
- We cannot find two linearly independent eigenvectors corresponding to eigenvalue λ_1 .

The case of the possibility of us being able to find two linearly independent eigenvectors K_1, K_2 corresponding to the eigenvalue λ_1 is clear. In this case the general solution of the system contains the linear combination

$$c_1 K_1 t e^{\lambda_1 t} + c_2 K_2 e^{\lambda_1 t}$$

Therefore, we suppose that there is only one eigenvector K_1 associated with this eigenvalue and hence only one solution vector X_1. Then, a second solution can be found of the following form:

$$X_2 = Kte^{\lambda_1 t} + Pe^{\lambda_1 t}$$

In this expression for a second solution, K and P are column vectors

$$K = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix}, \quad P = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}$$

We substitute the expression for X_2 into the system $X' = AX$ and simplify to obtain

$$(AK - \lambda_1 K)\ t\ e^{\lambda_1 t} + (AP - \lambda_1 P - K)\ e^{\lambda_1 t} = 0$$

Since this last equation is to hold for all values of t, we must have:

$$(A - \lambda_1 I)K = 0, \quad (A - \lambda_1 I)P = K$$

First equation does not tell anything new and simply states that K must be an eigenvector of the coefficient matrix A associated with the eigenvalue λ_1. Therefore, by solving this equation we find one solution

$$X_1 = Ke^{\lambda_1 t}$$

To find the second solution X_2, we only need to solve, for the vector P, the additional system

© Copyright Virtual University of Pakistan

$$(A - \lambda_1 I)P = K$$

First we solve a homogeneous system of differential equations having coefficient matrix for which we can find two distinct eigenvectors corresponding to a double eigenvalue and then in the second example we consider the case when cannot find two eigenvectors.

Example 1

Find general solution of the following system of linear differential equations

$$X' = \begin{pmatrix} 3 & -18 \\ 2 & -9 \end{pmatrix} X$$

Solution:
The coefficient matrix of the system is

$$A = \begin{pmatrix} 3 & -18 \\ 2 & -9 \end{pmatrix}$$

Thus $$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & -18 \\ 2 & -9-\lambda \end{vmatrix}$$

Therefore, the characteristic equation of the coefficient matrix A is

$$\det(A - \lambda I) = 0 = \begin{vmatrix} 3-\lambda & -18 \\ 2 & -9-\lambda \end{vmatrix}$$

or $$-(3-\lambda)(9+\lambda) + 36 = 0$$

or $$(\lambda + 3)^2 = 0 \Rightarrow \lambda = -3, -3$$

Therefore, the coefficient matrix A of the given system has an eigenvalue of multiplicity two. This means that

$$\lambda_1 = \lambda_2 = -3$$

Now $$(A - \lambda I)K = 0 \Rightarrow \begin{pmatrix} 3-\lambda & -18 \\ 2 & -9-\lambda \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

For $\lambda = -3$**,** this system of linear algebraic equations becomes

$$\begin{pmatrix} 6 & -18 \\ 2 & -6 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 6k_1 - 18k_2 = 0 \\ 2k_1 - 6k_2 = 0 \end{cases}$$

However

© Copyright Virtual University of Pakistan

$$\left.\begin{aligned} 6k_1 - 18k_2 &= 0 \\ 2k_1 - 6k_2 &= 0 \end{aligned}\right\} \Rightarrow k_1 - 3k_2 = 0$$

Thus $k_1 = 3k_2$

This means that the value of the constant k_2 can be chosen arbitrarily. If we choose $k_2 = 1$, we find the following single eigenvector for the eigenvalue $\lambda = -3$.

$$K = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

The corresponding one solution of the system of differential equations is given by

$$X_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{-3t}$$

But since we are interested in forming the general solution of the system, we need to pursue the question of finding a second solution. We identify the column vectors K and P as:

$$K = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad P = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$$

Then $$(A + 3I)P = K \Rightarrow \begin{pmatrix} 6 & -18 \\ 2 & -6 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Therefore, we need to solve the following system of linear algebraic equations to find P

$$\left.\begin{aligned} 6p_1 - 18p_2 &= 3 \\ 2p_1 - 6p_2 &= 1 \end{aligned}\right\} \Rightarrow 2p_1 - 6p_2 = 1$$

or $$p_2 = -(1 - 2p_1)/6$$

Therefore, the number p_1 can be chosen arbitrarily. So we have an infinite number of choices for p_1 and p_2. However, if we choose $p_1 = 1$, we find $p_2 = 1/6$. Similarly, if we choose the value of $p_1 = 1/2$ then $p_2 = 0$. Hence the column vector P is given by

$$P = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$$

Consequently, the second solution is given by

$$X_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} t e^{-3t} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} e^{-3t}$$

Hence the general solution of the given system of linear differential equations is then

© Copyright Virtual University of Pakistan

$$X = c_1 X_1 + c_2 X_2$$

$$X = c_1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{-3t} + c_2 \left[\begin{pmatrix} 3 \\ 1 \end{pmatrix} te^{-3t} + \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} e^{-3t} \right]$$

Example 2

Solve the homogeneous system

$$X' = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix} X$$

Solution:

The coefficient matrix of the system is:

$$A = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$$

To write the characteristic we find the expansion of the determinant:

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -2 & 2 \\ -2 & 1-\lambda & -2 \\ 2 & -2 & 1-\lambda \end{vmatrix}$$

The value of the determinant is

$$\det(A - \lambda I) = 5 + 9\lambda + 3\lambda^2 - \lambda^3$$

Therefore, the characteristic equation is

$$5 + 9\lambda + 3\lambda^2 - \lambda^3 = 0$$

or $$-(\lambda + 1)^2 (\lambda - 5) = 0$$

or $$\lambda = -1,\ -1,\ 5$$

Therefore, the eigenvalues of the coefficient matrix A are

$$\lambda_1 = \lambda_2 = -1,\ \lambda_3 = 5$$

Clearly -1 is a double root of the coefficient matrix A.

© Copyright Virtual University of Pakistan

Now $$(A-\lambda I)K=0 \Rightarrow \begin{pmatrix} 1-\lambda & -2 & 2 \\ -2 & 1-\lambda & -2 \\ 2 & -2 & 1-\lambda \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

For $\lambda_1=-1$, this system of the algebraic equations become

$$\begin{pmatrix} 2 & -2 & 2 \\ -2 & 2 & -2 \\ 2 & -2 & 2 \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The augmented matrix of the system is

$$(A+\mathrm{I}|0)=\left(\begin{array}{ccc|c} 2 & -2 & 2 & 0 \\ -2 & 2 & -2 & 0 \\ 2 & -2 & 2 & 0 \end{array}\right)$$

By applying the Gauss-Jordon method, the augmented matrix reduces to the reduced echelon form

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $$k_1-k_2+k_3=0 \Rightarrow k_1=k_2-k_3$$

By choosing $k_2=1$ and $k_3=0$ in $k_1=k_2-k_3$, we obtain $k_1=1$ and so one eigenvector is

$$K_1=\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

But the choice $k_2=1, k_3=1$ implies $k_1=0$. Hence, a second eigenvector is given by

$$K_2=\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Since neither eigenvector is a constant multiple of the other, we have found, corresponding to the same eigenvalue, two linearly independent solutions

$$X_1=\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}e^{-t}, \quad X_2=\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}e^{-t}$$

© Copyright Virtual University of Pakistan

Last for $\lambda_3 = 5$ we obtain the system of algebraic equations

$$\begin{pmatrix} -4 & -2 & 2 \\ -2 & -4 & -2 \\ 2 & -2 & -4 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The augmented matrix of the algebraic system is

$$\left(A - 5\mathrm{I} \middle| 0 \right) = \left(\begin{array}{ccc|c} -4 & -2 & 2 & 0 \\ -2 & -4 & -2 & 0 \\ 2 & -2 & -4 & 0 \end{array} \right)$$

By the elementary row operation we can transform the augmented matrix to the reduced echelon form

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

or $$k_1 = k_3,\ k_2 = -k_3$$

Picking $k_3 = 1$, we obtain $k_1 = 1,\ k_2 = -1$. Thus a third eigenvector is the following

$$K_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Hence, we conclude that the general solution of the system is

$$X = c_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} e^{5t}$$

Eigenvalues of Multiplicity Three

When a matrix A has only one eigenvector associated with an eigenvalue λ_1 of multiplicity three of the coefficient matrix A, we can find a second solution X_2 and a third solution X_3 of the following forms

$$X_2 = Kte^{\lambda_1 t} + Pe^{\lambda_1 t}$$

$$X_3 = K\frac{t^2}{2}e^{\lambda_1 t} + Pte^{\lambda_1 t} + Qe^{\lambda_1 t}$$

© Copyright Virtual University of Pakistan

The K, P and Q are vectors given by

$$K = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix}, \quad P = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix} \text{ and } \quad Q = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{pmatrix}$$

By substituting X_3 into the system $X' = AX$, we find the column vectors K, P and Q must satisfy the equations

$$(A - \lambda_1 I)K = 0$$
$$(A - \lambda_1 I)P = K$$
$$(A - \lambda_1 I)Q = P$$

The solutions of first and second equations can be utilized in the formulation of the solution X_1 and X_2.

Example

Find the general solution of the following homogeneous system

$$X' = \begin{pmatrix} 4 & 1 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{pmatrix} X$$

Solution

The coefficient matrix of the system is

$$A = \begin{pmatrix} 4 & 1 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{pmatrix}$$

Then

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 1 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{vmatrix}$$

Therefore, the characteristic equation is

$$\det(A - \lambda I) = 0 = \begin{vmatrix} 4-\lambda & 1 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{vmatrix}$$

Expanding the determinant in the last equation *w.r.to* the 3rd row to obtain

$$(-1)^{3+3}(4-\lambda)\begin{vmatrix} 4-\lambda & 1 \\ 0 & 4-\lambda \end{vmatrix} = 0$$

© Copyright Virtual University of Pakistan

or $$(4-\lambda)\ \left[(4-\lambda)(4-\lambda)-0\right]=0$$

or $$(4-\lambda)^3=0\Rightarrow\lambda=4,\ 4,\ 4$$

Thus, $\lambda=4$ is an eigenvalue of the coefficient matrix A of multiplicity three. For $\lambda=4$, we solve the following system of algebraic equations

$$(A-\lambda\mathrm{I})K=0$$

or $$\begin{pmatrix}4-\lambda & 1 & 0\\ 0 & 4-\lambda & 1\\ 0 & 0 & 4-\lambda\end{pmatrix}\begin{pmatrix}k_1\\ k_2\\ k_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

or $$\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}k_1\\ k_2\\ k_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

or $$\left.\begin{aligned}0k_1+1k_2+0k_3&=0\\ 0k_1+0k_2+1k_3&=0\\ 0k_1+0k_2+0k_3&=0\end{aligned}\right\}\Rightarrow\begin{aligned}k_2&=0\\ k_3&=0\end{aligned}$$

Therefore, the value of k_1 is arbitrary. If we choose $k_1=1$, then the eigen vector K is

$$K=\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}$$

Hence the first solution vector

$$X_1=Ke^{\lambda t}=\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}e^{4t}$$

Now for the second solution we solve the system

$$(A-\lambda I)P=K$$

or $$\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}p_1\\ p_2\\ p_3\end{pmatrix}=\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}$$

© Copyright Virtual University of Pakistan

$$\left.\begin{array}{l} 0p_1 + 1p_2 + 0p_3 = 1 \\ 0p_1 + 0p_2 + 1p_3 = 0 \\ 0p_1 + 0p_2 + 0p_3 = 0 \end{array}\right\} \Rightarrow \begin{array}{l} p_1 = 1 \\ p_2 = 1 \\ p_3 = 0 \end{array}$$

Hence, the vector P is given by

$$P = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Therefore, a second solution is

$$X_2 = Kte^{\lambda t} + Pe^{\lambda t}$$

$$X_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} te^{4t} + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^{4t}$$

$$X_2 = \left[\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} t + \begin{pmatrix} 1 \\ 1 \\ \end{pmatrix} \right] e^{4t}$$

Finally for the third solution we solve

$$(A - \lambda I)Q = P$$

or

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

or

$$\left.\begin{array}{l} 0q_1 + 1q_2 + 0q_3 = 1 \\ 0q_1 + 0q_2 + 1q_3 = 0 \\ 0q_1 + 0q_2 + 0q_3 = 0 \end{array}\right\} \Rightarrow \begin{array}{l} q_1 = 1 \\ q_2 = 1 \\ q_3 = 1 \end{array}$$

Hence, the vector Q is given by

$$Q = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Therefore, third solution vector is

© Copyright Virtual University of Pakistan

$$X_3 = K\frac{t^2}{2}e^{\lambda t} + Pte^{\lambda t} + Qe^{\lambda t}$$

$$X_3 = \begin{pmatrix}1\\0\\0\end{pmatrix}\frac{t^2}{2}e^{4t} + \begin{pmatrix}1\\1\\0\end{pmatrix}te^{4t} + \begin{pmatrix}1\\1\\1\end{pmatrix}e^{4t}$$

$$X_3 = \left[\begin{pmatrix}1\\0\\0\end{pmatrix}\frac{t^2}{2} + \begin{pmatrix}1\\1\\0\end{pmatrix}t + \begin{pmatrix}1\\1\\1\end{pmatrix}\right]e^{4t}$$

The general solution of the given system is

$$X = c_1X_1 + c_2X_2 + c_3X_3$$

$$X = c_1\begin{pmatrix}1\\0\\0\end{pmatrix}e^{4t} + c_2\left[\begin{pmatrix}1\\0\\0\end{pmatrix}t + \begin{pmatrix}1\\1\\0\end{pmatrix}\right]e^{4t} + \left[\begin{pmatrix}1\\0\\0\end{pmatrix}\frac{t^2}{2} + \begin{pmatrix}1\\1\\0\end{pmatrix}t + \begin{pmatrix}1\\1\\1\end{pmatrix}\right]e^{4t}$$

© Copyright Virtual University of Pakistan

Exercise

Find the general solution of the give systems

1. $\frac{dx}{dt} = -6x + 5y$

   $\frac{dy}{dt} = -5x + 4y$

2. $\frac{dx}{dt} = -x + 3y$

   $\frac{dy}{dt} = -3x + 5y$

3. $\frac{dx}{dt} = 3x - y - z$

   $\frac{dy}{dt} = x + y - z$

   $\frac{dz}{dt} = x - y + z$

4. $X' = \begin{pmatrix} 5 & -4 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 5 \end{pmatrix} X$

5. $X' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & -1 & 1 \end{pmatrix} X$

6. $X' = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 0 & 1 & 0 \end{pmatrix} X$

© Copyright Virtual University of Pakistan

Lecture 44

Non-Homogeneous System

Definition

Consider the system of linear first order differential equations

$$\begin{aligned}
\frac{dx_1}{dt} &= a_{11}(t)\ x_1 + a_{12}(t)\ x_2 + \cdots + a_{1n}(t)\ x_n + f_1(t)\\
\frac{dx_2}{dt} &= a_{21}(t)\ x_1 + a_{22}(t)\ x_2 \cdots + a_{2n}(t)\ x_n + \ f_2(t)\\
&\vdots \qquad\quad \vdots \qquad\quad \vdots \qquad\quad \vdots\\
\frac{dx_n}{dt} &= a_{n1}(t)\ x_1 + a_{n2}(t)\ x_2 + \cdots + a_{nn}(t)\ x_n + f_n(t)
\end{aligned}$$

where a_{ij} are coefficients and f_i are continuous on common interval I. The system is said to be non-homogeneous when $f_i(t) \neq 0, \forall i = 1, 2, ..., n$. Otherwise it is called a homogeneous system.

Matrix Notation

In the matrix notation we can write the above system of differential can be written as

$$\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}(t) & a_{12}(t) \ldots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) \ldots & a_{2n}(t) \\ \vdots & \vdots & \vdots \\ a_{n1}(t) & a_{n2}(t) \ldots & a_{nn}(t) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

Or $X' = AX + F(t)$

Method of Solution

To find general solution of the non-homogeneous system of linear differential equations, we need to find:

- The complementary function X_c, which is general solution of the corresponding homogeneous system $X' = AX$.
- Any particular solution X_p of the non-homogeneous system $X' = AX + F(t)$ by the method of undetermined coefficients and the variation of parameters.

The general solution X of the system is then given by sum of the complementary function and the particular solution.

© Copyright Virtual University of Pakistan

$$X = X_c + X_p$$

Method of Undetermined Coefficients

The form of $F(t)$

As mentioned earlier in the analogous case of a single *nth* order non-homogeneous linear differential equations. The entries in the matrix $F(t)$ can have one of the following forms:

- Constant functions.
- Polynomial functions
- Exponential functions
- $\sin(\beta x)$, $\cos(\beta x)$
- Finite sums and products of these functions.

Otherwise, we cannot apply the method of undetermined coefficients to find a particular solution of the non-homogeneous system.

Duplication of Terms

The assumption for the particular solution X_p has to be based on the prior knowledge of the complementary function X_c to avoid duplication of terms between X_c and X_p.

Example 1

Solve the system on the interval $(-\infty, \quad \infty)$

$$X' = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} X + \begin{pmatrix} -8 \\ 3 \end{pmatrix}$$

Solution

To find X_c, we solve the following homogeneous system

$$X' = \begin{pmatrix} -1 & 2 \\ - & \end{pmatrix} X$$

We find the determinant

$$\det\ (A - \lambda I) = \begin{vmatrix} -1-\lambda & 2 \\ -1 & 1-\lambda \end{vmatrix}$$

$$\det\ (A - \lambda I) = (-1-\lambda)\ (1-\lambda) + 2$$

$$\det\ (A - \lambda I) = \lambda^2 + \lambda - \lambda - 1 + 2 = \lambda^2 + 1$$

The characteristic equation is

$$\det\ (A - \lambda I) = 0 = \lambda^2 + 1$$

© Copyright Virtual University of Pakistan

or $\lambda^2 = -1 \Rightarrow \lambda = \pm i$

So that the coefficient matrix of the system has complex eigenvalues $\lambda_1 = i$ and $\lambda_2 = -i$ with $\alpha = 0$ and $\beta = \pm 1$.

To find the eigenvector corresponding to λ_1, we must solve the system of linear algebraic equations

$$\begin{pmatrix} -1-i & 2 \\ -1 & 1-i \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

or

$$\begin{aligned} -(1+i)\,k_1 + 2k_2 &= 0 \\ -k_1 + (1-i)\,k_2 &= 0 \end{aligned}$$

Clearly, the second equation of the system is $(1+i)$ times the first equation. So that both of the equations can be reduced to the following single equation

$$k_1 = (1-i)k_2$$

Thus, the value of k_2 can be chosen arbitrarily. Choosing $k_2 = 1$, we get $k_1 = 1-i$. Hence, the eigenvector corresponding to λ_1 is

$$K_1 = \begin{pmatrix} 1-i \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i\begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

Now we form the matrices B_1 and B_2

$$B_1 = \text{Re}(k_1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad B_2 = \text{Im}(k_1) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

Then, we obtain the following two linearly independent solutions from:

$$X_1 = (B_1 \cos \beta t - B_2 \sin \beta t)\, e^{\alpha t}$$

$$X_2 = (B_2 \cos \beta t + B_1 \sin \beta t) e^{\alpha t}$$

Therefore

$$X_1 = \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix}\cos t - \begin{pmatrix} -1 \\ 0 \end{pmatrix}\sin t\right] e^{0t}$$

$$X_2 = \left[\cos t\begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}\sin t\right] e^{0t}$$

or

$$X_1 = \begin{pmatrix} \cos t \\ \cos t \end{pmatrix} + \begin{pmatrix} \sin t \\ 0 \end{pmatrix} = \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix}$$

$$X_2 = \begin{pmatrix} -\cos t \\ 0 \end{pmatrix} + \begin{pmatrix} \sin t \\ \sin t \end{pmatrix} = \begin{pmatrix} -\cos t + \sin t \\ \sin t \end{pmatrix}$$

© Copyright Virtual University of Pakistan

Thus the complementary function is given by

$$X_c = c_1 X_1 + c_1 X_2$$

or

$$X_c = c_1 \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} + c_2 \begin{pmatrix} -\cos t + \sin t \\ \sin t \end{pmatrix}$$

Now since $F(t)$ is a constant vector, we assume a constant particular solution vector

$$X_p = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$$

Substituting this vector into the original system leads to

$$X_p' = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} + \begin{pmatrix} -8 \\ 3 \end{pmatrix}$$

Since

$$X_p' = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Thus

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -a_1 + 2b_1 \\ -a_1 + b_1 \end{pmatrix} + \begin{pmatrix} -8 \\ 3 \end{pmatrix}$$

or

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -a_1 + 2b_1 - 8 \\ -a_1 + b_1 + 3 \end{pmatrix}$$

This leads to the following pair of linear algebraic equations

$$-a_1 + 2b_1 - 8 = 0$$
$$-a_1 + b_1 + 3 = 0$$

Subtracting, we have

$$b_1 - 11 = 0 \Rightarrow b_1 = 11$$

Substituting this value of b_1 into the second equation of the above system of algebraic equations yields

$$a_1 = 11 + 3 = 14$$

Thus our particular solution is

$$X_p = \begin{pmatrix} 14 \\ 11 \end{pmatrix}$$

Hence, the general solution of the non-homogeneous system is

© Copyright Virtual University of Pakistan

$$X = c_1\begin{pmatrix}\cos t + \sin t\\ \cos t\end{pmatrix} + c_2\begin{pmatrix}-\cos t + \sin t\\ \sin t\end{pmatrix} + \begin{pmatrix}14\\ 11\end{pmatrix}$$

Note that

- In the above example the entries of the matrix $F(t)$ were constants and the complementary function X_c did not involve any constant vector. Thus there was no duplication of terms between X_c and X_p.
- However, if $F(t)$ were a constant vector and the coefficient matrix had an eigenvalue $\lambda = 0$. Then X_c contains a constant vector. In such a situation the assumption for the particular solution X_p would be

$$X_p = \begin{pmatrix}a_2\\ b_2\end{pmatrix}t + \begin{pmatrix}a_1\\ b_1\end{pmatrix}$$

instead of

$$X_p = \begin{pmatrix}a_1\\ b_1\end{pmatrix}$$

Example 2

Solve the system

$$\frac{dx}{dt} = 6x + y + 6t$$

$$\frac{dy}{dt} = 4x + 3y - 10t + 4$$

Solution

In the matrix notation

$$X' = \begin{pmatrix}6 & 1\\ 4 & 3\end{pmatrix}X + \begin{pmatrix}6\\ -10\end{pmatrix}t + \begin{pmatrix}0\\ 4\end{pmatrix}$$

or

$$X' = \begin{pmatrix}6 & 1\\ 4 & 3\end{pmatrix}X + F(t)$$

Where $F(t) = \begin{pmatrix}6\\ -10\end{pmatrix}t + \begin{pmatrix}0\\ 4\end{pmatrix}$

We first solve the homogeneous system

© Copyright Virtual University of Pakistan

$$X' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} X$$

Now, we use characteristic equation to find the eigen values

$$\det(A - \lambda I) = \begin{vmatrix} 6-\lambda & 1 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \quad (6-\lambda)(3-\lambda) - 4 = 0$$

$$\Rightarrow \quad \lambda^2 - 9\lambda + 14 = 0$$

So $\lambda_1 = 2$ and $\lambda_2 = 7$

The eigen vector corresponding to eigen value $\lambda = \lambda_1 = 2$, is obtained from

$$(A - \lambda I)K_1 = 0, \text{ Where } K_1 = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

Or $$(A - 2I)K_1 = 0,$$

Therefore

$$\begin{pmatrix} 6-2 & 1 \\ 4 & 3-2 \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 4k_1 + k_2 \\ 4k_1 + k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

or

$$\left.\begin{aligned} 4k_1 + k_2 = 0 \\ 4k_1 + k_2 = 0 \end{aligned}\right\} \Rightarrow 4k_1 + k_2 = 0$$

we choose $k_1 = 1$ arbitrarily then $k_2 = -4$

Hence the related corresponding eigen vector is

$$K_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

Now an eigen vector associated with $\lambda = \lambda_2 = 7$ is determined from the following system

$$(A - \lambda_2 I)K_2 = 0, \text{ where } K_2 = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

© Copyright Virtual University of Pakistan

or $$\begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

or $$\left.\begin{aligned} -k_1+k_2&=0 \\ 4k_1-4k_2&=0 \end{aligned}\right\}\Rightarrow -k_1+k_2=0$$

Therefore $$K_2=\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Consequently the complementary function is

$$X_c=c_1\begin{pmatrix} 1 \\ -4 \end{pmatrix}e^{2t}+c_2\begin{pmatrix} 1 \\ 1 \end{pmatrix}e^{7t}$$

Since $$F(t)=\begin{pmatrix} 6 \\ -10 \end{pmatrix}t+\begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

Now we find a particular solution of the system having the same form.

$$X_p=\begin{pmatrix} a_2 \\ b_2 \end{pmatrix}t+\begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$$

where a_1, a_2, b_1 and b_2 are constants to be determined.

in the matrix terms we must have

$$X_p'=\begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix}X_p+\begin{pmatrix} 6 \\ -10 \end{pmatrix}t+\begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} a_2 \\ b_2 \end{pmatrix}=\begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix}\left[\begin{pmatrix} a_2 \\ b_2 \end{pmatrix}t+\begin{pmatrix} a_1 \\ b_1 \end{pmatrix}\right]+\begin{pmatrix} 6 \\ -10 \end{pmatrix}t+\begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} a_2 \\ b_2 \end{pmatrix}=\begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} a_2t+a_1 \\ b_2t+b_1 \end{pmatrix}+\begin{pmatrix} 6t+0 \\ -10t+4 \end{pmatrix}$$

$$\begin{pmatrix} a_2 \\ b_2 \end{pmatrix}=\begin{pmatrix} 6a_2t+6a_1+b_2t+b_1 \\ 4a_2t+4a_1+3b_2t+3b_1 \end{pmatrix}+\begin{pmatrix} 6t+0 \\ -10t+4 \end{pmatrix}$$

$$\begin{pmatrix} a_2 \\ b_2 \end{pmatrix}=\begin{pmatrix} 6a_2t+b_2t+6t+6a_1+b_1 \\ 4a_2t+3b_2t-10t+4a_1+3b_1+4 \end{pmatrix}$$

© Copyright Virtual University of Pakistan

or
$$\begin{pmatrix} (6a_2 + b_2 + 6)t + (6a_1 + b_1 - a_2) \\ (4a_2 + 3b_2 - 10)t + (4a_1 + 3b_1 - b_2 + 4) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

from this last identity we conclude that

$$6a_2 + b_2 + 6 = 0 \qquad \text{And} \qquad 6a_1 + b_1 - a_2 = 0$$
$$4a_2 + 3b_2 - 10 = 0 \qquad 4a_1 + 3b_1 - b_2 + 4 = 0$$

Solving the first two equations simultaneously yields

$$a_2 = -2 \text{ and } b_2 = 6$$

Substituting these values into the last two equations and solving for a_1 and b_1 gives

$$a_1 = -\frac{4}{7}$$
$$b_1 = \frac{10}{7}$$

It follows therefore that a particular solution vector is

$$X_p = \begin{pmatrix} -2 \\ 6 \end{pmatrix} t + \begin{pmatrix} -4/7 \\ 10/7 \end{pmatrix}$$

and so the general solution of the system on $(-\infty, \infty)$ is

$$X = X_c + X_p$$
$$= c_1 \begin{pmatrix} 1 \\ -4 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{7t} + \begin{pmatrix} -2 \\ 6 \end{pmatrix} t + \begin{pmatrix} -4/7 \\ 10/7 \end{pmatrix}$$

Example 3

Determine the form of the particular solution vector X_p for

$$\frac{dx}{dt} = 5x + 3y - 2e^{-t} + 1$$
$$\frac{dy}{dt} = -x + y + e^{-t} - 5t + 7$$

Solution

First, we write the system in the matrix form

© Copyright Virtual University of Pakistan

$$\begin{pmatrix} dx/dt \\ dy/dt \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ -5 \end{pmatrix} t + \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

or $$X' = \begin{pmatrix} 5 & 3 \\ -1 & 1 \end{pmatrix} X + F(t)$$

where $$X' = \begin{pmatrix} dx/dt \\ dy/dt \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix} \text{ and } F(t) = \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ -5 \end{pmatrix} t + \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

Now we solve the homogeneous system $X' = \begin{pmatrix} 5 & 3 \\ -1 & 1 \end{pmatrix} X$ to determine the eigen values, we use the characteristic equation

$$\det(A - \lambda I) = 0$$

or $$\begin{vmatrix} 5-\lambda & 3 \\ -1 & 1-\lambda \end{vmatrix} = (5-\lambda)(1-\lambda) + 3 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda + 8 = 0$$

$$\Rightarrow \lambda = 2, 4$$

So the eigen values are $\lambda_1 = 2$ and $\lambda_2 = 4$

For $\lambda = \lambda_1 = 2$, an eigen vector corresponding to this eigen value is obtained from

$$(A - 2I)K_1 = 0$$

Where $$K_1 = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

$$\begin{pmatrix} 5-2 & 3 \\ -1 & 1-2 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 3 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left.\begin{aligned} 3k_1 + 3k_2 = 0 \\ -k_1 - k_2 = 0 \end{aligned}\right\} \Rightarrow -k_1 - k_2 = 0$$

We choose $k_2 = -1$ then $k_1 = 1$

© Copyright Virtual University of Pakistan

Therefore $K_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Similarly for $\lambda = \lambda_2 = 4$

$$\begin{pmatrix} 1 & 3 \\ -1 & -3 \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left.\begin{aligned} k_1 + 3k_2 &= 0 \\ -k_1 - 3k_2 &= 0 \end{aligned}\right\} \Rightarrow k_1 + 3k_2 = 0$$

Choosing $k_2 = -1$, we get $k_1 = 3$

Therefore $K_2 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

Hence the complementary solution is

$$X_c = c_1\begin{pmatrix} 1 \\ -1 \end{pmatrix}e^{2t} + c_2\begin{pmatrix} 3 \\ -1 \end{pmatrix}e^{4t}$$

Now since

$$F(t) = \begin{pmatrix} -2 \\ 1 \end{pmatrix}e^{-t} + \begin{pmatrix} 0 \\ -5 \end{pmatrix}t + \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

We assume a particular solution of the form

$$X_p = \begin{pmatrix} a_3 \\ b_3 \end{pmatrix}e^{-t} + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$$

Note:

If we replace e^{-t} in $F(t)$ on e^{2t} ($\lambda = 2$ an eigen value), then the correct form of the particular solution is

$$X_p = \begin{pmatrix} a_4 \\ b_4 \end{pmatrix}te^{2t} + \begin{pmatrix} a_3 \\ b_3 \end{pmatrix}e^{2t} + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$$

Variation of Parameters

Variation of parameters is more powerful technique than the method of undetermined coefficients.

We now develop a systematic produce for finding a solution of the non-homogeneous linear vector differential equation

$$\frac{dX}{dt} = AX + F(t) \qquad (1)$$

Assuming that we know the corresponding homogeneous vector differential equation

$$\frac{dX}{dt} = AX \qquad (2)$$

© Copyright Virtual University of Pakistan

Let $\phi(t)$ be a fundamental matrix of the homogeneous system (2), then we can express the general solution of (2) in the form

$$X_c = \phi(t)\, C$$

where C is an arbitrary n-rowed constant vector. We replace the constant vector C by a column matrix of functions

$$U(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_n(t) \end{pmatrix}$$

so that

$$X_p = \phi(t) U(t) \qquad (3)$$

is particular solution of the non-homogeneous system (1).
The derivative of (3) by the product rule is

$$X'_p = \phi(t)\, U'(t) + \phi'(t) U(t) \qquad (4)$$

Now we substitute equation (3) and (4) in the equation (1) then we have

$$\phi(t)\, U'(t) + \phi'(t) U(t) = A\phi(t) U(t) + F(t) \qquad (5)$$

Since $\phi'(t) = A\phi(t)$

On substituting this value of $\phi'(t)$ into (5),
We have

$$\phi(t) U'(t) + A\phi(t) U(t) = A\phi(t) U(t) + F(t)$$

Thus, equation (5) become s

or

$$\phi(t)\, U'(t) = F(t) \qquad (6)$$

Multiplying $\phi^{-1}(t)$ on both sides of equation (6), we get

$$\phi^{-1}(t)\phi(t)\, U'(t) = \phi^{-1}(t)\, F(t)$$

or

$$U'(t) = \phi^{-1}(t)\, F(t)$$

or

$$U(t) = \int \phi^{-1}(t)\, F(t) dt$$

Hence by equation (3)

$$X_p = \phi(t) \int \phi^{-1}(t)\, F(t) dt \qquad (7)$$

is particular solution of the non-homogeneous system (1).

To calculate the indefinite integral of the column matrix $\phi^{-1}(t)\ F(t)$ in (7), we integrate each entry. Thus the general solution of the system (1) is

$$X = X_c + X_p$$

or

$$X = \phi(t) C + \phi(t) \int \phi^{-1}(t)\, F(t) dt \qquad \textbf{(6)}$$

Example

© Copyright Virtual University of Pakistan

Find the general solution of the non-homogeneous system

$$X' = \begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix} X + \begin{pmatrix} 3t \\ e^{-t} \end{pmatrix}$$

on the interval $(-\infty, \infty)$

Solution

We first solve the corresponding homogeneous system

$$X' = \begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix} X$$

The characteristic equation of the coefficient matrix is

$$\det(A - \lambda I) = \begin{vmatrix} -3-\lambda & 1 \\ 2 & -4-\lambda \end{vmatrix} = 0$$

or

$$(-3-\lambda)(-4-\lambda) - 2 = 0$$

$$\Rightarrow \lambda^2 + 4\lambda + 3\lambda + 12 - 2 = 0$$

$$\Rightarrow \lambda^2 + 7\lambda + 10 = 0$$

$$\Rightarrow \lambda^2 + 5\lambda + 2\lambda + 10 = 0$$

$$\Rightarrow \lambda(\lambda+5) + 2(\lambda+5) = 0$$

$$\Rightarrow (\lambda+5)(\lambda+2) = 0$$

$$\Rightarrow \lambda_1 = -2, \quad \lambda_2 = -5$$

So the eigen values are $\lambda_1 = -2$ and $\lambda_2 = -5$

Now we find the eigen vectors corresponding to λ_1 and λ_2 respectively,

Therefore

$$(A - \lambda_1 I_2) K_1 = 0$$

$$(A - 2I_2) K_1 = 0$$

so

$$\begin{pmatrix} -3+2 & 1 \\ 2 & -4+2 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -k_1 + k_2 \\ 2k_1 - 2k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

or

$$\left.\begin{aligned} -k_1 + k_2 &= 0 \\ 2k_1 - 2k_2 &= 0 \end{aligned}\right\} \Rightarrow k_1 = k_2$$

We choose $k_2 = 1$ arbitrarily then $k_1 = 1$

Hence the eigen vector is

© Copyright Virtual University of Pakistan

$$K_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Now an eigen vector associated with $\lambda_2 = \lambda = -5$ is determined from the following system

$$(A - \lambda_2 I_2)K_2 = 0$$

or
$$\begin{pmatrix} -3+5 & 1 \\ 2 & -4+5 \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2k_1 + k_2 \\ 2k_1 + k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left.\begin{matrix} 2k_1 + k_2 = 0 \\ 2k_1 + k_2 = 0 \end{matrix}\right\} \Rightarrow k_2 = -2k_1$$

We choose arbitrarily $k_1 = 1$ then $k_2 = -2$

Therefore
$$K_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

The solution vectors of the homogeneous system are

$$X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} \text{ And } X_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-5t}$$

X_1 and X_2 can be written as

$$X_1 = \begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix}, \quad X_2 = \begin{pmatrix} e^{-5t} \\ -2e^{-5t} \end{pmatrix}$$

The complementary solution

$$X_c = c_1 X_1 + c_2 X_2$$
$$= c_1 \begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix} + c_2 \begin{pmatrix} e^{-5t} \\ -2e^{-5t} \end{pmatrix}$$

Next, we form the fundamental matrix

$$\phi(t) = \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}$$

and the inverse of this fundamental matrix is

© Copyright Virtual University of Pakistan

$$\phi^{-1}(t)=\begin{pmatrix} \frac{2}{3}e^{2t} & \frac{1}{3}e^{2t} \\ \frac{1}{3}e^{5t} & \frac{-1}{3}e^{5t} \end{pmatrix}$$

Now we find X_p by

$$X_p=\phi(t)\int\phi^{-1}(t)\,F(t)\,dt$$

$$X_p=\begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}\int\begin{pmatrix} \frac{2}{3}e^{2t} & \frac{1}{3}e^{2t} \\ \frac{1}{3}e^{5t} & \frac{-1}{3}e^{5t} \end{pmatrix}\begin{pmatrix} 3t \\ e^{-t} \end{pmatrix}dt$$

$$X_p=\begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}\int\begin{pmatrix} 2te^{2t}+\frac{1}{3}e^{t} \\ te^{5t}-\frac{1}{3}e^{4t} \end{pmatrix}dt=\begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}\begin{pmatrix} \int 2te^{2t}\,dt+\int\frac{1}{3}e^{t}\,dt \\ \int te^{5t}\,dt-\int\frac{1}{3}e^{4t}\,dt \end{pmatrix}$$

$$X_p=\begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}\begin{pmatrix} 2t\frac{e^{2t}}{2}-2\int\frac{e^{2t}}{2}dt+\frac{1}{3}e^{t} \\ t\frac{e^{5t}}{5}-\int\frac{e^{5t}}{5}dt-\frac{1}{3.4}e^{4t} \end{pmatrix}$$

$$X_p=\begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}\begin{pmatrix} 2t\frac{e^{2t}}{2}-\frac{e^{2t}}{2}+\frac{1}{3}e^{t} \\ \frac{te^{5t}}{5}-\frac{e^{5t}}{25}-\frac{1}{12}e^{4t} \end{pmatrix}$$

$$X_p=\begin{pmatrix} t-\frac{1}{2}+\frac{1}{3}e^{-t}+\frac{1}{5}-\frac{1}{25}-\frac{1}{12}e^{-t} \\ t-\frac{1}{2}+\frac{1}{3}e^{-t}-\frac{2t}{5}+\frac{2}{25}+\frac{1}{6}e^{-t} \end{pmatrix}$$

$$X_p=\begin{pmatrix} \frac{6}{5}t-\frac{27}{50}+\frac{1}{4}e^{-t} \\ \frac{3}{5}t-\frac{21}{50}+\frac{1}{2}e^{-t} \end{pmatrix}$$

Hence the general solution of the non-homogeneous system on the interval $(-\infty,\infty)$ is

$$X=X_c+X_p$$

© Copyright Virtual University of Pakistan

$$= \phi(t)C + \phi(t)\int \phi^{-1}(t)\, F(t)\, dt$$

or

$$= c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-5t} + \begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{21}{50} + \frac{1}{2}e^{-t} \end{pmatrix}$$

© Copyright Virtual University of Pakistan

Exercise

Use the method of undetermined coefficients to solve the given system on $(-\infty, \infty)$

1. $\frac{dx}{dt} = 5x + 9y + 2$
   $\frac{dy}{dt} = -x + 11y + 6$
2. $\frac{dx}{dt} = x + 3y - 2t^2$
   $\frac{dy}{dt} = 3x + y + t + 5$
3. $\frac{dx}{dt} = x - 4y + 4t + 9e^{6t}$
   $\frac{dy}{dt} = 4x + y - t + e^{6t}$
4. $X' = \begin{pmatrix} 4 & 1/3 \\ 9 & 6 \end{pmatrix} X + \begin{pmatrix} -3 \\ 10 \end{pmatrix} e^{t}$
5. $X' = \begin{pmatrix} -1 & 5 \\ -1 & 1 \end{pmatrix} X + \begin{pmatrix} \sin t \\ -2\cos t \end{pmatrix}$

Use variation of parameters to solve the given system

6. $\frac{dx}{dt} = 3x - 3y + 4$
   $\frac{dy}{dt} = 2x - 2y - 1$
7. $X' = \begin{pmatrix} 2 & -1 \\ 4 & 2 \end{pmatrix} X + \begin{pmatrix} \sin 2t \\ 2\cos t \end{pmatrix} e^{2t}$
8. $X' = \begin{pmatrix} 0 & 2 \\ -1 & 3 \end{pmatrix} X + \begin{pmatrix} 2 \\ e^{-3t} \end{pmatrix}$
9. $X' = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} X + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
10. $X' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} X + \begin{pmatrix} \sec t \\ 0 \end{pmatrix}$

© Copyright Virtual University of Pakistan

VIRTUAL Inspire! THE BEST LEADERS. TEAMS. ORGANISATIONS www.virtualinspire.com

MTH401

Final Term Examination – Spring 2006

Time Allowed: 150 Minutes

Question No. 1 **Marks : 5**

Determine the singular points of the differential equation and classify them as regular or irregular.

$$(x_2 - 9)_2 y'' + (x+3)y' + 2y =$$

Question No. 2 **Marks : 10**

Solve the given Bernoulli equation

$$\frac{dy}{dx} + y = y^2$$

Question No. 3 **Marks : 2**

Which of the following is singular point of the equation

$$2x_2 y'' + 10xy' + (x-1)y = 0$$

- $x = 0$
- $x = 2$
- $x = 1$
- None of other

Question No. 4 **Marks : 10**

Find the eigen value and eigen vector of the following system of linear differential equation

$$X' = \begin{pmatrix} 9 & -12 \\ 3 & -3 \end{pmatrix} X$$

Question No. 5 **Marks : 10**

Find solution of the differential equation

$$9y' + y = 0$$

$$x$$

Question No. 6 **Marks : 2**

The non-trivial solution of the system exists only when

$\det(A - \lambda I) = 0$

- True

Question No. 7 **Marks : 2**

The form of the particular solution for the differential equation $y' - y = x^2 e^{3x}$ is

- $y_p = {}^{x} + B_1 x + B_0$

  Ae
- $y_p = (Ax + B)e_x$
- $y_p = (Ax_2 + Bx + C)e_{3x}$
- None of above

Question No. 8 **Marks : 10**

Solve the following homogeneous system of differential equations

$$\frac{dx}{dt} - 7x + \frac{dy}{dt} = 3e^t$$

$$3\frac{dx}{dt} - 2x + \frac{dy}{dt} = 2e^t$$

(Just find the general solution of the equation).

Question No. 9 **Marks : 5**

Solve the differential equation



$y'' - 2y' + y = 0$

Question No. 10 **Marks : 2**

Which of the following is Legendre's Equation

$(1 - {}^{2})\frac{d^{2}y}{dx^{2}} - x\frac{dy}{dx} + 7y = 0$

x 2

- $(1 - {}^{2})\frac{d\ y}{dx^{2}} - 7\ x\frac{dy}{dx} + 35\ y = 0$

  x

- $\frac{=}{dx}d\left[(1 - x^{2})\frac{=}{dx}dy\right] + 6y = 0$

- *All of three equations are legendre's equation*
- *None of other*

Question No. 11 **Marks : 2**

The differential equation $(2x^{2}y - 2x^{3})dy + (4x^{3} - 6x^{2}y + 2xy^{2})dx = 0$ is

- Separable
- Exact
- Linear
- Bernoulli's

VIRTUAL Inspire! THE BEST LEADERS. TEAMS. ORGANISATIONS www.virtualinspire.com

	FINALTERM EXAMINATION SPRING 2007 MTH401 - DIFFERENTIAL EQUATIONS (Session - 4)	Marks: 55 Time: 150min

StudentID/LoginID: _____

Student Name: _____

Center Name/Code: _____

Exam Date: Tuesday, July 10, 2007

Please read the following instructions carefully before attempting any of the questions:

1. Symbols by using math type should be pasted on the paper direct from the math type not from the word document otherwise it would not be visible.

2. Do not ask any questions about the contents of this examination from anyone.

a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.

b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

c. Write all steps, missing steps may lead to deduction of marks.

3. This examination is closed book and closed notes.

4. Use of Calculator is allowed.

5. Attempt all questions. Marks are written adjacent to each question.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will be awarded grade `F` in this course.**

For Teacher's use only

Question	1	2	3	4	5	6	7	8	9	10	Total

Marks											
Question	11										
Marks											

Question No: 1 (Marks: 2) - Please choose one

The Wronskian of the function $W(e^x, e^{-x}) =$ _____ is

- 1
- 0
- -2
- None of these

Question No: 2 (Marks: 2) - Please choose one

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$$

The eigen values of matrix are

- $\lambda = 0, -1$
- $\lambda = 4, -1$
- $\lambda = 1, 5$
- None of these

Question No: 3 (Marks: 2) - Please choose one

Roots of the equation $y''' + y' = 0$ will be

- 0,1,2
- 0,+ i , - i
- 0,1,i
- None of these

Question No: 4 (Marks: 2) - Please choose one

$$y'' + 16y = 0 \ \ with \ y(0) = 0 \ , \ y(\frac{\pi}{8}) = 0$$

A differential equation is called

- Initial value problem
- Boundary value problem

▶ None of these

Question No: 5 (Marks: 2) - Please choose one

Suppose the functions $f_1(x), f_2(x), ..., f_n(x)$ possess at least n-1 derivatives on interval I , if $W(f_1, f_2, ..., f_n) \neq 0$ is called _____

▶ Linearly dependent

▶ Linearly independent

▶ None of these

Question No: 6 (Marks: 5)

Solve the initial value problem

$2(y-1)\,dy = (3x_2 + 4x + 2)\,dx \quad , \quad y(0) = -1$

Question No: 7 (Marks: 10)

$$\frac{dy}{dx} + \frac{xy}{1 - x_2} = x\,y^{\frac{1}{2}}$$

Solve

Question No: 8 (Marks: 5)

Find the general solution of the given differential equation on $(0, \infty)$

$25\,x_2\,y_{//} + 25\,xy_{/} + (25\,x_2 - 1)\,y = 0$

Question No: 9 (Marks: 10)

Use the Gauss-Jordan elimination method to solve the linear system

$5x - 2y + 4z = 10$

$x + y + z = 9$

$4x - 3y + 3z = 1$

Question No: 10 (Marks: 10)

$$J_n(x) \quad for \quad n = \frac{1}{2}$$

Derive the expression of

Question No: 11 (Marks: 5)

Solve by using Cauchy Euler method

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + 4y = 0$$

VIRTUAL Inspire!
THE BEST LEADERS. TEAMS. ORGANISATIONS
www.virtualinspire.com

Question No: 4 **Marks:2**

The given functions $f_1(x) = 1 + x, f_2(x) = x, f_2(x^2) = x$ are linearly independent.

T

F

Æ 02

Question No: 5 **Marks: 2**

A set of functions whose wronskian is zero guarantees that set of functions is linearly dependent.

T

F

Æ 02

Question No: 6 **Marks: 10**

(a) Define separable form. Just separate the variables of the given differential equation.

$$(r\theta - 4r + \theta - 4)dr - \left(r^2\theta + 20r^2 - \theta - 20\right) d\theta = 0$$

Solution:

The differential equation of the form d y / d x = f (x, y) is called **separable** if it can be written in the form

d y / d x = h(x)g(y)

$$(r\theta - 4r + \theta - 4)dr - \left(r_2\theta + 20r_2 - \theta - 20\right) d\theta = 0$$

$$(r\theta - 4r + \theta - 4)dr = \left(r^2\theta + 20r^2 - \theta - 20\right) d\theta$$

$$[r(\theta - 4) + 1(\theta - 4)] \quad r^2(\theta + 20) - 1(\theta + 20)\ d\theta$$

$$dr =$$

$$(r + 1)(\theta - 4)\, dr = \left(r^2 - 1\right)(\theta + 20)\, d\theta$$

$$\frac{(r +}{} dr = \frac{(\theta + 20)}{(\theta - 4)} d\theta$$

$$\left(r^2 - \frac{1)}{1)}\right.$$

$$\frac{1}{r -} dr = \frac{(\theta + 20)}{(\theta - 4)} d\theta$$

1

(b) Check whether the given differential equation is exact or not if not then make it exact also show that it is exact **(Just make the equation exact do not solve it further).**

$$\left(2y_2 + 3x\right) dx + 2xydy = 0$$

$\left(2y^2 + 3x\right)dx + 2xydy = 0$

$M(x,y) = 2y^2 + \quad N(x,y) = 2xy$

$3x,$

$M_y = 4 \quad N_x = 2y$

$y,$

$M_y \neq N_x$

Thus it is not exact now we apply techniques to make it exact

$\left(2y^2 + 3x\right)dx + 2xydy = 0$

$\frac{M_y - N_x}{N} = \frac{4y - 2y}{2xy} = \frac{2y}{2xy} = \frac{1}{x}$

$I.F = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$

$\left(2xy^2 + 3x^2\right)dx + 2x^2ydy = 0$

$M(x,y) = 2xy_2 + \quad N(x,y) = 2x_2y$

$3x_2,$

$M_y = 4xy, N_x =$

$4xy$

$M_y = N_x$

Which shows that now equation is exact

(a) Solve given Bernoulli equation $xy - \frac{dy}{dx} = y^3e^{-x^2}$ **(Just make the given equation linear in v, do not integrate)**

$xy - \frac{dy}{dx} = y^3e^{-x2}$

$xy^{-2} - \frac{dy}{dx}y^{-3} = e^{-x_2}$

$put\ y_{-2} = v$

$-2\frac{dy}{dx}y^{-3} = \frac{dv}{dx}$

$-\frac{dy}{dx}y^{-3} = \frac{1}{2}\frac{dv}{dx}$

Then

$$\frac{1}{2}\frac{dv}{dx} + vx = e^{-x^2}$$

$$\frac{dv}{dx} + 2vx = 2e^{-x^2}$$

Thus it is linear in “v”.

(b) The population of a town grows at rate proportional to the population at any time. Its initial population of 100 decreased by 20% in 20 years what will be the population in 30 years? **(Just make the model of the population dynamics as well as just describe the given conditions do not solve further)**

Solution

Suppose that P_0 is the initial population of the town, as given P_0 is 100 and P(t) the population at any time t then population growth by the differential equation

$$\frac{dP}{dt} \alpha P$$

$$\frac{dP}{dt} = kP$$

$$\frac{dP}{P} = kdt$$

Integrate both sides

$$\ln P = kt + c$$

$$P = e_{kt + c}$$

$$P = e^{kt} e^{c}$$

$$P = P_0 e^{kt} \qquad say\ P_0 = e^{c}$$

Where P_0 is the initial population of the town

P_0 =100 = P(0)

$$P(20)=100-\frac{20}{100}(100)$$

P(20)=80

P (30) =?

Question No: 8 **Marks: 10**

(a) Find a second solution of following differential equations where the first solution is given **(also write the formulae)**.

$$x^2y'' - 5xy' + 9y = \quad y_1 = {}^{3}\ln x$$

$$x^2 y'' - 5xy' + 9y = 0;\ y = \ln x$$

x^3 _ __

differential equation can be written as

$$x^2 y'' - \frac{5}{x} xy' + \frac{9}{x^2} = 0$$

the 2nd solution is given by

$$y_2 = y_1 \int \frac{e^{-\int pdx}}{y_1^2} dx$$

$$y_2 = x^3 \ln x \int \frac{e^{-\int -\frac{5}{x}\,dx}}{\left(x^3 \ln x\right)^2}\,dx$$

$$y_2 = x^3 \ln x \int \frac{e^{\int \frac{5}{x}\,dx}}{\left(x^3 \ln x\right)^2}\,dx$$

$$y_2 = x_3 \ln x \int \frac{x\, e^{5\ln x}}{\left(x_3 \ln x\right)}\,dx$$

$$y_2 = x_3 \ln x \int \frac{e^{\ln x^5}}{\left(x^3 \ln x\right)^2_5}\,dx$$

$$y_2 = x^3 \ln x \int \frac{x}{x_6 \left(\ln x\right)^2}\,dx$$

$$y_2 = x_3 \ln x \int \frac{(\ln x)^{-2}}{x}\,dx$$

$$y_2 = x^3 \ln x \frac{(\ln x)^{-2+1}}{-2+1}$$

[redacted]

$$y_2 = x_3 \ln \frac{}{\ln x}$$

x

$$y_2 = -x^3$$

(b) Solve the differential equation by the undetermined coefficient **(superposition approach)**

$$y'' - 2y' - 3y = 2\cos\theta$$

If complimentary solution is given below

$$y_c = c_1 e^{3x} + c_2 e^{-}$$

Then just find particular solution.

We find a particular solution of non-homogeneous differential equation.

Suppose input function

$$y_p = A\cos\theta + B\sin\theta$$

Then

$$y'_p = -A\sin\theta + B\cos\theta$$

$$y''_p = -A\cos\theta - B\sin\theta$$

Substituting in the given differential equation

$$-A\cos\theta - B\sin\theta - 2(-A\sin\theta + B\cos\theta) - 3(A\cos\theta + B\sin\theta) = 2\cos\theta$$

$$(-A - 2B - 3A)\cos\theta + (-B + 2A - 3B)\sin\theta = 2\cos\theta$$

From the resulting equations

$-A-2B-3A=$ $\quad -B+2A-3B=0$

$2;$

$-4A-2B=2;$ $\quad 2A-4B=0$

$2A+B=-1;$

$A-2B=0 \rightarrow A=2B$

$\rightarrow 2(2B)+B=-1$

$\rightarrow 5B=-1$

$\rightarrow B=\frac{-1}{5}$

$\rightarrow A=2\left(\frac{-1}{5}\right)=$

-2

$\frac{}{5}$

y_p

$\frac{}{5}$ $\frac{}{5}$

(a) Solve the differential equation by the undetermined coefficient **(annihilator operator).**

$y''-4y'=x_3-2x+1$

If complimentary solution is given below

$y_c=c_1+c_2e^{4x}$

Then find general solution.

Solution

In this case of input function is

g(x)= x^3-2x+1

further

$D_4(g(x))=D_3(x_3-2x+1)=0$

Therefore the differential operator D^4 annihilates the function g. operating on both sides

$D_4(D_2-4D)y=D_4(x_3-2x+1)$

$D^4(D^2-4D)y=0$

This is the homogeneous equation of order 5. Next we solve this higher order equation.

Thus auxiliary equation is

$m_4(m_2-4m)=0$

$m_5(m-4)=0$

$m=0,0,0,0,0,4$

Thus its general solution of the differential equation must be

$y=c_1+c_2x+c_3x^2+c_4x^3+c_5x^4+c_6e^{4x}$

(b) Solve the differential equation by the variation of parameters

$y''-2y'-3y=x_3$

If complimentary solution is given below

$y_c = c_1 e^{3x} + c_2 e^{-}$

Then just find particular solution **(do not integrate)**.

Solution

$y'' - 2y' - 3y = x^3$

This equation is already in standard form

y''+ P(x)y' +Q(x)y = f(x)

Therefore, we identify the function f(x) as

$f(x) = x^3$

We construct the determinants

Since $y_1 = e^{3x}$, $y_2 = e^{-x}$ so

$$W(y_1, y_2) = \begin{vmatrix} e^{3x} & e^{-x} \\ 3e^{3x} & -e^{-x} \end{vmatrix} = -e^{3x-x} - 3e^{3x-x} = -4e^{2x}$$

$$W_1 = \begin{vmatrix} 0 & e^{-x} \\ x^3 & -e_{-x} \end{vmatrix} = -x^3 e^{-x}$$

$$W_2 = \begin{vmatrix} e^{3x} & 0 \\ 3e^{3x} & x^3 \end{vmatrix} = x^3 e^{3x}$$

We determine the derivatives of the function u_1 and u_2

$$u'_1 = \frac{W_1}{W} = \frac{-x^3 e^{-x}}{-4e^{2x}} \rightarrow u_1 = \int \frac{x^3}{4} e^{-3x} dx$$

$$u'_2 = \frac{W_2}{W} = \frac{x^3 e^{3x}}{-4e^{2x}} \rightarrow u_2 = \int \frac{x^3}{4} e^{x} dx$$

$y_p = u_1 e^{3x} + u_2 e^{-x}$

is required particular solution

Population dynamics are not practical application of the first order differential equations.

T

F

Question No: 4 **Marks: 2**

A set

$$\{ y_1, y_2, \ldots, y_n \}$$

Of n linearly dependent solutions, on interval I, of the homogeneous linear nth-order differential equation

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$$

Is said to be a fundamental set of solutions on the interval I.

T

F

Question No: 5 **Marks: 2**

The differential operator that annihilates $10x^3 - 2x$ is D^4.

T

F

Question No: 6 **Marks:10**

(a) Define separable form. Just separate the variables of the given differential equation.

$$(3r\theta - 3\theta + r - 1)\,dr - (2r\theta + 4\theta - r - 2)\,d\theta = 0$$

Solution

The differential equation of the form d y / d x = f (x, y) is called **separable** if it can be written in the form

d y / d x = h(x)g(y)

$$(3r\theta - 3\theta + r - 1)\,dr - (2r\theta + 4\theta - r - 2)\,d\theta = 0$$

$$(3r\theta - 3\theta + r - 1)\,dr = (2r\theta + 4\theta - r - 2)\,d\theta$$

$$[3\theta(r-1) + 1(r-1)]\,dr = [2\theta(r+2) - 1(r+2)]\,d\theta$$

$$(r-1)(3\theta + 1)\,dr = (r+2)(2\theta - 1)\,d\theta$$

$$\frac{(r-1)}{(r+2)}dr = \frac{(2\theta - 1)}{(3\theta + 1)}d\theta$$

(b) Check whether the given differential equation is exact or not if not then make it exact also show that it is exact **(Just make the equation exact do not solve it further).**

$$\left(\frac{3y^2 - x^2}{y^5}\right)\frac{dy}{dx} + \frac{x}{2y^5} = 0$$

Solution

It can also be written as

$$xdx + 2\left(3y_2 - x_2\right)\,dy = 0$$

$$M(x,y) = x,\quad N(x,y) = 6y_2 - 2x_2$$

$$M_y = 0,\quad N_x = -4x$$

$$M_y \neq N_x$$

Thus it is not exact now we apply techniques to make it exact

$$xdx + \left(6y_2 - 2x_2\right)\,dy = 0$$

$$\frac{N_x - M_y}{M} = \frac{-4x}{x} = -4 = g(y)$$

$$I.F = e^{\int -4\,dy} = e^{-4y}$$

$$xe_{-4y}\,dx + e_{-4y}\left(3y_2 - 2x_2\right)\,dy = 0$$

$$M(x,y) = xe_{-4y},\quad N(x,y) = e_{-4y}\left(3y_2 - 2x_2\right)$$

$$M_y = -4xe^{-4y},\ N_x = -4xe^{-4y}$$

$$M_y = N_x$$

Which shows that equation is exact

Question No: 7 **Marks:10**

(a) Solve the Bernoulli equation $x^3\frac{dy}{dx} + 2xy = y^5$ **(Just make the given equation linear in v, do not integrate)**

Solution

$$x^3\frac{dy}{dx} + 2xy = y^5$$

$$\frac{dy}{dx}y^{-5} + \frac{2}{x^2}y^{-4} = \frac{1}{x^3}$$

$$put\ y_{-4} = v$$

$$-4\frac{dy}{dx}y^{-5} = \frac{dv}{dx}$$

$$\frac{dy}{dx}y^{-5} = -\frac{1}{4}\frac{dv}{dx}$$

Then

$$-\frac{1}{4}\frac{dv}{dx} + \frac{2v}{x^2} = \frac{1}{x^3}$$

$$\frac{dv}{dx} - \frac{8v}{x^2} = \frac{-4}{x^3}$$

Thus it is linear in "v".

(b) Initially there were 200 milligrams of a radioactive substance present. After 8 hours the mass increased by 4%. If the rate of decay is proportional to the amount of the substance present at any time, determine half-life of the radioactive substance? **(Just make the model of the radioactive decay as well as describe the given conditions do not solve further)**

Solution

Suppose that A_0 is the initial amount, as given A_0 is 200 and A(t) be the amount present at time t then its governed by the differential equation

$$\frac{dA}{dt} \alpha A$$

$$\frac{dA}{dt} = kA$$

$$\frac{dA}{A} = kdt$$

Integrate both sides

$$\ln A = kt + c$$

$$A = e_{kt + c}$$

$$A = e^{kt} e^{c}$$

$$A = A \quad say \; A = e^{c}$$
$$e^{kt} \quad 0 \qquad\qquad 0$$

Where A_0 is the initial amount

A_0=200 = A(0)

$$A(8)=200+\frac{4}{100}\left(\; 200\right)$$

A(8)=208

A (T) =100

(a) Find a second solution of following differential equations where the first solution is given **(also write the formulae)**.

$$\left(1 + 2x\right) y''+ 2xy'- 4 = 0 \quad = e^{-2x}$$
$$, y_1$$

$$\left(1 + 2x\right) y''+ 2xy'- 4 = 0 \quad , \; = e^{-2x}$$

$$y_1$$

differential equation can be written as

$$y''+ \frac{2x}{1 + 2x} y'- \frac{4}{1 + 2x} = 0$$

the 2nd solution is given by

$$y_2 = y_1 \int \frac{e^{-\int p dx}}{y_1^2} dx$$

$$y_2 = e^{-2x} \int \frac{e^{-\int \frac{2x}{1+2x} dx}}{\left(e_{-2x}\right)_2} dx$$

$$y_2 = e^{-2x} \int \frac{e^{-\int \frac{1+2x}{-1} dx}}{e_{-4x}} dx$$

$$y_2 = e^{-2x} \int \frac{e^{\int (\frac{1}{1+2} - 1) dx}}{e^{-4x}} dx$$

$$y_2 = e^{-2x} \int \frac{e^{-x} e^{\ln(1+2x)}}{e^{-4x}} dx$$

$$y_2 = e^{-2x} \int \frac{e^{2\ln(1+2x)}}{e^{-3x}} dx$$

$$y_2 = e^{-2x} \int \frac{e^{\ln(1+2x)}}{e^{-3x}} dx$$

$$y_2 = e^{-2x} \int (1+2x) e^{3x} dx$$

$$y_2 = e^{-2x} \left[(1+2x) \frac{e^{3x}}{3} - \int \frac{(1+2x)}{3} e^{3x} dx \right]$$

$$y_2 = e^{-2x} \left[(1+2x) \frac{e^{3x}}{3} - \int \frac{2e^{3x}}{3} dx \right]$$

$$y_2 = (1+2x) \frac{e^x}{3} - \frac{2e^x}{9}$$

(b) Solve the differential equation by the undetermined coefficient **(superposition approach)**

$$y'' - 2y' - 3y = e_{4x}$$

Solution

We find a particular solution of non-homogeneous differential equation.

Suppose input function

$y_p = Ae^{4x}$

Then

$y'_p = 4Ae^{4x}$

$y''_p = 16Ae^{4x}$

Substituting in the given differential equation

$16Ae_{4x} - 2\left(4Ae_{4x}\right) - 3Ae_{4x} = e_{4x}$

$e_{4x}\left(16A - 8A - 3A\right) = e_{4x}$

From the resulting equations

$5A = 1$

$A = \frac{1}{5}$

$y_p = \frac{1}{5}e^{4x}$

Question No: 9 **Marks: 10**

(a) Solve differential equation by the undetermined coefficient **(annihilator operator).**

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} = x\cos x$

If complimentary solution is given below

$y_c = c_1 + c_2 e^{4x}$

Then just find general solution.

Solution

In this case of input function is

g(x)= $x\cos x$

further

$(D_2 + 1)_2(g(x)) = (D_2 + 1)_2(x\cos x) = 0$

Therefore the differential operator $(D_2 + 1)_2$ annihilates the function g. operating on both sides

$(D_2 + 1)_2(D_2 - 4D)y = (D_2 + 1)_2(x\cos x)$

$(D_2 + 1)_2(D_2 - 4D)y = 0$

This is the homogeneous equation of order 6. Next we solve this higher order equation.

Thus auxiliary equation is

$(m_2 + 1)_2 (m_2 - 4m) y = 0$

$m(m-4)(m_2 + 1)_2 = 0$

$m = 0, 4, i, i, -i, -i$

Thus its general solution of the differential equation must be

$y = c_1 + c_2 e^{4x} + (c_3 + c_4 x)\cos x + (c_5 + c_6 x)\sin x$

(b) Solve the differential equation by the variation of parameters

$y'' - y = x_2$

Complimentary solution is given below If

$y_c = c_1 e^{x} + c_2 e^{-x}$

Then just find particular solution **(do not integrate).**

Solution

$y'' - y = x_2$

This equation is already in standard form

y''+ P(x)y' +Q(x)y = f(x)

Therefore, we identify the function f(x) as

$f(x) = x^2$

We construct the determinants

Since $y_1 = e^{x}$, $y_2 = e^{-x}$ so

$$W(y_1, y_2) = \begin{vmatrix} e^{x} & e^{-x} \\ e^{x} & -e^{-x} \end{vmatrix} = -e^{x-x} - e^{x-x} = -2$$

$$W_1 = \begin{vmatrix} 0 & e^{-x} \\ x_2 & -e^{-x} \end{vmatrix} = -x^2 e^{-x}$$

$$W_2 = \begin{vmatrix} e^{x} & 0 \\ e^{x} & x^2 \end{vmatrix} = x^2 e^{x}$$

We determine the derivatives of the function u_1 and u_2

$$u'_1 = \frac{W_1}{W} = \frac{-x^2 e^{-x}}{2} \rightarrow u_1 = \int \frac{x^2}{2} e^{-x} dx$$

$$u'_2 = \frac{W_2}{W} = \frac{x^2 e^{-x}}{2} \rightarrow u_2 = -\int \frac{x^2}{2} e^{-x} dx$$

$y_p = u_1 e^{x} + u_2 e^{-x}$

is required particular solution

[redacted]

[redacted]

[redacted]

[redacted]

The given functions $f_1(x) = 5, f_2(x) = \cos^2 x, f_3(x) = \sin^2 x$ are linearly independent.

T

F

Question No: 5 **Marks: 2**

A set of functions whose wronskian is zero then set of functions may or may not be dependent.

T

F

Question No: 6 **Marks: 10**

(a) Define separable form. Just separate the variables of the given differential equation.

$$\sec y \frac{dy}{dx} + \sin(x - y) = \sin(x + y)$$

Solution

The differential equation of the form d y / d x = f (x, y) is called **separable** if it can be written in the form

d y / d x = h(x)g(y)

$$\sec y \frac{dy}{dx} + \sin(x - y) = \sin(x + y)$$

$$\sec y \frac{dy}{dx} = \sin(x + y) - \sin(x - y)$$

$$\sec y \frac{dy}{dx} = \sin(x + y) - \sin(x - y)$$

$$\sec y \frac{dy}{dx} = \sin x \cos y + \cos x \sin y - \sin x \cos y + \cos x$$

$$\sec y \frac{dy}{dx} = 2 \cos x \sin$$

$$\frac{dy}{2 \cos y \sin y} = \cos x dx$$

$$\frac{dy}{\sin 2y} = \cos x dx$$

$$\overline{\cos ec} 2y dy = \cos x dx$$

(b) Check whether the given differential equation is exact or not if not then make it exact also show that it is exact **(Just make the equation exact do not solve it further).**

$$e^x dx + \left(e^x \cot y + 2y \cos ecy\right) dy = 0$$

Solution

$$e_x dx + \left(e_x \cot y + 2y \operatorname{cosec} y\right) dy = 0$$

$$M(x, y) = e_x, \quad N(x, y) = e_x \cot y + 2y \operatorname{cosec} y$$

$$M_y = 0, \quad N_x = e^x \cot y$$

$$M_y \neq N_x$$

Thus it is not exact now we apply techniques to make it exact

$$e_x dx + \left(e_x \cot y + 2y \operatorname{cosec} y\right) dy = 0$$

$$\frac{N_x - M_y}{} = \frac{e \cot y - 0}{} = \cot y$$

$$x$$

$$M \qquad e^x$$

$$I.F = e^{\int \cot y dy} = e_{\ln \sin y} = \sin y$$

$$e_x \sin y dx + \left(e_x \cos y + 2y\right) dy = 0$$

$$M(x, y) = e_x \sin y, \quad N(x, y) = e_x \cos y + 2y$$

Question No: 7 **Marks: 10**

(a) Solve Bernoulli equation $y + 2\frac{dy}{dx} = y^3(x - 1)$ (**Just make the given equation linear in v, do not integrate**)

Solution

$$xy - \frac{dy}{dx} = y^3(x - 1)$$

$$xy^{-2} - \frac{dy}{dx} y^{-3} = (x - 1)$$

$$put\ y_{-2} = v$$

$$-2\frac{dy}{dx} y^{-3} = \frac{dv}{dx}$$

$$-\frac{dy}{dx} y^{-3} = \frac{1}{2}\frac{dv}{dx}$$

Then

$$\frac{1}{2}\frac{dv}{dx} + vx = (x - 1)$$

$$\frac{dv}{dx} + 2vx = 2(x - 1)$$

(b) The radioactive isotope of the lead, Pb-209, decay at a rate proportional to the amount present at any time and has a half-life of 4 hours. If 2 grams of the lead is present initially, how long will it take for 80% of the lead to decay? **(Just make the model of the radioactive decay as well as describe the given conditions do not solve further)**

Solution

Suppose that A_0 is the initial amount of isotope, as given A_0 is 100 and A (t) be the amount present at time t it governed by the differential equation.

$$\frac{dA}{dt} \alpha A$$

$$\frac{dA}{dt} = kA$$

$$\frac{dA}{A} = kdt$$

Integrate both sides

$$\ln A = kt + c$$

$$A = e_{kt + c}$$

$$A = e^{kt} e^{c}$$

$$A = P_0 e^{kt} \qquad say\ A_0 = e^{c}$$

Where A_0 is the initial amount of isotope

A_0 =2 = A(0)

A(4)=2/2=1

Then we have to find time when radioactive isotope will take 80% decay. So as A initially given 2 and 80% of 2 is 8/5 so decay would be 1-8/5 = -3/5

P (t) = -3/5 , t=?

Question No: 8 **Marks:10**

(a) Find a second solution of following differential equations where the first solution is given **(also write the formulae)**. $x^2 y'' - 4xy' + 6y = \quad y_1 = x^2$

Solution

0;

$x_2 y'' - 4xy' + 6y = 0_1; y = x_2$

differential equation can be written as

$$y'' - \frac{4}{x} xy' + \frac{6}{x^2} = 0$$

the 2nd solution is given by

$$y_2 = y_1 \int \frac{e^{-\int p dx}}{y_1^2} dx$$

$$y_2 = x^2 \int \frac{e^{-\int -4/x\, dx}}{(x^2)^2}\, dx$$

$$y_2 = x^3 \ln x \int \frac{e^{\int 4/x\, dx}}{(x^3 \ln x)^2}\, dx$$

$$y_2 = x^2 \int \frac{e^{4\ln x}}{x^4}\, dx$$

$$y_2 = x^3 \ln x \int \frac{e^{\ln x^4}}{x^4}\, dx$$

$$y_2 = x^2 \int \frac{x^4}{x^4}\, dx$$

$$y_2 = x^2 \int dx$$

$$y_2 = x^3$$

(b) Solve the differential equation by the undetermined coefficient **(superposition approach)**

$$y'' - 2y' - 3y = 4\sin\theta$$

If complimentary solution is given below

$$y_c = c_1 e^{3x} + c_2 e^{-x}$$

Solution

We find a particular solution of non-homogeneous differential equation.

Suppose input function

$$y_p = A\cos\theta + B\sin\theta$$

Then

$$y'_p = -A\sin\theta + B\cos\theta$$

$$y''_p = -A\cos\theta - B\sin\theta$$

Substituting in the given differential equation

$$-A\cos\theta - B\sin\theta - 2(-A\sin\theta + B\cos\theta) - 3(A\cos\theta + B\sin\theta) = 4\sin\theta$$

$$(-A - 2B - 3A)\cos\theta + (-B + 2A - 3B)\sin\theta = 4\sin\theta$$

From the resulting equations

$-A-2B-3A=0$;

$-B+2A-3B=4$

$-4A-2B=0$;

$2A+B=0$;

$2A-4B=4$

$A-2B=2 \rightarrow A=2+2B$

$\rightarrow 2(2+2B)+B=0$

$\rightarrow 4+5B=0$

$\rightarrow B=\frac{-4}{5}$

$\rightarrow A=2+2\left(\frac{-4}{5}\right)=\frac{2}{5}$

$y_p=\frac{2}{5}\cos\theta-\frac{4}{5}\sin\theta$

[redacted]

(a) Solve differential equation by the undetermined coefficient **(annihilator operator).**

$$\frac{d^2y}{dx^2}-4\frac{dy}{dx}=e^{2x}$$

If complimentary solution is given below

[redacted]

e

Then just find general solution.

In this case of input function is

g(x)= e^{2x}

further

$(D-2)(g(x))=(D-2)(e_{2x})=0$ annihilates the function g. operating on both sides

Therefore the differential operator D^3

$(D-2)(D_2-4D)y=(D-2)(e_{2x})$

$(D-2)(D^2-4D)y=0$

This is the homogeneous equation of order 3. Next we solve this higher order equation.

Thus auxiliary equation is

$(m-2)(m_2-4m)=0$

$m(m-2)(m-4)=0$

$m=0,2,4$

Thus its general solution of the differential equation must be

$y=c_1+c_2e^{4x}+\varsigma\, e^{2x}$

(b) Solve the differential equation by the variation of parameters

$y'' - 4y + 3 = \cos x$

If complimentary solution is given below

$y_c = c_1 e^{3x} + c_2 e^{x}$

Then just find particular solution **(do not integrate).**

Solution

$y'' - 4y + 3 = \cos x$

This equation is already in standard form

y''+ P(x)y' +Q(x)y = f(x)

Therefore, we identify the function f(x) as

f(x) = cos x

We construct the determinants

Since $y_1 = e^{3x}$, $y_2 = e^{x}$ so

$$W(y_1, y_2) = \begin{vmatrix} e^{3x} & e^{x} \\ 3e^{3x} & e^{x} \end{vmatrix} = e^{3x+x} - 3e^{3x+x} = -2e^{4x}$$

$$W_1 = \begin{vmatrix} 0 & e^{x} \\ \cos x & e^{x} \end{vmatrix} = \cos x e^{x}$$

$$W_2 = \begin{vmatrix} e^{3x} & 0 \\ 3e^{3x} & \cos x \end{vmatrix} = \cos x e^{3x}$$

We determine the derivatives of the function u_1 and u_2

$$u'_1 = \frac{W_1}{W} = \frac{\cos x e^{x}}{-2e_{4x}} \rightarrow u_1 = \int \frac{\cos x}{-2} e^{-3x} dx$$

$$u'_2 = \frac{W_2}{W} = \frac{\cos x e^{3x}}{-2e_{4x}} \rightarrow u_2 = \int \frac{\cos x}{-2} e^{-x} dx$$

$y_p = u_1 e^{3x} + u_2 e^{x}$

is required particular solution

$f_1(x) = 1 + x, f(x) = x, f(x) = x^2$

Question No: 5 **Marks:2**

The differential operator that annihilates $10x - 2x$ is:

D^4

Question No: 6 **Marks:10**

Solve the following differential equation by using an appropriate substitution.

$$\frac{dy}{dx} = \frac{y}{x} + \frac{x}{y}$$

Solution

$$\frac{dy}{dx} = \frac{y}{x} + \frac{x}{y}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y^2 + x^2}{xy}$$

$$\Rightarrow \quad \text{Homogeneous equation, so put } y = vx, \frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{v^2x^2 + x^2}{x^2v}$$

$$\Rightarrow \quad v + x\frac{dv}{dx} = v + \frac{1}{v}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{1}{v} \quad \Rightarrow \quad vdv = \frac{1}{x}dx$$

$$\Rightarrow \quad \int vdv = \int \frac{1}{x}dx \quad \Rightarrow \quad \frac{v^2}{2} = \ln x + \ln C$$

$$\Rightarrow \quad \frac{y^2}{x^2} = 2\ln xC$$

Question No: 7 **Marks:10**

The population of a town grows at a rate proportional to the population at any time. Its initial population of 500 increases by 15% in 10 years. What will be the population in 30 years?

Solution:

Let $P(t)$ be the population at any time t, then rate of grows will be

$$\frac{dP}{dt} \propto P$$

$$\Rightarrow \quad \frac{dP}{dt} = kP$$

Here k is constant of proportionality. Since initially population was 500, therefore $P(0) = 500$. Also this population increases by 15% in 10 years. The 15% of 500 is $\frac{15}{100}(500) = 75$, therefore population after 10 years is (initial population + increase in 10 years) = 500+75 = 575 i.e. $P(10) = 575$. So we have the boundary value problem

$\frac{dP}{dt} = kP$ subject to boundary conditions $P(0) = 500, P(10) = 575$.

This first order differential equation. Its solution is given by

$P = Ce^{kt}$ where C is constant of integration.

Applying boundary conditions, we get $C = 500,\ k = 0.0139$. So the solution is

$$P(t) = 500e^{(0.0139)t}$$

Thus population after 30 years is obtained by putting $t = 30$ in above equation i.e.

$$P(30) = 500e_{(0.0139)30}$$
$$\approx 760.$$

Question No: 8 **Marks:10**

Find a second solution of following differential equations where the first solution is given. You can use any method (reduction of order or formula given in handouts).

$x^2y'' + 2xy' - 6y = 0;$ $\quad y_1 = x^2$

Solution: Comparing with $y'' + P(x)y' + Q(x)y = 0$

Here $P(x) = \frac{2}{x}$. But second solution is given be

$$y_2 = y_1 \int \frac{e^{-\int P(x)\,dx}}{y_1^2}\,dx$$

$$\Rightarrow \quad = x^2 \int \frac{e^{-\int \frac{2}{x}\,dx}}{x^4}\,dx$$

$$\Rightarrow \quad = x_2 \int \frac{e^{-2\ln x}}{x^4}\,dx = x^2 \int \frac{e^{\ln x^{-2}}}{x^4}\,dx$$

$$\Rightarrow \quad = x^2 \int \frac{x^{-2}}{x^4}\,dx = x_2 \int \frac{1}{x^6}\,dx$$

$$\Rightarrow \quad = x^2 \left(\frac{x^{-5}}{-5}\right)$$

$$\Rightarrow \quad = -\frac{1}{5x^3}$$

Note: This question can also solve using other method.

Question No: 5 **Marks:2**

The differential operator that annihilates $4e^{x/2}$ is:

$2D-1$

Question No: 6 **Marks:10**

Solve the following differential equations.

$$\left(1+\ln x+\frac{y}{x}\right)dx=(1-\ln x)\,dy$$

Solution:

Here

$$M=1+\ln x+\frac{y}{x},\quad N=-(1-\ln x)$$

$$\Rightarrow \quad M_y=\frac{\delta M}{\delta y}=\frac{1}{x},\quad N_x=\frac{\delta N}{\delta x}=\frac{1}{x}$$

$$\Rightarrow \quad M_y=N_x$$

So the given equation is an exact equation. Thus there exists a function $f(x,y)$ such that

$$\frac{\delta f}{\delta x}=M \qquad and \qquad \frac{\delta f}{\delta y}=N$$

$$\Rightarrow \quad \frac{\delta f}{\delta x}=1+\ln x+\frac{y}{x}\quad --(1) \quad and \quad \frac{\delta f}{\delta y}=\ln x-1\quad --(2)$$

$$(1)\Rightarrow f=x+x\ln x-x+y\ln x+H(y)=x\ln x+y\ln x+H(y)$$

$$\Rightarrow \quad \frac{\delta f}{\delta y}=\ln x+H'(y)$$

$$(2)\Rightarrow \ln x-1=\ln x+H'(y)$$

$$\Rightarrow \quad -1=H'(y)$$

$$\Rightarrow \quad H(y)=-y$$

$$Hence \quad f(x,y)=x\ln x+y\ln x-y$$

Question No: 7 **Marks:10**

Initially there were 100 milligrams of a radioactive substance present. After 6 hours the mass decreased by 3%. If the rate of decay is proportional to the amount of the substance present at any time, find the amount remaining after 24 hours.

Solution:

Let $A(t)$ be amount present at any time t. Then by given conditions, we have

$$\frac{dA}{dt} \propto A$$

$$\Rightarrow \quad \frac{dA}{dt} = kA$$

Initially there were 100 milligrams, therefore $A(0) = 100$. Moreover, decreased by 3% will give us $100 - \frac{3}{100}(100) = 97$ milligrams after 6 hours i.e. $A(6) = 97$. So we have boundary value problem

$$\frac{dA}{dt} = kA \text{ subject to boundary conditions } A(0) = 100,\ A(6) = 97$$

The solution of this equation is given by

$$A(t) = Ce^{kt} \text{ where C is constant of integration.}$$

Applying boundary conditions, we get

$$C = 100, \quad k = -0.005076$$

$$\Rightarrow \quad A(t) = 100e^{-0.005076t}$$

Amount remaining after 24 hours is obtained by putting $t = 24$ in above equation i.e.

$$\Rightarrow \quad A(t) = 100e^{-0.005076(24)}$$

$$= 88.529 \ mg.$$

Question No: 8 **Marks:10**

Find a second solution of following differential equations where the first solution is given. You can use any method (reduction of order or formula given in handouts).

$x^2 y'' + y' = 0;$ $\quad y_1 = \ln x$

Solution:

Comparing this equation with $y'' + P(x)y' + Q(x)y = 0$, we get

$$P(x) = \frac{1}{x^2}.$$

But second solution is given by

$$y_2 = y_1 \int \frac{e^{-\int P(x)\,dx}}{y_1^2}\,dx$$

$$\Rightarrow \quad y_2 = \ln x \int \frac{e^{-\int \frac{1}{x^2}\,dx}}{(\ln x)}\,dx$$

$$= \ln x \int \frac{e^{\frac{2}{\frac{1}{x}}}}{(\ln x)^2}\,dx$$

This is the answer.

Note: This question can also solve using second method.

Question No: 5 **Marks:2**

The differential operator that annihilates $4e^{2x}$ is:

$$(D-2)(D+5)$$

Question No: 6 **Marks:10**

Find the general solution of the given differential equation.

$$\frac{dy}{dx} + 2xy = x^3$$

Solution:

It is of the form $\frac{dy}{dx} + P(x)\,y = Q(x)$ i.e. Linear First Order Differential Equation with

$$P(x) = 2x, \quad Q(x) = x_3$$

Thus integration factor is given by

$$I.F = u(x) = e^{\int P(x)dx}$$

$$\Rightarrow \quad = e^{\int 2xdx} = e_{x_2}$$

But the solution in this case is

$$y = \frac{\int u(x)\,Q(x)\,dx + C}{u(x)} \text{ ----------(1.}$$

)

Now –

$$\int u(x)\,Q(x)\,dx = \int x_3 e_x$$

$$= \frac{1}{2}\int \left(e^{x_2} 2x\right) x^2 dx$$

$$= \frac{1}{2}\left\{e^{x_2}x^2 - \int e^{x_2} 2xdx\right\} \quad \text{int } egration\ by\ parts$$

$$\frac{1}{2}\{$$

$$-\frac{1}{2}\left\{x^2 - 1\right\} e^{x_2}$$

$$y = \frac{+ C}{e^{x^2}}$$

Question No: 7 **Marks:10**

A thermometer is taken from an inside room to the outside where the air temperature is $5^\circ F$. After 1 minute the thermometer reads $55^\circ F$, and after 5 minutes the reading is $30^\circ F$. What is the initial temperature of the room?

Solution:

Let $T(t)$ be temperature at any time t and T_0 be the temperature of the surroundings. Then by Newton's Method, we know that

$$\frac{dT}{dt} = k\left(T - T_0\right)$$

Where k is constant of proportionality. Here we are given $T_0 = 5$ and $T(1) = 55, T(5) = 30$. Solving above equation we get

$$T = T_0 + Ce^{kt}$$

$$\Rightarrow \quad T = 5 + Ce_{kt}$$

Using above conditions we get

$$k = -0.173\,,\ C \approx 59.44\,.$$

So the initial temperature is given by

$$T = 5 + Ce0$$
$$= 5 + C$$

$$\approx 5 + 59.44 = 64.44_{\circ} \quad F\,.$$

Question No: 8 **Marks:10**

Find a second solution of following differential equations where the first solution is given. You can use any method (reduction of order or formula given in handouts).

$$4x^2 y'' + y = 0;\ y_1 = x^{1/2} \ln x$$

Solution:

Comparing this equation with $y'' + P(x)\,y' + Q(x)\,y = 0$, we get

$$P(x) = 0.$$

As second solution is given by

$$y_2 = y_1 \int \frac{e^{-\int P(x)\,dx}}{y_1^2}\,dx$$

Using given conditions, we get

$$y_2 = x^{1/2} \ln x \int \frac{e^0}{\left(x^{1/2} \ln x\right)^2}\,dx$$

$$= x^{1/2} \ln x \int \frac{1}{x (\ln x)^2}\,dx$$

$$= x^{1/2} \ln x \int \frac{1}{(\ln x)^2}\, x\,dx$$

$$= x^{1/2} \ln x \left| \frac{(\ln x)^{-1}}{-1} \right|$$

$$= -x_{1/2}$$

MTH401 Deferential Equations

Mid Term Examination – Spring 2006

Time Allowed: 90 Minutes

1. The duration of this examination is 90 minutes.
2. Symbols by using math type should be pasted on the paper direct from the math type not from the word document otherwise it would not be visible.
3. Do not ask any questions about the contents of this examination from anyone.

   a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.

   b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

   c. Write all steps, missing steps may lead to deduction of marks.
4. This examination is closed book and closed notes.
5. Use of Calculator is allowed.
6. Attempt all questions. Marks are written adjacent to each question.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will be awarded grade `F` in this course.**

Question No. 1	**Marks : 1**

The method of undetermined coefficient is limited to homogeneous linear differential equation

- True
- False

Question No. 2 **Marks : 1**

In the homogeneous differential equation after substitution v=y/x the equation reduces to.

- Separable differential equation.
- Exact differential equation.
- Remain homogeneous equation.
- None of the other

Question No. 3 **Marks : 10**

Solve the differential equation by the variation of parameters

$$y'' - 9y' + 9y = xe^{3x}$$

If complimentary solution is given below

$$y_c = c_1 e^{3x} + c_2 x e^{3x}$$

Then just find the particular solution.

Question No. 4 **Marks : 5**

Determine whether the functions are linearly independent or dependent on $(-\infty, \infty)$

$f_1(x) = \quad f_2(x) = x, f_3(x) = e$
$0,$

Question No. 5 **Marks : 10**

Solve

$$\frac{dy}{dx} + xy = xy^2$$

Question No. 6 **Marks : 10**

Solve the differential equation by integrating factor technique

$$y^2 dx + xydy = 0$$

Question No. 7 **Marks : 1**

If the Wronskian W of three function f(x),g(x),h(x) is zero, what can be said about the dependency of the functions

May or may not be dependent
Always dependent
Never dependent
None of the other

Question No. 8 **Marks : 1**

If $a_n(x) = 0$ in the differential equation

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + a_{n-2}(x)\frac{d^{n-2} y}{dx^{n-2}} + ... + a_1(x)\frac{dy}{dx} + a_0(x)\,y = g(x)$$

for some $x \in I$ then

I. Solution of initial value problem may not unique.
II. Solution of initial value problem may not even exist.
III. Solution of initial value problem should exist.
IV. Solution of initial value problem is unique.

I is correct only
I and II are correct
I and III are correct
IV is correct only

Question No. 9 **Marks : 1**

Equation of the form $\frac{dy}{dx} + y = x^2 y^2$ is called

First order linear differential equation
Bernoulli equation
Separable equation
None of the other.

	FINALTERM EXAMINATION FALL 2006 MTH401 - DIFFERENTIAL EQUATIONS (Session - 1)	Marks: 53 Time: 120min

StudentID/LoginID: _____

Student Name: _____

Center Name/Code: _____

Exam Date: Monday, February 12, 2007

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.

2. Do not ask any question about the contents of this examination from anyone.

a. If you think that there is something wrong with any of the question, attempt it to the best of your understanding.

b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

c. Write all steps, missing steps may lead to deduction of marks.

3. Calculator is allowed.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

For Teacher's use only

Question	1	2	3	4	5	6	7	8	9	10	Total
Marks											
Question	11										
Marks											

Question No: 1 (Marks: 2) - Please choose one

If the variation of the path of the curves can be describrd by the concept of differential equations then which of the following differential equation describe the path for $y-axis$.

- $\frac{dy}{dx} = 1$
- $\frac{dy}{dx} = 0$
- $\frac{dy}{dx} = -1$
- $\frac{dy}{dx} = \infty$

Question No: 2 (Marks: 2) - Please choose one

Suggestive form of the constant input function for the non homogeneous differential equation under the method entitled as "**Method of the undetermined coefficient**" is

- $f(x) = e_x$
- $f(x) = a$
- $f(x) = e_{ax}\left(ACosx + BSinx \right)$
- Suggestive form is impossible.

Question No: 3 (Marks: 2) - Please choose one

Which of the following function is linearly dependant to the exponential function e^x ?

- $-e_x$
- e_{-x}
- xe^x
- $-xe_{-x}$

Question No: 4 (Marks: 2) - Please choose one

Eigen values for the system of the differential equations $X' = AX$ are evaluated for the

- Solution vector X
- Coefficient matrix A
- Differentiated solution vector X'
- Transpose of the Coefficient matrix A

Question No: 5 (Marks: 2) - Please choose one

$$X_1, X_2, \cdots, X_n$$

Fundamental set of the solution vectors for any system of the differential equations are obtained by

- $\{X\} = \{c_1 X_1, c_2 X_2, \ldots\ldots, c_n X_n\}$

  Developing the singleton set of the linear combinitions of the solution vectors.

- Taking derivative of the each solution vector and forming the set $\lceil X', X', \cdots, X' \rceil$

- Taking Integral of the each solution vector and forming the set

$$\left\{\int X_1 dx, \int X_2 dx, \cdots, \int X_n dx\right\}$$

- Just verifying their linear independance and establishing the set

$$\{X_1, X_2, \cdots, X_n\}$$

.

Question No: 6 (Marks: 5)

For the family with parameter '**a**', of rectangular hyperbola $y = \frac{a}{x}$, find its corresponding orthogonal trajectorie and induce the parameter of the new family.

Question No: 7 (Marks: 5)

(a) Justify whether $D^2 + 9$ annihilates the function $f(x) = \sin 3x$ **(3)**

(b) Evaluate Wronskian of functions $Sin|x|$ and $|x|$ for $x < 0$ provided that $|x| = -x$ for $x < 0$ **(2)**

Question No: 8 (Marks: 8)

$$\sum_{k=0}^{\infty} \frac{2^k}{k} x^k$$

(a) Determine the interval of convergence of power series **(5)**

$$\sum_{n=0}^{\infty} nc_n x \quad and \quad \sum_{n=1}^{\infty} nc_{n-1} x^{n+1}$$

(b) If are power series solutions for a differential equation then find their sum

by introducing a single summation symbol

Question No: 9 (Marks: 10)

(a) Develop a recurrence relation for the differential equation $y' - xy = 0$ by applying power series method. **(8)**

(b) Discuss shortly the linear independence of the power series solutions

$$y_1 = A\sqrt{x}\left[2 + \sum_{k=1}^{\infty} \frac{x^k}{(k+1)!5.8.11.....(3k+2)}\right]$$

and

$$y_2 = Bx\left[3 + \sum_{k=1}^{\infty} \frac{x^k}{k!1.4.7.....(3k-2)}\right]$$

Question No: 10 (Marks: 10)

$$y'' - \frac{1}{x}y' + \frac{1}{(x-1)^3}y = 0$$

(a) Determine the singular points of differential equation .Also classify each singular point as regular or irregular.

(5)

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}\left(x^2-1\right)^n$$

(b) Using Rodrigues Formula; to generate fourth Legendre's polynomial $P_3(x)$. **(5)**

Question No: 11 (Marks: 5)

$$\frac{dx}{dt} = 3x - 4y$$
$$\frac{dy}{dt} = 4x - 7y$$

$$X = \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-5t}$$

Verify that the vector is a solution of the system **;** .

	MIDTERM EXAMINATION SPRING 2007 MTH401 - DIFFERENTIAL EQUATIONS (Session - 4)	Marks: 40 Time: 90min

StudentID/LoginID: _____

Student Name: _____

Center Name/Code: _____

Exam Date: Friday, May 18, 2007

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.

2. Do not ask any questions about the contents of this examination from anyone.

a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.

b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

c. Write all steps, missing steps may lead to deduction of marks.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

For Teacher's use only

Question	1	2	3	4	5	6	7	8	9		Total
Marks											

Question No: 1 (Marks: 1) - Please choose one

The differential equation

$(3x^2 y + 2)dx + (x^3 \quad y)dy = 0$ is
+

- Exact
- Linear
- Homogenous
- Separable

Question No: 2 (Marks: 1) - Please choose one

The assumed particular solution for the U.C(Undetermined Coefficient) differential equation $y' - y = x^2 e^{2x}$ is

- $y_p = c_1 e^{x^2} + c_2 x^2$
- $y_p = (Ax + B)e^{2x}$
- $y_{x_p} = (Ax^2 + Bx + c)e^2$
- None of these.

Question No: 3 (Marks: 1) - Please choose one

The differential equation $x\frac{dy}{dx} + y = y^2 \ln x$ is an example of

- Separable
- Homogenous
- Exact
- None of these.

Question No: 4 (Marks: 1) - Please choose one

For the differential equation
$y' - 2xy = x$
Integrating factor is

- $-x^2$
- e^{x^2}

- e^{-x^2}
- x^2

Question No: 5 (Marks: 1) - Please choose one

$$\frac{dy}{dx} = \frac{x + 3y - 5}{x - y - 1}$$

Identify the ordinary differential equation

- Homogenous
- Separable
- Exact
- None of these.

Question No: 6 (Marks: 5)

Solve the differential equation

$$\frac{dy}{dx} + \sqrt{\frac{1 - y^2}{1 - x^2}} = 0$$

Question No: 7 (Marks: 10)

Solve

$$(y\,Sec^2 x + Sec x \tan x)\,dx + (\tan x + 2y)\,dy = 0$$

Question No: 8 (Marks: 10)

Find the equation of orthogonal trajectories of the curve

$$x^2 + y^2 = cx$$

Question No: 9 (Marks: 10)

Solve the differential equation by method of variations of parameters

$$\frac{d^2 y}{dx^2} + y = \tan x \sec x$$


VIRTUAL
Inspire!
THE BEST
LEADERS. TEAMS. ORGANISATIONS
www.virtualinspire.com